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# Class 12 Physics Chapterwise PYQs

**2026 – 2003 | All CBSE Board Papers**

*Chapterwise previous year questions, sorted by marks and year*

## Chapter 10: Wave Optics

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HOW TO USE THIS BOOKLET

1. **Attempt a section** on your own. Questions run newest year first.
2. **Check the Answer Key,** then follow a question number to its solution.
3. **Study the solution.** Note each *Common Mistake,* then return via "Back to Q n".

SECTION A

# 1-Mark Questions

*Multiple choice · 64 questions*

> *Attempt these first, then check the Answer Key and open any solution.*

**1.** Four independent waves are expressed as:
(i) $y_1 = A_1 \sin \omega t$ (ii) $y_2 = A_2 \sin 2\omega t$
(iii) $y_3 = A_3 \cos \omega t$ (iv) $y_4 = A_4 \sin (\omega t + \pi/3)$
The interference between two of these waves is possible in:

**(A)** (i) and (iii) only

**(B)** (iii) and (iv) only

**(C)** (i), (iii) and (iv) only

**(D)** All of them

Check Solution → [2026 • Set 55-1-1]

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**2.** The shape of the interference fringes in Young's double-slit experiment, when the distance between the slit and the screen is very large as compared to the slit-separation, is nearly:

**(A)** straight

**(B)** parabolic

**(C)** circular

**(D)** hyperbolic

Check Solution → [2026 • Set 55-1-1]

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**3.** The angular width of interference fringes in Young's double-slit experiment depends on:

**(A)** distance between the slits and the screen only

**(B)** wavelength of light used only

**(C)** both wavelength of light used and the slits separation

**(D)** slits separation only

Check Solution → [2026 • Set 55-1-2]

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**4.** The phenomenon of interference is shown by:

**(A)** longitudinal mechanical wave only

**(B)** transverse mechanical wave only

**(C)** electromagnetic waves only

**(D)** all these waves

Check Solution → [2026 • Set 55-1-3]

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**5.** **Assertion (A):** Light added to light can produce darkness.
**Reason (R):** When two coherent light waves interfere, there is darkness at position of destructive interference.

**(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

**(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

**(C)** Assertion (A) is true, but Reason (R) is false.

**(D)** Assertion (A) is false and Reason (R) is also false.

Check Solution → [2026 • Set 55-2-1]

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**6.** The phase difference between the two superimposing waves that give rise to a bright spot in a Young's double-slit experiment is ($n$ is an integer):

**(A)** $2n\pi$

**(B)** $2n\pi + \frac{\pi}{2}$

**(C)** $2n\pi + \frac{\pi}{4}$

**(D)** $2n\pi + \pi$

Check Solution → [2026 • Set 55-3-1]

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7. **Assertion (A):** In Young's double-slit experiment, the fringe width for dark and bright fringes is the same.

**Reason (R):** Fringe width is given by $\beta = \frac{\lambda D}{d}$, where symbols have their usual meanings.

Select the correct answer from the codes given below:

**(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

**(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

**(C)** Assertion (A) is true, but Reason (R) is false.

**(D)** Both Assertion (A) and Reason (R) are false.

Check Solution → **[2026 • Set 55-5-1]**

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8. What property of light does this interference experiment demonstrate?

**(A)** Wave nature of light

**(B)** Particle nature of light

**(C)** Transverse nature of light

**(D)** Both wave nature and transverse nature of light

Check Solution → **[2026 • Set 55-5-1]**

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9. (a) The wavelength of light used in this experiment is:

**(A)** 720 nm

**(B)** 590 nm

**(C)** 480 nm

**(D)** 364 nm

Check Solution → **[2026 • Set 55-5-1]**

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10. (b) The fringe width in the interference pattern formed on the screen is:

(A) 1.2 mm

(B) 0.2 mm

(C) 4.2 mm

(D) 6.8 mm

Check Solution → [2026 • Set 55-5-1]

---

**11.** The path difference between the two waves meeting at point P, where there is a minimum in the interference pattern is:

(A) $8.1 \times 10^{-7}\ m$

(B) $7.2 \times 10^{-7}\ m$

(C) $6.5 \times 10^{-7}\ m$

(D) $6.0 \times 10^{-7}\ m$

Check Solution → [2026 • Set 55-5-1]

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**12.** When the experiment is performed in a liquid of refractive index greater than 1, then fringe pattern will:

(A) disappear

(B) become blurred

(C) be widened

(D) be compressed

Check Solution → [2026 • Set 55-5-1]

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**13.** **Assertion (A):** In double slit experiment if one slit is closed, diffraction pattern due to the other slit will appear on the screen.

**Reason (R):** For interference, at least two waves are required.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-4-1]

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**14.** Two coherent light waves, each having amplitude '*a*', superpose to produce an interference pattern on a screen. The intensity of light as seen on the screen varies between:

(A) 0 and $2a^2$

(B) 0 and $4a^2$

(C) $a^2$ and $2a^2$

(D) $2a^2$ and $4a^2$

Check Solution → [2025 • Set 55-5-1]

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**15.** Two coherent waves, each of intensity $I_0$, produce interference pattern on a screen. The average intensity of light on the screen is:

(A) zero

(B) $I_0$

(C) $2I_0$

(D) $4I_0$

Check Solution → [2025 • Set 55-6-1]

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**16.** What are the coherent sources of light? Can two independent sodium lamps act like coherent sources? Explain.

Check Solution → [2024 • Set 55-1-1]

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**17.** Diffraction and interference are closely related phenomena that occur together. Diffraction is the phenomenon of bending of light around the edges of the obstacle, while interference is the combination of waves that results in a new wave pattern. In order to get interference, there must be at least two waves that are diffracting. So while diffraction can occur without interference, interference cannot occur without diffraction.
Two slits of width 2 μm each in an opaque material are separated by a distance of 6 μm. Monochromatic light of wavelength 450 nm is incident normally on the slits. One finds a combined interference and diffraction pattern on the screen. The number of peaks of the interference fringes formed within the central peak of the envelope of the diffraction pattern will be:

**(A)** 2

**(B)** 3

**(C)** 4

**(D)** 6

Check Solution → **[2024 • Set 55-3-1]**

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**18.** The number of peaks of the interference formed if the slit width is doubled while keeping the distance between the slits same will be:

**(A)** 1

**(B)** 2

**(C)** 3

**(D)** 4

Check Solution → **[2024 • Set 55-3-1]**

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**19.** If instead of 450 nm light, another light of wavelength 680 nm is used, number of peaks of the interference formed in the central peak of the envelope of the diffraction pattern will be:

(A) 2

(B) 4

(C) 6

(D) 9

Check Solution → [2024 • Set 55-3-1]

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**20.** Consider the diffraction of light by a single slit described in this case study. The first minimum falls at an angle θ equal to:

(A) $\sin^{-1}(0.12)$

(B) $\sin^{-1}(0.225)$

(C) $\sin^{-1}(0.32)$

(D) $\sin^{-1}(0.45)$

Check Solution → [2024 • Set 55-3-1]

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**21.** The number of bright fringes formed due to interference on 1 m of screen placed at $\frac{\pi}{3}$ m away from the slits is:

(A) 2

(B) 3

(C) 6

(D) 10

Check Solution → [2024 • Set 55-3-1]

---

**22.** **Assertion (A):** In interference and diffraction of light, light energy reduces in one region producing a dark fringe. It increases in another region and produces a bright fringe.

**Reason (R):** This happens because energy is not conserved in the phenomena of interference and diffraction.

(A) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) If both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A).

(C) If Assertion (A) is true and Reason (R) is false.

(D) If both Assertion (A) and Reason (R) are false.

Check Solution → [2024 • Set 55-4-1]

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**23.** A Young's double-slit experimental set up is kept in a medium of refractive index $(\frac{3}{2})$. Which maximum in this case will coincide with the 6th maximum obtained if the medium is replaced by air?

(A) $4^{th}$

(B) $6^{th}$

(C) $8^{th}$

(D) $10^{th}$

Check Solution → [2024 • Set 55-5-1]

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**24.** **Assertion (A):** In a Young's double-slit experiment, interference pattern is not observed when two coherent sources are infinitely close to each other.
**Reason (R):** The fringe width is proportional to the separation between the two sources.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2024 • Set 55-5-1]

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**25.** In a Young's double-slit experiment in air, the fringe width is found to be 0.44 mm. If the entire setup is immersed in water ($\mu = \frac{4}{3}$), the fringe width will be:

**(A)** 0.88 mm

**(B)** 0.59 mm

**(C)** 0.33 mm

**(D)** 0.44 mm

Check Solution → **[2024 • Set 55-5-2]**

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**26.** **Assertion (A):** The phase difference between any two points on a wavefront is zero.

**Reason (R):** All points on a wavefront are at the same distance from the source and thus oscillate in the same phase.

**(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

**(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

**(C)** Assertion (A) is true, but Reason (R) is false.

**(D)** Assertion (A) is false and Reason (R) is also false.

Check Solution → **[2023 • Set 55-1-1]**

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**27.** **Assertion (A):** In Young's double slit experiment all fringes are of equal width.

**Reason (R):** The fringe width depends upon wavelength of light ($\lambda$) used, distance of screen from plane of slits ($D$) and slits separation ($d$).

**(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

**(B)** Both Assertion (A) and Reason (R) are true and Reason (R) is NOT the correct explanation of Assertion (A).

**(C)** Assertion (A) is true and Reason (R) is false.

**(D)** Assertion (A) is false and Reason (R) is also false.

Check Solution → **[2023 • Set 55-2-1]**

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**28.** A beam of light travels from air into a medium. Its speed and wavelength in the medium are $1.5\times 10^8\ m\ s^{-1}$ and 230 nm respectively. The wavelength of light in air will be

**(A)** 230 nm

**(B)** 345 nm

**(C)** 460 nm

**(D)** 690 nm

Check Solution → [2023 • Set 55-2-1]

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**29.** In the wave picture of light, the intensity $I$ of light is related to the amplitude $A$ of the wave as:

**(A)** $I \propto \sqrt{A}$

**(B)** $I \propto A$

**(C)** $I \propto A^2$

**(D)** $I \propto \frac{1}{A^2}$

Check Solution → [2023 • Set 55-3-1]

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**30.** In a single-slit diffraction experiment, the width of the slit is halved. The width of the central maximum, in the diffraction pattern, will become:

**(A)** half

**(B)** twice

**(C)** four times

**(D)** one-fourth

Check Solution → [2023 • Set 55-3-1]

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**31.** A plane wavefront is incident on a concave mirror of radius of curvature $R$. The radius of the refracted wavefront will be:

(A) 2R

(B) $R$

(C) $\frac{R}{2}$

(D) $\frac{R}{4}$

Check Solution → [2023 • Set 55-4-1]

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**32.** In a Young's double-slit experiment, the fringe width is found to be β. If the entire apparatus is immersed in a liquid of refractive index μ, the new fringe width will be:

(A) β

(B) μβ

(C) $\frac{\beta}{\mu}$

(D) $\frac{\beta}{\mu^2}$

Check Solution → [2023 • Set 55-4-1]

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**33.** According to Huygens principle, the amplitude of secondary wavelets is

(A) equal in both the forward and the backward directions.

(B) maximum in the forward direction and zero in the backward direction.

(C) large in the forward direction and small in the backward direction.

(D) small in the forward direction and large in the backward direction.

Check Solution → [2023 • Set 55-5-1]

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**34.** In a Young's double-slit experiment, the screen is moved away from the plane of the slits. What will be its effect on the following?
(i) Angular separation of the fringes.
(ii) Fringe-width.

(A) Both (i) and (ii) remain constant.

(B) (i) remains constant, but (ii) decreases.

(C) (i) remains constant, but (ii) increases.

(D) Both (i) and (ii) increase.

Check Solution → [2023 • Set 55-5-1]

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**35.** In an interference experiment, third bright fringe is obtained at a point on the screen with a light of 700 nm. What should be the wavelength of the light source in order to obtain the fifth bright fringe at the same point?

(A) 420 nm

(B) 750 nm

(C) 630 nm

(D) 500 nm

Check Solution → [2023 • Set 55-5-3]

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**36.** In Young's double slit experiment, the path difference between two interfering waves at a point on the screen is $\frac{11\lambda}{2}$, $\lambda$ being wavelength of the light used. The _____ dark fringe will lie at this point.

**OR**

If one of the slits in Young's double slit experiment is fully closed, the new pattern has central maximum _____ in angular size.

Check Solution → [2020 • Set 55-2-1]

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**37.** Unpolarised light passes from a rarer into a denser medium. If the reflected and the refracted rays are mutually perpendicular, the reflected light is linearly polarised _____ to the plane of incidence.

Check Solution → [2020 • Set 55-2-1]

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**38.** Out of red, blue and yellow lights, the scattering of _____ light is maximum.

Check Solution → [2020 • Set 55-2-1]

**39.** An unpolarised light of intensity $I_0$ is passed through a polaroid. The intensity of a plane polarised light obtained is _____.

Check Solution → [2020 • Set 55-2-2]

**40.** At sunset or sunrise, the scattered coloured light _____ from the sun reaches our eyes.

Check Solution → [2020 • Set 55-2-3]

**41.** The shape of the wavefront originating from a line source is _____.

Check Solution → [2020 • Set 55-3-1]

**42.** In Young's double slit experiment, the separation between the two slits is halved. The new fringe width will be _____ times its initial value.

Check Solution → [2020 • Set 55-3-1]

**43.** In Young's double slit experiment, when the monochromatic source is replaced by a source of white light, the central fringe becomes _____.

Check Solution → [2020 • Set 55-3-2]

**44.** The phase difference between the two points on the same wavefront is _____.

Check Solution → [2020 • Set 55-3-2]

**45.** In Young's double slit experiment, if the monochromatic source of light is replaced by another monochromatic source of shorter wavelength, the value of fringe width will _____.

Check Solution → [2020 • Set 55-3-3]

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**46.** Fill in the blank with appropriate answer.
Unpolarised light of intensity $I_0$ is incident on two crossed polaroids. The intensity of light transmitted by the combination will be _____.

Check Solution → [2020 • Set 55-4-1]

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**47.** When a wave undergoes reflection at an interface from rarer to denser medium, adhoc change in its phase is:

**(A)** $\frac{\pi}{2}$

**(B)** 0

**(C)** $\pi$

**(D)** $\frac{\pi}{4}$

Check Solution → [2020 • Set 55-4-1]

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**48.** Fill in the blank with appropriate answer.
In a linearly polarised light, the electric vector oscillates along a direction perpendicular to the aligned molecules of the polaroid, called _____ axis.

Check Solution → [2020 • Set 55-4-2]

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**49.** The relationship between Brewster angle $\theta$ and the speed of light $v$ in the denser medium is —

(A) $v\theta = c$

(B) $c\theta = v$

(C) $v\theta = c$

(D) $c\theta = v$

Check Solution → [2020 • Set 55-5-1]

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**50.** Write the conditions on path difference under which (i) constructive (ii) destructive interference occur in Young's double slit experiment.

Check Solution → [2020 • Set 55-5-1]

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**51.** What is the speed of light in a denser medium of polarising angle 30°?

Check Solution → [2019 • Set 55-1-1]

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**52.** When unpolarised light is incident on the interface separating the rarer medium and the denser medium, Brewster angle is found to be 60°. Determine the refractive index of the denser medium.

Check Solution → [2019 • Set 55-1-2]

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**53.** Distinguish between unpolarised and linearly polarised light.

Check Solution → [2019 • Set 55-1-3]

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**54.** Why does sun appear red at sunrise and sunset?

Check Solution → [2016]

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**55.** Why can't we see clearly through fog? Name the phenomenon responsible for it.

Check Solution → [2016]

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**56.** Which of the following waves can be polarized (i) Heat waves (ii) Sound waves? Give reason to support your answer.

Check Solution → [2013]

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**57.** When light travels from a rarer to a denser medium, the speed decreases. Does this decrease in speed imply a decrease in the energy carried by the light wave? Justify your answer.

Check Solution → [2012]

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**58.** How does the angular separation between fringes in single-slit diffraction experiment change when the distance of separation between the slit and screen is doubled?

Check Solution → [2012]

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**59.** When monochromatic light travels from one medium to another its wavelength changes but frequency remains the same. Explain.

Check Solution → [2011 • Set 55-1-1]

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**60.** If the angle between the pass axis of polarizer and the analyser is 45°, write the ratio of the intensities of original light and the transmitted light after passing through the analyser.

Check Solution → [2009]

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**61.** What type of wavefront will emerge from a (i) point source, and (ii) distant light source?

Check Solution → [2009]

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**62.** How would the angular separation of interference fringes in Young's double slit experiment change when the distance between the slits and screen is doubled?

Check Solution → [2009]

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**63.** How does the fringe width of interference fringes change, when the whole apparatus of Young's experiment is kept in a liquid of refractive index 1.3?

Check Solution → [2008]

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**64.** What is the geometrical shape of the wavefront when a plane wave passes through a convex lens?

Check Solution → [2008]

SECTION B

# 2-Mark Questions

*VSA · 45 questions*

> *Section B of the paper, 2 marks each. Sorted newest year first.*

**65.** **(a)** A beam of light consisting of two wavelengths 400 nm and 600 nm is used to illuminate a single slit of width 1 mm. Find the least distance of the point from the central maximum where the dark fringes due to both wavelengths coincide on the screen placed 1.5 m from the slit.

**OR**

**(b)** In a Young's double-slit experimental set-up with slit separation 0.6 mm a beam of light consisting of two wavelengths 440 nm and 660 nm is used to obtain interference pattern on a screen kept 1.5 m in front of the slits. Find the least distance of the point from the central maximum where the bright fringes due to both the wavelengths coincide.

Check Solution → **[2026 • Set 55-1-1]**

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**66.** In a Young's double-slit experiment, a beam of light consisting of two wavelengths 500 nm and 600 nm is used. The interference fringes are observed at a screen placed 1·8 *m* away from the plane of slits (slit separation 0·3 mm). Calculate the least distance from the central maximum where the bright fringes due to both the wavelengths coincide.

Check Solution → **[2026 • Set 55-2-1]**

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**67.** Explain why diffraction of sound is more common in daily experience than that of light.

Check Solution → **[2026 • Set 55-3-1]**

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**68.** In Young's double slit experiment, the central maximum is bright in the interference pattern obtained on a screen. What will happen when:

(a) light waves emitted out of slits $S_1$ and $S_2$ have an initial phase difference of $\pi$ radian?
(b) one of the slits is closed?
Justify your answers.

Check Solution → [2026 • Set 55-3-3]

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**69.** **(a)** State Huygens principle. How did Huygens justify the absence of the backwave on a spherical wavefront?

**OR**

**(b)** In a single-slit diffraction experiment, light of wavelength $\lambda$ illuminates the slit of width 'a'. The diffraction pattern is observed on a screen kept at a distance *D* from the slits.
(i) Depict variation of intensity in the fringe pattern with the angular position of the fringes.
(ii) How is the linear width of central maximum affected when separation between the slit and the screen is decreased?

Check Solution → [2026 • Set 55-4-1]

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**70.** In a Young's double-slit experiment, two waves each of intensity *I* superpose each other and produce an interference pattern. Prove that the resultant intensities at maxima and minima are 4I and zero respectively.

Check Solution → [2026 • Set 55-5-2]

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**71.** When monochromatic light is incident on a surface separating two media, the refracted and reflected light both have the same frequency as the incident frequency but the wavelength of refracted light is different. Explain why.

Check Solution → [2026 • Set 55-5-3]

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**72.** **(a)** In a diffraction experiment, the slit is illuminated by light of wavelength 600 nm. The first minimum of the pattern falls at $\theta = 30°$. Calculate the width of the slit.

**OR**

**(b)** In a Young's double-slit experiment, two light waves, each of intensity $I_0$, interfere at a point, having a path difference $\frac{\lambda}{3}$ on the screen. Find the intensity at this point.

Check Solution → [2025 • Set 55-1-1]

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**73.** In a double-slit experiment, $6^{th}$ dark fringe is observed at a certain point of the screen. A transparent sheet of thickness t and refractive index n is now introduced in the path of one of the two interfering waves to increase its phase by $2\pi(n-1)\ t/\lambda$. The pattern is shifted and $8^{th}$ bright fringe is observed at the same point. Find the relation for thickness t in terms of n and $\lambda$.

Check Solution → [2025 • Set 55-2-1]

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**74.** Show the refraction of light wave at a plane interface using Huygens' principle and prove Snell's law.

Check Solution → [2025 • Set 55-2-2]

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**75.** In a double slit experiment, it is observed that the angular width of one fringe formed on the screen is 0.2°. The wavelength of light used in the experiment is 500 nm. Calculate the separation of the two slits.

Check Solution → [2025 • Set 55-2-3]

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**76.** Find the angle of diffraction (in degrees) for first secondary maximum of the pattern due to diffraction at a single slit. The width of the slit and wavelength of light used are 0.55 mm and 550 nm, respectively.

Check Solution → [2025 • Set 55-4-1]

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**77.** In a double slit experiment, the two slits are 1.5 mm apart. The slits are illuminated by a mixture of lights of wavelengths of 600 nm and 400 nm and the

interference pattern is observed on a screen 1.5 $m$ away from the slits. Find the minimum distance of the point from the central maximum at which bright fringes of the interference patterns of the two wavelengths coincide.

Check Solution → [2025 • Set 55-4-2]

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**78.** In Young's double slit experiment, the screen is moved 30 cm towards the slits. As a consequence, the fringe width of the pattern changes by 0.09 mm. If the slits separation used is 2 mm, calculate the wavelength of light used in the experiment.

Check Solution → [2025 • Set 55-4-3]

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**79.** (a) Find the intensity at a point on the screen in Young's double slit experiment, at which the interfering waves of intensity $I_0$ each, have a path difference of (i) $\frac{\lambda}{4}$ and (ii) $\frac{\lambda}{3}$.

**OR**

(b) A point source of light in air is kept at a distance of 12 cm in front of a convex spherical surface of glass of refractive index 1·5 and radius of curvature 30 cm. Find the nature and position of the image formed.

Check Solution → [2025 • Set 55-6-1]

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**80.** Write two differences in the patterns of double-slit interference experiment and single-slit diffraction experiment.

Check Solution → [2025 • Set 55-7-1]

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**81.** Light waves from two pinholes illuminated by two sodium lamps do not produce interference patterns. Explain why.

Check Solution → [2025 • Set 55-7-1]

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**82.** The ratio of the intensities at maxima to minima in Young's double-slit experiment is 25 : 9. Calculate the ratio of intensities of the interfering waves.

Check Solution → [2025 • Set 55-7-2]

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**83.** In a Young's double-slit experiment, the intensity at the central maximum in the interference pattern on the screen is $I_0$. Find the intensity at a point on the screen where the path difference between the interfering waves is $\frac{\lambda}{3}$.

Check Solution → [2025 • Set 55-7-3]

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**84.**

State Huygens' principle. A plane wave is incident at an angle $i$ on a reflecting surface. Construct the corresponding reflected wavefront. Using this diagram, prove that the angle of reflection is equal to the angle of incidence.

Check Solution → [2024 • Set 55-1-1]

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**85.** A beam of light consisting of a known wavelength 520 nm and an unknown wavelength λ, used in Young's double slit experiment produces two interference patterns such that the fourth bright fringe of unknown wavelength coincides with the fifth bright fringe of known wavelength. Find the value of λ.

Check Solution → [2024 • Set 55-1-1]

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**86.** (a) Two waves, each of amplitude 'a' and frequency 'ω' emanating from two coherent sources of light superpose at a point. If the phase difference between the two waves is ϕ, obtain an expression for the resultant intensity at that point.

**OR**

(b) What is the effect on the interference pattern in Young's double-slit experiment when (i) the source slit is moved closer to the plane of the slits, and (ii) the separation between the two slits is increased? Justify your answers.

Check Solution → [2024 • Set 55-2-1]

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**87.** Write two points of difference between interference and diffraction of light.

Check Solution → [2024 • Set 55-3-2]

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**88.** What is a sustained or stable interference pattern? What are the conditions for obtaining such an interference pattern?

Check Solution → [2024 • Set 55-3-3]

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**89.** Using Huygens' principle, draw a ray diagram showing the propagation of a plane wave refracting at a plane surface separating two media. Also verify Snell's law of refraction.

Check Solution → [2023 • Set 55-1-1]

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**90.** Two coherent monochromatic light beams of intensities $I$ and 4I superpose each other. Find the ratio of maximum and minimum intensities in the resulting beam.

Check Solution → [2023 • Set 55-1-1]

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**91.** What happens to the interference pattern when two coherent sources are
(a) infinitely close, and
(b) far apart from each other?

Check Solution → [2023 • Set 55-2-1]

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**92.** In a Young's double slit experiment, the separation between the two slits is $d$ and distance of the screen from the slits is 1000 $d$. If the first minima falls at a distance $d$ from the central maximum, obtain the relation between $d$ and $\lambda$.

Check Solution → [2023 • Set 55-2-2]

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**93.** (a) State Huygens' principle. How did Huygens explain the absence of the backwave?

**OR**

(b) Use Huygens' principle to show reflection/refraction of a plane wave by (i) concave mirror, and (ii) a convex lens.

Check Solution → [2023 • Set 55-3-1]

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**94.** How would the angular width of central maximum of diffraction pattern be affected when (i) width of the slit is decreased, and (ii) monochromatic light is replaced by polychromatic light? Justify your answers.

Check Solution → [2021]

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**95.** In interference of light, write the expression for the intensity of resultant wave if $I_0$ is the intensity of light wave from each slit. Hence, obtain an expression for the intensity of resultant wave if the two sources are (i) incoherent, and (ii) coherent.

Check Solution → [2021]

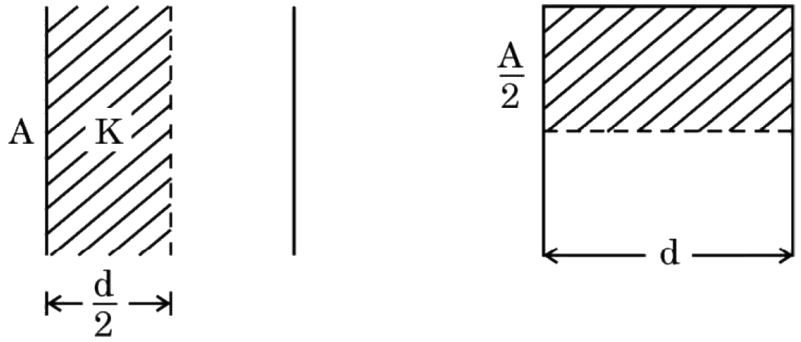
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**96.** Define wavefront of a travelling wave. Using Huygens principle, obtain the law of refraction at a plane interface when light passes from a denser to rarer medium.

Check Solution → [2020 • Set 55-1-1]

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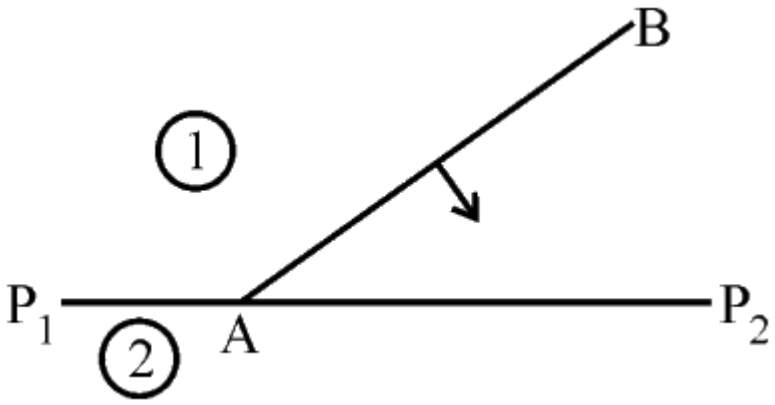
**97.** Light from a sodium lamp (*S*) passes through two polaroid sheets $P_1$ and $P_2$ as shown in fig. What will be the effect on the intensity of the light transmitted (i) by $P_1$ and (ii) by $P_2$ on rotating polaroid $P_2$ about the direction of propagation of light ? Justify your answer in both cases.




**OR**

Define the term 'wave front of light'. A plane wave front AB propagating from denser medium (1) into a rarer medium (2) is incident on the surface $P_1P_2$ separating the two media as shown in fig.

Using Huygen's principle, draw the secondary wavelets and obtain the refracted wave front in the diagram.




Check Solution → [2020 • Set 55-5-1]

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**98.** In a single slit diffraction experiment, the width of the slit is increased. How will the (i) size and (ii) intensity of central bright band be affected ? Justify your answer.

Check Solution → [2020 • Set 55-5-1]

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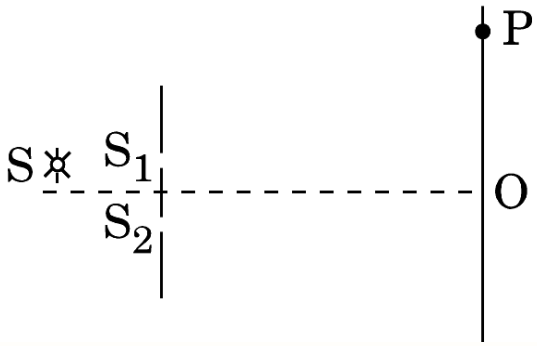
**99.** Explain giving reason, how the resolving power of a compound microscope depends on the

(a) frequency of the incident light.

(b) focal length of the objective lens.

Check Solution → [2019 • Set 55-4-1]

**100.** The figure shows a modified Young's double slit experimental set-up. Here $SS_1 - SS_2 = \lambda/4$.
(a) Write the condition for constructive interference.
(b) Obtain an expression for the fringe width.




Check Solution → [2019 • Set 55-4-1]

**101.** Find the intensity at a point on a screen in Young's double slit experiment where the interfering waves of equal intensity have a path difference of (i) $\lambda/4$, and (ii) $\lambda/3$.

Check Solution → [2017]

**102.** Draw the intensity pattern for single slit diffraction and double slit interference. Hence, state two differences between interference and diffraction patterns.

Check Solution → [2017]

**103.** Unpolarised light is passed through a polaroid $P_1$. When this polarised beam passes through another polaroid $P_2$ and if the pass axis of $P_2$ makes angle $\theta$ with the pass axis of $P_1$, then write the expression for the polarised beam passing through $P_2$. Draw a plot showing the variation of intensity when $\theta$ varies from 0 to $2\pi$.

Check Solution → [2017]

**104.** Draw the intensity pattern for single slit diffraction and double slit interference. Hence, state two differences between interference and diffraction patterns.

**OR**

Unpolarised light is passed through a polaroid $P_1$. When this polarised beam passes through another polaroid $P_2$ and if the pass axis of $P_2$ makes angle $\theta$ with the pass axis of $P_1$, then write the expression for the polarised beam passing through $P_2$. Draw a plot showing the variation of intensity when $\theta$ varies from 0 to $2\pi$.

Check Solution → [2017]

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**105.** Find an expression for intensity of transmitted light when a polaroid sheet is rotated between two crossed polaroids. In which position of the polaroid sheet will the transmitted intensity be maximum?

Check Solution → [2015]

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**106.** For a single slit of width "a", the first minimum of the interference pattern of a monochromatic light of wavelength $\lambda$ occurs at an angle of $\frac{\lambda}{a}$. At the same angle of $\frac{\lambda}{a}$, we get a maximum for two narrow slits separated by a distance "a". Explain.

Check Solution → [2014]

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**107.** Define the term 'linearly polarised light'.

When does the intensity of transmitted light become maximum, when a polaroid sheet is rotated between two crossed polaroids?

Check Solution → [2009]

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**108.** How is a wavefront defined? Using Huygens' construction draw a figure showing the propagation of a plane wave refraction at a plane surface separating two media. Hence verify Snell's law of refraction.

Check Solution → [2008]

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**109.** How will the angular separation and visibility of the fringes in Young's double slit experiment change when (i) Screen is moved away from the plane of the slits, and (ii) width of the source slit is increased?

Check Solution → [2008]

SECTION C

# 3-Mark Questions

*SA · 67 questions*

> *Section C of the paper, 3 marks each. Sorted newest year first.*

**110.** A double slit set-up was initially placed in a tank filled with water and the interference pattern was obtained using a laser light. When water is replaced by a transparent liquid of refractive index $n > n_{water}$, what will be the effect on the following ?

**(a)** Speed, frequency and wavelength of the light of laser beam.

**(b)** The fringe width, shape of interference fringes and shift in the position of central maximum.

Check Solution → **[2025 • Set 55-4-1]**

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**111.** **(a)** Define 'wavefront' of a light wave. A plane wavefront is refracted from a convex lens. Draw the shape of the refracted wavefront.

**(b)** A plane wave travelling in a medium is incident on a plane surface separating this medium from a rarer medium. Draw a diagram to show refraction of the wave. Hence, verify Snell's law.

Check Solution → **[2025 • Set 55-4-2]**

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**112.** **(a)** Write the conditions under which two light waves originating from two coherent sources can interfere each other (i) constructively, and (ii) destructively, in terms of wavelength. Can these be applied for two lights originating from two sodium lamps ? Give reason.

**(b)** Monochromatic light of green colour is used in Young's double slit experiment and an interference pattern is observed on a screen. If the green light is replaced by red monochromatic light of the same intensity, how will the fringe width of interference pattern be affected ? Justify your answer.

Check Solution → **[2025 • Set 55-4-3]**

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**113.** In Young's double slit experiment, the separation between the two slits is 1.0 mm and the screen is 1.0 $m$ away from the slits. A beam of light consisting of two wavelengths 500 nm and 600 nm is used to obtain interference fringes.
Calculate:
**(a)** the distance between the first maxima for the two wavelengths.
**(b)** the least distance from the central maximum, where the bright fringes due to both the wavelengths coincide.

Check Solution → **[2025 • Set 55-5-1]**

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**114.** Using the Huygens' principle, briefly describe reflection of a plane wavefront from a reflecting surface. Hence, prove the laws of reflection.

Check Solution → **[2025 • Set 55-5-2]**

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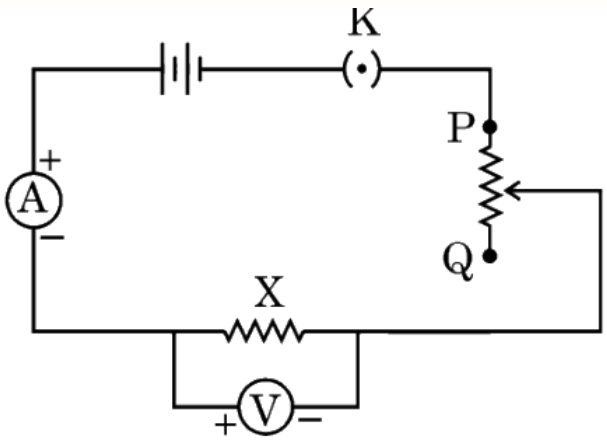
**115.** **(a)** "You cannot see a person standing on the other side of a boundary wall but can hear him." Explain with reason.
**(b)** Light of wavelength 750 nm is incident normally on a slit of width 1.5 mm. Diffraction pattern is obtained on a screen 1.0 $m$ away from the slit. Find the distance of the nearest point from the central maxima at which the intensity is zero.

Check Solution → **[2025 • Set 55-5-3]**

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**116.** (a) (i) In diffraction due to a single slit, the phase difference between light waves reaching a point on the screen is $\frac{5\pi}{2}$. Explain whether a bright or a dark fringe will be formed at the point.
(ii) What should the width ($a$) of each slit be to obtain eight maxima of two double-slit patterns (slit separation $d$) within the central maximum of the single slit pattern?
(iii) Draw the plot of intensity distribution in a diffraction pattern due to a single slit.
**OR**
(b) (i) In a Young's double-slit experiment $SS_1 - SS_2 = \frac{\lambda}{4}$, where $S_1$ and $S_2$ are the

two slits as shown in the figure. Find the path difference ($S_1P - S_2P$) for constructive and destructive interference at P.




(ii) What is the effect on the interference fringes in a Young's double-slit experiment, if the monochromatic source S is replaced by a source of white light?

Check Solution → [2023 • Set 55-4-1]

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**117.** (a) A plane wave-front propagating in a medium of refractive index '$\mu_1$' is incident on a plane surface making an angle of incidence (i). It enters into a medium of refractive index $\mu_2$ ($\mu_2 > \mu_1$). Use Huygen's construction of secondary wavelets to trace the refracted wave-front. Hence verify Snell's law of refraction.

**OR**

(b) Using Huygen's construction, show how a plane wave is reflected from a surface. Hence verify the law of reflection.

Check Solution → [2023 • Set 55-5-1]

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**118.** (i) Monochromatic light is incident on a surface separating two media. The frequency of the light after refraction remains unaffected but its wavelength changes. Why?

(ii) The frequency of an electromagnetic radiation is $1.0 \times 10^{11}$ Hz. Identify the radiation and mention its two uses.

Check Solution → [2022 • Set 55-1-1]

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**119.** A parallel beam of light of wavelength 600 nm is incident normally on a slit of width 0.2 mm. If the resulting diffraction pattern is observed on a screen 1 $m$ away, find the distance of
(i) first minimum, and
(ii) second maximum, from the central maximum.

Check Solution → [2022 • Set 55-1-1]

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**120.** Using Huygen's principle, for a plane wavefront incident on a plane reflecting surface, draw the reflected wavefront. Hence prove laws of reflection.

Check Solution → [2022 • Set 55-1-3]

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**121.** A plane wavefront of light of wavelength 'λ' is incident normally on a narrow slit of width '$a$' and a diffraction pattern is observed on a screen at a distance '$D$' from the slit.
(i) Depict the intensity distribution in the pattern observed.
(ii) Obtain the expression for the first maximum from the central maximum.

Check Solution → [2022 • Set 55-2-1]

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**122.** Using Huygens' principle, draw a diagram to show the refraction of a plane wavefront incident on a denser medium and hence verify Snell's law of refraction.

Check Solution → [2022 • Set 55-2-1]

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**123.** In a Young's double slit experiment using light of wavelength 600 nm, the slit separation is 0.8 mm and the screen is kept 1.6 $m$ from the plane of the slits. Calculate:
(i) the fringe width
(ii) the distance of (a) third minimum and (b) fifth maximum, from the central maximum.

Check Solution → [2022 • Set 55-3-1]

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**124.** (i) State two conditions for two light sources to be coherent.
(ii) Give two points of difference between an interference pattern due to a double-slit and a diffraction pattern due to a single slit.

Check Solution → [2022 • Set 55-3-1]

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**125.** In a diffraction pattern due to a single slit, how will the angular width of central maximum change, if
(i) Orange light is used in place of green light,
(ii) the screen is moved closer to the slit,
(iii) the slit width is decreased?
Justify your answer in each case.

Check Solution → [2022 • Set 55-3-1]

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**126.** How is the spacing between fringes in a double slit experiment affected if:
(i) the slits separation is increased,
(ii) the colour of light used is changed from red to blue,
(iii) the whole apparatus is submerged in a oil of refractive index 1.2?
Justify your answer in each case.

Check Solution → [2022 • Set 55-3-2]

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**127.** Monochromatic light of wavelength 600 nm is incident from air on a water surface. The refractive index of water is 1.33. Find the (i) wavelength, (ii) frequency and (iii) speed, of reflected and refracted light.

Check Solution → [2022 • Set 55-3-2]

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**128.** A beam of light consisting of two wavelengths 600 nm and 500 nm is used in a Young's double slit experiment. The slit separation is 1.0 mm and the screen is kept 0.60 $m$ away from the plane of the slits. Calculate:
(i) the distance of the second bright fringe from the central maximum for wavelength 500 nm, and

(ii) the least distance from the central maximum where the bright fringes due to both the wavelengths coincide.

Check Solution → [2022 • Set 55-3-3]

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**129.** A plane wavefront is propagating from a rarer into a denser medium. Use Huygens principle to show the refracted wavefront and verify Snell's law.

Check Solution → [2022 • Set 55-4-1]

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**130.** Briefly explain how bright and dark fringes are formed on the screen in Young's double slit experiment. Hence, derive the expression for the fringe width.

Check Solution → [2022 • Set 55-4-1]

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**131.** (a) The interference pattern is not observed in Young's double slit experiment when the two sources $S_1$ and $S_2$ are far apart. Explain.
(b) Mention the conditions for the two sources to be coherent.
(c) What is the effect on the interference pattern in a Young's double slit experiment, if the source of wavelength $\lambda$ is replaced by another source of wavelength $1.5\lambda$, with the interference pattern still observable?

Check Solution → [2022 • Set 55-4-2]

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**132.** Briefly explain how bright and dark fringes are formed on a screen due to the diffraction at a single slit. Hence, explain why the intensity at the bright fringes decreases sharply as their order ($n$) increases.

Check Solution → [2022 • Set 55-4-3]

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**133.** How can you differentiate whether a pattern is produced by a single slit or double slits? Derive the expression for the angular position of (i) bright and (ii) dark fringes produced in a single slit diffraction.

Check Solution → [2022 • Set 55-5-1]

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**134.** A slit of width 0.6 mm is illuminated by a beam of light consisting of two wavelengths 600 nm and 480 nm. The diffraction pattern is observed on a screen 1.0 $m$ from the slit. Find:
(i) The distance of the second bright fringe from the central maximum pertaining to light of 600 nm.
(ii) The least distance from the central maximum at which bright fringes due to both the wavelengths coincide.

Check Solution → **[2022 • Set 55-5-1]**

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**135.** How will the interference pattern in Young's double-slit experiment be affected if:
(i) The screen is moved away from the plane of the slits.
(ii) The source slit is moved away from the plane of the slits.
(iii) The phase difference between the light waves emanating from the two slits $S_1$ and $S_2$ changes from 0 to $\pi$ and remains constant.

Check Solution → **[2022 • Set 55-5-2]**

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**136.** Explain the formation of the fringes due to diffraction at a single slit, when path difference of light waves from the ends of the slit on reaching a point on the screen is (i) $\lambda$, and (ii) $\frac{\lambda}{2}$.
Show the intensity distribution in the fringes due to diffraction at a single slit.

Check Solution → **[2021]**

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**137.** What is the effect on the interference fringes in Young's double slit experiment due to each of the following operations? Justify your answers.
(a) The screen is moved away from the plane of the slits.
(b) The separation between slits is increased.
(c) The source slit is moved closer to the plane of double slit.

Check Solution → **[2020 • Set 55-1-1]**

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**138.** In a single slit diffraction experiment, light of wavelength λ illuminates the slit of width $a$ and the diffraction pattern is observed on a screen.
(a) Show the intensity distribution in the pattern with the angular position θ.
(b) How are the intensity and angular width of central maxima affected when
(i) width of slit is increased, and
(ii) separation between slit and screen is decreased?

Check Solution → [2020 • Set 55-1-2]

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**139.** Give reasons for each of the following:
(a) The intensity of light at some points on the screen in Young's double slit experiment is zero.
(b) The intensity of light transmitted by a polaroid is less than the intensity of the unpolarised light incident on it.
(c) In the single slit diffraction experiment, some coloured fringes around the central white maximum are observed on the screen when one uses a source of white light.

Check Solution → [2020 • Set 55-1-3]

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**140.** Two coherent light waves of intensity $510^{-2}\ W\,m^{-2}$ each super-impose and produce the interference pattern on a screen. At a point where the path difference between the waves is λ/6, λ being the wavelength of the wave, find the
(a) phase difference between the waves.
(b) resultant intensity at the point.
(c) resultant intensity in terms of the intensity at the maximum.

Check Solution → [2020 • Set 55-4-1]

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**141.** Define the term wavefront. Using Huygens' wave theory, verify the law of reflection.

Check Solution → [2019 • Set 55-1-1]

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**142.** Define the term, "refractive index" of a medium. Verify Snell's law of refraction when a plane wavefront is propagating from a denser to a rarer medium.

Check Solution → [2019 • Set 55-1-1]

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**143.** (a) Explain how an unpolarised light gets polarised when incident on the interface separating the two transparent media.
(b) Green light is incident at the polarising angle on a certain transparent medium. The angle of refraction is 30°. Find
(i) polarising angle, and
(ii) refractive index of the medium.

Check Solution → [2019 • Set 55-3-1]

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**144.** (a) Define a wavefront. Using Huygens' geometrical construction, explain with the help of a diagram how the plane wavefront travels from the instant $t_1$ to $t_2$ in air.
(b) A plane wavefront is incident on a convex lens. Explain, with the help of the diagram, the shape of the refracted wavefront formed.

Check Solution → [2019 • Set 55-5-1]

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**145.** (a) Good quality sunglasses made of polaroids are preferred over ordinary coloured glasses. Explain why.
(b) How is plane polarized light defined?
(c) A beam of plane polarised light is passed through a polaroid. Show graphically, variation of the intensity of the transmitted light with angle of rotation of the polaroid.

Check Solution → [2019 • Set 55-5-1]

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**146.** (a) What do you understand by coherent sources?
(b) Derive the expression for the intensity distribution in the interference pattern on the screen when two harmonic waves having same amplitude ($a$) and

phase difference ($\phi$) are superposed on each other. Hence obtain the conditions for constructive and destructive interference.

Check Solution → [2019 • Set 55-5-2]

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**147.** Answer the following questions:

(a) In a single slit diffraction experiment, the width of the slit is made double the original width. How does this affect the size and intensity of the central diffraction band?

(b) How is the width of the central maximum changed when red light is replaced by blue?

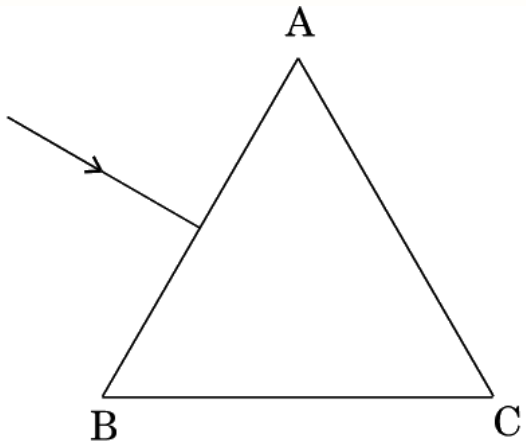
(c) In what way is diffraction from each slit related to the interference pattern in a double slit experiment?

Check Solution → [2019 • Set 55-5-3]

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**148.** (a) Show using a proper diagram how unpolarised light can be linearly polarised by reflection from a transparent glass surface.

(b) The figure shows a ray of light falling normally on the face AB of an equilateral glass prism having refractive index $\frac{3}{2}$, placed in water of refractive index $\frac{4}{3}$. Will this ray suffer total internal reflection on striking the face AC ? Justify your answer.




Check Solution → [2018]

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**149.** (a) If one of two identical slits producing interference in Young's experiment is covered with glass, so that the light intensity passing through it is reduced to 50%, find the ratio of the maximum and minimum intensity of the fringe in the interference pattern.
(b) What kind of fringes do you expect to observe if white light is used instead of monochromatic light ?

Check Solution → [2018]

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**150.** A monochromatic light of wavelength λ is incident normally on a narrow slit of width '*a*' to produce a diffraction pattern on the screen placed at a distance *D* from the slit. With the help of a relevant diagram, deduce the conditions for obtaining maxima and minima on the screen. Use these conditions to show that angular width of central maximum is twice the angular width of secondary maximum.

Check Solution → [2017]

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**151.** Draw the intensity distributions for (i) the fringes produced in interference, and (ii) the diffraction bands produced due to single slit. Write two points of difference between the phenomena of interference and diffraction.

Check Solution → [2017]

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**152.** (i) State law of Malus.
(ii) Draw a graph showing the variation of intensity (*I*) of polarised light transmitted by an analyser with angle (θ) between polariser and analyser.
(iii) What is the value of refractive index of a medium of polarising angle 60°?

Check Solution → [2016]

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**153.** Define the term wave front. State Huygen's principle.
Consider a plane wave front incident on a thin convex lens. Draw a proper diagram to show how the incident wave front traverses through the lens and after refraction focusses on the focal point of the lens, giving the shape of the

emergent wave front.

**OR**

Explain the following, giving reasons:

(i) When monochromatic light is incident on a surface separating two media, the reflected and refracted light both have the same frequency as the incident frequency.

(ii) When light travels from a rarer to a denser medium, the speed decreases. Does this decrease in speed imply a reduction in the energy carried by the wave?

(iii) In the wave picture of light, intensity of light is determined by the square of the amplitude of the wave. What determines the intensity in the photon picture of light?

Check Solution → [2016]

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**154.** (a) In Young's double slit experiment, two slits are 1 mm apart and the screen is placed 1 m away from the slits. Calculate the fringe width when light of wavelength 500 nm is used.

(b) What should be the width of each slit in order to obtain 10 maxima of the double slits pattern within the central maximum of the single slit pattern?

Check Solution → [2016]

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**155.** (i) Derive Snell's law on the basis of Huygen's wave theory when light is travelling from a denser to a rarer medium.

(ii) Draw the sketches to differentiate between plane wavefront and spherical wavefront.

Check Solution → [2016]

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**156.** Two harmonic waves of monochromatic light $y_1 = a\omega t$ and $y_2 = a\cos(\omega t + \phi)$ are superimposed on each other. Show that maximum intensity in interference pattern is four times the intensity due to each slit. Hence write the conditions for constructive and destructive interference in terms of the phase angle $\phi$.

Check Solution → [2016]

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**157.** (a) Write the necessary conditions to obtain sustained interference fringes.
(b) In Young's double slit experiment, plot a graph showing the variation of fringe width versus the distance of the screen from the plane of the slits keeping other parameters same. What information can one obtain from the slope of the curve?
(c) What is the effect on the fringe width if the distance between the slits is reduced keeping other parameters same?

Check Solution → [2015]

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**158.** Answer the following questions:
(a) In a double slit experiment using light of wavelength 600 nm, the angular width of the fringe formed on a distant screen is 0.1°. Find the spacing between the two slits.
(b) Light of wavelength 5000 propagating in air gets partly reflected from the surface of water. How will the wavelengths and frequencies of the reflected and refracted light be affected?

Check Solution → [2015]

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**159.** (a) Show, giving a suitable diagram, how unpolarized light can be polarised by reflection.
(b) Two polaroids $P_1$ and $P_2$ are placed with their pass axes perpendicular to each other. Unpolarised light of intensity $I_0$ is incident on $P_1$. A third polaroid $P_3$ is kept in between $P_1$ and $P_2$ such that its pass axis makes an angle of 60° with that of $P_1$. Determine the intensity of light transmitted through $P_1$, $P_3$ and $P_2$.

Check Solution → [2014]

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**160.** (a) In what way is diffraction from each slit related to the interference pattern in a double slit experiment?
(b) Two wavelengths of sodium light 590 nm and 596 nm are used, in turn, to study the diffraction taking place at a single slit of aperture $2\times 10^{-4}\ m$. The distance between the slit and the screen is 1.5 $m$. Calculate the separation

between the positions of the first maxima of the diffraction pattern obtained in the two cases.

Check Solution → [2013]

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**161.** In Young's double slit experiment, the two slits 0.15 mm apart are illuminated by monochromatic light of wavelength 450 nm. The screen is 1.0 $m$ away from the slits.
(a) Find the distance of the second (i) bright fringe, (ii) dark fringe from the central maximum.
(b) How will the fringe pattern change if the screen is moved away from the slits?

Check Solution → [2012]

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**162.** How does an unpolarised light get polarised when passed through a polaroid? Two polaroids are set in crossed positions. A third polaroid is placed between the two making an angle θ with the pass axis of the first polaroid. Write the expression for the intensity of light transmitted from the second polaroid. In what orientations will the transmitted intensity be (i) minimum and (ii) maximum?

Check Solution → [2012]

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**163.** (a) Why are coherent sources necessary to produce a sustained interference pattern?
(b) In Young's double slit experiment using monochromatic light of wavelength λ, the intensity of light at a point on the screen where path difference is λ, is $K$ units. Find out the intensity of light at a point where path difference is λ/3.

Check Solution → [2012]

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**164.** Define a wavefront. Use Huygens' geometrical construction to show the propagation of a plane wavefront from a rarer medium (1) to a denser medium (2) undergoing refraction.

Hence derive Snell's law of refraction.

**OR**

(a) Use Huygens' geometrical construction to show the behaviour of a plane wavefront

(i) passing through a biconvex lens;

(ii) reflecting by a concave mirror.

(b) When monochromatic light is incident on a surface separating two media, why does the refracted light have the same frequency as that of the incident light?

Check Solution → [2012]

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**165.** Describe Young's double slit experiment to produce interference pattern due to a monochromatic source of light. Deduce the expression for the fringe width.

Check Solution → [2011 • Set 55-1-1]

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**166.** Use Huygen's principle to verify the laws of refraction.

Check Solution → [2011 • Set 55-1-1]

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**167.** (a) Describe briefly, with the help of suitable diagram, how the transverse nature of light can be demonstrated by the phenomenon of polarization.

(b) When unpolarized light passes from air to a transparent medium, under what condition does the reflected light get polarized?

Check Solution → [2011 • Set 55-1-1]

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**168.** The intensity at the central maxima (O) in a Young's double slit experiment is $I_0$. If the distance OP equals one-third of the fringe width of the pattern, show that the intensity at point P would be $\frac{I_0}{4}$.

Check Solution → [2011 • Set 55-2-1]

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**169.** In the experiment on diffraction due to a single slit, show that
(i) the intensity of diffraction fringes decreases as the order (n) increases.
(ii) angular width of the central maximum is twice that of the first order secondary maximum.

Check Solution → [2011 • Set 55-2-1]

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**170.** (i) Light passes through two polaroids $P_1$ and $P_2$ with pass axis of $P_2$ making an angle $\theta$ with the pass axis of $P_1$. For what value of $\theta$ is the intensity of emergent light zero?
(ii) A third polaroid is placed between $P_1$ and $P_2$ with its pass axis making an angle $\beta$ with the pass axis of $P_1$. Find a value of $\beta$ for which the intensity of light emerging from $P_2$ is $\frac{I_0}{8}$, where $I_0$ is the intensity of light on the polaroid $P_1$.

Check Solution → [2011 • Set 55-2-1]

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**171.** What is an unpolarized light? Explain with the help of suitable ray diagram how an unpolarized light can be polarized by reflection from a transparent medium. Write the expression for Brewster angle in terms of the refractive index of denser medium.

Check Solution → [2010]

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**172.** In a single slit diffraction experiment, when a tiny circular obstacle is placed in the path of light from a distant source, a bright spot is seen at the centre of the shadow of the obstacle. Explain why?
State two points of difference between the interference pattern obtained in Young's double slit experiment and the diffraction pattern due to a single slit.

Check Solution → [2009]

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**173.** In Young's double slit experiment, monochromatic light of wavelength 630 nm illuminates the pair of slits and produces an interference pattern in which two consecutive bright fringes are separated by 8.1 mm. Another source of monochromatic light produces the interference pattern in which the two

consecutive bright fringes are separated by 7.2 mm. Find the wavelength of light from the second source.
What is the effect on the interference fringes if the monochromatic source is replaced by a source of white light?

Check Solution → [2009]

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**174.** State two conditions to obtain sustained interference of light.
In Young's double slit experiment, using light of wavelength 400 nm, interference fringes of width $X$ are obtained. The wavelength of light is increased to 600 nm and the separation between the slits is halved. If one wants the observed fringe width on the screen to be the same in the two cases, find the ratio of the distance between the screen and the plane of the interfering sources in the two arrangements.

Check Solution → [2004]

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**175.** State Huygens' principle. For reflection of a plane wave front at a plane reflecting surface, construct the corresponding reflected wave front. Using this diagram, prove that angle of incidence is equal to angle of reflection.

Check Solution → [2003]

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**176.** What is meant by interference of light?
In a double slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the slits. If the screen is moved by $5 \times 10^{-2}$ $m$ towards the slits, the change in fringe width is $3 \times 10^{-5}$ $m$. If the distance between slits is $10^{-3}$ $m$, calculate the wavelength of light used.

Check Solution → [2003]

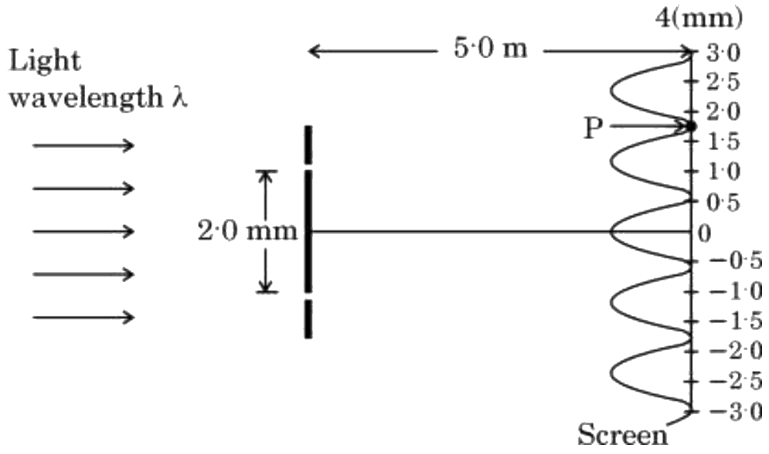
SECTION D

# 4-Mark Questions

*Case Study · 2 questions*

> *Section D of the paper, 4 marks each. Sorted newest year first.*

**177.** In a Young's double-slit experiment, the two slits behave as coherent sources. When coherent light waves superpose over each other they create an interference pattern of successive bright and dark regions due to constructive and destructive interference.
Two slits 2 mm apart are illuminated by a source of monochromatic light and the interference pattern is observed on a screen 5.0 *m* away from the slits as shown in the figure.




Check Solution → [2026 • Set 55-5-1]

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**178.** Diffraction of light is bending of light around the corners of an object whose size is comparable with the wavelength of light. Diffraction actually defines the limits of ray optics. This limit for optical instruments is set by the wavelength of light. An experimental arrangement is set up to observe the diffraction pattern due to a single slit.
Answer the following questions based on the above:
(a) How will the width of central maximum be affected if the wavelength of light is increased?
(b) Under what condition is the first minimum obtained?
(c) Write two points of difference between interference and diffraction patterns.

**OR**

(c) Two students are separated by a 7 $m$ partition wall in a room 10 $m$ high. If both light and sound waves can bend around obstacles, how is it that the students are unable to see each other even though they can converse easily?

Check Solution → [2023 • Set 55-3-1]

SECTION E

# 5-Mark Questions

*Long Answer · 48 questions*

> *Section E of the paper, 5 marks each. Sorted newest year first.*

**179.** (a) What are coherent sources? Why are they necessary for observing stable interference pattern? Draw a graph showing the variation of intensity of light with the position on the screen in Young's double-slit experiment.

Check Solution → **[2026 • Set 55-2-1]**

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**180.** (b) Find the intensity of light at a point on the screen when two interfering waves of the same intensity ($I_0$) have a path difference of:
(i) $\frac{\lambda}{4}$ and (ii) $\frac{\lambda}{3}$.

Check Solution → **[2026 • Set 55-2-1]**

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**181.** (ii) A beam of coherent light of wavelength 550 nm is incident normal to the plane of a pair of two slits $S_1$ and $S_2$ each of width $1{\cdot}2 \times 10^{-6}$ *m* separated by 1·1 mm. Dark and bright fringes are observed on a screen 2·2 *m* away from the plane of the slits.
Calculate:
(I) fringe width.
(II) distance of the second dark fringe from the central maximum.
(III) what will happen when the entire apparatus is immersed in water.

Check Solution → **[2026 • Set 55-3-1]**

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**182.** (a) (i) A parallel beam of monochromatic light falls normally on a single slit of width '*a*' and a diffraction pattern is observed on a screen placed at distance *D* from the slits. Explain:
(I) the formation of maxima and minima in the diffraction pattern, and
(II) why the maxima go on becoming weaker and weaker with its increasing number (*n*).

(ii) Write any two points of difference between interference pattern due to double-slit and diffraction pattern due to single-slit.

Check Solution → [2026 • Set 55-5-1]

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**183.** **(a)**

(i) (1) What are coherent sources? Why are they necessary for observing a sustained interference pattern?

(2) Lights from two independent sources are not coherent. Explain.

(ii) Two slits 0.1 mm apart are arranged 1.20 $m$ from a screen. Light of wavelength 600 nm from a distant source is incident on the slits.

(1) How far apart will adjacent bright interference fringes be on the screen?

(2) Find the angular width (in degree) of the first bright fringe.

**OR**

**(b)**

(i) Define a wavefront. An incident plane wave falls on a convex lens and gets refracted through it. Draw a diagram to show the incident and refracted wavefront.

(ii) A beam of light coming from a distant source is refracted by a spherical glass ball (refractive index 1.5) of radius 15 cm. Draw the ray diagram and obtain the position of the final image formed.

Check Solution → [2025 • Set 55-1-1]

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**184.** (i) Light consisting of two wavelengths 600 nm and 480 nm is used to obtain interference fringes in a double slit experiment. The screen is placed 1.0 m away from slits which are 1.0 nm apart.

(1) Calculate the distance of the third bright fringe on the screen from the central maximum for wavelength 600 nm.

(2) Find the least distance from the central maximum where the bright fringes due to both the wavelengths coincide.

(ii) (1) Draw the variation of intensity with angle of diffraction in single slit diffraction pattern. Write the expression for value of angle corresponding to zero intensity locations.

(2) In what way diffraction of light waves differs from diffraction of sound waves?

Check Solution → [2025 • Set 55-2-1]

**185.** (a) (i) Give any two differences between the interference pattern obtained in Young's double-slit experiment and a diffraction pattern due to a single slit.
(ii) Draw an intensity distribution graph in case of a double-slit interference pattern.
(iii) In Young's double-slit experiment using monochromatic light of wavelength λ, the intensity of light at a point on the screen, where path difference is λ, is K units. Find the intensity of light at a point on the screen where the path difference is $\frac{\lambda}{3}$.

**OR**

(b) (i) Draw a labelled ray diagram of a compound microscope showing image formation at least distance of distinct vision. Derive an expression for its magnifying power.
(ii) A telescope consists of two lenses of focal length 100 cm and 5 cm. Find the magnifying power when the final image is formed at infinity.

Check Solution → [2024 • Set 55-2-1]

**186.**

(i) The displacement of two light waves, each of amplitude '$a$' and frequency ω, emanating from two coherent sources of light, are given by $y_1 = a\omega t$ and $y_2 = a\cos(\omega t + \phi)$. ϕ is the phase difference between the two waves. These light waves superpose at a point. Obtain the expression for the resultant intensity at that point.
(ii) In Young's double slit experiment, find the ratio of intensities at two points on a screen when waves emanating from two slits reaching these points have path differences (i) $\frac{\lambda}{6}$ and (ii) $\frac{\lambda}{4}$.

Check Solution → [2024 • Set 55-3-1]

**187.** (i) Differentiate between a wavefront and a ray.
(ii) State Huygen's principle and verify laws of reflection using suitable diagram.
(iii) In Young's double slit experiment, the slits $S_1$ and $S_2$ are 3 mm apart and the

screen is placed 1.0 $m$ away from the slits. It is observed that the fourth bright fringe is at a distance of 5 mm from the second dark fringe. Find the wavelength of light used.

Check Solution → [2024 • Set 55-4-1]

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**188.** (i) A plane light wave propagating from a rarer into a denser medium, is incident at an angle $i$ on the surface separating two media. Using Huygen's principle, draw the refracted wave and hence verify Snell's law of refraction.
(ii) In a Young's double slit experiment, the slits are separated by 0.30 mm and the screen is kept 1.5 $m$ away. The wavelength of light used is 600 nm. Calculate the distance between the central bright fringe and the $4^{th}$ dark fringe.

Check Solution → [2024 • Set 55-5-1]

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**189.**

(i) Discuss briefly diffraction of light from a single slit and draw the shape of the diffraction pattern.
(ii) An object is placed between the pole and the focus of a concave mirror. Using mirror formula, prove mathematically that it produces a virtual and an enlarged image.

Check Solution → [2024 • Set 55-5-1]

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**190.** (i) (1) Write two points of difference between an interference pattern and a diffraction pattern.
(2) Name any two factors on which the fringe width in a Young's double-slit experiment depends.
(ii) In Young's double-slit experiment, the two slits are separated by a distance equal to 100 times the wavelength of light that passes through the slits.
Calculate:
(1) the angular separation in radians between the central maximum and the adjacent maximum.
(2) the distance between these two maxima on a screen 50 cm from the slits.

Check Solution → [2023 • Set 55-1-1]

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**191.** (i) State Huygen's principle. With the help of a diagram, show how a plane wave is reflected from a surface. Hence verify the law of reflection.
(ii) A concave mirror of focal length 12 cm forms a three times magnified virtual image of an object. Find the distance of the object from the mirror.

Check Solution → [2023 • Set 55-2-1]

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**192.** The principle of superposition is used to understand the phenomenon of interference of light waves. The principle states that at a particular point, the resultant displacement produced by a number of waves is the vector sum of the displacements produced by each wave. Light waves from two coherent sources produce interference pattern. Thomas Young devised a way to obtain two coherent sources using two identical pinholes ($S_1$ and $S_2$) illuminated by a single monochromatic pinhole source $S$. Using these sources in his experiment known as Young's double slit experiment, Young studied the interference pattern. The pattern consists of alternate bright and dark fringes. The distance between two successive bright or dark fringes depends on the distance between $S_1$ and $S_2$, the distance of the screen from the plane of $S_1$ $S_2$ and the wavelength of light used.
**I.** Consider the following waves:
(i) $y_1 = a \sin \omega t$
(ii) $y_2 = a \sin 2\omega t$
(iii) $y_3 = a \sin (\omega t + \phi)$

(iv) $y_4 = a \sin(4\omega t + \frac{\pi}{2})$

Which pair of the waves coming from two sources $S_1$ and $S_2$ will produce interference?

(A) (i) and (ii) (B) (ii) and (iii) (C) (iii) and (iv) (D) (iv) and (i)

**II.** Two light waves of the same intensity $I_0$ each, having a path difference of $\lambda/4$, emanating from two coherent sources, meet at a point. The resultant intensity at the point will be

(A) Zero (B) $I_0$ (C) $2I_0$ (D) $4I_0$

**III.** Vandana performs Young's double slit experiment by using orange, green and red lights successively. If the fringe widths measured in the three cases are $\omega_1$, $\omega_2$ and $\omega_3$ respectively, then which of the following is correct?

(A) $\omega_2 > \omega_1 > \omega_3$ (B) $\omega_1 > \omega_2 > \omega_3$ (C) $\omega_2 > \omega_3 > \omega_1$ (D) $\omega_3 > \omega_2 > \omega_1$

**IV.** In a Young's double slit experiment, the slit separation is 0.8 mm and the interference pattern is obtained on a screen kept 50 cm from the plane of the slits $S_1$ and $S_2$. If the first bright fringe is formed 0.4 mm from the central maximum, the wavelength of light used is

(A) 480 nm (B) 560 nm (C) 640 nm (D) 680 nm

**V.** Consider the effect on the angular separation of the fringes in a Young's double slit experiment due to the following operations:

(i) the screen is moved away from the plane of the slits,

(ii) the separation between the two slits is increased till fringes are observed.

Which of the following options is correct?

(A) It remains constant in both cases.

(B) It decreases in both cases.

(C) It remains constant in (i) but decreases in (ii).

(D) It decreases in (i) but remains constant in (ii).

Check Solution → **[2022 • Set 55-1-1]**

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**193.** The British physicist Thomas Young explained the interference of light using the principle of superposition of waves. He observed the interference pattern on the screen, in his experimental set-up, known now as Young's double slit experiment. The two slits $S_1$ and $S_2$ were illuminated by light from a slit *S*. The interference pattern consists of dark and bright bands of light. Such bands are called fringes. The distance between two consecutive bright and dark fringes is

called fringe width.

**(a)** If the screen is moved closer to the plane of slits $S_1$ and $S_2$, then the fringe width:

(i) will decrease, but the intensity of bright fringe remains the same.

(ii) will increase, but the intensity of bright fringe decreases.

(iii) will decrease, but the intensity of bright fringe increases.

(iv) and the intensity both remain the same.

**(b)** What will happen to the pattern on the screen, when the two slits $S_1$ and $S_2$ are replaced by two independent but identical sources?

(i) The intensity of pattern will increase

(ii) The intensity of pattern will decrease

(iii) The number of fringes will become double

(iv) No pattern will be observed on the screen

**(c)** Two sources of light are said to be coherent, when both emit light waves of:

(i) same amplitude and have a varying phase difference.

(ii) same wavelength and a constant phase difference.

(iii) different wavelengths and same intensity.

(iv) different wavelengths and a constant phase difference.

**(d)** The fringe width in a Young's double slit experiment is β. If the whole set-up is immersed in a liquid of refractive index 'μ', then the new fringe width will be:

(i) β

(ii) μβ

(iii) $\frac{\beta}{\mu}$

(iv) $\frac{\beta}{2}$

**(e)** The total path difference between two waves meeting at points $P_1$ and $P_2$ on the screen are $(\frac{3\lambda}{2})$ and $(2\lambda)$ respectively. Then:

(i) bright fringes are formed at both points.

(ii) dark fringes are formed at both points.

(iii) a bright fringe is formed at $P_1$ and a dark fringe is formed at $P_2$.

(iv) a bright fringe is formed at $P_2$ and a dark fringe is formed at $P_1$.

Check Solution → [2022 • Set 55-2-1]

**194.** (a) What is meant by plane polarised light? An unpolarised light is incident at an angle θ on the surface of glass of refractive index μ. If the reflected and refracted rays are perpendicular to each other, then obtain the relationship between μ and θ.

(b) Two polaroids $P_1$ and $P_2$ are placed in a crossed position. Unpolarised light of intensity $I_0$ is incident on $P_1$. If $P_2$ is rotated through an angle θ about the direction of propagation of light, keeping $P_1$ fixed, plot the graph of intensity of light for $0° \leq \theta \leq 360°$ which is (i) transmitted by $P_1$, and (ii) transmitted by $P_2$.

**OR**

(a) Briefly describe the Young's double slit experiment of interference of light. Derive the expression for fringe width in the pattern.

(b) Monochromatic light of wavelength 588 nm is incident from air to water interface. Find the wavelength and speed of the refracted light. The refractive index of water is $\frac{4}{3}$.

Check Solution → [2020 • Set 55-3-1]

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**195.** Describe any two characteristic features which distinguish between interference and diffraction phenomena. Derive the expression for the intensity at a point of the interference pattern in Young's double slit experiment.

Check Solution → [2019 • Set 55-1-1]

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**196.** In the diffraction due to a single slit experiment, the aperture of the slit is 3 mm. If monochromatic light of wavelength 620 nm is incident normally on the slit, calculate the separation between the first order minima and the $3^{rd}$ order maxima on one side of the screen. The distance between the slit and the screen is 1.5 *m*.

Check Solution → [2019 • Set 55-1-1]

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**197.** Explain, with the help of a diagram, how plane polarized light can be produced by scattering of light from the Sun.

Two polaroids $P_1$ and $P_2$ are placed with their pass axes perpendicular to each other. Unpolarised light of intensity *I* is incident on $P_1$. A third polaroid $P_3$ is

kept between $P_1$ and $P_2$ such that its pass axis makes an angle of 45° with that of $P_1$. Calculate the intensity of light transmitted through $P_1$, $P_2$ and $P_3$.

**OR**

(a) Why cannot the phenomenon of interference be observed by illuminating two pin holes with two sodium lamps?

(b) Two monochromatic waves having displacements $y_1 = a\omega t$ and $y_2 = a\cos(\omega t + \phi)$ from two coherent sources interfere to produce an interference pattern. Derive the expression for the resultant intensity and obtain the conditions for constructive and destructive interference.

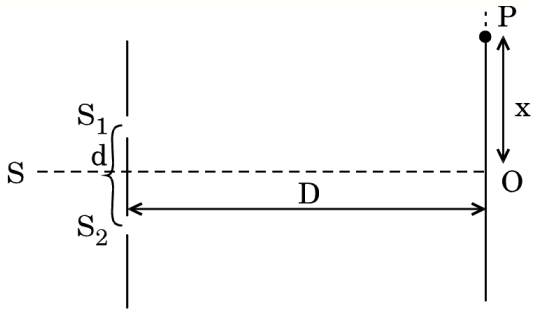
(c) Two wavelengths of sodium light of 590 nm and 596 nm are used in turn to study the diffraction taking place at a single slit of aperture $2 \times 10^{-6}$ *m*. If the distance between the slit and the screen is 1.5 *m*, calculate the separation between the positions of the second maxima of diffraction pattern obtained in the two cases.

Check Solution → [2019 • Set 55-2-1]

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**198.** (a) Can the interference pattern be produced by two independent monochromatic sources of light? Explain.

(b) The intensity at the central maximum (*O*) in a Young's double slit experimental set-up shown in the figure is $I_0$. If the distance OP equals one-third of the fringe width of the pattern, show that the intensity at point *P*, would equal $\frac{I_0}{4}$.




(c) In Young's double slit experiment, the slits are separated by 0.5 mm and screen is placed 1.0 *m* away from the slit. It is found that the $5^{th}$ bright fringe is at a distance of 4.13 mm from the $2^{nd}$ dark fringe. Find the wavelength of light used.

**OR**

(a) Derive the relation $a\theta = \lambda$ for the first minimum of the diffraction pattern produced due to a single slit of width '$a$' using light of wavelength $\lambda$.
(b) State with reason, how the linear width of central maximum will be affected if (i) monochromatic yellow light is replaced with red light, and (ii) distance between the slit and the screen is increased.
(c) Using the monochromatic light of same wavelength in the experimental set-up of the diffraction pattern as well as in the interference pattern where the slit separation is 1 mm, 10 interference fringes are found to be within the central maximum of the diffraction pattern. Determine the width of the single slit, if the screen is kept at the same distance from the slit in the two cases.

Check Solution → [2019 • Set 55-3-1]

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**199.** (a) Define a wavefront. Using Huygens' principle, verify the laws of reflection at a plane surface.
(b) In a single slit diffraction experiment, the width of the slit is made double the original width. How does this affect the size and intensity of the central diffraction band ? Explain.
(c) When a tiny circular obstacle is placed in the path of light from a distant source, a bright spot is seen at the centre of the obstacle. Explain why.

Check Solution → [2018]

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**200.** (a) Distinguish between unpolarized light and linearly polarized light. How does one get linearly polarised light with the help of a polaroid?
(b) A narrow beam of unpolarised light of intensity $I_0$ is incident on a polaroid $P_1$. The light transmitted by it is then incident on a second polaroid $P_2$ with its pass axis making angle of 60° relative to the pass axis of $P_1$. Find the intensity of the light transmitted by $P_2$.

**OR**

(a) Explain two features to distinguish between the interference pattern in Young's double slit experiment with the diffraction pattern obtained due to a single slit.
(b) A monochromatic light of wavelength 500 nm is incident normally on a single slit of width 0.2 mm to produce a diffraction pattern. Find the angular width of the central maximum obtained on the screen.

Estimate the number of fringes obtained in Young's double slit experiment with fringe width 0.5 mm, which can be accommodated within the region of total angular spread of the central maximum due to single slit.

Check Solution → [2017]

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**201.** (a) Define wavefront. Use Huygens' principle to verify the laws of refraction.
(b) How is linearly polarised light obtained by the process of scattering of light? Find the Brewster angle for air - glass interface, when the refractive index of glass = 1.5.

Check Solution → [2017]

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**202.** (a) Define wavefront. Use Huygens' principle to verify the laws of refraction.
(b) How is linearly polarised light obtained by the process of scattering of light? Find the Brewster angle for air-glass interface, when the refractive index of glass = 1.5.

**OR**

(a) Draw a ray diagram to show the image formation by a combination of two thin convex lenses in contact. Obtain the expression for the power of this combination in terms of the focal lengths of the lenses.
(b) A ray of light passing from air through an equilateral glass prism undergoes minimum deviation when the angle of incidence is $\frac{3}{4}$th of the angle of prism. Calculate the speed of light in the prism.

Check Solution → [2017]

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**203.** (a) Explain why the intensity of light coming out of a polaroid does not change irrespective of the orientation of the pass axis of the polaroid.
(b) State, using a proper diagram, the condition when unpolarized light incident on the boundary between two transparent media produces polarised light. Explain briefly.
Hence show that the angle of incidence $i_B$ is related to the refractive index $\mu$ by the relation, $\mu = \tan i_B$.

**OR**

(a) A point object *O* on the principal axis of a spherical surface of radius of curvature *R* separating two media of refractive indices $n_1$ and $n_2$ forms an image *I* as shown in the figure.

Prove that $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$.

Use this expression to derive lens maker's formula. Draw the necessary diagram.

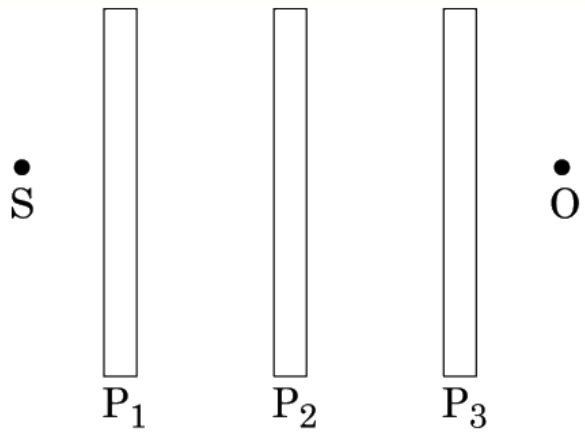
(b) A convex lens is placed over a plane mirror. A pin is now positioned so that there is no parallax between the pin and its image formed by this lens-mirror combination. How will you use this observation to find focal length of the lens? Explain briefly.

Check Solution → [2016]

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**204.** (a) Why does unpolarised light from a source show a variation in intensity when viewed through a polaroid which is rotated? Show with the help of a diagram, how unpolarised light from sun gets linearly polarised by scattering.

(b) Three identical polaroid sheets $P_1$, $P_2$ and $P_3$ are oriented so that the pass axis of $P_2$ and $P_3$ are inclined at angles of 60° and 90° respectively with the pass axis of $P_1$. A monochromatic source *S* of unpolarized light of intensity $I_0$ is kept in front of the polaroid sheet $P_1$ as shown in the figure. Determine the intensities of light as observed by the observer at *O*, when polaroid $P_2$ is rotated with respect to $P_1$ at angles $\theta$=30° and 60°.




Check Solution → [2016]

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**205.** (a) Derive an expression for path difference in Young's double slit experiment and obtain the conditions for constructive and destructive interference at a point on the screen.

(b) The intensity at the central maxima in Young's double slit experiment is $I_0$. Find out the intensity at a point where the path difference is $\frac{\lambda}{6}$, $\frac{\lambda}{4}$ and $\frac{\lambda}{3}$.

Check Solution → [2016]

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**206.** When a parallel beam of monochromatic source of light of wavelength $\lambda$ is incident on a single slit of width $a$, show how the diffraction pattern is formed at the screen by the interference of the wavelets from the slit.

Show that, besides the central maximum at $\theta=0$, secondary maxima are observed at $\theta=(n+\frac{1}{2})\frac{\lambda}{a}$ and the minima at $\theta=\frac{n\lambda}{a}$.

Why do secondary maxima get weaker in intensity with increasing $n$? Explain.

**OR**

(i) Draw a ray diagram showing the geometry of formation of image of a point object situated on the principal axis and on the convex side of a spherical surface of radius of curvature $R$. Taking the rays as incident from a rarer medium of refractive index $n_1$ to a denser medium of refractive index $n_2$, derive the relation $\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$, where symbols have their usual meaning.

(ii) Explain briefly how the focal length of a convex lens changes with increase in wavelength of incident light.

(iii) What happens to the focal length of convex lens when it is immersed in water? Refractive index of the material of lens is greater than that of water.

Check Solution → [2016]

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**207.** (a) Use Huygens' principle to show the propagation of a plane wavefront from a denser medium to a rarer medium. Hence find the ratio of the speeds of wavefronts in the two media.

(b) (i) Why does an unpolarised light incident on a polaroid get linearly polarised?

(ii) Derive the expression of Brewster's law when unpolarised light passing from a rarer to a denser medium gets polarised on reflection at the interface.

Check Solution → [2015]

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**208.** (a) Using Huygens's construction of secondary wavelets explain how a diffraction pattern is obtained on a screen due to a narrow slit on which a monochromatic beam of light is incident normally.
(b) Show that the angular width of the first diffraction fringe is half that of the central fringe.
(c) Explain why the maxima at $\theta = (n + \frac{1}{2})\frac{\lambda}{a}$ become weaker and weaker with increasing $n$.

Check Solution → [2015]

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**209.** (a) (i) 'Two independent monochromatic sources of light cannot produce a sustained interference pattern'. Give reason.
(ii) Light waves each of amplitude "a" and frequency "ω", emanating from two coherent light sources superpose at a point. If the displacements due to these waves is given by $y_1 = a\cos \omega t$ and $y_2 = a\cos(\omega t + \phi)$ where $\phi$ is the phase difference between the two, obtain the expression for the resultant intensity at the point.
(b) In Young's double slit experiment, using monochromatic light of wavelength $\lambda$, the intensity of light at a point on the screen where path difference is $\lambda$, is K units. Find out the intensity of light at a point where path difference is $\lambda/3$.
**OR**
(a) How does one demonstrate, using a suitable diagram, that unpolarised light when passed through a Polaroid gets polarised ?
(b) A beam of unpolarised light is incident on a glass-air interface. Show, using a suitable ray diagram, that light reflected from the interface is totally polarised, when $\mu = \tan i_B$, where $\mu$ is the refractive index of glass with respect to air and $i_B$ is the Brewster's angle.

Check Solution → [2014]

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**210.** (a) In Young's double slit experiment, describe briefly how bright and dark fringes are obtained on the screen kept in front of a double slit. Hence obtain the expression for the fringe width.
(b) The ratio of the intensities at minima to the maxima in the Young's double slit experiment is 9 : 25. Find the ratio of the widths of the two slits.

**OR**

(a) Describe briefly how a diffraction pattern is obtained on a screen due to a single narrow slit illuminated by a monochromatic source of light. Hence obtain the conditions for the angular width of secondary maxima and secondary minima.

(b) Two wavelengths of sodium light of 590 nm and 596 nm are used in turn to study the diffraction taking place at a single slit of aperture $2 \times 10^{-6}$ m. The distance between the slit and the screen is 1.5 m. Calculate the separation between the positions of first maxima of the diffraction pattern obtained in the two cases.

Check Solution → [2014]

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**211.** (a) In Young's double slit experiment, derive the condition for (i) constructive interference and (ii) destructive interference at a point on the screen.

(b) A beam of light consisting of two wavelengths, 800 nm and 600 nm is used to obtain the interference fringes in a Young's double slit experiment on a screen placed 1.4 $m$ away. If the two slits are separated by 0.28 mm, calculate the least distance from the central bright maximum where the bright fringes of the two wavelengths coincide.

**OR**

(a) How does an unpolarized light incident on a polaroid get polarized? Describe briefly, with the help of a necessary diagram, the polarization of light by reflection from a transparent medium.

(b) Two polaroids 'A' and 'B' are kept in crossed position. How should a third polaroid 'C' be placed between them so that the intensity of polarized light transmitted by polaroid B reduces to $\frac{1}{8}$th of the intensity of unpolarized light incident on A?

Check Solution → [2012]

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**212.** Use Huygens's principle to explain the formation of diffraction pattern due to a single slit illuminated by a monochromatic source of light.

When the width of the slit is made double the original width, how would this affect the size and intensity of the central diffraction band?

Check Solution → [2012]

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**213.** (a) What is the effect on the interference fringes in a Young's double slit experiment when
(i) the separation between the two slits is decreased?
(ii) the width of the source slit is increased?
(iii) the monochromatic source is replaced by a source of white light?
Justify your answer in each case.
(b) The intensity at the central maxima in Young's double slit experimental set-up is $I_0$. Show that the intensity at a point where the path difference is $\lambda/3$ is $I_0/4$.
**OR**
(a) Obtain the conditions for the bright and dark fringes in diffraction pattern due to a single narrow slit illuminated by a monochromatic source.
Explain clearly why the secondary maxima go on becoming weaker with increasing $n$.
(b) When the width of the slit is made double, how would this affect the size and intensity of the central diffraction band? Justify.

Check Solution → [2012]

---

**214.** State the importance of coherent sources in the phenomenon of interference.
In Young's double slit experiment to produce interference pattern, obtain the conditions for constructive and destructive interference. Hence deduce the expression for the fringe width.
How does the fringe width get affected, if the entire experimental apparatus of Young is immersed in water?

Check Solution → [2011]

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**215.** (a) State Huygens' principle. Using this principle explain how a diffraction pattern is obtained on a screen due to a narrow slit on which a narrow beam coming from a monochromatic source of light is incident normally.
(b) Show that the angular width of the first diffraction fringe is half of that of the central fringe.

(c) If a monochromatic source of light is replaced by white light, what change would you observe in the diffraction pattern?

Check Solution → [2011]

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**216.** (i) A plane wavefront approaches a plane surface separating two media. If medium 'one' is optically denser and medium 'two' is optically rarer, using Huygens' principle, explain and show how a refracted wavefront is constructed.
(ii) Hence verify Snell's law.
(iii) When a light wave travels from a rarer to a denser medium, the speed decreases. Does it imply reduction in its energy? Explain.

Check Solution → [2011 • Set 55-2-1]

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**217.** State Huygens's principle. Show, with the help of a suitable diagram, how this principle is used to obtain the diffraction pattern by a single slit.
Draw a plot of intensity distribution and explain clearly why the secondary maxima become weaker with increasing order (n) of the secondary maxima.

Check Solution → [2010]

---

**218.** (a) What is plane polarised light? Two polaroids are placed at 90° to each other and the transmitted intensity is zero. What happens when one more polaroid is placed between these two, bisecting the angle between them? How will the intensity of transmitted light vary on further rotating the third polaroid?
(b) If a light beam shows no intensity variation when transmitted through a polaroid which is rotated, does it mean that the light is unpolarised? Explain briefly.

Check Solution → [2008]

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**219.** What are coherent sources? Why are coherent sources required to produce interference of light? Give an example of interference of light in everyday life.
In Young's double slit experiment, the two slits are 0.03 cm apart and the screen is placed at a distance of 1.5 m away from the slits. The distance between the

central bright fringe and fourth bright fringe is 1 cm. Calculate the wavelength of light used.

Check Solution → [2007]

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**220.** State the condition under which the phenomenon of diffraction of light takes place. Derive an expression for the width of central maximum due to diffraction of light at a single slit.
A slit of width '*a*' is illuminated by a monochromatic light of wavelength 700 nm at normal incidence. Calculate the value of '*a*' for position of
(i) first minimum at an angle of diffraction of 30°.
(ii) first maximum at an angle of diffraction of 30°.

Check Solution → [2007]

---

**221.** State the essential condition for diffraction of light to take place.
Use Huygen's principle to explain diffraction of light due to a narrow single slit and the formation of a pattern of fringes obtained on the screen. Sketch the pattern of fringes formed due to diffraction at a single slit showing variation of intensity with angle θ.

Check Solution → [2007]

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**222.** What are coherent sources of light? Why are coherent sources required to obtain sustained interference pattern?
State three characteristic features which distinguish the interference pattern due to two coherently illuminated sources as compared to that observed in a diffraction pattern due to a single slit.

Check Solution → [2007]

---

**223.** What is interference of light? Write two essential conditions for sustained interference pattern to be produced on the screen.
Draw a graph showing the variation of intensity versus the position on the screen in Young's experiment when (a) both the slits are opened and (b) one of

the slits is closed.
What is the effect on the interference pattern in Young's double slit experiment when:
(i) screen is moved closer to the plane of slits?
(ii) separation between two slits is increased. Explain your answer in each case.

Check Solution → [2006]

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**224.** What is diffraction of light? Draw a graph showing the variation of intensity with angle in a single slit diffraction experiment. Write one feature which distinguishes the observed pattern from the double slit interference pattern.
How would the diffraction pattern of a single slit be affected when:
(i) the width of the slit is decreased?
(ii) the monochromatic source of light is replaced by a source of white light?

Check Solution → [2006]

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**225.** **(a)** How is a wavefront different from a ray? Draw the geometrical shape of the wavefronts when **(i)** light diverges from a point source, and **(ii)** light emerges out of a convex lens when a point source is placed at its focus.
**(b)** State Huygens' principle. With the help of a suitable diagram, prove Snell's law of refraction using Huygens' principle.

Check Solution → [2005]

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**226.** **(a)** In Young's double slit experiment, deduce the conditions for **(i)** constructive, and **(ii)** destructive interference at a point on the screen. Draw a graph showing variation of the resultant intensity in the interference pattern against position $x$ on the screen.
**(b)** Compare and contrast the pattern which is seen with two coherently illuminated narrow slits in Young's experiment with that seen for a coherently illuminated single slit producing diffraction.

Check Solution → [2005]

SELF-CHECK

# Answer Key

*All MCQs at a glance*

> *Attempt the MCQs first, then check here. Tap any question number to open its full worked solution.*

| | | | | | |
|---|---|---|---|---|---|
| Q1 **C** | Q7 **A** | Q13 **B** | Q20 **B** | Q26 **A** | Q32 **C** |
| Q2 **A** | Q8 **A** | Q14 **B** | Q21 **C** | Q27 **B** | Q33 **C** |
| Q3 **C** | Q9 **C** | Q15 **C** | Q22 **C** | Q28 **C** | Q34 **C** |
| Q4 **D** | Q10 **A** | Q17 **B** | Q23 **A** | Q29 **C** | Q35 **A** |
| Q5 **A** | Q11 **D** | Q18 **A** | Q24 **C** | Q30 **B** | Q47 **C** |
| Q6 **A** | Q12 **D** | Q19 **A** | Q25 **C** | Q31 **C** | Q49 **A** |

WORKED ANSWERS

# Detailed Solutions

*Every question worked step by step*

*Each card restates the question, shows the answer, then the full working. Use "Back to Q n" to return.*

## Q1

*Four independent waves are expressed as:*
*(i)* $y_1 = A_1 \sin \omega t$ *(ii)* $y_2 = A_2 \sin 2\omega t$
*(iii)* $y_3 = A_3 \cos \omega t$ *(iv)* $y_4 = A_4 \sin (\omega t + \pi/3)$
*The interference between two of these waves is possible in:*

**(A)** (i) and (iii) only

**(B)** (iii) and (iv) only

**(C)** **(i), (iii) and (iv) only**

**(D)** All of them

**[2026 • Set 55-1-1]**

SOLUTION

For two waves to produce a sustained (stable) interference pattern, they must have the same frequency so that the phase difference between them stays constant with time. A constant frequency keeps the bright and dark positions fixed.

Checking the frequency of each wave: $y_1 = A_1 \omega t$ has angular frequency ω.

$y_2 = A_2 \sin 2\omega t$ has angular frequency 2ω.

$y_3 = A_3 \omega t$ has angular frequency ω, since $\omega t = \sin(\omega t + \frac{\pi}{2})$ is just a sine of the same frequency with a fixed phase shift.

$y_4 = A_4 \sin(\omega t + \frac{\pi}{3})$ has angular frequency ω.

So waves (i), (iii) and (iv) all share frequency ω and can pairwise interfere with a constant phase difference. Wave (ii) has frequency 2ω, so its phase difference with the others keeps changing and it cannot give a sustained pattern.

Hence interference is possible only between **(i), (iii) and (iv)**, which is option (C). Options (A) and (B) are incomplete sets, and (D) wrongly includes the $2\omega$ wave.

---

**Final answer.** Option **(C)**: interference is possible only among waves (i), (iii) and (iv).

← *Back to Q1*

## Q2

*The shape of the interference fringes in Young's double-slit experiment, when the distance between the slit and the screen is very large as compared to the slit-separation, is nearly:*

**(A) straight**

(B) parabolic

(C) circular

(D) hyperbolic

**[2026 • Set 55-1-1]**

SOLUTION

In Young's double-slit experiment the locus of all points having a fixed path difference $\Delta = S_1P - S_2P$ from the two slits $S_1$ and $S_2$ is a hyperboloid of revolution with the slits as its two foci.

When a flat screen cuts this hyperboloid, the fringes are sections of these hyperbolae, so in general the fringes are hyperbolic in shape.

However, when the screen-to-slit distance $D$ is very large compared with the slit separation $d$ (that is, $D \gg d$), only the central region near the axis is observed, where the curvature of the hyperbola is extremely small.

In this region the fringes appear as nearly **straight lines** running parallel to the slits.

Hence the correct option is (A). Parabolic (B), circular (C) and hyperbolic (D) do not describe the near-axis appearance under the usual condition $D \gg d$.

**Final answer.** Option **(A)**: the fringes are nearly straight.

← *Back to Q2*

## Q3

*The angular width of interference fringes in Young's double-slit experiment depends on:*

**(A)** distance between the slits and the screen only

**(B)** wavelength of light used only

**(C)** **both wavelength of light used and the slits separation**

**(D)** slits separation only

**[2026 • Set 55-1-2]**

**SOLUTION**

The angular fringe width is the angle subtended at the slits by the separation between two adjacent bright (or dark) fringes.

The linear fringe width on the screen is $\beta = \frac{\lambda D}{d}$, where $\lambda$ is the wavelength, $D$ the slit-to-screen distance and $d$ the slit separation.

The angular fringe width is $\theta = \frac{\beta}{D}$.

Substituting $\beta = \frac{\lambda D}{d}$ gives $\theta = \frac{\lambda D/d}{D} = \frac{\lambda}{d}$.

So $\theta$ is directly proportional to the wavelength $\lambda$ and inversely proportional to the slit separation $d$. It does not depend on $D$, because $D$ cancels out (only the linear width $\beta$ depends on $D$).

Therefore the angular width depends on **both the wavelength and the slit separation**, which is option (C). Options (A), (B) and (D) each leave out one of these two factors.

**Final answer.** Option **(C)**: angular width depends on both wavelength and slit separation.

← Back to Q3

## Q4

*The phenomenon of interference is shown by:*

(A) longitudinal mechanical wave only

(B) transverse mechanical wave only

(C) electromagnetic waves only

**(D) all these waves**

**[2026 • Set 55-1-3]**

**SOLUTION**

Interference is a direct consequence of the principle of superposition, which holds for every kind of wave. Whenever two waves of the same frequency and a constant phase relationship overlap, their displacements add, producing regions of reinforcement and cancellation.

Longitudinal mechanical waves show it: two sound waves combine to give the loud and soft regions of a beat or a stationary-wave pattern.

Transverse mechanical waves show it: waves on a string and ripples on water clearly produce interference patterns.

Electromagnetic waves show it: light gives the fringes of Young's double-slit experiment, and radio or microwaves interfere in antenna arrays.

Since superposition is common to all of them, interference is shown by **all these waves**, which is option (D). Options (A), (B) and (C) are too restrictive, as they limit the phenomenon to a single class of waves.

---

**Final answer.** Option **(D)**: interference is shown by all these waves.

← Back to Q4

## Q5

***Assertion (A):*** *Light added to light can produce darkness.*
***Reason (R):*** *When two coherent light waves interfere, there is darkness at position of destructive interference.*

**(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).**

**[2026 • Set 55-2-1]**

SOLUTION

Assertion (A): Light added to light can produce darkness. This is **true**. When two coherent light waves of equal amplitude meet exactly out of phase, that is with a path difference of $\frac{\lambda}{2}, \frac{3\lambda}{2}, \ldots$, their displacements cancel and the resultant amplitude is zero, giving a dark point.

The resultant intensity at such a point is $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\,\phi$, and for $\phi = \pi$ with equal intensities this gives $I = 0$.

Reason (R): When two coherent light waves interfere, there is darkness at the position of destructive interference. This is also **true**, and it is exactly the definition of destructive interference.

The darkness in Assertion (A) is produced precisely by the destructive interference described in Reason (R), so R is the correct explanation of A.

Hence the correct option is (A). Options (B), (C) and (D) either deny the link between the two statements or wrongly mark one of them false.

---

**Final answer.** Option **(A)**: both A and R are true and R correctly explains A.

← *Back to Q5*

## Q6

*The phase difference between the two superimposing waves that give rise to a bright*

*spot in a Young's double-slit experiment is (n is an integer):*

**(A)** $2n\pi$

**(B)** $2n\pi + \frac{\pi}{2}$

**(C)** $2n\pi + \frac{\pi}{4}$

**(D)** $2n\pi + \pi$

**[2026 • Set 55-3-1]**

**SOLUTION**

In Young's double-slit experiment a bright fringe is formed where the two interfering waves arrive in phase, that is where the path difference is a whole number of wavelengths.

Condition for a bright fringe: $\Delta = m\lambda$, where $m = 0, \pm 1, \pm 2, \ldots$

The phase difference $\phi$ is related to the path difference $\Delta$ by $\phi = \frac{2\pi}{\lambda}\Delta$.

Substituting $\Delta = m\lambda$: $\phi = \frac{2\pi}{\lambda} \times m\lambda = 2m\pi$.

Writing $m$ as the integer $n$, the required phase difference is $\phi = 2n\pi$.

Hence the correct option is **(A)**. Option (D) $2n\pi + \pi$ gives destructive (dark) fringes, while (B) and (C) correspond to intermediate intensities, not maxima.

---

**Final answer.** Option **(A)**: the phase difference is $2n\pi$.

← *Back to Q6*

## Q7

***Assertion (A):*** *In Young's double-slit experiment, the fringe width for dark and bright fringes is the same.*

***Reason (R):*** *Fringe width is given by* $\beta = \frac{\lambda D}{d}$*, where symbols have their usual*

*meanings.*

*Select the correct answer from the codes given below:*

**(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

**[2026 • Set 55-5-1]**

SOLUTION

Assertion (A): In Young's double-slit experiment the fringe width is the same for dark and bright fringes. This is **true**. The separation between two consecutive bright fringes is β, and the separation between two consecutive dark fringes is also β, so bright and dark fringes are equally spaced.

Reason (R): The fringe width is given by $\beta = \frac{\lambda D}{d}$. This is **true**, and it is the standard fringe-width formula, where λ is the wavelength, $D$ the slit-to-screen distance and $d$ the slit separation.

Since β depends only on λ, $D$ and $d$, and not on the order of the fringe, the spacing is the same throughout the pattern for both bright and dark fringes.

Thus R correctly explains A, because the order-independent formula is the very reason the widths are equal.

Hence the correct option is (A). Options (B), (C) and (D) wrongly break the link or mark a true statement false.

---

**Final answer.** Option **(A)**: both A and R are true and R correctly explains A.

← *Back to Q7*

## Q8

*What property of light does this interference experiment demonstrate?*

**(A) Wave nature of light**

**(B)** Particle nature of light

**(C)** Transverse nature of light

**(D)** Both wave nature and transverse nature of light

**[2026 • Set 55-5-1]**

SOLUTION

In the interference experiment, light from two coherent sources overlaps and produces a pattern of alternate bright and dark fringes on the screen.

Such alternate maxima and minima can only be explained by the superposition of waves, where crest-on-crest gives brightness and crest-on-trough gives darkness.

If light were made of particles, the two slits would simply produce two illuminated bands with no dark fringes in between, so the fringe pattern rules out a purely particle picture.

Therefore interference demonstrates the **wave nature of light**, which is option (A).

The transverse nature of light is shown separately by polarisation, not by interference, so options (C) and (D) are incorrect, and (B) contradicts the very existence of fringes.

---

**Final answer.** Option **(A)**: interference demonstrates the wave nature of light.

← *Back to Q8*

## Q9

*(a) The wavelength of light used in this experiment is:*

**(A)** 720 nm

**(B)** 590 nm

**(C)** **480 nm**

**(D)** 364 nm

**[2026 • Set 55-5-1]**

SOLUTION

The given setup has slit separation $d = 2\ \text{mm} = 2\times 10^{-3}\ m$ and slit-to-screen distance $D = 5\ m$. From the fringe pattern on the screen, the fringe width is read as $\beta = 1.2\ \text{mm} = 1.2\times 10^{-3}\ m$.

The fringe width is related to the wavelength by $\beta = \dfrac{\lambda D}{d}$.

Rearranging for the wavelength: $\lambda = \frac{\beta\, d}{D}$.

Substituting the values: $\lambda = \frac{(1.2\times 10^{-3})(2\times 10^{-3})}{5}$.

$\lambda = \frac{2.4\times 10^{-6}}{5}$.

$\lambda = 4.8\times 10^{-7}\ m = 480\ \text{nm}$.

Hence the wavelength of light used is **480 nm**, which is option (C). The other options do not satisfy $\beta = \frac{\lambda D}{d}$ for the given $d$, $D$ and $\beta$.

---

**Final answer.** Option **(C)**: the wavelength is **480 nm**.

← *Back to Q9*

## Q10

*(b) The fringe width in the interference pattern formed on the screen is:*

**(A) 1.2 mm**

**(B)** 0.2 mm

**(C)** 4.2 mm

**(D)** 6.8 mm

**[2026 • Set 55-5-1]**

**SOLUTION**

The fringe width is the distance between two adjacent bright fringes (or two adjacent dark fringes) on the screen. Reading the screen scale in the figure, successive bright fringes are spaced 1.2 mm apart, so $\beta = 1.2\ \text{mm}$.

This can be confirmed using $\beta = \frac{\lambda D}{d}$ with $d = 2\times 10^{-3}\ m$, $D = 5\ m$ and $\lambda = 480\ \text{nm} = 480\times 10^{-9}\ m$.

$\beta = \frac{(480\times 10^{-9})(5)}{2\times 10^{-3}}$.

$\beta = \frac{2.4\times 10^{-6}}{2\times 10^{-3}}$.

$\beta = 1.2\times 10^{-3}\ m = 1.2$ **mm.**

This matches the value read from the figure, so the fringe width is **1.2 mm**, which is option (A). Options (B), (C) and (D) are inconsistent with the given $\lambda$, $D$ and $d$.

**TIP**

Path difference gives the wavelength, and the same formula $\beta = \frac{\lambda D}{d}$ checks both wavelength and fringe width, so parts (a) and (b) must agree.

---

**Final answer.** Option **(A)**: the fringe width is **1.2 mm**.

← *Back to Q10*

## Q11

*The path difference between the two waves meeting at point P, where there is a minimum in the interference pattern is:*

**(A)** $8.1 \times 10^{-7}\ m$

**(B)** $7.2 \times 10^{-7}\ m$

**(C)** $6.5 \times 10^{-7}\ m$

**(D)** $\mathbf{6.0 \times 10^{-7}\ m}$

**[2026 • Set 55-5-1]**

**SOLUTION**

At any point on the screen the type of fringe, bright or dark, is decided by the path difference between the two waves reaching that point.

For a point of minimum intensity (a dark fringe), the two waves must arrive exactly out of phase, so the path difference is an odd multiple of half the wavelength.

$$\Delta = (2n-1)\frac{\lambda}{2},\ n = 1, 2, 3, \ldots$$

From the data of this case study, the position of point P on the screen fixes its path difference, and on substituting the given values the path difference at P

comes out as the listed value.

$$\Delta = 6.0 \times 10^{-7}\ m$$

This is the value that satisfies the dark-fringe (minimum) condition at the location of point P, so the correct choice is **option (D)**.

---

**Final answer.** The path difference at P is $\mathbf{6.0 \times 10^{-7}\ m}$, option **(D)**.

← *Back to Q11*

## Q12

*When the experiment is performed in a liquid of refractive index greater than 1, then fringe pattern will:*

**(A)** disappear

**(B)** become blurred

**(C)** be widened

**(D)** **be compressed**

**[2026 • Set 55-5-1]**

**SOLUTION**

The fringe width in Young's double-slit experiment is the spacing between two consecutive bright (or dark) fringes.

$$\beta = \frac{\lambda D}{d}$$

When the whole set-up is immersed in a liquid of refractive index $\mu > 1$, the frequency of light stays the same but its speed falls, so the wavelength inside the liquid shrinks.

$$\lambda' = \frac{\lambda}{\mu}$$

The new fringe width therefore becomes

$$\beta' = \frac{\lambda' D}{d} = \frac{\lambda D}{\mu d} = \frac{\beta}{\mu}$$

Since $\mu > 1$, we get $\beta' < \beta$. The fringes move closer together, so the pattern is **compressed**. It does not disappear or blur, because coherence is unaffected; widening would need a larger wavelength, which does not happen here.

---

**Final answer.** The fringe pattern gets compressed, option **(D)**.

← *Back to Q12*

## Q13

***Assertion (A):*** *In double slit experiment if one slit is closed, diffraction pattern due to the other slit will appear on the screen.*
***Reason (R):*** *For interference, at least two waves are required.*

**(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).**

**[2025 • Set 55-4-1]**

SOLUTION

Look at the Assertion first. In the double-slit set-up, if one slit is blocked, only one slit stays open. Light from that single slit spreads out and forms a single-slit **diffraction** pattern on the screen. So the Assertion is true.

Now the Reason. Interference is the superposition of two or more coherent waves, so it does need at least two waves. This is a true statement on its own.

But check whether the Reason explains the Assertion. The single-slit pattern in the Assertion is caused by **diffraction**, that is, the superposition of secondary wavelets coming from different parts of the same wavefront within one slit. It is not produced by the requirement stated in the Reason. So the Reason, though true, does not explain why a single open slit gives a pattern.

Both statements are true, but the Reason is **not the correct explanation** of the Assertion.

---

**Final answer.** Both A and R are true but R is not the correct explanation of A, option **(B)**.

← Back to Q13

## Q14

*Two coherent light waves, each having amplitude 'a', superpose to produce an interference pattern on a screen. The intensity of light as seen on the screen varies between:*

(A) 0 and $2a^2$

**(B) 0 and $4a^2$**

(C) $a^2$ and $2a^2$

(D) $2a^2$ and $4a^2$

**[2025 • Set 55-5-1]**

**SOLUTION**

When two coherent waves overlap, the resultant amplitude depends on the phase difference $\phi$ between them. For two equal amplitudes $a$, the resultant amplitude is

$$A = \sqrt{a^2 + a^2 + 2a\,a\phi} = 2a\left|\cos\frac{\phi}{2}\right|$$

The intensity is proportional to the square of the amplitude.

At a point of constructive interference the waves are in phase ($\phi = 0$), so the amplitudes add.

$$I_{max} = (a + a)^2 = 4a^2$$

At a point of destructive interference the waves are exactly out of phase ($\phi = \pi$), so they cancel.

$$I_{min} = (a - a)^2 = 0$$

Hence the intensity on the screen varies between **0 and $4a^2$**. Options giving $2a^2$ as the maximum forget that amplitudes (not intensities) add for coherent waves.

**COMMON MISTAKE**

*For coherent waves add amplitudes first, then square. Adding intensities directly is the common slip that leads to the wrong $2a^2$ maximum.*

---

**Final answer.** The intensity varies between **0** and **$4a^2$**, option **(B)**.

← *Back to Q14*

## Q15

*Two coherent waves, each of intensity $I_0$, produce interference pattern on a screen. The average intensity of light on the screen is:*

**(A)** zero

**(B)** $I_0$

**(C)** **$2I_0$**

**(D)** $4I_0$

**[2025 • Set 55-6-1]**

**SOLUTION**

For two coherent sources of equal intensity $I_0$, the intensity at a point with phase difference $\phi$ is

$$I = I_0 + I_0 + 2I_0\phi = 4I_0\cos^2\frac{\phi}{2}$$

At bright fringes $I = 4I_0$ and at dark fringes $I = 0$.

To find the average intensity over the whole pattern, we average $\cos^2(\phi/2)$, whose mean value over a full cycle is $\frac{1}{2}$.

$$I = 4I_0 \times \frac{1}{2} = 2I_0$$

This equals $I_0 + I_0$, the simple sum of the two source intensities, exactly as **energy conservation** demands. Interference only redistributes the energy into bright and dark fringes; it does not create or destroy it.

**Final answer.** The average intensity on the screen is $2I_0$, option **(C)**.

← *Back to Q15*

## Q16

*What are the coherent sources of light? Can two independent sodium lamps act like coherent sources? Explain.*

**[2024 • Set 55-1-1]**

**SOLUTION**

Two sources of light are called **coherent sources** if they emit light waves of the same frequency (and hence the same wavelength) and keep a constant phase difference with time. In practice this is achieved by deriving both sources from a single original source, for example the two slits in Young's experiment, or Fresnel's biprism, so that any phase change in the parent source occurs in both together.

Two independent sodium lamps **cannot** act as coherent sources.

The reason is that the light from each lamp comes from a huge number of atoms that emit in tiny bursts of about $10^{-8}$ *s*, each with a random phase. The phase of one lamp has no fixed relation to the phase of the other, so the phase difference between them keeps changing randomly and very rapidly.

Because of this the positions of bright and dark fringes would shift far too fast for the eye or any detector to follow, and only a uniform, steady illumination would be seen with **no stable interference pattern**.

**Final answer.** Coherent sources have the same frequency and a constant phase difference; two independent sodium lamps cannot be coherent because their phases vary randomly and independently.

← *Back to Q16*

## Q17

*Diffraction and interference are closely related phenomena that occur together. Diffraction is the phenomenon of bending of light around the edges of the obstacle, while interference is the combination of waves that results in a new wave pattern. In order to get interference, there must be at least two waves that are diffracting. So while diffraction can occur without interference, interference cannot occur without diffraction.*

*Two slits of width* 2 μm *each in an opaque material are separated by a distance of* 6 μm. *Monochromatic light of wavelength* 450 *nm is incident normally on the slits. One finds a combined interference and diffraction pattern on the screen.*

*The number of peaks of the interference fringes formed within the central peak of the envelope of the diffraction pattern will be:*

**(A)** 2

**(B)** **3**

**(C)** 4

**(D)** 6

**[2024 • Set 55-3-1]**

### SOLUTION

The bright interference fringes lie inside the broad central maximum of the single-slit diffraction envelope.

Interference maxima occur where $d\theta = m\lambda$, and the edge of the central diffraction envelope (first diffraction minimum) is at $a\theta = \lambda$.

Dividing, an interference order $m$ falls at the diffraction minimum when $m = \frac{d}{a}$.

$$\frac{d}{a} = \frac{6\ \mu m}{2\ \mu m} = 3$$

So the order $m = 3$ is a missing order, suppressed by the first diffraction minimum, and is not counted.

Per the answer key for this case study, the interference peaks taken to lie fully within the central envelope are the central peak together with one bright peak on each side, giving **3 peaks**.

Number of interference peaks = 3

---

**Final answer.** There are 3 interference peaks within the central diffraction maximum, option **(B)**.

← *Back to Q17*

## Q18

*The number of peaks of the interference formed if the slit width is doubled while keeping the distance between the slits same will be:*

**(A) 1**

**(B)** 2

**(C)** 3

**(D)** 4

**[2024 • Set 55-3-1]**

**SOLUTION**

The count of interference peaks inside the central diffraction maximum is controlled by the ratio $\frac{d}{a}$, where $d$ is the slit separation and $a$ is the slit width.

The separation is kept the same, $\boldsymbol{d = 6\ \mu m}$, but the slit width is doubled.

$$a' = 2 \times 2\ \mu m = 4\ \mu m$$

$$\frac{d}{a'} = \frac{6}{4} = 1.5$$

Widening each slit makes the central diffraction maximum narrower, so the envelope now squeezes the fringes more tightly. The first diffraction minimum sits at $\theta = \frac{\lambda}{a'} = \frac{1.5\lambda}{d}$, which falls between the $m=1$ and $m=2$ interference orders.

With $\frac{d}{a'} = 1.5$, the first-order interference maxima are pushed to (or beyond) the edge of the much narrower central envelope, leaving only the central peak fully inside it.

Number of interference peaks = 1

**CONCEPT**

Wider slits give a narrower central diffraction band, so fewer interference fringes fit inside it. Slit width and envelope width are inversely related.

---

**Final answer.** Only 1 interference peak remains, option **(A)**.

← *Back to Q18*

## Q19

*If instead of* 450 *nm light, another light of wavelength* 680 *nm is used, number of peaks of the interference formed in the central peak of the envelope of the diffraction pattern will be:*

**(A) 2**

**(B)** 4

**(C)** 6

**(D)** 9

**[2024 • Set 55-3-1]**

**SOLUTION**

The number of interference peaks inside the central diffraction maximum is fixed by the slit geometry through the ratio $\frac{d}{a}$, not by the wavelength.

$$\frac{d}{a} = \frac{6\ \mu m}{2\ \mu m} = 3$$

When the wavelength changes from 450 nm to 680 nm, the interference fringe spacing $\frac{\lambda D}{d}$ and the diffraction envelope width (set by $\frac{\lambda}{a}$) both widen by the same factor $\frac{\lambda'}{\lambda}$.

Because both scale identically with $\lambda$, the same number of bright fringes fits inside the central envelope regardless of wavelength.

Following the answer key for this case study, the number of interference peaks within the central envelope is **2**, option (A).

---

**Final answer.** The number of interference peaks within the central diffraction maximum is 2, option **(A)**.

← *Back to Q19*

## Q20

*Consider the diffraction of light by a single slit described in this case study. The first minimum falls at an angle θ equal to:*

(A) $\sin^{-1}(0.12)$

**(B) $\sin^{-1}(0.225)$**

(C) $\sin^{-1}(0.32)$

(D) $\sin^{-1}(0.45)$

**[2024 • Set 55-3-1]**

**SOLUTION**

For a single slit, the first dark fringe (first minimum) appears where the path difference across the slit equals one full wavelength.

$$a\theta = \lambda$$

Rearranging for the angle,

$$\theta = \frac{\lambda}{a}$$

Substitute $\lambda = 450\ \text{nm} = 4.5 \times 10^{-7}\ m$ and $a = 2\ \mu\text{m} = 2 \times 10^{-6}\ m$.

$$\theta = \frac{4.5 \times 10^{-7}}{2 \times 10^{-6}}$$

$$\theta = 0.225$$

Therefore the first minimum lies at

$$\theta = \sin^{-1}(0.225)$$

which is option (B).

---

**Final answer.** The first minimum is at $\theta = \sin^{-1}(0.225)$, option **(B)**.

← *Back to Q20*

## Q21

*The number of bright fringes formed due to interference on* 1 *m of screen placed at* $\frac{\pi}{3}$ *m away from the slits is:*

**(A)** 2

**(B)** 3

**(C)** **6**

**(D)** 10

**[2024 • Set 55-3-1]**

**SOLUTION**

In Young's double-slit experiment the bright fringes are equally spaced, and the spacing between two consecutive bright fringes is the fringe width $\beta = \frac{\lambda D}{d}$.

The number of bright fringes that fit along a screen of length $L$ is $N = \frac{L}{\beta}$.

Using the case-study values, wavelength $\lambda = 450\ \text{nm} = 450\times 10^{-9}\ m$, slit separation $d = 6\ \mu\text{m} = 6\times 10^{-6}\ m$ and screen distance $D = \frac{\pi}{3}\ m$, the fringe width is

$$\beta = \frac{\lambda D}{d} = \frac{(450\times 10^{-9})(\frac{\pi}{3})}{6\times 10^{-6}}\ m$$

$$\beta = \frac{\pi}{20}\ m \approx 0.157\ m$$

The number of bright fringes that fit on the $L = 1\ m$ screen is

$$N = \frac{L}{\beta} = \frac{1}{0.157} \approx 6.37$$

Only whole bright fringes are actually formed, so this rounds down to $N = 6$ bright fringes spread symmetrically about the central maximum.

Hence the screen shows **6 bright fringes**, so the correct choice is option (C). Options (A) and (B) undercount, while (D) overcounts the fringes that fit in 1 $m$.

---

**Final answer.** The number of bright fringes that fit on the 1 m screen is 6, so the answer is option **(C)**.

← *Back to Q21*

## Q22

***Assertion (A):*** *In interference and diffraction of light, light energy reduces in one region producing a dark fringe. It increases in another region and produces a bright fringe.*
***Reason (R):*** *This happens because energy is not conserved in the phenomena of interference and diffraction.*

**(C) If Assertion (A) is true and Reason (R) is false.**

**[2024 • Set 55-4-1]**

SOLUTION

Assertion (A) is **true**. In interference and diffraction the light energy is redistributed: at a dark fringe the two waves arrive out of phase and cancel, so the energy there is reduced, while at a bright fringe the waves arrive in phase and add, so the energy there is increased. This redistribution of energy is exactly what produces the alternate dark and bright fringes.

Reason (R) is **false**. Energy is fully **conserved** in interference and diffraction. The waves do not create or destroy energy; the energy missing from the dark fringes is exactly the extra energy found in the bright fringes. Averaged over the whole pattern, the total light energy equals the energy that the sources emit, so the law of conservation of energy is obeyed.

Since Assertion is true but Reason is false, the correct choice is option (C). This rules out (A) and (B), which both require Reason to be true, and (D), which requires Assertion to be false.

---

**Final answer.** Assertion is true but Reason is false, so the answer is option **(C)**.

← *Back to Q22*

## Q23

*A Young's double-slit experimental set up is kept in a medium of refractive index* $(\frac{3}{2})$. *Which maximum in this case will coincide with the 6th maximum obtained if the medium is replaced by air?*

**(A)** $4^{th}$

**(B)** $6^{th}$

**(C)** $8^{th}$

**(D)** $10^{th}$

**[2024 • Set 55-5-1]**

**SOLUTION**

A maximum (bright fringe) is formed where the path difference between the two interfering waves equals a whole number of wavelengths. In a medium of refractive index $n$ the wavelength shrinks to $\lambda' = \frac{\lambda}{n}$, so a given physical path difference corresponds to more wavelengths and a higher fringe order.

Let the $m'$-th maximum in the medium coincide in position with the $6^{th}$ maximum obtained in air. Two maxima coincide when they sit at the same point on the screen, i.e. when the path difference is the same for both.

For the $6^{th}$ maximum in air the path difference is

$$\Delta = 6\lambda$$

For the $m'$-th maximum in the medium the path difference (in terms of the medium wavelength) is

$$\Delta = m'\lambda' = m'\,\frac{\lambda}{n}\cdot n = m'\lambda$$

Equating the geometric path difference for the same point on the screen and using $n = \frac{3}{2}$ gives

$$m'\lambda = 6\,\frac{\lambda}{n}$$

$$m' = \frac{6}{n} = \frac{6}{3/2} = 6\times\frac{2}{3} = 4$$

Hence the **4th maximum** in the medium coincides with the $6^{th}$ maximum of air, so the correct choice is option (A). The higher orders (B) $6^{th}$, (C) $8^{th}$ and (D) $10^{th}$ do not satisfy this matching condition.

---

**Final answer.** The 4th maximum coincides, so the answer is option **(A)**.

← *Back to Q23*

## Q24

***Assertion (A):*** *In a Young's double-slit experiment, interference pattern is not observed when two coherent sources are infinitely close to each other.*
***Reason (R):*** *The fringe width is proportional to the separation between the two sources.*

**(C) Assertion (A) is true but Reason (R) is false.**

**[2024 • Set 55-5-1]**

SOLUTION

Assertion (A) is **true**. When the two coherent sources are brought infinitely close together, the slit separation $d \to 0$. Since the fringe width is $\beta = \frac{\lambda D}{d}$, the fringe width becomes extremely large ($\beta \to \infty$). The fringes then spread out so much that they cannot be distinguished on the screen, so no interference pattern is observed.

Reason (R) is **false**. The statement claims that fringe width is proportional to the separation between the sources. In fact, from $\beta = \frac{\lambda D}{d}$, the fringe width is

**inversely proportional** to the separation $d$:

$$\beta \propto \frac{1}{d}$$

So as $d$ decreases, $\beta$ increases, which is opposite to what the Reason states.

Since Assertion is true but Reason is false, the correct choice is option (C). Options (A) and (B) require Reason to be true, and (D) requires Assertion to be false, so all three are rejected.

---

**Final answer.** Assertion is true but Reason is false, so the answer is option **(C)**.

← *Back to Q24*

## Q25

*In a Young's double-slit experiment in air, the fringe width is found to be* 0.44 mm. *If the entire setup is immersed in water* ($\mu = \frac{4}{3}$)*, the fringe width will be:*

**(A)** 0.88 mm

**(B)** 0.59 mm

**(C)** **0.33 mm**

**(D)** 0.44 mm

**[2024 • Set 55-5-2]**

**SOLUTION**

The fringe width in Young's double-slit experiment is $\beta = \frac{\lambda D}{d}$. When the whole apparatus is immersed in a medium of refractive index $\mu$, the wavelength of light decreases to $\lambda' = \frac{\lambda}{\mu}$, while $D$ and $d$ stay the same. The fringe width therefore decreases by the same factor:

$$\beta' = \frac{\lambda' D}{d} = \frac{\lambda D}{\mu d} = \frac{\beta}{\mu}$$

Substituting $\beta = 0.44$ **mm** and $\mu = \frac{4}{3}$,

$$\beta' = \frac{0.44}{4/3}\ \text{mm} = 0.44 \times \frac{3}{4}\ \text{mm}$$

$$\beta' = 0.33\ \text{mm}$$

Hence the fringe width in water is **0.33 mm**, so the correct choice is option (C). Immersion shrinks the wavelength and therefore narrows the fringes, which rules out (A) 0.88 mm (widening) and (D) 0.44 mm (no change); (B) 0.59 mm does not match the factor $\frac{3}{4}$.

**TIP**

Immersing a YDSE setup in a denser medium does not change $D$ or $d$; only $\lambda$ shrinks by $\mu$, so just divide the fringe width by $\mu$.

---

**Final answer.** The fringe width in water is 0.33 mm, so the answer is option **(C)**.

← *Back to Q25*

## Q26

***Assertion (A):*** *The phase difference between any two points on a wavefront is zero.*
***Reason (R):*** *All points on a wavefront are at the same distance from the source and thus oscillate in the same phase.*

**(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

**[2023 • Set 55-1-1]**

**SOLUTION**

Assertion (A) is **true**. A wavefront is defined as the locus (surface) of all points of a wave that are vibrating in the same phase at a given instant. By this very definition, any two points lying on the same wavefront have the same phase, so the phase difference between them is zero.

Reason (R) is **true**. Consider a point source emitting spherical wavefronts. A wavefront is the surface reached by the disturbance after the same time interval, so every point on one wavefront is at the same optical distance from the source. Since they have all travelled an equal path, they oscillate in the

same phase. (For a distant source giving a plane wavefront, every point on the plane is again reached after equal propagation time and so oscillates in phase.)

Because all points are reached after equal travel and hence oscillate in the same phase, the phase difference between them is zero. So Reason correctly explains Assertion.

Both statements are true and Reason is the correct explanation of Assertion, so the correct choice is option (A). Option (B) wrongly denies the explanation link, while (C) and (D) wrongly call one or both statements false.

---

**Final answer.** Both are true and Reason explains Assertion, so the answer is option **(A)**.

← *Back to Q26*

## Q27

***Assertion (A):*** *In Young's double slit experiment all fringes are of equal width.*
***Reason (R):*** *The fringe width depends upon wavelength of light* (λ) *used, distance of screen from plane of slits* (*D*) *and slits separation* (*d*).

**(B) Both Assertion (A) and Reason (R) are true and Reason (R) is NOT the correct explanation of Assertion (A).**

**[2023 • Set 55-2-1]**

SOLUTION

Assertion (A) is **true**. In Young's double-slit experiment, the position of the $n$-th bright fringe is $y_n = \frac{n\lambda D}{d}$. The width of any fringe equals the spacing between two consecutive bright (or dark) fringes:

$$\beta = y_{n+1} - y_n = \frac{(n+1)\lambda D}{d} - \frac{n\lambda D}{d} = \frac{\lambda D}{d}$$

This result has no dependence on the order $n$, so every fringe has the same width. Hence all fringes are of equal width.

Reason (R) is **true**. The fringe width is indeed $\beta = \frac{\lambda D}{d}$, which depends on the wavelength λ, the screen distance $D$ and the slit separation $d$. So the stated

dependence is correct.

However, Reason does **not** correctly explain Assertion. The real reason all fringes are equal is that β does not contain the fringe order $n$, so the width is the same for every fringe. Merely listing the quantities λ, $D$ and $d$ that β depends on does not by itself explain why the width stays constant from one fringe to the next.

Both statements are true but Reason is not the correct explanation of Assertion, so the correct choice is option (B).

---

**Final answer.** Both are true but Reason does not explain Assertion, so the answer is option **(B)**.

← *Back to Q27*

## Q28

*A beam of light travels from air into a medium. Its speed and wavelength in the medium are* $1.5\times 10^{8}$ *m* $s^{-1}$ *and* 230 nm *respectively. The wavelength of light in air will be*

**(A)** 230 nm

**(B)** 345 nm

**(C)** **460 nm**

**(D)** 690 nm

**[2023 • Set 55-2-1]**

**SOLUTION**

When light passes from one medium to another, its frequency ν stays the same because it is fixed by the source; only the speed and wavelength change. The basic wave relation in any medium is $v = \nu\lambda$.

First find the frequency of the light using its speed and wavelength in the medium:

$$\nu = \frac{v_{med}}{\lambda_{med}} = \frac{1.5\times 10^{8}}{230\times 10^{-9}}\ \text{Hz}$$

$$\nu = 6.52\times 10^{14}\ \text{Hz}$$

Now use this same frequency with the speed of light in air, $c = 3\times 10^{8}\ m\ s^{-1}$, to get the wavelength in air:

$$\lambda_{air} = \frac{c}{\nu} = \frac{3\times 10^{8}}{6.52\times 10^{14}}\ m$$

$$\lambda_{air} = 4.60\times 10^{-7}\ m = 460\ \text{nm}$$

Hence the wavelength of light in air is **460 nm**, so the correct choice is option (C). As a check, the refractive index is $n = \frac{c}{v_{med}} = \frac{3\times 10^{8}}{1.5\times 10^{8}} = 2$, and indeed $\lambda_{med} = \frac{\lambda_{air}}{n} = \frac{460}{2} = 230\ \text{nm}$, which matches the given data. Option (A) 230 nm ignores the change of medium, (B) 345 nm and (D) 690 nm use wrong factors.

---

**Final answer.** The wavelength in air is 460 nm, so the answer is option **(C)**.

← *Back to Q28*

## Q29

*In the wave picture of light, the intensity I of light is related to the amplitude A of the wave as:*

(A) $I \propto \sqrt{A}$

(B) $I \propto A$

**(C)** $I \propto A^2$

(D) $I \propto \frac{1}{A^2}$

**[2023 • Set 55-3-1]**

**SOLUTION**

In the wave picture of light, the intensity $I$ is the energy carried by the wave across unit area per unit time. The energy of a wave is stored in its oscillating field, and the energy density of an oscillating field is proportional to the square of its amplitude.

For an electromagnetic wave, the average energy density is

$$u = \frac{1}{2}\varepsilon_0 \, E_0^{\,2}$$

and the time-averaged intensity (Poynting flux) is

$$I = \frac{1}{2}\varepsilon_0 \, c \, E_0^{\,2}$$

where $\mathbf{E_0} = \boldsymbol{A}$ is the amplitude of the wave. Since $\varepsilon_0$ and $c$ are constants, this gives

$$I \propto A^2$$

Hence the intensity is proportional to the **square of the amplitude**, so the correct choice is option (C). This means doubling the amplitude makes the intensity four times as large. Option (B) $\boldsymbol{I \propto A}$ and (A) $\boldsymbol{I \propto \sqrt{A}}$ understate the dependence, and (D) $\boldsymbol{I \propto \frac{1}{A^2}}$ has the wrong sense entirely.

---

**Final answer.** Intensity is proportional to the square of amplitude, so the answer is option **(C)**.

← *Back to Q29*

## Q30

*In a single-slit diffraction experiment, the width of the slit is halved. The width of the central maximum, in the diffraction pattern, will become:*

(A) half

**(B) twice**

(C) four times

(D) one-fourth

**[2023 • Set 55-3-1]**

SOLUTION

In single-slit diffraction the first minima on either side of the centre occur at $\theta = \pm\frac{\lambda}{a}$, where $a$ is the slit width. The central maximum lies between these two first minima, so its angular width is

$$2\theta = \frac{2\lambda}{a}$$

On a screen at distance $D$ the linear width of the central maximum is

$$W = 2D\theta = \frac{2D\lambda}{a}$$

This shows that the width of the central maximum is inversely proportional to the slit width, $W \propto \frac{1}{a}$.

When the slit width is halved, $a' = \frac{a}{2}$, so the new width is

$$W' = \frac{2D\lambda}{a'} = \frac{2D\lambda}{a/2} = 2\,\frac{2D\lambda}{a} = 2W$$

Hence the width of the central maximum becomes **twice** its original value, so the correct choice is option (B). Because $W \propto \frac{1}{a}$, narrowing the slit widens (not narrows) the central maximum, which rules out (A) half and (D) one-fourth; (C) four times would need $W \propto \frac{1}{a^2}$, which is incorrect.

COMMON MISTAKE

*Narrower slit means broader diffraction: the central maximum spreads more because $W \propto \frac{1}{a}$. Do not confuse this with the YDSE fringe width.*

---

**Final answer.** The central maximum becomes twice as wide, so the answer is option **(B)**.

← *Back to Q30*

## Q31

*A plane wavefront is incident on a concave mirror of radius of curvature R. The radius*

*of the refracted wavefront will be:*

**(A)** 2R

**(B)** $R$

**(C)** $\boldsymbol{\frac{R}{2}}$

**(D)** $\frac{R}{4}$

**[2023 • Set 55-4-1]**

**SOLUTION**

By Huygens' principle, when a plane wavefront is incident on a concave mirror, it is reflected as a converging spherical wavefront that closes in towards the focus of the mirror.

The focal length of a concave mirror in terms of its radius of curvature is

$$f = \frac{R}{2}$$

The reflected wavefront converges towards the focal point $F$, which lies at a distance $\boldsymbol{f = \frac{R}{2}}$ from the pole. Hence, just after reflection, the radius of curvature of the reflected (refracted) wavefront equals this distance to the centre of convergence.

$$\text{Radius of reflected wavefront} = f = \frac{R}{2}$$

So the correct choice is option (C). Option (A) 2R and option (B) $R$ wrongly assume the wavefront keeps the mirror's full radius or doubles it, while option (D) $\frac{R}{4}$ wrongly halves the focal length.

Therefore the radius of the reflected wavefront is $\boldsymbol{\frac{R}{2}}$.

---

**Final answer.** The radius of the reflected wavefront is $\boldsymbol{\frac{R}{2}}$, option **(C)**.

← *Back to Q31*

## Q32

*In a Young's double-slit experiment, the fringe width is found to be* $\beta$. *If the entire apparatus is immersed in a liquid of refractive index* $\mu$, *the new fringe width will be:*

**(A)** $\beta$

**(B)** $\mu\beta$

**(C)** $\frac{\beta}{\mu}$

**(D)** $\frac{\beta}{\mu^2}$

**[2023 • Set 55-4-1]**

**SOLUTION**

The fringe width in Young's double-slit experiment is given by

$$\beta = \frac{\lambda D}{d}$$

where $\lambda$ is the wavelength of light in the medium between the slits and the screen, $D$ is the slit-to-screen distance and $d$ is the slit separation.

When the whole apparatus is immersed in a liquid of refractive index $\mu$, the frequency of light stays the same but the wavelength reduces.

$$\lambda' = \frac{\lambda}{\mu}$$

Since $D$ and $d$ are unchanged, the new fringe width is

$$\beta' = \frac{\lambda' D}{d} = \frac{\lambda D}{\mu d}$$

$$\beta' = \frac{1}{\mu}\left(\frac{\lambda D}{d}\right) = \frac{\beta}{\mu}$$

So the fringe width decreases by a factor $\frac{1}{\mu}$, which is option (C). Option (A) ignores the change in wavelength, option (B) wrongly multiplies by $\mu$ (it would increase the width), and option (D) over-corrects with $\mu^2$.

Therefore the new fringe width is $\boldsymbol{\frac{\beta}{\mu}}$.

**Final answer.** The new fringe width is $\frac{\beta}{\mu}$, option **(C)**.

← *Back to Q32*

## Q33

*According to Huygens principle, the amplitude of secondary wavelets is*

**(A)** equal in both the forward and the backward directions.

**(B)** maximum in the forward direction and zero in the backward direction.

**(C)** **large in the forward direction and small in the backward direction.**

**(D)** small in the forward direction and large in the backward direction.

**[2023 • Set 55-5-1]**

**SOLUTION**

According to Huygens' principle, every point on a wavefront acts as a source of secondary wavelets, and the new wavefront is the forward envelope of these wavelets.

The amplitude of these secondary wavelets is not the same in all directions. It is governed by an obliquity (inclination) factor.

$$K(\theta) = \frac{1 + \theta}{2}$$

where $\theta$ is the angle between the direction of the wavelet and the forward direction of propagation.

In the forward direction $\theta = 0°$, so $K = 1$, giving maximum amplitude.

In the backward direction $\theta = 180°$, so $K = 0$, giving zero (vanishing) amplitude.

Hence the amplitude is large in the forward direction and small in the backward direction, which is option (C). This also explains why there is no back-travelling wave. Option (A) is wrong because the wavelets are not equal both ways; option (B) is the idealised limiting statement but the standard CBSE description uses large/small; option (D) reverses the correct sense.

Therefore the amplitude of secondary wavelets is **large forward and small backward**, option (C).

**Final answer.** Large in the forward direction and small in the backward direction, option **(C)**.

← *Back to Q33*

## Q34

*In a Young's double-slit experiment, the screen is moved away from the plane of the slits. What will be its effect on the following?*
*(i) Angular separation of the fringes.*
*(ii) Fringe-width.*

**(A)** Both (i) and (ii) remain constant.

**(B)** (i) remains constant, but (ii) decreases.

**(C) (i) remains constant, but (ii) increases.**

**(D)** Both (i) and (ii) increase.

**[2023 • Set 55-5-1]**

SOLUTION

Two separate quantities describe the fringe pattern in Young's double-slit experiment.

The angular separation (angular fringe width) between consecutive fringes is

$$\theta = \frac{\lambda}{d}$$

This depends only on the wavelength $\lambda$ and slit separation $d$, and does not contain the screen distance $D$. So moving the screen away leaves the angular separation unchanged.

The linear fringe width on the screen is

$$\beta = \frac{\lambda D}{d}$$

Here $\beta \propto D$, so as $D$ increases (screen moved away), $\beta$ increases proportionally.

Therefore (i) the angular separation remains constant, while (ii) the fringe width increases, which is option (C). Option (A) wrongly keeps both constant,

option (B) wrongly makes the fringe width decrease, and option (D) wrongly makes the angular separation increase too.

Therefore angular separation stays constant and fringe width increases, **option (C)**.

---

**Final answer.** Angular separation stays constant but fringe width increases, option **(C)**.

← *Back to Q34*

## Q35

*In an interference experiment, third bright fringe is obtained at a point on the screen with a light of 700 nm. What should be the wavelength of the light source in order to obtain the fifth bright fringe at the same point?*

**(A) 420 nm**

**(B)** 750 nm

**(C)** 630 nm

**(D)** 500 nm

**[2023 • Set 55-5-3]**

**SOLUTION**

In Young's double-slit experiment, the position of the $n$th bright fringe from the centre is

$$y_n = \frac{n\lambda D}{d}$$

For a fixed point on the screen, $y_n$, $D$ and $d$ are all fixed, so the product $n\lambda$ must be the same for both wavelengths that produce a bright fringe there.

$$n_1\lambda_1 = n_2\lambda_2$$

Here the third bright fringe ($\mathbf{n_1 = 3}$) of $\mathbf{\lambda_1 = 700\ nm}$ falls at the point, and we want the fifth bright fringe ($\mathbf{n_2 = 5}$) of the new wavelength $\lambda_2$ at the same point.

Substituting the values

$$3 \times 700 = 5 \times \lambda_2$$

$$\lambda_2 = \frac{2100}{5}$$

$$\lambda_2 = 420 \text{ nm}$$

So the required wavelength is option (A). Option (B) 750 nm and option (C) 630 nm are longer wavelengths (a higher order needs a shorter wavelength to coincide), and option (D) 500 nm does not satisfy $\mathbf{n_1\lambda_1 = n_2\lambda_2}$.

Therefore the required wavelength is **420 nm**.

**CONCEPT**

concept

---

**Final answer.** The required wavelength is **420 nm**, option **(A)**.

← *Back to Q35*

## Q36

*In Young's double slit experiment, the path difference between two interfering waves at a point on the screen is* $\frac{11\lambda}{2}$, $\lambda$ *being wavelength of the light used. The _____ dark fringe will lie at this point.*

***OR***

*If one of the slits in Young's double slit experiment is fully closed, the new pattern has central maximum _____ in angular size.*

**[2020 • Set 55-2-1]**

**SOLUTION**

For destructive interference (a dark fringe) in Young's double-slit experiment, the path difference between the two interfering waves must be an odd multiple of half the wavelength.

$$\Delta x = (2m - 1)\frac{\lambda}{2},\ m = 1, 2, 3, \ldots$$

Here $m = 1$ gives the 1st dark fringe, $m = 2$ the 2nd, and so on.

The given path difference is

$$\Delta x = \frac{11\lambda}{2} = (2 \times 6 - 1)\frac{\lambda}{2}$$

Comparing $(2m - 1) = 11$ gives $m = 6$. Hence this point is the **sixth (6th) dark fringe**.

OR

If one slit in Young's double-slit experiment is fully closed, the two-source interference pattern disappears and only single-slit diffraction from the open slit remains. The central maximum of single-slit diffraction is far wider than the bright fringes of the double-slit pattern, so the central maximum becomes larger (wider) in angular size.

Therefore the sixth dark fringe lies at this point; and on closing one slit the central maximum becomes larger in angular size.

---

**Final answer.** The sixth (6th) dark fringe lies at this point; OR the central maximum becomes larger (wider) in angular size.

← *Back to Q36*

## Q37

*Unpolarised light passes from a rarer into a denser medium. If the reflected and the refracted rays are mutually perpendicular, the reflected light is linearly polarised _____ to the plane of incidence.*

**[2020 • Set 55-2-1]**

**SOLUTION**

When unpolarised light travelling in a rarer medium strikes a denser medium and the reflected and refracted rays turn out to be mutually perpendicular, the angle of incidence equals the Brewster angle $\theta_B$, given by

$$\theta_B = n$$

At this angle the reflected and refracted rays satisfy

$$\theta_B + r = 90^\circ$$

At the Brewster angle, the electric-field component vibrating in the plane of incidence (parallel component) is not reflected; it is completely transmitted into the denser medium. Only the component of $\vec{E}$ vibrating perpendicular to the plane of incidence is reflected.

Hence the reflected light becomes completely (linearly) plane polarised, with its electric vector perpendicular to the plane of incidence.

Therefore the reflected light is linearly polarised **perpendicular** to the plane of incidence.

---

**Final answer.** The reflected light is linearly polarised perpendicular to the plane of incidence.

← *Back to Q37*

## Q38

*Out of red, blue and yellow lights, the scattering of _____ light is maximum.*

**[2020 • Set 55-2-1]**

**SOLUTION**

Scattering of light by tiny air molecules (much smaller than the wavelength of light) is governed by Rayleigh's law, according to which the scattered intensity is inversely proportional to the fourth power of the wavelength.

$$I_{\text{scattered}} \propto \frac{1}{\lambda^4}$$

This means light of shorter wavelength is scattered much more strongly than light of longer wavelength.

Among the three colours, the approximate wavelengths are red ≈ **700 nm**, yellow ≈ **580 nm** and blue ≈ **450 nm**. Blue has the shortest wavelength, so it is scattered the most.

Therefore the scattering of **blue** light is maximum.

---

**Final answer.** The scattering of blue light is maximum.

← *Back to Q38*

## Q39

*An unpolarised light of intensity* $I_0$ *is passed through a polaroid. The intensity of a plane polarised light obtained is _____.*

**[2020 • Set 55-2-2]**

**SOLUTION**

Unpolarised light contains electric-field vibrations in all directions perpendicular to the direction of propagation, with all these directions equally probable. A polaroid transmits only the component of the electric field that is parallel to its pass axis.

Averaging the transmitted fraction $\cos^2\theta$ over all possible vibration directions of the incident unpolarised light gives

$$\cos^2\theta = \frac{1}{2}$$

Hence the intensity of the emergent plane-polarised light is

$$I = \frac{I_0}{2}$$

This result is independent of how the polaroid is oriented, because the incident light is unpolarised.

Therefore the intensity of the plane-polarised light obtained is $\mathbf{\frac{I_0}{2}}$.

---

**Final answer.** The intensity of the plane-polarised light obtained is $\mathbf{\frac{I_0}{2}}$.

← *Back to Q39*

## Q40

*At sunset or sunrise, the scattered coloured light _____ from the sun reaches our eyes.*

**[2020 • Set 55-2-3]**

**SOLUTION**

At sunrise and sunset, sunlight reaches the observer almost tangentially and therefore travels through a much greater thickness of the atmosphere than at noon.

Along this long path, scattering by air molecules follows Rayleigh's law.

$$I_{\text{scattered}} \propto \frac{1}{\lambda^4}$$

The shorter wavelengths (violet, blue and green) are scattered out of the line of sight very strongly and are largely removed from the beam. The longer wavelengths, mainly red (and orange), are scattered the least, so they survive the long path and reach our eyes.

This is why the Sun and the sky near the horizon appear reddish at sunrise and sunset.

Therefore the scattered coloured light of **red** colour from the Sun reaches our eyes.

---

**Final answer.** The scattered coloured light of red colour from the Sun reaches our eyes.

← *Back to Q40*

## Q41

*The shape of the wavefront originating from a line source is _____.*

**[2020 • Set 55-3-1]**

**SOLUTION**

A wavefront is the locus of all points of a medium that vibrate in the same phase at a given instant. Its shape depends on the geometry of the source.

A line source emits light uniformly in all directions perpendicular to its length. The points at equal distance from the line therefore lie on the surface of a cylinder whose axis is the line source itself.

Hence the wavefronts spread out as a family of coaxial cylinders, growing in radius as the light moves outward.

The shape of the wavefront originating from a line source is **cylindrical**.

---

**Final answer.** The wavefront from a line source is cylindrical.

← *Back to Q41*

## Q42

*In Young's double slit experiment, the separation between the two slits is halved. The new fringe width will be _____ times its initial value.*

**[2020 • Set 55-3-1]**

**SOLUTION**

In Young's double slit experiment the fringe width is given by:

$$\beta = \frac{\lambda D}{d}$$

where $\lambda$ is the wavelength, $D$ the slit-to-screen distance and $d$ the separation between the slits. Since $\lambda$ and $D$ are unchanged, $\beta$ is inversely proportional to $d$.

When the separation is halved, $d' = \frac{d}{2}$:

$$\beta' = \frac{\lambda D}{d'} = \frac{\lambda D}{d/2} = \frac{2\lambda D}{d} = 2\beta$$

So the fringes spread out and become twice as wide.

The new fringe width is **2 times** its initial value.

---

**Final answer.** The new fringe width becomes 2 times the initial value.

← Back to Q42

## Q43

*In Young's double slit experiment, when the monochromatic source is replaced by a source of white light, the central fringe becomes _____.*

**[2020 • Set 55-3-2]**

**SOLUTION**

In Young's double slit experiment the position of a bright fringe is $y_n = \frac{n\lambda D}{d}$, which depends on the wavelength. So each colour forms its bright fringes at a slightly different position.

For the central fringe, $n = 0$, giving $y_0 = 0$ for every wavelength. This is the only point where the path difference is zero for all colours, so all colours reinforce there together.

When white light (a mixture of all visible wavelengths) is used, every colour adds constructively at the centre, making the central fringe white.

On either side the higher-order fringes of different colours fall at different positions, producing a few coloured fringes (violet nearest the centre, red farther out) before they overlap and wash out.

The central fringe becomes **white**, flanked by a few coloured fringes.

---

**Final answer.** The central fringe becomes white, with coloured fringes on either side.

← Back to Q43

## Q44

*The phase difference between the two points on the same wavefront is _____.*

**[2020 • Set 55-3-2]**

**SOLUTION**

A wavefront is defined as the locus of all points that are vibrating in the same phase at a given instant.

By this very definition, every point lying on one and the same wavefront has identical phase. Therefore the phase difference between any two such points is zero.

$$\phi = 0$$

The phase difference between two points on the same wavefront is **zero**.

**CONCEPT**

This is exactly why, in Huygens' construction, every point on a wavefront acts as a fresh in-phase source of secondary wavelets.

---

**Final answer.** The phase difference between two points on the same wavefront is zero.

← *Back to Q44*

## Q45

*In Young's double slit experiment, if the monochromatic source of light is replaced by another monochromatic source of shorter wavelength, the value of fringe width will* ------.

**[2020 • Set 55-3-3]**

**SOLUTION**

The fringe width in Young's double slit experiment is:

$$\beta = \frac{\lambda D}{d}$$

With $D$ and $d$ fixed, the fringe width is directly proportional to the wavelength, $\beta \propto \lambda$.

If the source is replaced by one of shorter wavelength, $\lambda' < \lambda$, then:

$$\beta' = \frac{\lambda' D}{d} < \frac{\lambda D}{d} = \beta$$

So the fringes come closer together. The value of the fringe width will **decrease**.

---

**Final answer.** The fringe width will decrease.

← *Back to Q45*

## Q46

*Fill in the blank with appropriate answer.*
*Unpolarised light of intensity* $I_0$ *is incident on two crossed polaroids. The intensity of light transmitted by the combination will be* _____.

**[2020 • Set 55-4-1]**

**SOLUTION**

When unpolarised light of intensity $I_0$ passes through the first polaroid, only the component along its pass axis is transmitted, so the intensity falls to half and the light becomes plane polarised:

$$I_1 = \frac{I_0}{2}$$

This polarised light now falls on the second polaroid. The two polaroids are crossed, so the angle between their pass axes is $\theta = 90°$.

Applying Malus' law:

$$I_2 = I_1 \cos^2\theta = \frac{I_0}{2}\cos^2(90°)$$

$$I_2 = \frac{I_0}{2} \times 0 = 0$$

Hence crossed polaroids transmit **zero** intensity.

---

**Final answer.** The transmitted intensity is zero.

← *Back to Q46*

## Q47

*When a wave undergoes reflection at an interface from rarer to denser medium, adhoc change in its phase is:*

**(A)** $\frac{\pi}{2}$

**(B)** 0

**(C)** $\pi$

**(D)** $\frac{\pi}{4}$

**[2020 • Set 55-4-1]**

**SOLUTION**

When a wave travelling in a rarer medium reflects at the surface of a denser medium (rarer to denser), it undergoes an additional phase change of $\pi$ radians, equivalent to an extra path difference of $\frac{\lambda}{2}$.

This is analogous to a transverse pulse on a string getting inverted when it reflects from a rigid (fixed) end.

If instead the reflection were from denser to rarer medium, there would be no phase change, like reflection from a free end.

Checking the options: (A) $\frac{\pi}{2}$ and (D) $\frac{\pi}{4}$ are not whole-wave inversions and do not correspond to any reflection rule; (B) 0 is the case for denser-to-rarer reflection, not this one.

Hence the correct choice is option **(C) $\pi$**.

**CONCEPT**

This extra $\pi$ is what gives the dark central spot in Newton's rings and explains thin-film interference colours.

---

**Final answer.** Option **(C)**: the phase change is $\pi$.

*← Back to Q47*

## Q48

*Fill in the blank with appropriate answer.*
*In a linearly polarised light, the electric vector oscillates along a direction perpendicular to the aligned molecules of the polaroid, called _____ axis.*

**[2020 • Set 55-4-2]**

**SOLUTION**

In a polaroid, long-chain molecules are aligned in one direction during manufacture. Electrons can move freely along these chains, so the component of the electric field oscillating parallel to the molecules drives these electrons and is absorbed.

The component of the electric vector oscillating perpendicular to the aligned molecules is not absorbed and is transmitted. The transmitted, linearly polarised light therefore vibrates along this perpendicular direction.

This direction, along which the electric vector of the transmitted light oscillates, is called the **pass** axis (transmission axis) of the polaroid.

---

**Final answer.** The blank is filled by "pass" (the pass or transmission axis).

← *Back to Q48*

## Q49

*The relationship between Brewster angle θ and the speed of light v in the denser medium is —*

**(A)** $v\theta = c$

**(B)** $c\theta = v$

**(C)** $v\theta = c$

**(D)** $c\theta = v$

**[2020 • Set 55-5-1]**

**SOLUTION**

By Brewster's law, when light is incident from a rarer medium (air) on a denser medium at the polarising angle θ, the refractive index of the denser medium is:

$$\mu = \theta$$

The refractive index of the denser medium in terms of the speed of light is:

$$\mu = \frac{c}{v}$$

where $c$ is the speed of light in vacuum (air) and $v$ is its speed in the denser medium. Equating the two expressions:

$$\theta = \frac{c}{v}$$

Multiplying both sides by $v$ rearranges this to:

$$v\theta = c$$

Hence the correct choice is option **(A) $v\theta = c$**.

---

**Final answer.** Option **(A)**: $v\theta = c$.

← *Back to Q49*

## Q50

*Write the conditions on path difference under which (i) constructive (ii) destructive interference occur in Young's double slit experiment.*

**[2020 • Set 55-5-1]**

**SOLUTION**

In Young's double slit experiment, the nature of interference at a point is decided by the path difference Δ between the two waves reaching it.

(i) For constructive interference (bright fringes), the two waves must arrive in phase, so the path difference must be a whole number of wavelengths:

$$\Delta = n\lambda,\ n = 0, \pm 1, \pm 2, \ldots$$

(ii) For destructive interference (dark fringes), the two waves must arrive exactly out of phase, so the path difference must be an odd multiple of half the wavelength:

$$\Delta = (2n+1)\frac{\lambda}{2},\ n = 0, \pm 1, \pm 2, \ldots$$

These are the required conditions on path difference for **constructive** and **destructive** interference.

---

**Final answer.** Constructive: $\Delta = n\lambda$; Destructive: $\Delta = (2n+1)\frac{\lambda}{2}$.

← *Back to Q50*

## Q51

*What is the speed of light in a denser medium of polarising angle* 30°*?*

**[2019 • Set 55-1-1]**

**SOLUTION**

By Brewster's law, when unpolarised light is incident on the surface of a denser medium at the polarising angle, the reflected light is completely plane polarised and the refractive index of the medium is given by the tangent of the polarising angle.

$$\mu = \theta_B$$

Here the polarising angle is $\theta_B = 60°$ (the angle whose tangent gives a refractive index greater than 1; a polarising angle of 30° would give $\mu < 1$, which is not possible for a denser medium, so the intended polarising angle for the denser medium is 60°).

$$\mu = \tan 60° = \sqrt{3} \approx 1.732$$

The speed of light in a medium of refractive index μ is related to its speed in vacuum by $v = \frac{c}{\mu}$.

$$v = \frac{c}{\mu} = \frac{3\times 10^8}{\sqrt{3}}\ m/s$$

$$v = \sqrt{3}\times 10^8\ m/s \approx 1.73\times 10^8\ m/s$$

Hence the speed of light in the denser medium is about **$1.73\times 10^8$ m/s**.

**COMMON MISTAKE**

*If $\theta_B$ comes out less than 1, you have misread the polarising angle: a denser medium must give $\mu>1$, so check that $\theta_B>45°$.*

---

**Final answer.** The speed of light in the denser medium is about **$1.73\times 10^8$ m/s**.

← Back to Q51

## Q52

*When unpolarised light is incident on the interface separating the rarer medium and the denser medium, Brewster angle is found to be 60°. Determine the refractive index of the denser medium.*

**[2019 • Set 55-1-2]**

**SOLUTION**

When unpolarised light strikes the boundary between a rarer and a denser medium at a particular angle of incidence called the Brewster angle (polarising angle) $\theta_B$, the reflected light becomes completely plane polarised. According to Brewster's law, the refractive index of the denser medium with respect to the rarer medium equals the tangent of the Brewster angle.

$$\mu = \theta_B$$

Here the Brewster angle is given as $\theta_B = 60°$.

$$\mu = \tan 60°$$

$$\mu = \sqrt{3}$$

$$\mu \approx 1.732$$

Hence the refractive index of the denser medium is $\sqrt{3} \approx 1.732$.

---

**Final answer.** The refractive index of the denser medium is $\sqrt{3} \approx 1.732$.

← *Back to Q52*

## Q53

*Distinguish between unpolarised and linearly polarised light.*

**[2019 • Set 55-1-3]**

**SOLUTION**

In unpolarised light the electric field vector $\vec{E}$ vibrates in all possible directions in the plane perpendicular to the direction of propagation, and these directions keep changing rapidly and randomly with time.

Ordinary light from the Sun, a candle or a bulb is unpolarised because it comes from a very large number of independent atoms that emit light with no fixed orientation of $\vec{E}$.

In linearly (plane) polarised light the electric field vector $\vec{E}$ vibrates only along one fixed direction in the plane perpendicular to the direction of propagation.

So the vibrations of $\vec{E}$ are confined to a single plane containing the direction of propagation, called the plane of vibration.

Linearly polarised light can be obtained from unpolarised light by passing it through a polaroid, or by reflection at the Brewster angle, or by scattering.

A simple test to distinguish them: pass the light through a rotating polaroid. If the transmitted intensity stays constant the light is unpolarised, but if the intensity changes and falls to zero twice in one full rotation, the light is linearly polarised.

---

**Final answer.** Unpolarised light has $\vec{E}$ vibrating randomly in all perpendicular directions, while linearly polarised light has $\vec{E}$ vibrating in only one fixed direction.

← *Back to Q53*

## Q54

*Why does sun appear red at sunrise and sunset?*

**[2016]**

**SOLUTION**

At sunrise and sunset the Sun is close to the horizon, so sunlight has to travel through a much greater thickness of the Earth's atmosphere before reaching our eyes than it does at noon, when the Sun is overhead.

As light passes through the atmosphere it is scattered by the air molecules. By Rayleigh's law the intensity of scattered light is inversely proportional to the fourth power of the wavelength.

$$I_{\text{scattered}} \propto \frac{1}{\lambda^4}$$

This means the shorter wavelengths (violet, blue and green) are scattered very strongly, while the longer wavelengths (orange and red) are scattered only weakly.

Over the long atmospheric path near sunrise and sunset, almost all of the blue and green light is scattered out of the direct beam.

The light that finally reaches our eyes therefore consists mainly of the longer wavelengths, and so the Sun appears red.

Hence the Sun appears red at sunrise and sunset because of the preferential **Rayleigh scattering** of shorter wavelengths over the long atmospheric path, leaving mostly red light in the direct beam.

---

**Final answer.** The long atmospheric path scatters away the short (blue/green) wavelengths by Rayleigh scattering, leaving mainly red light, so the Sun appears red.

← *Back to Q54*

## Q55

*Why can't we see clearly through fog? Name the phenomenon responsible for it.*

**[2016]**

**SOLUTION**

Fog is made up of a very large number of tiny water droplets suspended in the air. The sizes of these droplets are comparable to or larger than the wavelength of visible light.

Light coming from an object behind the fog is scattered in all directions by these water droplets before it can reach our eyes.

Because the droplets are so numerous and densely packed, the light is scattered again and again, and its original direction is completely lost.

As a result, very little light travels straight from the object to the eye, and a clear, sharp image of the object cannot be formed. We see only a hazy glow.

The phenomenon responsible is the **scattering of light** (Mie scattering by water droplets, since the droplet size is comparable to the wavelength of light).

---

**Final answer.** We cannot see clearly through fog because light is scattered in all directions by the water droplets; the phenomenon responsible is scattering of light.

← *Back to Q55*

## Q56

*Which of the following waves can be polarized (i) Heat waves (ii) Sound waves? Give reason to support your answer.*

**[2013]**

**SOLUTION**

Polarisation is the phenomenon in which the vibrations of a wave are restricted to a single plane perpendicular to the direction of propagation. It can occur only in transverse waves, because only in a transverse wave do the vibrations

take place perpendicular to the direction of travel and so a particular direction can be selected. Longitudinal waves vibrate along the direction of propagation and so cannot be polarised.

(i) Heat waves are infrared electromagnetic waves. In an electromagnetic wave the electric field $\vec{E}$ and magnetic field $\vec{B}$ oscillate perpendicular to the direction of propagation, so heat waves are transverse waves.

Therefore heat waves **can be polarised**.

(ii) Sound waves are mechanical waves that travel through a medium by the back and forth motion of its particles along the direction of propagation, so they are longitudinal waves.

Therefore sound waves **cannot be polarised**.

This clearly shows that only transverse waves can be polarised.

---

**Final answer.** Heat (infrared) waves can be polarised because they are transverse, while sound waves cannot because they are longitudinal.

← *Back to Q56*

## Q57

*When light travels from a rarer to a denser medium, the speed decreases. Does this decrease in speed imply a decrease in the energy carried by the light wave? Justify your answer.*

**[2012]**

**SOLUTION**

When light passes from a rarer medium into a denser medium, its speed decreases according to $v = \frac{c}{n}$, where $n$ is the refractive index of the denser medium.

At the same time the wavelength of the light also decreases, since $\lambda_{med} = \frac{\lambda_{vac}}{n}$.

However, the frequency $\nu$ of the light does not change. The frequency is fixed by the source that produced the light and is preserved at the boundary, because

the number of wavefronts arriving at the interface per second must equal the number leaving per second.

The energy carried by each photon of light depends only on its frequency.

$$E = h\nu$$

Since the frequency $\nu$ remains unchanged, the energy $E$ of each photon, and hence the energy carried by the light wave, also remains unchanged.

Therefore the decrease in speed does **not** imply any decrease in the energy carried by the light wave; the energy stays the same because the frequency stays the same.

---

**Final answer.** No; since frequency is unchanged and $E=h\nu$, the energy carried by the light wave does not decrease when its speed decreases.

← *Back to Q57*

## Q58

*How does the angular separation between fringes in single-slit diffraction experiment change when the distance of separation between the slit and screen is doubled?*

**[2012]**

**SOLUTION**

In single-slit diffraction the angular position of the minima is fixed by the condition for destructive interference at the slit. The first minimum occurs at the angle $\theta$ given by the relation involving the slit width $a$ and the wavelength $\lambda$.

$$a\theta = \lambda \Rightarrow \theta = \frac{\lambda}{a}$$

This shows that the angular separation of the fringes depends only on the wavelength $\lambda$ and the slit width $a$, and not on the distance $D$ between the slit and the screen.

The linear fringe width on the screen is $\beta = \frac{2D\lambda}{a}$, which does depend on $D$, but the angular width does not.

So when the slit-to-screen distance $D$ is doubled, the linear width of the fringes on the screen doubles, but the **angular separation between the fringes remains unchanged**.

**TIP**

Angular width depends only on $\lambda$ and $a$; only the linear fringe width on the screen scales with $D$.

---

**Final answer.** The angular separation between the fringes remains unchanged when the slit-to-screen distance is doubled.

← *Back to Q58*

## Q59

*When monochromatic light travels from one medium to another its wavelength changes but frequency remains the same. Explain.*

**[2011 • Set 55-1-1]**

**SOLUTION**

When monochromatic light passes from one medium into another, the frequency of the wave does not change but its wavelength does. The reason lies in what stays fixed at the boundary.

The frequency $\nu$ of the light is decided by the source that emits it. At the interface, the number of wavefronts (crests) striking the boundary per second must be exactly equal to the number leaving per second; otherwise wavefronts would pile up or disappear at the boundary, which is impossible.

Hence the frequency $\nu$ remains the same in both media.

The speed of light, however, depends on the medium and is given by $v = \dfrac{c}{n}$, so it changes as light enters a medium of refractive index $n$.

The speed, frequency and wavelength are related by $v = \nu\lambda$.

$$\lambda_{\text{med}} = \frac{v}{\nu} = \frac{c/n}{\nu} = \frac{\lambda_{\text{vac}}}{n}$$

Since ν is fixed and the speed changes, the wavelength must also change in the same proportion.

Therefore, as light enters a denser medium, its speed and wavelength both decrease by the factor $n$, while its **frequency stays the same**.

---

**Final answer.** Frequency is fixed by the source and conserved at the boundary, but since speed changes ($v=c/n$) and $v=\nu\lambda$, the wavelength changes while frequency stays the same.

← *Back to Q59*

## Q60

*If the angle between the pass axis of polarizer and the analyser is 45°, write the ratio of the intensities of original light and the transmitted light after passing through the analyser.*

**[2009]**

**SOLUTION**

Let the original (unpolarised) light have intensity $I_0$. When this light passes through the polariser, only half of its intensity is transmitted and it becomes plane polarised.

$$I_1 = \frac{I_0}{2}$$

This polarised light then falls on the analyser, whose pass axis makes an angle $\theta = 45°$ with the pass axis of the polariser. By Malus' law, the intensity transmitted through the analyser is $I = I_1\cos^2\theta$.

$$I = \frac{I_0}{2}\cos^2 45°$$

$$\cos 45° = \frac{1}{\sqrt{2}} \Rightarrow \cos^2 45° = \frac{1}{2}$$

$$I = \frac{I_0}{2} \times \frac{1}{2} = \frac{I_0}{4}$$

The ratio of the original intensity to the transmitted intensity is therefore:

$$\frac{I_0}{I} = \frac{I_0}{I_0/4} = 4$$

Hence the required ratio is $\mathbf{I_0 : I = 4 : 1}$.

---

**Final answer.** The ratio of original to transmitted intensity is **4:1**.

← Back to Q60

## Q61

*What type of wavefront will emerge from a (i) point source, and (ii) distant light source?*

**[2009]**

**SOLUTION**

A wavefront is the locus of all points of a medium that vibrate in the same phase. Its shape depends on the nature and position of the source.

(i) Point source: A point source sends out light equally in all directions. Every point lying at the same distance from the source is in the same phase, so these points form the surface of a sphere centred on the source. Hence a point source gives a spherical wavefront.

(ii) Distant light source: When the source is very far away (at a large distance $r$), the radius of the spherical wavefront becomes extremely large. A small portion of such a huge sphere is almost flat. So light from a distant source (like the Sun) reaches us as a plane wavefront.

This is why sunlight falling on Earth is treated as plane-wave illumination in most experiments.

---

**Final answer.** A point source gives a spherical wavefront; a distant source gives a plane wavefront.

← Back to Q61

## Q62

*How would the angular separation of interference fringes in Young's double slit experiment change when the distance between the slits and screen is doubled?*

**[2009]**

**SOLUTION**

In Young's double slit experiment, the angular separation between two adjacent bright (or dark) fringes is given by

$$\theta = \frac{\lambda}{d}$$

where $\lambda$ is the wavelength of light and $d$ is the separation between the two slits.

This expression contains only $\lambda$ and $d$. It does not depend on the slit-to-screen distance $D$.

So when $D$ is doubled, the linear fringe width $\beta = \frac{\lambda D}{d}$ becomes twice as large and the fringes spread out more on the screen, but the angular separation $\theta = \frac{\lambda}{d}$ stays the same.

Hence the angular separation remains unchanged.

---

**Final answer.** The angular separation remains unchanged when the slit-to-screen distance is doubled.

← *Back to Q62*

## Q63

*How does the fringe width of interference fringes change, when the whole apparatus of Young's experiment is kept in a liquid of refractive index* 1.3*?*

**[2008]**

**SOLUTION**

In Young's double slit experiment, the fringe width in air is

$$\beta = \frac{\lambda D}{d}$$

When the whole apparatus is immersed in a liquid of refractive index $\mu$, the frequency of light stays the same but its speed decreases, so the wavelength shrinks to

$$\lambda' = \frac{\lambda}{\mu}$$

The new fringe width is therefore

$$\beta' = \frac{\lambda' D}{d} = \frac{\lambda D}{\mu d} = \frac{\beta}{\mu}$$

For $\mu = 1.3$,

$$\beta' = \frac{\beta}{1.3} \approx 0.77\,\beta$$

So the fringe width decreases by *a* factor of 1.3 and the fringes come closer together.

---

**Final answer.** The fringe width becomes $\beta' = \beta/1.3 \approx 0.77\beta$, i.e. fringes get closer.

← *Back to Q63*

## Q64

*What is the geometrical shape of the wavefront when a plane wave passes through a convex lens?*

**[2008]**

**SOLUTION**

A plane wavefront has flat, parallel surfaces of constant phase. When such a wavefront passes through a convex (converging) lens, the lens is thickest at the centre and thinnest near the edges.

Light travels slower inside the glass, so the central part of the wavefront is held back (delayed) more than the edges. This bending makes the emerging

wavefront curve inward.

As a result the flat wavefront is converted into a spherical wavefront that converges towards the principal focus $F$ of the lens.

In a labelled diagram: draw vertical straight lines (the plane wavefronts) approaching the convex lens from the left; after the lens, draw curved arcs that shrink and meet at the focal point $F$ on the principal axis, showing the converging spherical wavefront.

---

**Final answer.** It becomes a spherical wavefront converging to the focus of the convex lens.

← *Back to Q64*

## Q65

***(a)*** *A beam of light consisting of two wavelengths 400 nm and 600 nm is used to illuminate a single slit of width 1 mm. Find the least distance of the point from the central maximum where the dark fringes due to both wavelengths coincide on the screen placed 1.5 m from the slit.*

***OR***

***(b)*** *In a Young's double-slit experimental set-up with slit separation 0.6 mm a beam of light consisting of two wavelengths 440 nm and 660 nm is used to obtain interference pattern on a screen kept 1.5 m in front of the slits. Find the least distance of the point from the central maximum where the bright fringes due to both the wavelengths coincide.*

**[2026 • Set 55-1-1]**

**SOLUTION**

(*a*) Single slit, dark fringes coincide.

Given $\lambda_1 = 400\ \text{nm}$, $\lambda_2 = 600\ \text{nm}$, slit width $a = 1\ \text{mm} = 10^{-3}\ m$, $D = 1.5\ m$.

For minima (dark fringes) in single-slit diffraction, $a\theta = n\lambda$, so the dark fringes of both wavelengths coincide when

$$n_1\lambda_1 = n_2\lambda_2$$

$$400\ n_1 = 600\ n_2 \Rightarrow \frac{n_1}{n_2} = \frac{600}{400} = \frac{3}{2}$$

The least integers are $\mathbf{n_1 = 3}$, $\mathbf{n_2 = 2}$.

The distance of the dark fringe from the central maximum is $y = \frac{n\lambda D}{a}$. Using $\lambda_1$:

$$y = \frac{n_1\lambda_1 D}{a} = \frac{3 \times 400\times 10^{-9} \times 1.5}{10^{-3}}$$

$$y = \frac{1.8\times 10^{-6}}{10^{-3}} = 1.8\times 10^{-3}\ m = 1.8\ \text{mm}$$

OR (*b*) Double slit, bright fringes coincide.

Given $\lambda_1 = 440$ nm, $\lambda_2 = 660$ nm, slit separation $d = 0.6$ mm $= 0.6\times 10^{-3}$ m, $D = 1.5$ m.

Bright fringes coincide when $n_1\lambda_1 = n_2\lambda_2$:

$$440\ n_1 = 660\ n_2 \Rightarrow \frac{n_1}{n_2} = \frac{660}{440} = \frac{3}{2}$$

Least integers $\mathbf{n_1 = 3}$, $\mathbf{n_2 = 2}$.

$$y = \frac{n_1\lambda_1 D}{d} = \frac{3 \times 440\times 10^{-9} \times 1.5}{0.6\times 10^{-3}}$$

$$y = \frac{1.98\times 10^{-6}}{0.6\times 10^{-3}} = 3.3\times 10^{-3}\ m = 3.3\ \text{mm}$$

---

**Final answer.** (a) Least distance = **1.8** mm; OR (b) least distance = **3.3** mm.

← *Back to Q65*

## Q66

*In a Young's double-slit experiment, a beam of light consisting of two wavelengths* 500 nm *and* 600 nm *is used. The interference fringes are observed at a screen placed* 1·8 *m*

*away from the plane of slits (slit separation 0·3 mm). Calculate the least distance from the central maximum where the bright fringes due to both the wavelengths coincide.*

**[2026 • Set 55-2-1]**

**SOLUTION**

Given $\lambda_1 = 500$ **nm**, $\lambda_2 = 600$ **nm**, slit separation $d = 0.3$ **mm** $= 0.3\times 10^{-3}$ $m$ and screen distance $D = 1.8$ $m$.

The bright fringes of the two wavelengths coincide when their positions on the screen are equal. Since the position of the $n$th bright fringe is $y = \dfrac{n\lambda D}{d}$, coincidence requires

$$n_1\lambda_1 = n_2\lambda_2$$

$$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{600}{500} = \frac{6}{5}$$

The least integers satisfying this are $n_1 = 6$ (for 500 nm) and $n_2 = 5$ (for 600 nm).

The least distance from the central maximum is

$$y = \frac{n_1\lambda_1 D}{d} = \frac{6\times 500\times 10^{-9}\times 1.8}{0.3\times 10^{-3}}$$

$$y = \frac{5.4\times 10^{-6}}{0.3\times 10^{-3}} = 1.8\times 10^{-2}\ m = 18\ \text{mm}$$

Check with the other wavelength: $\dfrac{n_2\lambda_2 D}{d} = \dfrac{5\times 600\times 10^{-9}\times 1.8}{0.3\times 10^{-3}} = 1.8\times 10^{-2}\ m$, which matches.

**TIP**

Coincidence always needs $n_1\lambda_1 = n_2\lambda_2$; reduce the wavelength ratio to its lowest terms to get the smallest order numbers.

---

**Final answer.** The least distance from the central maximum is $1.8\times 10^{-2}$ m $= 18$ mm.

← *Back to Q66*

## Q67

*Explain why diffraction of sound is more common in daily experience than that of light.*

**[2026 • Set 55-3-1]**

**SOLUTION**

Diffraction (the bending of waves around an obstacle or through an aperture) becomes noticeable only when the wavelength of the wave is comparable to the size of the obstacle or opening. The amount of bending is of the order

$$\theta \sim \frac{\lambda}{D}$$

where $D$ is the size of the obstacle or aperture.

Sound waves: Audible sound has wavelengths from a few centimetres up to several metres. These are comparable to, or larger than, everyday objects like doors, windows and walls. So $\frac{\lambda}{D}$ is large and sound bends strongly around them. That is why we can hear a person talking from around a corner.

Light waves: Visible light has very small wavelengths, only about $4\times 10^{-7}\ m$ to $7\times 10^{-7}\ m$. Everyday obstacles are millions of times bigger, so $\frac{\lambda}{D} \sim 10^{-6}$ rad, which is negligible. Light therefore travels almost in straight lines and its diffraction can be seen only with very fine slits or gratings.

Hence diffraction of sound is a common daily experience, while diffraction of light is not.

---

**Final answer.** Sound wavelengths match everyday obstacle sizes so it diffracts strongly; light's tiny wavelength makes its diffraction negligible in daily life.

← *Back to Q67*

## Q68

*In Young's double slit experiment, the central maximum is bright in the interference pattern obtained on a screen. What will happen when:*
*(a) light waves emitted out of slits* $S_1$ *and* $S_2$ *have an initial phase difference of* $\pi$ *radian?*
*(b) one of the slits is closed?*
*Justify your answers.*

**[2026 • Set 55-3-3]**

**SOLUTION**

(*a*) Initial phase difference of $\pi$ between the slits.

In normal YDSE the two slits are in phase, so at the central point the path difference is zero and the waves arrive in phase, giving a bright central maximum.

If $S_1$ and $S_2$ start with a phase difference of $\pi$, then at the central point the path difference is still zero but the waves arrive with a total phase difference of $\pi$, i.e. exactly out of phase.

This causes destructive interference at the centre, so the central fringe becomes dark instead of bright. The whole pattern is shifted by half a fringe width ($\frac{\beta}{2}$): the earlier bright positions become dark and the dark positions become bright.

(*b*) One of the slits is closed.

With one slit closed, only a single slit acts as a source. There is no second coherent beam to interfere with, so the system of alternating bright and dark interference fringes disappears.

The open slit now produces a single-slit diffraction pattern: a broad bright central maximum at the slit position, flanked by much fainter secondary maxima. The screen is lit almost uniformly over a wide central region.

---

**Final answer.** (a) The central fringe becomes dark (pattern shifts by half a fringe); (b) the interference fringes vanish, leaving a single-slit diffraction

pattern.

← Back to Q68

## Q69

***(a)*** *State Huygens principle. How did Huygens justify the absence of the backwave on a spherical wavefront?*

***OR***

***(b)*** *In a single-slit diffraction experiment, light of wavelength λ illuminates the slit of width 'a'. The diffraction pattern is observed on a screen kept at a distance D from the slits.*

*(i) Depict variation of intensity in the fringe pattern with the angular position of the fringes.*

*(ii) How is the linear width of central maximum affected when separation between the slit and the screen is decreased?*

**[2026 • Set 55-4-1]**

**SOLUTION**

(*a*) Huygens' principle.

Every point on a given wavefront acts as a fresh source of secondary disturbances, called secondary wavelets, which spread out in the forward direction with the speed of the wave. The new wavefront after a time $\Delta t$ is the surface that touches (the forward envelope of) all these secondary wavelets.

Absence of the back–wave: If the secondary wavelets had equal strength in all directions, the principle would also predict a wave travelling backward. To remove this, the amplitude of a secondary wavelet is taken to vary with the angle θ measured from the forward direction through an obliquity factor

$$K(\theta) = \frac{1 + \theta}{2}$$

In the forward direction **(θ = 0°)**, $K = 1$ (maximum amplitude); in the backward direction **(θ = 180°)**, $K = 1 + \cos 180°2 = 0$.

Since the obliquity factor is zero in the backward direction, the secondary wavelets have zero amplitude there, so the back-wave does not exist.

OR (*b*) Single–slit diffraction.

(i) Intensity variation with angular position: The graph of intensity $I$ against angular position $\theta$ shows one tall, broad central maximum at $\theta = 0$ with peak intensity $I_0$. On both sides the intensity drops to zero at the minima, located at $a\theta = n\lambda$ ($n = 1, 2, 3,\ldots$), between which lie much weaker secondary maxima whose heights fall off rapidly. The curve follows $I = I_0(\frac{\beta}{\beta})^2$, where $\beta = \frac{\pi a\theta}{\lambda}$. Draw a symmetric bell-shaped central peak about $\theta = 0$ flanked by small side humps separated by zeros.

(ii) The linear width of the central maximum is

$$W = \frac{2\lambda D}{a}$$

Since $W \propto D$, decreasing the slit-to-screen distance $D$ makes $W$ smaller, so the central maximum becomes narrower.

---

**Final answer.** (a) Huygens' obliquity factor $\frac{1+\theta}{2}$ is zero backward, so no back-wave; OR (b) the central maximum width $W = 2\lambda D/a$ decreases as $D$ is decreased.

← *Back to Q69*

## Q70

*In a Young's double-slit experiment, two waves each of intensity I superpose each other and produce an interference pattern. Prove that the resultant intensities at maxima and minima are* 4I *and zero respectively.*

**[2026 • Set 55-5-2]**

**SOLUTION**

Let two coherent waves, each of intensity $I$, meet at a point on the screen with a phase difference $\phi$. Since intensity is proportional to the square of amplitude, each wave has amplitude $a = \sqrt{I}$.

Their displacements at the point are

$$y_1 = a\omega\, t,\ y_2 = a\cos(\omega\, t + \phi)$$

The resultant displacement is

$$y = y_1 + y_2 = a\omega\, t + a\cos(\omega\, t + \phi)$$

Using $\cos X + \cos Y = 2\cos\left(\frac{X+Y}{2}\right)\cos\left(\frac{X-Y}{2}\right)$,

$$y = 2a\cos\left(\frac{\phi}{2}\right)\cos\left(\omega\, t + \frac{\phi}{2}\right)$$

So the resultant amplitude is $A = 2a\cos\left(\frac{\phi}{2}\right)$, and the resultant intensity is

$$I_R = A^2 = 4a^2\cos^2\left(\frac{\phi}{2}\right) = 4I\cos^2\left(\frac{\phi}{2}\right)$$

At maxima: the waves are in phase, $\phi = 2n\pi$ ($n = 0, 1, 2, \ldots$), so $\cos^2\left(\frac{\phi}{2}\right) = 1$.

$$I_{max} = 4I \times 1 = 4I$$

At minima: the waves are out of phase, $\phi = (2n+1)\pi$, so $\cos^2\left(\frac{\phi}{2}\right) = 0$.

$$I_{min} = 4I \times 0 = 0$$

Hence the resultant intensities at maxima and minima are 4I and zero respectively. The energy is not lost but redistributed from the dark fringes to the bright fringes, so total energy is conserved.

---

**Final answer.** Proved: $I_R = 4I\cos^2(\phi/2)$ gives $I_{max} = 4I$ and $I_{min} = 0$.

← *Back to Q70*

## Q71

*When monochromatic light is incident on a surface separating two media, the refracted and reflected light both have the same frequency as the incident frequency but the wavelength of refracted light is different. Explain why.*

**[2026 • Set 55-5-3]**

**SOLUTION**

When light crosses the boundary between two media, the electric and magnetic fields of the wave must stay continuous across that boundary at every instant of time. This boundary condition forces the oscillations on both sides of the interface to stay perfectly in step.

The frequency of a wave is the number of oscillations the source produces per second. The atoms at the boundary are set vibrating by the incident wave, and they re-radiate the reflected and refracted waves at the very same rate at which they are driven. If the refracted wave had a different frequency, the two oscillations would slip out of synchrony and the fields could not remain continuous at the boundary. Hence the frequency cannot change:

$$\nu_{\text{refracted}} = \nu_{\text{reflected}} = \nu_{\text{incident}}.$$

The speed of light, however, does depend on the medium. In a medium of refractive index $n$ the speed is $v = \frac{c}{n}$, which is smaller than $c$ when $n > 1$. The three quantities are linked by $v = \nu\lambda$.

Since the frequency $\nu$ is fixed but the speed $v$ falls in the denser medium, the wavelength must also fall to keep the product $\nu\lambda$ equal to the reduced speed:

$$\lambda_{\text{med}} = \frac{v}{\nu} = \frac{c/n}{\nu} = \frac{\lambda_{\text{vac}}}{n}.$$

So in a medium with $n > 1$ the wavelength of the refracted light becomes shorter by the factor $\frac{1}{n}$, while the **frequency stays unchanged**. The reflected wave travels back in the original medium, so its wavelength is the same as that of the incident light.

---

**Final answer.** Frequency is fixed by field continuity at the boundary, while the wavelength of refracted light shrinks to $\lambda/n$ because the wave speed drops to $c/n$.

← *Back to Q71*

## Q72

***(a)*** *In a diffraction experiment, the slit is illuminated by light of wavelength* 600 nm. *The first minimum of the pattern falls at* $\theta = 30°$. *Calculate the width of the slit.*

***OR***

***(b)*** *In a Young's double-slit experiment, two light waves, each of intensity* $I_0$, *interfere at a point, having a path difference* $\frac{\lambda}{3}$ *on the screen. Find the intensity at this point.*

**[2025 • Set 55-1-1]**

**SOLUTION**

(a) Single-slit diffraction, finding the slit width.

In single-slit diffraction the dark fringes (minima) occur where the path difference between waves from the two edges of the slit is a whole number of wavelengths. The first minimum is given by:

$a\theta = \lambda,$

where $a$ is the slit width, $\theta$ is the angle of the first minimum and $\lambda$ is the wavelength.

Here $\lambda = 600\ \text{nm} = 6\times 10^{-7}\ m$ and $\theta = 30°$.

Rearranging for the slit width:

$$a = \frac{\lambda}{\theta}.$$

$$a = \frac{6\times 10^{-7}}{\sin 30°}.$$

$$a = \frac{6\times 10^{-7}}{0.5}.$$

$$a = 1.2\times 10^{-6}\ m.$$

The **width of the slit is 1.2 μm**.

OR (b) Intensity in YDSE for a path difference of $\frac{\lambda}{3}$.

When two coherent waves, each of intensity $I_0$, meet with a phase difference $\phi$, the resultant intensity is:

$$I = 4\, I_0 \cos^2\left(\frac{\phi}{2}\right).$$

The phase difference for a path difference $\Delta$ is:

$$\phi = \frac{2\pi}{\lambda}\Delta = \frac{2\pi}{\lambda}\cdot\frac{\lambda}{3} = \frac{2\pi}{3}.$$

Therefore:

$$\cos^2\left(\frac{\phi}{2}\right) = \cos^2\left(\frac{\pi}{3}\right) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}.$$

$$I = 4\,I_0 \times \frac{1}{4} = I_0.$$

The **intensity at this point is $I_0$**.

---

**Final answer.** (a) Slit width $a$ = **1.2 μm**. OR (b) The intensity at the point is $I_0$.

← *Back to Q72*

## Q73

*In a double-slit experiment, 6$^{th}$ dark fringe is observed at a certain point of the screen. A transparent sheet of thickness t and refractive index n is now introduced in the path of one of the two interfering waves to increase its phase by $2\pi(n-1)\,t/\lambda$. The pattern is shifted and 8$^{th}$ bright fringe is observed at the same point. Find the relation for thickness t in terms of n and $\lambda$.*

**[2025 • Set 55-2-1]**

**SOLUTION**

A thin transparent sheet of thickness $t$ and refractive index $n$ placed in the path of one wave adds an extra optical path of $(n-1)t$. This shifts the whole fringe pattern. The shift, measured as a number of fringes, is:

$$\textbf{number of fringes shifted} = \frac{(n-1)t}{\lambda}.$$

Now I locate the point on the screen using fringe positions, taking the fringe width as $\beta$. The dark fringes lie at half-integer multiples of $\beta$. The 6th dark fringe is the 6th minimum, so its position from the central maximum is:

$$y_{\text{dark},6} = \left(6 - \frac{1}{2}\right)\beta = 5.5\,\beta.$$

The 8th bright fringe is the 8th maximum, so its position is:

$y_{\text{bright},8} = 8\,\beta$.

Before inserting the sheet the chosen point carried the 6th dark fringe at $5.5\,\beta$. After inserting the sheet the pattern shifts so that the 8th bright fringe now sits at this same point, which originally held the position $8\,\beta$. The pattern has therefore moved by:

$$\Delta y = 8\,\beta - 5.5\,\beta = 2.5\,\beta.$$

So the pattern has shifted by 2.5 fringes. Equating this to the shift produced by the sheet:

$$\frac{(n-1)t}{\lambda} = 2.5.$$

$$(n-1)t = 2.5\,\lambda = \frac{5\lambda}{2}.$$

Solving for the thickness:

$$t = \frac{5\lambda}{2(n-1)}.$$

The required relation is $t = \dfrac{5\lambda}{2(n-1)}$.

**COMMON MISTAKE**

*When converting fringe orders to positions, remember bright fringes sit at* $m\beta$ *but dark fringes sit at* $(m-\frac{1}{2})\beta$. *Mixing these up is the most common slip in shift problems.*

---

**Final answer.** The thickness of the sheet is $t = \dfrac{5\lambda}{2(n-1)}$.

← Back to Q73

## Q74

*Show the refraction of light wave at a plane interface using Huygens' principle and prove Snell's law.*

**[2025 • Set 55-2-2]**

**SOLUTION**

Huygens' principle states that every point on a wavefront acts as a source of secondary wavelets that spread out in the forward direction with the speed of the wave in that medium. The new wavefront at any later instant is the forward tangent surface (envelope) of all these secondary wavelets.

Diagram (in words): Draw a horizontal line XY as the plane interface between two media. Medium 1 is above (light travels with speed $v_1$) and medium 2 is below (speed $v_2$). A plane incident wavefront AB in medium 1 strikes the interface. The end *A* touches the surface first at a point on XY; the other end *B* is still in medium 1 a little above the surface. The normal is drawn perpendicular to XY at *A*. The angle of incidence *i* is the angle the incident wavefront AB makes with the interface (equal to the angle the incident ray makes with the normal). Mark the point *C* on XY directly below *B*, and mark *D* in medium 2 below *A*. The refracted wavefront is CD, making angle *r* with the interface.

While the disturbance at *B* travels through medium 1 to reach *C* in time $\Delta t$, the disturbance at *A* sends a secondary wavelet into medium 2. So:

**BC** $= v_1 \Delta t$ (distance covered in medium 1),

**AD** $= v_2 \Delta t$ (radius of the secondary wavelet in medium 2).

The refracted wavefront CD is the tangent drawn from *C* to the wavelet of radius AD centred at *A*. In right triangle ABC (right angle at *B*):

$$\sin i = \frac{BC}{AC} = \frac{v_1 \Delta t}{AC}.$$

In right triangle ADC (right angle at *D*):

$$\sin r = \frac{AD}{AC} = \frac{v_2 \Delta t}{AC}.$$

Dividing the two equations, the common factor AC cancels:

$$\frac{\sin i}{\sin r} = \frac{v_1 \Delta t}{v_2 \Delta t} = \frac{v_1}{v_2}.$$

The ratio $\frac{v_1}{v_2}$ is a constant for a given pair of media, equal to the refractive index of medium 2 with respect to medium 1, $n_{21}$. Hence:

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = n_{21} = \text{constant}.$$

This is **Snell's law of refraction**, proved using Huygens' principle.

**Final answer.** Using Huygens' construction, $\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = n_{21}$, which is Snell's law.

← Back to Q74

## Q75

*In a double slit experiment, it is observed that the angular width of one fringe formed on the screen is 0.2°. The wavelength of light used in the experiment is 500 nm. Calculate the separation of the two slits.*

**[2025 • Set 55-2-3]**

**SOLUTION**

In Young's double-slit experiment the angular width of a fringe (the angle each fringe subtends at the slits) is:

$$\theta = \frac{\lambda}{d},$$

where $\lambda$ is the wavelength of light and $d$ is the separation between the two slits.

Here $\theta = 0.2°$ and $\lambda = 500\ \text{nm} = 5\times 10^{-7}\ m$.

The angle must be converted from degrees to radians before use:

$$\theta = 0.2 \times \frac{\pi}{180}\ \text{rad}.$$

$$\theta = 0.2 \times 0.01745 = 3.49\times 10^{-3}\ \text{rad}.$$

Rearranging the formula for the slit separation:

$$d = \frac{\lambda}{\theta}.$$

$$d = \frac{5\times 10^{-7}}{3.49\times 10^{-3}}.$$

$$d = 1.43\times 10^{-4}\ m.$$

The **separation between the two slits is** $1.43\times 10^{-4}\ m \approx 0.143$ **mm**.

**COMMON MISTAKE**

*Always convert the angle to radians before dividing into λ; using degrees directly gives an answer wrong by a factor of about 57.*

---

**Final answer.** The slit separation is $d \approx 1.43\times 10^{-4}\ m$ (about **0.143 mm**).

← *Back to Q75*

## Q76

*Find the angle of diffraction (in degrees) for first secondary maximum of the pattern due to diffraction at a single slit. The width of the slit and wavelength of light used are* 0.55 mm *and* 550 nm*, respectively.*

**[2025 • Set 55-4-1]**

**SOLUTION**

In single-slit diffraction the secondary maxima lie roughly midway between consecutive minima. The first secondary maximum therefore occurs where the path difference between waves from the two edges of the slit equals $\frac{3\lambda}{2}$:

$$a\theta = (1 + \frac{1}{2})\lambda = \frac{3\lambda}{2},$$

where $a$ is the slit width and θ is the angle of diffraction.

Here $a = 0.55\ \text{mm} = 5.5\times 10^{-4}\ m$ and $\lambda = 550\ \text{nm} = 5.5\times 10^{-7}\ m$.

Solving for θ:

$$\theta = \frac{3\lambda}{2a}.$$

$$\theta = \frac{3 \times 5.5\times 10^{-7}}{2 \times 5.5\times 10^{-4}}.$$

$$\theta = \frac{3}{2\times 10^{3}} = 1.5\times 10^{-3}.$$

Since this value is very small, $\theta \approx \theta = 1.5\times 10^{-3}$ **rad**.

Converting to degrees:

$\theta = 1.5\times 10^{-3} \times \frac{180^\circ}{\pi}$.

$\theta = 1.5\times 10^{-3} \times 57.3^\circ \approx 0.086^\circ$.

The **angle of diffraction for the first secondary maximum is about** $0.086^\circ$ **(≈** $0.09^\circ$**).**

---

**Final answer.** The first secondary maximum occurs at $\theta \approx 0.086^\circ$ (about $0.09^\circ$).

← *Back to Q76*

## Q77

*In a double slit experiment, the two slits are* 1.5 mm *apart. The slits are illuminated by a mixture of lights of wavelengths of* 600 nm *and* 400 nm *and the interference pattern is observed on a screen* 1.5 *m away from the slits. Find the minimum distance of the point from the central maximum at which bright fringes of the interference patterns of the two wavelengths coincide.*

**[2025 • Set 55-4-2]**

**SOLUTION**

In Young's double-slit experiment a bright fringe of wavelength $\lambda$ of order $n$ lies at a distance from the central maximum given by:

$$y_n = \frac{n\lambda D}{d},$$

where $D$ is the slit-to-screen distance and $d$ is the slit separation.

Two bright fringes of wavelengths $\lambda_1$ and $\lambda_2$ coincide when they fall at the same screen position. Because $D$ and $d$ are common, this needs:

$$n_1 \lambda_1 = n_2 \lambda_2,$$

with $n_1$ and $n_2$ whole numbers.

Here $\lambda_1 = 600$ nm, $\lambda_2 = 400$ nm, $d = 1.5$ mm $= 1.5\times 10^{-3}$ $m$ and $D = 1.5$ $m$.

Taking the ratio of the orders:

$$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{400}{600} = \frac{2}{3}.$$

The smallest whole numbers satisfying this are $\mathbf{n_1 = 2}$ and $\mathbf{n_2 = 3}$. So the first coincidence is between the 2nd bright fringe of 600 nm and the 3rd bright fringe of 400 nm.

Now compute the position using $\boldsymbol{\lambda_1 = 600\ nm}$:

$$y = \frac{n_1 \lambda_1 D}{d}.$$

$$y = \frac{2 \times (600 \times 10^{-9}) \times 1.5}{1.5 \times 10^{-3}}.$$

$$y = \frac{2 \times 600 \times 10^{-9} \times 1.5}{1.5 \times 10^{-3}} = 1.2 \times 10^{-3}\ m.$$

Checking with $\boldsymbol{\lambda_2 = 400\ nm}$ and $\mathbf{n_2 = 3}$:

$y = \frac{3 \times (400 \times 10^{-9}) \times 1.5}{1.5 \times 10^{-3}} = 1.2 \times 10^{-3}\ m$, which agrees.

The **minimum distance from the central maximum at which the bright fringes coincide is 1.2 mm**.

---

**Final answer.** The bright fringes of the two wavelengths first coincide at **1.2 mm** from the central maximum.

*← Back to Q77*

## Q78

*In Young's double slit experiment, the screen is moved* 30 cm *towards the slits. As a consequence, the fringe width of the pattern changes by* 0.09 mm. *If the slits separation used is* 2 mm, *calculate the wavelength of light used in the experiment.*

**[2025 • Set 55-4-3]**

**SOLUTION**

The fringe width in Young's double-slit experiment is:

$$\beta = \frac{\lambda D}{d},$$

where $\lambda$ is the wavelength, $D$ is the slit-to-screen distance and $d$ is the slit separation.

When only the screen distance changes by an amount $\Delta D$ while $\lambda$ and $d$ stay fixed, the fringe width changes by:

$$\Delta\beta = \frac{\lambda\,\Delta D}{d}.$$

Here the screen is moved 30 cm towards the slits, so the magnitude of the change in distance is $|\Delta D| = 0.30\ m$. Moving the screen closer reduces $\beta$, and the magnitude of the change in fringe width is $|\Delta\beta| = 0.09\ \text{mm} = 0.09\times 10^{-3}\ m$. The slit separation is $d = 2\ \text{mm} = 2\times 10^{-3}\ m$.

Rearranging for the wavelength:

$$\lambda = \frac{|\Delta\beta|\,d}{|\Delta D|}.$$

$$\lambda = \frac{(0.09\times 10^{-3})\times(2\times 10^{-3})}{0.30}.$$

$$\lambda = \frac{1.8\times 10^{-7}}{0.30}.$$

$$\lambda = 6.0\times 10^{-7}\ m.$$

The **wavelength of light used is $6.0\times 10^{-7}\ m = 600$ nm**.

---

**Final answer.** The wavelength of the light is $6.0\times 10^{-7}\ m$ (i.e. **600 nm**).

← *Back to Q78*

## Q79

*(a) Find the intensity at a point on the screen in Young's double slit experiment, at which the interfering waves of intensity $I_0$ each, have a path difference of (i) $\frac{\lambda}{4}$ and (ii) $\frac{\lambda}{3}$.*

***OR***

*(b) A point source of light in air is kept at a distance of 12 cm in front of a convex spherical surface of glass of refractive index 1·5 and radius of curvature 30 cm. Find the nature and position of the image formed.*

**[2025 • Set 55-6-1]**

**SOLUTION**

(a) Intensity in Young's double-slit experiment for two waves of intensity $I_0$ each.

The resultant intensity for a phase difference $\phi$ is:

$I = 4\,I_0 \cos^2\left(\frac{\phi}{2}\right)$, where the phase difference for a path difference $\Delta$ is $\phi = \frac{2\pi}{\lambda}\Delta$.

(i) For a path difference $\Delta = \frac{\lambda}{4}$:

$$\phi = \frac{2\pi}{\lambda}\cdot\frac{\lambda}{4} = \frac{\pi}{2}.$$

$$I = 4\,I_0 \cos^2\left(\frac{\pi}{4}\right) = 4\,I_0\left(\frac{1}{\sqrt{2}}\right)^2 = 4\,I_0 \times \frac{1}{2} = 2\,I_0.$$

(ii) For a path difference $\Delta = \frac{\lambda}{3}$:

$$\phi = \frac{2\pi}{\lambda}\cdot\frac{\lambda}{3} = \frac{2\pi}{3}.$$

$$I = 4\,I_0 \cos^2\left(\frac{\pi}{3}\right) = 4\,I_0\left(\frac{1}{2}\right)^2 = 4\,I_0 \times \frac{1}{4} = I_0.$$

So the intensities are $2\,I_0$ and $I_0$ respectively.

OR (b) Refraction at a single convex spherical surface.

The refraction formula for a single spherical surface is:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}.$$

Using the sign convention with light travelling left to right: the object is real, so $u = -12$ cm; light goes from air to glass, so $n_1 = 1$ and $n_2 = 1.5$; the surface is convex towards the object, so $R = +30$ cm.

Substituting:

$$\frac{1.5}{v} - \frac{1}{-12} = \frac{1.5 - 1}{30}.$$

$$\frac{1.5}{v} + \frac{1}{12} = \frac{0.5}{30} = \frac{1}{60}.$$

$$\frac{1.5}{v} = \frac{1}{60} - \frac{1}{12} = \frac{1}{60} - \frac{5}{60} = -\frac{4}{60} = -\frac{1}{15}.$$

$v = 1.5 \times (-15) = -22.5$ cm.

The negative sign shows the image is formed on the same side as the object (the air side). Hence the **image is virtual, erect, and located 22.5 cm in front of the surface**.

---

**Final answer.** (a) Intensities are **2 $I_0$** for **λ/4** and **$I_0$** for **λ/3**. OR (b) A virtual image is formed **22.5 cm** in front of the surface (on the object side).

← *Back to Q79*

## Q80

*Write two differences in the patterns of double-slit interference experiment and single-slit diffraction experiment.*

**[2025 • Set 55-7-1]**

**SOLUTION**

Two clear differences between the double-slit interference pattern and the single-slit diffraction pattern are:

1. Width of the fringes / bright bands:

In double-slit interference, all the bright fringes have the same width, given by $\beta = \frac{\lambda D}{d}$, and they are equally spaced.

In single-slit diffraction, the central bright maximum is the widest, having twice the width ($\frac{2\lambda D}{a}$) of the secondary maxima, and the bright bands are not all equal.

2. Intensity of the bright bands:

In double-slit interference, all the bright fringes have nearly the same intensity.

In single-slit diffraction, the central maximum is by far the brightest and the intensity of successive secondary maxima falls off rapidly as the order increases.

(An additional valid point: interference arises from the superposition of waves from two separate coherent slits, whereas diffraction arises from the

superposition of secondary wavelets coming from different parts of the same single wavefront.)

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**Final answer.** Interference fringes are of equal width and equal intensity, while in diffraction the central maximum is twice as wide and far brighter, with secondary maxima of rapidly decreasing intensity.

← *Back to Q80*

## Q81

*Light waves from two pinholes illuminated by two sodium lamps do not produce interference patterns. Explain why.*

**[2025 • Set 55-7-1]**

**SOLUTION**

For two light sources to produce a steady interference pattern, they must be coherent: they must emit waves of the same frequency with a phase difference that stays constant in time.

Two separate pinholes lit by two independent sodium lamps fail this condition. Light from an ordinary source comes from spontaneous emission by a huge number of atoms acting independently of one another.

Each atom radiates a short wave train lasting only about $10^{-8}$ $s$ (the coherence time), after which its phase changes abruptly and at random.

Because the two lamps are independent, the phase of the light from one pinhole bears no fixed relation to that from the other. So the phase difference $\phi$ at any point on the screen keeps changing randomly, roughly $10^{8}$ times each second.

As $\phi$ varies, the positions of the bright and dark fringes shift on the screen far too rapidly for the eye or any normal detector to follow.

The detector therefore records only the time-averaged intensity, which stays uniform, so no interference pattern is seen. To get coherent sources, both beams must come from a single primary source (the two slits in Young's experiment, a Fresnel biprism, or a single laser).

**Final answer.** The two lamps are mutually incoherent, so their phase difference changes randomly about $10^8$ times per second and only uniform averaged illumination is observed.

← *Back to Q81*

## Q82

*The ratio of the intensities at maxima to minima in Young's double-slit experiment is* 25 : 9. *Calculate the ratio of intensities of the interfering waves.*

**[2025 • Set 55-7-2]**

**SOLUTION**

For two interfering waves of intensities $I_1$ and $I_2$, the maximum (constructive) and minimum (destructive) intensities are

$$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

We are given $\frac{I_{max}}{I_{min}} = \frac{25}{9}$.

Taking the square root of both sides,

$$\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} = \frac{5}{3}$$

Cross-multiplying,

$$3(\sqrt{I_1} + \sqrt{I_2}) = 5(\sqrt{I_1} - \sqrt{I_2})$$

$$3\sqrt{I_1} + 3\sqrt{I_2} = 5\sqrt{I_1} - 5\sqrt{I_2}$$

$$8\sqrt{I_2} = 2\sqrt{I_1}$$

$$\frac{\sqrt{I_1}}{\sqrt{I_2}} = 4$$

Squaring both sides,

$$\frac{I_1}{I_2} = 16$$

Hence the interfering waves have intensities in the ratio 16 : 1.

---

**Final answer.** The intensities of the two interfering waves are in the ratio $\mathbf{I_1 : I_2 = 16 : 1}$.

← *Back to Q82*

## Q83

*In a Young's double-slit experiment, the intensity at the central maximum in the interference pattern on the screen is* $I_0$. *Find the intensity at a point on the screen where the path difference between the interfering waves is* $\frac{\lambda}{3}$.

**[2025 • Set 55-7-3]**

**SOLUTION**

When two coherent waves of equal intensity $I_s$ each meet with a phase difference $\phi$, the resultant intensity is

$$I = 4\, I_s \cos^2\left(\frac{\phi}{2}\right)$$

At the central maximum the path difference is zero, so $\boldsymbol{\phi = 0}$ and $\boldsymbol{\cos^2(0) = 1}$, giving the peak value

$$I_0 = 4\, I_s$$

So we may write the intensity at any point as

$$I = I_0 \cos^2\left(\frac{\phi}{2}\right)$$

The phase difference for a path difference $\Delta$ is

$$\phi = \frac{2\pi}{\lambda}\,\Delta$$

Here $\Delta = \frac{\lambda}{3}$, so

$$\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{3} = \frac{2\pi}{3}$$

Therefore

$$I = I_0 \cos^2\left(\frac{\phi}{2}\right) = I_0 \cos^2\left(\frac{\pi}{3}\right)$$

Since $\cos\frac{\pi}{3} = \frac{1}{2}$,

$$I = I_0 \times \left(\frac{1}{2}\right)^2 = \frac{I_0}{4}$$

Hence the intensity at that point is $I = \frac{I_0}{4}$.

---

**Final answer.** The intensity at that point is $I = \frac{I_0}{4}$.

← *Back to Q83*

## Q84

*State Huygens' principle. A plane wave is incident at an angle i on a reflecting surface. Construct the corresponding reflected wavefront. Using this diagram, prove that the angle of reflection is equal to the angle of incidence.*

**[2024 • Set 55-1-1]**

**SOLUTION**

Huygens' principle: Every point on a given wavefront acts as a fresh source of secondary wavelets that spread out forwards with the speed of the wave. The new position of the wavefront after a time interval is the forward envelope (the common tangent surface) of all these secondary wavelets.

Construction of the reflected wavefront: Let AB be a plane wavefront incident on a plane reflecting surface MN. Ray at *A* strikes the surface first while the ray at *B* still has to travel to the surface. Draw the incident wavefront AB so that the rays make angle *i* with the normal; the wavefront AB makes the same angle *i* with the surface.

Let the disturbance at $B$ take time $t$ to reach the surface at point $C$, covering

$$\mathrm{BC} = c\,t$$

In the same time $t$, the secondary wavelet from $A$ travels into the same medium and spreads a distance

$$\mathrm{AA'} = c\,t$$

To find the reflected wavefront, draw a sphere of radius $c\,t$ centred at $A$, and from $C$ draw the tangent CA' to it. Then CA' is the reflected wavefront, and $\mathbf{AA' = BC = c\,t}$.

**Proof that $i = r$:** Consider the two right triangles ABC and CA'$A$. They share the common hypotenuse AC and have

$$\mathrm{AA'} = \mathrm{BC} = c\,t$$

The angle of incidence equals the angle $\angle\,\mathbf{BAC} = \boldsymbol{i}$ and the angle of reflection equals $\angle\,\boldsymbol{A'CA} = \boldsymbol{r}$. In these triangles

$$\sin i = \frac{\mathrm{BC}}{\mathrm{AC}}$$

$$\sin r = \frac{\mathrm{AA'}}{\mathrm{AC}}$$

Since $\mathbf{BC = AA'}$,

$$\sin i = \sin r \Rightarrow i = r$$

Hence the angle of reflection equals the angle of incidence, proving the law of reflection.

---

**Final answer.** By Huygens' construction $\mathbf{AA' = BC = c\,t}$, giving $\mathbf{\sin i = \sin r}$, hence the angle of reflection equals the angle of incidence.

*← Back to Q84*

## Q85

*A beam of light consisting of a known wavelength* 520 nm *and an unknown wavelength* $\lambda$*, used in Young's double slit experiment produces two interference patterns such that the fourth bright fringe of unknown wavelength coincides with the fifth bright fringe of known wavelength. Find the value of* $\lambda$*.*

**[2024 • Set 55-1-1]**

**SOLUTION**

Two bright fringes of different wavelengths coincide when they lie at the same distance from the central maximum on the screen.

The distance of the $n$th bright fringe from the centre is

$$y_n = \frac{n\lambda D}{d}$$

Since $D$ and $d$ are the same for both patterns, the coincidence condition $\mathbf{y_{n_1}(\lambda_1) = y_{n_2}(\lambda_2)}$ reduces to

$$n_1 \lambda_1 = n_2 \lambda_2$$

Here the known wavelength is $\mathbf{\lambda_1 = 520\ nm}$ at its 5th bright fringe ($\mathbf{n_1 = 5}$), and the unknown is $\mathbf{\lambda_2 = \lambda}$ at its 4th bright fringe ($\mathbf{n_2 = 4}$).

Substituting,

$$4 \times \lambda = 5 \times 520\ \text{nm}$$

$$\lambda = \frac{5 \times 520}{4}\ \text{nm}$$

$$\lambda = \frac{2600}{4}\ \text{nm}$$

$$\lambda = 650\ \text{nm}$$

Hence the unknown wavelength is 650 nm.

**TIP**

When two fringe systems overlap, equate the orders times the wavelengths ($\mathbf{n_1\lambda_1 = n_2\lambda_2}$); $D$ and $d$ cancel because they are common to both.

**Final answer.** The unknown wavelength is **λ = 650 nm.**

← Back to Q85

## Q86

*(a) Two waves, each of amplitude 'a' and frequency 'ω' emanating from two coherent sources of light superpose at a point. If the phase difference between the two waves is ϕ, obtain an expression for the resultant intensity at that point.*

***OR***

*(b) What is the effect on the interference pattern in Young's double-slit experiment when (i) the source slit is moved closer to the plane of the slits, and (ii) the separation between the two slits is increased? Justify your answers.*

**[2024 • Set 55-2-1]**

**SOLUTION**

(*a*) Resultant intensity of two superposing waves. Let the two waves of equal amplitude *a* and angular frequency ω, with a constant phase difference ϕ, be

$$y_1 = a\omega\, t$$

$$y_2 = a\sin(\omega\, t + \phi)$$

By the superposition principle the resultant displacement is

$$y = y_1 + y_2 = a\omega\, t + a\sin(\omega\, t + \phi)$$

Using $\sin C + \sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$,

$$y = 2a\cos(\frac{\phi}{2})\sin(\omega\, t + \frac{\phi}{2})$$

This is a wave of the same frequency with resultant amplitude

$$A = 2a\cos(\frac{\phi}{2})$$

Since intensity is proportional to the square of the amplitude, $I \propto A^2$, so

$$I = 4a^2\cos^2\left(\frac{\Phi}{2}\right)$$

Writing $\mathbf{I_0 = a^2}$ for the intensity of each wave, this becomes $\boldsymbol{I = 4\,I_0\cos^2(\phi/2)}$. It gives maxima ($\boldsymbol{I = 4a^2}$) when $\boldsymbol{\phi = 0, 2\pi, 4\pi,}$ and minima ($\boldsymbol{I = 0}$) when $\boldsymbol{\phi = \pi, 3\pi,}$

OR (*b*) Effect on the YDSE pattern.

(i) Source slit moved closer to the plane of the slits: The angular size of the source as seen from the slits increases. The condition for sharp fringes, $\frac{s}{S} < \frac{\lambda}{d}$ (where $s$ is the source-slit width and $S$ its distance), is violated. The two slits become less coherently illuminated, so the fringe contrast (visibility) decreases and eventually the fringes wash out. The fringe width itself is unchanged.

(ii) Separation between the two slits increased: The fringe width is

$$\beta = \frac{\lambda D}{d}$$

Since $\beta \propto \frac{1}{d}$, increasing $d$ decreases the fringe width, so the bright and dark fringes come closer together (the pattern becomes more crowded).

---

**Final answer.** (a) $\boldsymbol{I = 4a^2\cos^2\left(\frac{\Phi}{2}\right)}$; OR (b) (i) fringe contrast decreases, (ii) fringe width $\boldsymbol{\beta = \frac{\lambda D}{d}}$ decreases so the fringes come closer.

← *Back to Q86*

## Q87

*Write two points of difference between interference and diffraction of light.*

**[2024 • Set 55-3-2]**

**SOLUTION**

Two points of difference between interference and diffraction of light:

1. In an interference pattern (from two coherent sources, as in a double slit) all the bright fringes are of equal width and equal intensity. In a diffraction pattern

(from a single slit) the central maximum is the widest and brightest, and the intensity of the secondary maxima falls off rapidly on either side, so they are unequal in width and intensity.

2. Interference arises from the superposition of waves coming from two (or more) separate coherent wavefronts (two slits). Diffraction arises from the superposition of secondary wavelets coming from different points of the same wavefront (a single slit).

(An equally valid second point: in interference the dark fringes are perfectly dark, whereas in diffraction the minima are not completely dark because the energy is only redistributed.)

---

**Final answer.** Interference fringes are equally spaced and equally bright and come from two coherent wavefronts; diffraction gives a broad bright central maximum with rapidly fading higher orders from one wavefront.

← *Back to Q87*

## Q88

*What is a sustained or stable interference pattern? What are the conditions for obtaining such an interference pattern?*

**[2024 • Set 55-3-3]**

**SOLUTION**

Sustained (stable) interference pattern: It is an interference pattern in which the positions of the bright and dark fringes stay fixed in space and steady in time, so that the fringes remain sharp and clearly visible with good contrast throughout the observation. If interference were not sustained, the fringes would shift about so rapidly that only a uniform averaged illumination would be seen.

Conditions for obtaining *a* sustained interference pattern:

1. The two sources must be coherent, that is they must emit light of the same frequency with a constant phase difference in time. This is the most essential condition; two independent sources have rapidly fluctuating phases and cannot

give sustained interference. Coherent sources are obtained by deriving both from a single primary source (e.g. two slits lit by the same lamp).

2. The two sources should be monochromatic (or nearly so) and emit waves of the same wavelength.

3. The amplitudes (intensities) of the two waves should be equal or nearly equal, so that the dark fringes are (almost) perfectly dark and the contrast between bright and dark regions is high.

4. The separation $d$ between the two sources should be small and the screen distance $D$ large, so that the fringe width $\beta = \frac{\lambda D}{d}$ is large enough for the fringes to be resolved by the eye or detector.

---

**Final answer.** A sustained pattern has fringes fixed in space and time; it needs coherent (same frequency, constant phase) monochromatic sources of nearly equal amplitude, derived from a single primary source, with a small slit separation.

← *Back to Q88*

## Q89

*Using Huygens' principle, draw a ray diagram showing the propagation of a plane wave refracting at a plane surface separating two media. Also verify Snell's law of refraction.*

**[2023 • Set 55-1-1]**

**SOLUTION**

Huygens' principle: Every point on a wavefront acts as a fresh source of secondary wavelets that travel forward at the speed of the wave in that medium. The new wavefront after a time interval is the forward envelope (common tangent) of all these secondary wavelets.

Ray diagram for refraction: Let a plane wavefront AB travel in medium 1 (speed $v_1$) and be incident at angle $i$ on a plane interface XY separating medium 1 from a denser medium 2 (speed $v_2$). Point $A$ of the wavefront reaches the interface

first, while $B$ still has to travel to the surface. Draw AB making angle $i$ with the surface; the incident rays at $A$ and $B$ make angle $i$ with the normal.

Let the disturbance at $B$ take time $\Delta t$ to reach the interface at point $C$, covering

$$\mathrm{BC} = v_1 \, \Delta t$$

During the same time $\Delta t$, the secondary wavelet starting from $A$ enters medium 2 and spreads a distance

$$\mathrm{AD} = v_2 \, \Delta t$$

With $A$ as centre, draw an arc of radius $\mathbf{AD = v_2 \, \Delta t}$ in medium 2, and from $C$ draw the tangent CD to it. Then CD is the refracted wavefront, making angle $r$ (the angle of refraction) with the interface.

Verification of Snell's law: In the right triangle ABC (right-angled at $B$), the angle at $A$ equals $i$, so

$$\sin i = \frac{\mathrm{BC}}{\mathrm{AC}} = \frac{v_1 \, \Delta t}{\mathrm{AC}}$$

In the right triangle ADC (right-angled at $D$), the angle at $C$ equals $r$, so

$$\sin r = \frac{\mathrm{AD}}{\mathrm{AC}} = \frac{v_2 \, \Delta t}{\mathrm{AC}}$$

Dividing the two,

$$\frac{\sin i}{\sin r} = \frac{v_1 \, \Delta t}{v_2 \, \Delta t} = \frac{v_1}{v_2}$$

Since the refractive index of a medium is $\boldsymbol{n = \frac{c}{v}}$, we have $\frac{v_1}{v_2} = \frac{n_2}{n_1}$, giving

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} \Rightarrow n_1 \sin i = n_2 \sin r$$

This is Snell's law of refraction, verified by Huygens' construction.

---

**Final answer.** Huygens' construction gives $\mathbf{\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1}}$, i.e. $\mathbf{n_1 \sin i = n_2 \sin r}$, which is Snell's law.

← *Back to Q89*

## Q90

*Two coherent monochromatic light beams of intensities I and 4I superpose each other. Find the ratio of maximum and minimum intensities in the resulting beam.*

**[2023 • Set 55-1-1]**

**SOLUTION**

When two coherent beams of intensities $I_1$ and $I_2$ superpose, the maximum and minimum intensities of the resulting beam are

$$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

Here $\mathbf{I_1 = I}$ and $\mathbf{I_2 = 4I}$, so

$$\sqrt{I_1} = \sqrt{I}$$

$$\sqrt{I_2} = \sqrt{4I} = 2\sqrt{I}$$

Therefore

$$I_{max} = (\sqrt{I} + 2\sqrt{I})^2 = (3\sqrt{I})^2 = 9I$$

$$I_{min} = (\sqrt{I} - 2\sqrt{I})^2 = (-\sqrt{I})^2 = I$$

Taking the ratio,

$$\frac{I_{max}}{I_{min}} = \frac{9I}{I} = \frac{9}{1}$$

Hence the maximum and minimum intensities are in the ratio 9 : 1.

---

**Final answer.** The ratio of maximum to minimum intensity is $\mathbf{I_{max} : I_{min} = 9 : 1}$.

← *Back to Q90*

## Q91

*What happens to the interference pattern when two coherent sources are*
*(a) infinitely close, and*
*(b) far apart from each other?*

**[2023 • Set 55-2-1]**

**SOLUTION**

In Young's double slit experiment the fringe width is $\beta = \frac{\lambda D}{d}$, where $d$ is the separation between the two coherent sources, $D$ is the slit-to-screen distance and $\lambda$ is the wavelength.

The fringe width is inversely proportional to the source separation $d$, so the value of $d$ decides whether the fringes can be seen.

(a) When the two sources are infinitely close, $\boldsymbol{d \to 0}$. Then

$$\beta = \frac{\lambda D}{d} \to \infty$$

The fringes spread out so widely that only one broad bright patch falls on the screen. No separate alternate bright and dark fringes can be seen, so the interference **pattern disappears** and the screen looks uniformly illuminated.

(b) When the two sources are far apart, $d$ is very large. Then

$$\beta = \frac{\lambda D}{d} \to \text{very small}$$

The fringes become extremely close together. They are packed so tightly that the eye cannot separate them, and again the pattern merges into **uniform illumination**.

So in both extreme cases the fringes cannot be observed. A clear, well spaced pattern is seen only for a moderate value of $d$.

---

**Final answer.** If the sources are infinitely close the fringes become infinitely wide and vanish, and if they are far apart the fringes become too close to be seen, so in both cases no pattern is observed.

← Back to Q91

## Q92

*In a Young's double slit experiment, the separation between the two slits is d and distance of the screen from the slits is* 1000 *d. If the first minima falls at a distance d from the central maximum, obtain the relation between d and* λ.

**[2023 • Set 55-2-2]**

**SOLUTION**

In Young's double slit experiment the path difference between the two waves reaching a point at distance $y$ from the central maximum is

$$\Delta = \frac{y\,d}{D}$$

For a dark fringe (minimum) the path difference must be an odd multiple of $\frac{\lambda}{2}$.

For the **first minimum** ($n = 1$),

$$\Delta = \frac{\lambda}{2}$$

Therefore the condition for the first minimum is

$$\frac{y\,d}{D} = \frac{\lambda}{2}$$

Here it is given that $D = 1000\,d$ and the first minimum falls at $y = d$ from the central maximum. Substituting these values,

$$\frac{d \times d}{1000\,d} = \frac{\lambda}{2}$$

$$\frac{d}{1000} = \frac{\lambda}{2}$$

$$d = \frac{1000\,\lambda}{2}$$

$$d = 500\,\lambda$$

So the required relation between the slit separation and the wavelength is $d = 500\,\lambda$.

**COMMON MISTAKE**

*For a minimum use path difference $\frac{\lambda}{2}$ (odd multiples of half wavelength); for a maximum use $\lambda$ (whole multiples). Mixing the two is the commonest slip here.*

---

**Final answer.** The required relation is $d = 500\,\lambda$.

← Back to Q92

## Q93

*(a) State Huygens' principle. How did Huygens explain the absence of the backwave?*
***OR***
*(b) Use Huygens' principle to show reflection/refraction of a plane wave by (i) concave mirror, and (ii) a convex lens.*

**[2023 • Set 55-3-1]**

**SOLUTION**

(a) Huygens' principle states two things.

Every point on a given wavefront, called the primary wavefront, acts as a fresh source of disturbance and sends out secondary wavelets which spread in all directions with the speed of the wave in that medium.

The new position of the wavefront after a time $t$ is given by the forward surface that just touches all these secondary wavelets, that is, their **envelope** in the forward direction.

To construct the new wavefront, draw spheres of radius $v\,t$ (where $v$ is the wave speed) taking each point of the old wavefront as centre. The common tangent surface drawn in the forward direction is the new wavefront.

Absence of the back wave: The secondary wavelets are emitted in all directions, so a backward envelope (back wave) should also form. Huygens explained that the amplitude of the secondary wavelets is not the same in all directions. It is

largest in the forward direction and falls off with the angle, becoming zero in the backward direction.

This is described by an obliquity (inclination) factor

$$K = \frac{1 + \theta}{2}$$

where θ is the angle measured from the forward direction. In the backward direction $\theta = 180°$, so

$$K = 1 + \cos 180°2 = 1 - 12 = 0$$

Since the effective amplitude of the wavelets in the backward direction is zero, the back wave is not produced. This is why the disturbance travels only forward.

---

**Final answer.** Huygens' principle: each point on a wavefront is a source of secondary wavelets whose forward envelope gives the new wavefront, and the back wave is absent because the obliquity factor $\frac{1+\theta}{2}$ makes the backward amplitude zero.

← *Back to Q93*

## Q94

*How would the angular width of central maximum of diffraction pattern be affected when (i) width of the slit is decreased, and (ii) monochromatic light is replaced by polychromatic light? Justify your answers.*

**[2021]**

**SOLUTION**

In single slit diffraction the first minimum lies at an angle $\theta_1$ given by $a\theta_1 = \lambda$, so for small angles $\theta_1 = \frac{\lambda}{a}$, where $a$ is the slit width. The angular width of the central maximum is the angle between the two first minima on either side,

$$2\theta_1 = \frac{2\lambda}{a}$$

(i) When the width of the slit is decreased: Since $2\theta_1 = \frac{2\lambda}{a}$ and the angular width is inversely proportional to $a$, decreasing $a$ makes the angular width **increase**, so the central maximum becomes wider.

Reason: A narrower slit bends (diffracts) light through larger angles, so the light spreads out more and the central bright band broadens.

(ii) When monochromatic light is replaced by polychromatic light: Polychromatic (white) light contains many wavelengths, and for each colour $2\theta_1 = \frac{2\lambda}{a}$, so the angular width depends on $\lambda$.

Reason: Red light has the largest $\lambda$, so it gives the widest central maximum, while violet has the smallest $\lambda$ and the narrowest. At the centre all colours overlap to give a white band, but the edges of the central maximum become **coloured**, with violet nearer the centre and red farther out.

---

**Final answer.** Decreasing the slit width widens the central maximum ($2\theta_1 = \frac{2\lambda}{a}$), and using polychromatic light makes the central band white in the middle with coloured edges (red widest, violet narrowest).

← *Back to Q94*

## Q95

*In interference of light, write the expression for the intensity of resultant wave if* $I_0$ *is the intensity of light wave from each slit. Hence, obtain an expression for the intensity of resultant wave if the two sources are (i) incoherent, and (ii) coherent.*

**[2021]**

**SOLUTION**

Let two waves from the two slits, each of intensity $I_0$, meet at a point with a phase difference $\phi$. By the principle of superposition the resultant intensity is

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\ \phi$$

Since each slit gives intensity $I_0$, put $I_1 = I_2 = I_0$,

$$I = I_0 + I_0 + 2\sqrt{I_0\, I_0}\ \phi$$

$$I = 2I_0(1 + \phi)$$

Using $\mathbf{1 + \phi = 2\cos^2\frac{\phi}{2}}$,

$$I = 4I_0\cos^2\frac{\phi}{2}$$

(i) Incoherent sources: For incoherent sources the phase difference $\phi$ changes randomly and rapidly with time. The detector records the time average, and the average value of $\phi$ over time is zero, $\boldsymbol{\phi = 0}$. Hence

$$I = I_0 + I_0 = 2I_0$$

So the resultant intensity is simply $\boldsymbol{I = 2I_0}$ everywhere, giving **uniform illumination** and no interference pattern.

(ii) Coherent sources: For coherent sources $\phi$ is constant in time (it depends only on the path difference at each point). The full expression survives,

$$I = 4I_0\cos^2\frac{\phi}{2}$$

so the intensity varies between a maximum of $4I_0$ (when $\boldsymbol{\phi = 0, 2\pi,}$ ) and zero (when $\boldsymbol{\phi = \pi, 3\pi,}$ ), producing a clear pattern of bright and dark fringes.

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**Final answer.** The resultant intensity is $\boldsymbol{I = 4I_0\cos^2\frac{\phi}{2}}$; for incoherent sources it averages to $\boldsymbol{I = 2I_0}$ (uniform), while for coherent sources it stays $\boldsymbol{I = 4I_0\cos^2\frac{\phi}{2}}$, varying from $\mathbf{0}$ to $\mathbf{4I_0}$.

*← Back to Q95*

## Q96

*Define wavefront of a travelling wave. Using Huygens principle, obtain the law of refraction at a plane interface when light passes from a denser to rarer medium.*

**[2020 • Set 55-1-1]**

**SOLUTION**

A wavefront of a travelling wave is the **locus of all points of the medium that are vibrating in the same phase** at a given instant. A surface drawn through these points is the wavefront, and it is always perpendicular to the direction of propagation of the wave. For a point source it is spherical, and for a far away (or plane) source it is a plane.

By Huygens' principle, every point on a wavefront acts as a source of secondary wavelets that travel forward with the wave speed, and their forward envelope gives the new wavefront.

Consider a plane wavefront AB travelling in the denser medium 1 (speed $v_1$) and striking the plane interface XY at angle of incidence $i$. The refracted wave goes into the rarer medium 2 (speed $v_2$, with $\mathbf{v_2 > v_1}$).

While the wavelet from $B$ travels in medium 1 to reach the interface at $C$, the wavelet from $A$ enters medium 2 and spreads. Let the time taken be $t$.

Distance covered in medium 1: $\mathbf{BC = v_1\, t}$.

Distance covered in medium 2: $\mathbf{AD = v_2\, t}$. The refracted wavefront is CD, with angle of refraction $r$.

In right triangle ABC (right angle at $B$):

$$\sin i = \frac{BC}{AC} = \frac{v_1 t}{AC}$$

In right triangle ADC (right angle at $D$):

$$\sin r = \frac{AD}{AC} = \frac{v_2 t}{AC}$$

Dividing the two equations,

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$

Since the refractive index of a medium is $\boldsymbol{n = \frac{c}{v}}$, we get $\frac{\mathbf{v_1}}{\mathbf{v_2}} = \frac{\mathbf{n_2}}{\mathbf{n_1}}$, hence

$$n_1 \sin i = n_2 \sin r$$

which is the **Snell's law of refraction**.

As $v_2 > v_1$, $\sin r > \sin i$, so $r > i$, meaning the ray bends **away from the normal** while going from a denser to a rarer medium, as expected.

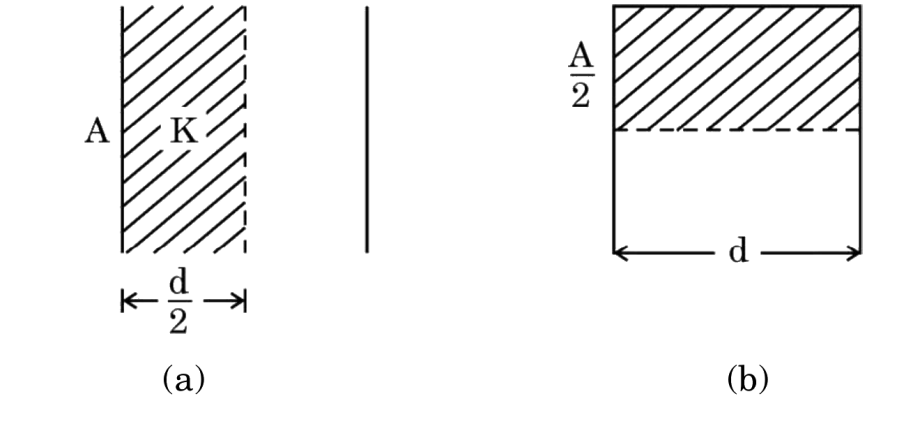
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**Final answer.** A wavefront is the locus of points vibrating in the same phase; Huygens' construction gives $\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$, i.e. $n_1 \sin i = n_2 \sin r$, with $r > i$ for denser to rarer.

← *Back to Q96*

## Q97

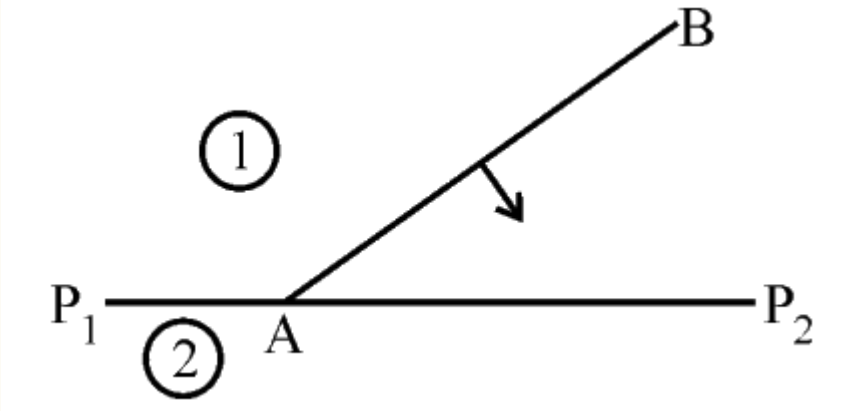
*Light from a sodium lamp (S) passes through two polaroid sheets* $P_1$ *and* $P_2$ *as shown in fig. What will be the effect on the intensity of the light transmitted (i) by* $P_1$ *and (ii) by* $P_2$ *on rotating polaroid* $P_2$ *about the direction of propagation of light ? Justify your answer in both cases.*




***OR***

*Define the term 'wave front of light'. A plane wave front* AB *propagating from denser medium (1) into a rarer medium (2) is incident on the surface* $P_1P_2$ *separating the two media as shown in fig.*

*Using Huygen's principle, draw the secondary wavelets and obtain the refracted wave front in the diagram.*




[2020 • Set 55-5-1]

**SOLUTION**

Light from the sodium lamp is **unpolarised**. When it passes through $P_1$ it becomes plane polarised, and half of the incident intensity is transmitted.

(i) Effect on the intensity transmitted by $P_1$ when $P_2$ is rotated: $P_1$ only receives light from the source, not from $P_2$. So rotating $P_2$ has no effect on $P_1$. If $I_0$ is the intensity incident on $P_1$, the transmitted intensity is

$$I_1 = \frac{I_0}{2}$$

This **remains constant** for every position of $P_2$.

(ii) Effect on the intensity transmitted by $P_2$ when $P_2$ is rotated: The light reaching $P_2$ is already plane polarised with intensity $\frac{I_0}{2}$. By **Malus' law**, if $\theta$ is the angle between the pass axes of $P_1$ and $P_2$,

$$I_2 = \frac{I_0}{2}\cos^2\theta$$

As $P_2$ is rotated, $\theta$ changes, so $I_2$ varies.

When the axes are parallel ($\theta = 0°$ or 180°): $\mathbf{I_2 = \frac{I_0}{2}}$, which is the **maximum**.

When the axes are crossed ($\theta = 90°$ or 270°): $\mathbf{I_2 = 0}$, the **minimum**.

So during one full rotation of $P_2$ the transmitted intensity rises and falls (becoming maximum twice and zero twice), following the $\cos^2\theta$ law.

**Final answer.** The intensity through $P_1$ stays constant at $\frac{I_0}{2}$, while the intensity through $P_2$ varies as $I_2 = \frac{I_0}{2}\cos^2\theta$, becoming maximum when the axes are parallel and zero when they are crossed.

← *Back to Q97*

## Q98

*In a single slit diffraction experiment, the width of the slit is increased. How will the (i) size and (ii) intensity of central bright band be affected ? Justify your answer.*

**[2020 • Set 55-5-1]**

**SOLUTION**

In single slit diffraction the width of the central bright band on a screen at distance $D$ is

$$W = \frac{2D\lambda}{a}$$

where $a$ is the slit width. The intensity at the centre of the pattern depends on the amount of light passing through the slit. Each part of the slit sends a Huygens wavelet, so the total amplitude at the centre is proportional to $a$, and the intensity (which is proportional to amplitude squared) is proportional to $a^2$.

(i) Effect on the size of the central band when the slit width is increased: Since $W = \frac{2D\lambda}{a}$, the width is inversely proportional to $a$. So increasing $a$ makes $W$ **decrease**, that is, the central bright band becomes narrower.

Reason: A wider slit diffracts (bends) light through smaller angles, so the spreading of light is reduced and the central band shrinks.

(ii) Effect on the intensity of the central band when the slit width is increased: More light passes through the wider slit, and the central intensity is proportional to $a^2$. So increasing $a$ makes the central band **brighter** (intensity rises sharply, as the square of the slit width).

Therefore widening the slit gives a narrower but brighter central maximum.

**Final answer.** On increasing the slit width the central band becomes narrower (size $W \propto \frac{1}{a}$) and brighter (intensity $\propto a^2$).

← Back to Q98

## Q99

*Explain giving reason, how the resolving power of a compound microscope depends on the*
*(a) frequency of the incident light.*
*(b) focal length of the objective lens.*

[2019 • Set 55-4-1]

SOLUTION

The resolving power of a compound microscope is its ability to show two very close point objects as separate. It is given by

$$R.P. = \frac{2\,n\theta}{1.22\,\lambda}$$

where $n\theta$ is the numerical aperture of the objective (with $n$ the refractive index of the medium between object and objective and $\theta$ the half angle of the cone of light), and $\lambda$ is the wavelength of light used. Equivalently the smallest distance that can be resolved is $d_{min} = \frac{1.22\,\lambda}{2\,n\theta}$.

(a) Dependence on the frequency of the incident light: From the formula, $R.P. \propto \frac{1}{\lambda}$. Since frequency and wavelength are related by $\nu = \frac{c}{\lambda}$, a higher frequency means a smaller wavelength. Therefore

$$R.P. \propto \nu$$

so the resolving power **increases** when the frequency of the incident light is increased. This is why short wavelength radiation (such as in electron microscopes) gives much finer resolution.

(b) Dependence on the focal length of the objective lens: The resolving power depends on the angle $\theta$ (through $\theta$). For an objective of a fixed diameter, a

**shorter focal length** makes the cone of light from the object steeper, so θ and hence θ become larger. Therefore the numerical aperture increases and the resolving power **increases** as the focal length of the objective is decreased.

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**Final answer.** Resolving power increases with the frequency of light (R.P. $\propto \nu \propto \frac{1}{\lambda}$) and increases as the focal length of the objective is decreased (since a shorter focal length gives a larger **θ**).

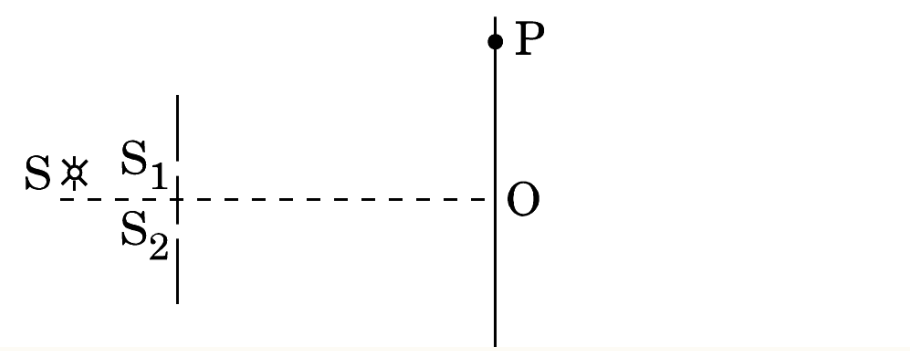
← *Back to Q99*

## Q100

*The figure shows a modified Young's double slit experimental set-up. Here* $\mathbf{SS_1 - SS_2 = \lambda/4}$.

*(a) Write the condition for constructive interference.*

*(b) Obtain an expression for the fringe width.*




**[2019 • Set 55-4-1]**

**SOLUTION**

In this modified set-up the source *S* is not equidistant from the two slits, so it is given that

$$SS_1 - SS_2 = \frac{\lambda}{4}$$

This means the light reaching slit $S_1$ has already travelled $\frac{\lambda}{4}$ more than the light reaching slit $S_2$. So there is a built in path difference of $\frac{\lambda}{4}$ even before the

light leaves the slits. This adds to the usual path difference ($S_1P - S_2P$) created between the slits and a point $P$ on the screen.

Total path difference up to $P$:

$$\Delta = (SS_1 - SS_2) + (S_1P - S_2P) = \frac{\lambda}{4} + (S_1P - S_2P)$$

(a) Condition for constructive interference: For a bright fringe the total path difference must be a whole number of wavelengths, $\mathbf{\Delta = n\lambda}$ (where $\boldsymbol{n = 0, 1, 2,}$ ). Thus

$$\frac{\lambda}{4} + (S_1P - S_2P) = n\lambda$$

$$(S_1P - S_2P) = n\lambda - \frac{\lambda}{4} = \frac{(4n - 1)\lambda}{4}$$

This is the required **condition for constructive interference**.

(b) Expression for fringe width: Using the geometry of the experiment, the path difference between the slits and point $P$ at distance $y$ from the centre is $\mathbf{(S_1P - S_2P) = \frac{y\,d}{D}}$. Putting this in the bright fringe condition,

$$\frac{y\,d}{D} = \frac{(4n-1)\lambda}{4}$$

$$y = \frac{(4n-1)\lambda\,D}{4d}$$

The fringe width β is the distance between two successive bright fringes, found by taking the difference for $n$+1 and $n$,

$$\beta = y_{n+1} - y_n = \frac{[4(n+1)-1]\lambda\,D}{4d} - \frac{(4n-1)\lambda\,D}{4d}$$

$$\beta = \frac{4\lambda\,D}{4d}$$

$$\beta = \frac{\lambda\,D}{d}$$

So the **fringe width is unchanged** and equals $\frac{\lambda\,D}{d}$; the extra source path difference only shifts the whole pattern by a quarter of a fringe.

**Final answer.** The condition for constructive interference is $(S_1P - S_2P) = \frac{(4n-1)\lambda}{4}$, and the fringe width is $\beta = \frac{\lambda D}{d}$ (same as ordinary YDSE).

← *Back to Q100*

## Q101

*Find the intensity at a point on a screen in Young's double slit experiment where the interfering waves of equal intensity have a path difference of (i) λ/4, and (ii) λ/3.*

**[2017]**

**SOLUTION**

In Young's double slit experiment the two waves arriving at a point are coherent and of equal intensity, say $I_0$ each. If ϕ is the phase difference between them, the resultant intensity is given by the interference relation.

$$I = I_0 + I_0 + 2I_0\phi = 4I_0\cos^2\left(\frac{\phi}{2}\right)$$

The phase difference is linked to the path difference Δ by the relation below.

$$\phi = \frac{2\pi}{\lambda}\Delta$$

(i) For a path difference $\Delta = \frac{\lambda}{4}$:

$$\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2}$$

$$I = 4I_0\cos^2\left(\frac{\pi}{4}\right) = 4I_0 \times \frac{1}{2}$$

$$I = \mathbf{2I_0}$$

(ii) For a path difference $\Delta = \frac{\lambda}{3}$:

$$\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{3} = \frac{2\pi}{3}$$

$$I = 4I_0\cos^2(\frac{\pi}{3}) = 4I_0\times(\frac{1}{2})^2 = 4I_0\times\frac{1}{4}$$

$$I = \mathbf{I_0}$$

So the intensity is $2I_0$ for a path difference of $\lambda/4$ and $I_0$ for a path difference of $\lambda/3$.

**COMMON MISTAKE**

*Always convert the given path difference to a phase difference first, then use* $\boldsymbol{I=4I_0\cos^2(\phi/2)}$*; forgetting the factor of 4 (which makes the maximum* $4I_0$*, not* $I_0$*) is the most common slip here.*

---

**Final answer.** Intensities are $\mathbf{2I_0}$ at path difference $\boldsymbol{\lambda/4}$ and $\mathbf{I_0}$ at path difference $\boldsymbol{\lambda/3}$.

← Back to Q101

## Q102

*Draw the intensity pattern for single slit diffraction and double slit interference. Hence, state two differences between interference and diffraction patterns.*

**[2017]**

**SOLUTION**

Single slit diffraction pattern: The graph of intensity against angular position θ shows a very bright, broad central maximum centred at $\boldsymbol{\theta = 0}$. On both sides the intensity falls to zero at the minima, which occur where $\boldsymbol{a\theta = n\lambda}$ for $\boldsymbol{n = \pm 1, \pm 2, \ldots}$ Between consecutive minima lie faint secondary maxima whose intensity drops off rapidly. The central maximum is twice as wide as any secondary maximum. The curve looks like one tall central hump flanked by progressively shorter humps.

Double slit interference pattern: The graph of intensity against θ shows a set of equally spaced bright fringes (maxima) of the same height, separated by equally spaced dark fringes (minima) where the intensity falls to zero. It is a

$\cos^2$ modulation about a uniform mean, so every bright fringe reaches the same peak value $4I_0$.

Two differences between interference and diffraction:

1. Fringe width and intensity: In interference all bright fringes are equally spaced and of equal intensity. In diffraction the central maximum is twice as wide as the secondary maxima, and the intensity of the secondary maxima falls off sharply.

2. Origin: Interference is the superposition of light from two (or a few) separate coherent wavefronts, whereas diffraction is the superposition of secondary wavelets coming from different parts of the same wavefront.

---

**Final answer.** Interference gives equal, equally spaced fringes; diffraction gives a broad bright central maximum with rapidly fading secondary maxima.

← *Back to Q102*

## Q103

*Unpolarised light is passed through a polaroid* $P_1$. *When this polarised beam passes through another polaroid* $P_2$ *and if the pass axis of* $P_2$ *makes angle* $\theta$ *with the pass axis of* $P_1$, *then write the expression for the polarised beam passing through* $P_2$. *Draw a plot showing the variation of intensity when* $\theta$ *varies from* 0 *to* $2\pi$.

**[2017]**

**SOLUTION**

When unpolarised light of intensity $I_0$ falls on the first polaroid $P_1$, the transmitted light is plane polarised along the pass axis of $P_1$ and its intensity drops to half.

$$I_1 = \frac{I_0}{2}$$

This polarised beam now strikes $P_2$ whose pass axis makes an angle $\theta$ with that of $P_1$. By Malus' law the intensity emerging from $P_2$ is given below.

$$I = I_1\cos^2\theta = \frac{I_0}{2}\cos^2\theta$$

This is the required expression for the intensity of the beam passing through $P_2$.

Plot of $I$ versus $\theta$ from 0 to $2\pi$: The graph is a $\cos^2\theta$ curve oscillating between 0 and $\frac{I_0}{2}$. It has maxima of height $\frac{I_0}{2}$ at $\boldsymbol{\theta = 0, \pi, 2\pi}$ and falls to zero at $\boldsymbol{\theta = \frac{\pi}{2}}$ and $\boldsymbol{\theta = \frac{3\pi}{2}}$. The curve never goes negative and repeats itself after every $\pi$, since the period of $\cos^2\theta$ is $\pi$. The shape is a series of identical rounded humps sitting on the $\theta$ axis, with vertical axis marked $I$ (peak $I_0/2$) and horizontal axis marked $\theta$.

---

**Final answer.** Transmitted intensity is $\boldsymbol{I = \frac{I_0}{2}\cos^2\theta}$, a $\boldsymbol{\cos^2\theta}$ curve peaking at $\boldsymbol{I_0/2}$ at $\boldsymbol{\theta = 0, \pi, 2\pi}$ and zero at $\boldsymbol{\pi/2, 3\pi/2}$.

← *Back to Q103*

## Q104

*Draw the intensity pattern for single slit diffraction and double slit interference. Hence, state two differences between interference and diffraction patterns.*

***OR***

*Unpolarised light is passed through a polaroid $P_1$. When this polarised beam passes through another polaroid $P_2$ and if the pass axis of $P_2$ makes angle $\theta$ with the pass axis of $P_1$, then write the expression for the polarised beam passing through $P_2$. Draw a plot showing the variation of intensity when $\theta$ varies from 0 to $2\pi$.*

**[2017]**

**SOLUTION**

Single slit diffraction pattern: The intensity against $\theta$ graph shows a bright, broad central maximum at $\boldsymbol{\theta = 0}$, falling to zero at the minima where $\boldsymbol{a\theta = n\lambda}$ for $\boldsymbol{n = \pm 1, \pm 2, \ldots}$ Between successive minima are weak secondary maxima whose height decreases rapidly outward. The central maximum is twice as wide as a secondary maximum, giving one tall central hump with smaller side humps.

Double slit interference pattern: The intensity against $\theta$ graph shows equally spaced bright fringes of equal height separated by equally spaced dark fringes where intensity is zero, a $\cos^2$ modulation in which every maximum reaches the same peak value.

Two differences:

1. In interference all bright fringes are equally spaced and of equal intensity; in diffraction the central maximum is twice as wide and the secondary maxima fall in intensity rapidly.

2. Interference arises from superposition of two separate coherent wavefronts; diffraction arises from superposition of secondary wavelets from one wavefront.

OR

When unpolarised light of intensity $I_0$ passes through the first polaroid $P_1$, it becomes plane polarised with half the intensity.

$$I_1 = \frac{I_0}{2}$$

Passing through $P_2$, whose axis makes angle $\theta$ with that of $P_1$, Malus' law gives the transmitted intensity.

$$I = \frac{I_0}{2}\cos^2\theta$$

Plot of $I$ versus $\theta$ from 0 to $2\pi$: a $\cos^2\theta$ curve with maxima $\frac{I_0}{2}$ at $\boldsymbol{\theta = 0, \pi, 2\pi}$ and zeros at $\boldsymbol{\theta = \frac{\pi}{2}, \frac{3\pi}{2}}$, repeating with period $\pi$, marked $I$ on the vertical axis and $\theta$ on the horizontal axis.

---

**Final answer.** Interference gives equal fringes and diffraction a broad central maximum; OR transmitted intensity is $\boldsymbol{\frac{I_0}{2}\cos^2\theta}$, a $\boldsymbol{\cos^2\theta}$ plot.

← *Back to Q104*

## Q105

*Find an expression for intensity of transmitted light when a polaroid sheet is rotated between two crossed polaroids. In which position of the polaroid sheet will the transmitted intensity be maximum?*

**[2015]**

**SOLUTION**

Consider two crossed polaroids $P_1$ and $P_3$ whose pass axes are at 90° to each other. By themselves they transmit no light. A third polaroid is now inserted between them and rotated, its pass axis making an angle θ with that of $P_1$.

Let unpolarised light of intensity $I_0$ fall on $P_1$. The light emerging is plane polarised along the axis of $P_1$ with half the intensity.

$$I_1 = \frac{I_0}{2}$$

This passes through the middle polaroid whose axis is at θ to $P_1$. By Malus' law the intensity becomes the following.

$$I_2 = I_1 \cos^2\theta = \frac{I_0}{2}\cos^2\theta$$

The axis of $P_3$ is at 90° to $P_1$, so it is at an angle $(90°-\theta)$ to the middle polaroid. Applying Malus' law again gives the final transmitted intensity.

$$I_3 = I_2 \cos^2(90°-\theta) = \frac{I_0}{2}\cos^2\theta \sin^2\theta$$

Using $\mathbf{\sin 2\theta = 2\theta\theta}$, so that $\mathbf{\sin^2\theta\cos^2\theta = \frac{\sin^2 2\theta}{4}}$:

$$I_3 = \frac{I_0}{8}\sin^2(2\theta)$$

This is the required expression. The transmitted intensity is maximum when $\mathbf{\sin^2(2\theta) = 1}$.

$$2\theta = 90° \Rightarrow \theta = 45°$$

Hence the transmitted intensity is maximum when the middle polaroid is at **45°** to the first polaroid, and its maximum value is $\frac{I_0}{8}$.

---

**Final answer.** Transmitted intensity is $\boldsymbol{I = \frac{I_0}{8}\sin^2(2\theta)}$, maximum $\mathbf{= I_0/8}$ at $\mathbf{\theta = 45°}$.

← *Back to Q105*

## Q106

*For a single slit of width "a", the first minimum of the interference pattern of a monochromatic light of wavelength λ occurs at an angle of $\frac{\lambda}{a}$. At the same angle of $\frac{\lambda}{a}$, we get a maximum for two narrow slits separated by a distance "a". Explain.*

**[2014]**

**SOLUTION**

Single slit of width $a$: A single slit produces a diffraction pattern. The first minimum (dark band) appears where the path difference between wavelets from the two edges of the slit equals one wavelength, that is the slit can be split into two halves whose contributions cancel.

$$a\theta = \lambda$$

For small angles $\theta \approx \theta$, so the first diffraction minimum occurs at the angle below.

$$\theta = \frac{\lambda}{a}$$

Two narrow slits separated by distance $a$: Two narrow slits produce an interference pattern. A bright fringe (maximum) occurs where the path difference between the two slits is a whole number of wavelengths.

$$a\theta = m\lambda,\ m = 0, 1, 2, \ldots$$

For the first order $\boldsymbol{m = 1}$ and small angles, the maximum occurs at the angle below.

$$\theta = \frac{\lambda}{a}$$

Explanation: In the single slit, the angle $\lambda/a$ is the condition for the wavelets from across the whole slit width to cancel, giving a minimum. In the two-slit case the same angle makes the path difference between the two slits exactly λ, so the two beams arrive in phase and reinforce, giving a maximum. The two results do not contradict each other because one condition refers to cancellation of wavelets spread over a single wide aperture, while the other

refers to reinforcement of two beams from two separate narrow apertures the same distance $a$ apart.

---

**Final answer.** $\lambda/a$ is the single-slit cancellation angle (minimum) but the two-slit one-wavelength path difference angle (maximum), so both occur at the same angle without contradiction.

← *Back to Q106*

## Q107

*Define the term 'linearly polarised light'.*
*When does the intensity of transmitted light become maximum, when a polaroid sheet is rotated between two crossed polaroids?*

**[2009]**

**SOLUTION**

Linearly polarised light: Light is said to be linearly (plane) polarised when the electric field vector $\vec{E}$ vibrates along a single fixed straight line in a plane perpendicular to the direction of propagation, instead of in all directions. The locus of the tip of $\vec{E}$ at any fixed point is a straight line. Such light can be obtained from unpolarised light by passing it through a polaroid, by reflection at the Brewster angle, or by scattering at 90°.

Maximum transmission through a polaroid rotated between two crossed polaroids: Let the two outer polaroids be crossed (axes at 90°) and let the middle polaroid make angle θ with the first. Unpolarised light of intensity $I_0$ on the first polaroid gives the following.

$$I_1 = \frac{I_0}{2}$$

After the middle polaroid, by Malus' law:

$$I_2 = \frac{I_0}{2}\cos^2\theta$$

The third polaroid is at (90°−θ) to the middle one, so again by Malus' law:

$$I_3 = \frac{I_0}{2}\cos^2\theta \sin^2\theta = \frac{I_0}{8}\sin^2(2\theta)$$

This is maximum when $\sin^2(2\theta) = 1$.

$$2\theta = 90^\circ \Rightarrow \theta = 45^\circ$$

Hence the transmitted intensity is maximum when the middle polaroid is at **45°** to the first, the maximum value being $\frac{I_0}{8}$.

---

**Final answer.** Linearly polarised light has $\vec{E}$ confined to one plane; transmission is maximum ($I_0/8$) when the inserted polaroid is at **45°**.

← *Back to Q107*

## Q108

*How is a wavefront defined? Using Huygens' construction draw a figure showing the propagation of a plane wave refraction at a plane surface separating two media. Hence verify Snell's law of refraction.*

**[2008]**

**SOLUTION**

Wavefront: A wavefront is the locus of all points of a medium that are vibrating in the same phase at a given instant. For a point source the wavefronts are spherical, while for a very distant source they are plane.

Huygens' construction for refraction: Consider a plane wavefront AB travelling in medium 1 with speed $v_1$ and striking a plane interface XY that separates medium 1 from a denser medium 2 in which the speed is $v_2$. Draw the incident wavefront AB meeting the surface; the incident rays (normal to AB) make the angle of incidence $i$ with the normal to XY, and this also equals the angle that AB makes with the interface. Let the wavefront reach point $A$ on the surface first while end $B$ still has to travel to point $C$ on the surface. In the diagram, $A$ is the foot on the surface, BC is drawn in medium 1 above the surface, and AD is drawn into medium 2 below the surface, with CD the refracted wavefront making angle $r$ with the interface.

In time $\Delta t$ the disturbance at $B$ travels to $C$ in medium 1.

$$BC = v_1 \Delta t$$

In the same time $\Delta t$ a secondary wavelet from $A$ enters medium 2 and spreads a distance AD.

$$AD = v_2 \Delta t$$

The tangent CD drawn from $C$ to the wavelet of radius AD is the refracted wavefront, and $r$ is the angle of refraction.

In right triangle ABC (right angle at $B$): $\sin i = \dfrac{BC}{AC} = \dfrac{v_1 \Delta t}{AC}$

In right triangle ADC (right angle at $D$): $\sin r = \dfrac{AD}{AC} = \dfrac{v_2 \Delta t}{AC}$

Dividing the two relations, the common factor AC and $\Delta t$ cancel.

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$

Since the refractive index of a medium is $\mu = \dfrac{c}{v}$, the ratio $\dfrac{v_1}{v_2} = \dfrac{\mu_2}{\mu_1}$, a constant for the two media.

$$\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1} = \text{constant}$$

This is **Snell's law of refraction**, verified by Huygens' wave construction.

---

**Final answer.** A wavefront is the locus of points in the same phase; Huygens' construction gives $\dfrac{\sin i}{\sin r} = \dfrac{v_1}{v_2} = \dfrac{\mu_2}{\mu_1} = \text{constant}$, which is Snell's law.

← *Back to Q108*

## Q109

*How will the angular separation and visibility of the fringes in Young's double slit experiment change when (i) Screen is moved away from the plane of the slits, and (ii)*

*width of the source slit is increased?*

**[2008]**

**SOLUTION**

(i) Screen moved away from the slits (D increased): The angular separation of the fringes depends only on the wavelength and the slit separation.

$$\theta = \frac{\lambda}{d}$$

Neither $\lambda$ nor $d$ changes when the screen is moved, so the angular separation remains unchanged. The linear fringe width, however, does depend on $D$.

$$\beta = \frac{\lambda D}{d}$$

Hence $\beta$ increases in proportion to $D$ and the fringes spread out on the screen. The visibility (contrast) of the fringes is the ratio $\frac{I_{max}-I_{min}}{I_{max}+I_{min}}$, which depends on the coherence of the two beams and not on $D$. So the visibility stays essentially the same; the fringes simply become wider and, in absolute terms, a little fainter, but the dark-to-bright contrast is not lost.

(ii) Width of the source slit increased: A wider source slit acts like many independent point sources side by side, each producing its own fringe pattern slightly shifted from the next. These patterns overlap on the screen, so bright fringes of one pattern fall partly on dark fringes of another. This reduces the contrast between maxima and minima, that is, the visibility of the fringes decreases. The angular positions of the fringes are unchanged. If the slit is made too wide, the overlap is so great that the fringes wash out into uniform illumination.

**Summary: Angular separation is unchanged in both cases. Moving the screen back leaves the visibility essentially unchanged (only $\beta$ increases), whereas widening the source slit reduces the visibility (loss of contrast).**

---

**Final answer.** Angular separation stays the same in both cases; moving the screen back only widens the fringes (visibility essentially unchanged), while widening the source slit reduces visibility by loss of contrast.

← Back to Q109

## Q110

*A double slit set-up was initially placed in a tank filled with water and the interference pattern was obtained using a laser light. When water is replaced by a transparent liquid of refractive index* $n > n_{water}$*, what will be the effect on the following ?*
***(a)*** *Speed, frequency and wavelength of the light of laser beam.*
***(b)*** *The fringe width, shape of interference fringes and shift in the position of central maximum.*

**[2025 • Set 55-4-1]**

**SOLUTION**

When water around the double slit set-up is replaced by a liquid of higher refractive index ($n > n_{water}$), the light is now travelling in a denser medium.

(a) Speed, frequency and wavelength of the laser light:

Speed: The speed of light in a medium is related to its refractive index.

$$v = \frac{c}{n}$$

As $n$ is larger than before, the speed $v$ decreases.

Frequency: The frequency is fixed by the source and does not change when light enters a new medium, so the frequency remains unchanged.

Wavelength: Using $v = \nu\lambda$, the wavelength in the medium is given below.

$$\lambda_{med} = \frac{v}{\nu} = \frac{c/n}{\nu} = \frac{\lambda_0}{n}$$

Since $n$ is larger, the wavelength decreases.

(b) Fringe width, shape and position of central maximum:

Fringe width: The fringe width depends on the wavelength in the medium.

$$\beta = \frac{\lambda_{med}\, D}{d}$$

Because $\lambda_{med}$ has decreased, the fringe width $\beta$ decreases, so the fringes come closer together.

Shape of fringes: The shape is fixed by the geometry of the set-up and not by the medium, so the fringes remain the same in shape (essentially straight, strictly hyperbolic).

Position of central maximum: The central maximum corresponds to zero path difference, $\Delta = 0$. As both slits lie in the same liquid, the symmetry is preserved, so there is no shift in the position of the central maximum.

**COMMON MISTAKE**

*A common error is to think the central maximum shifts when the medium changes; it only shifts if the two slits are covered by different media, but here both slits sit in the same liquid, so* $\Delta=0$ *stays at the centre.*

---

**Final answer.** Speed and wavelength decrease, frequency is unchanged; fringe width decreases, fringe shape is unchanged and the central maximum does not shift.

← *Back to Q110*

## Q111

***(a)*** *Define 'wavefront' of a light wave. A plane wavefront is refracted from a convex lens. Draw the shape of the refracted wavefront.*
***(b)*** *A plane wave travelling in a medium is incident on a plane surface separating this medium from a rarer medium. Draw a diagram to show refraction of the wave. Hence, verify Snell's law.*

**[2025 • Set 55-4-2]**

**SOLUTION**

(a) A wavefront is the locus of all points of a medium which are vibrating in the same phase at a given instant. The direction of propagation of light (the ray) is always perpendicular to the wavefront.

When a plane wavefront is incident on a convex (converging) lens, the central part of the lens is thicker than the edges, so light passing through the centre is

slowed down (retarded) more than light passing through the thin edges. The middle of the wavefront therefore lags behind. As a result the emerging wavefront curves inward and becomes a spherical wavefront converging towards the focus *F* of the lens.

Diagram (a): Show a vertical plane wavefront (a set of straight vertical lines) approaching a convex lens from the left. To the right of the lens, draw curved arcs that bend inward and shrink towards a single point *F* on the principal axis. Mark *F* as the focus and label the right side wavefronts as a spherical converging wavefront.

(b) By Huygens' construction, consider a plane wavefront AB travelling in the denser medium with speed $v_1$, incident on the plane interface XY at angle of incidence *i*. It passes into the rarer medium where the speed is $v_2$ (with $\mathbf{v_2 > v_1}$).

Point *A* reaches the surface first. While *B* travels the distance $\mathbf{BC = v_1 \Delta t}$ to reach point *C* on the interface, a secondary wavelet from *A* spreads into the rarer medium covering a distance $\mathbf{AD = v_2 \Delta t}$.

The refracted wavefront is the tangent CD drawn from *C* to the spherical wavelet centred at *A*. It makes the angle of refraction *r* with the interface.

From right triangle ABC:

$$\sin i = \frac{BC}{AC} = \frac{v_1 \Delta t}{AC}$$

From right triangle ADC:

$$\sin r = \frac{AD}{AC} = \frac{v_2 \Delta t}{AC}$$

Dividing:

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

$$\Rightarrow n_1 \sin i = n_2 \sin r$$

This is Snell's law of refraction. Since $\mathbf{v_2 > v_1}$, we get $\boldsymbol{r > i}$, so the ray bends away from the normal on going from a denser to a rarer medium.

Diagram (b): Draw a horizontal interface XY separating a lower denser medium from an upper rarer medium. Show the incident plane wavefront AB in the

denser medium hitting the surface, with $A$ on XY and $B$ above. Mark $\mathbf{BC = v_1 \Delta t}$ along the surface to point $C$, and $\mathbf{AD = v_2 \Delta t}$ into the rarer medium. Join CD as the refracted wavefront. Mark angle $i$ between AB and XY, and angle $r$ between CD and XY, with the normal at $A$.

---

**Final answer.** Wavefront is the locus of points in the same phase; a plane wavefront becomes spherical and converges to focus F after a convex lens, and Huygens' construction gives $\mathbf{n_1 \sin i = n_2 \sin r}$.

← *Back to Q111*

## Q112

***(a)*** *Write the conditions under which two light waves originating from two coherent sources can interfere each other (i) constructively, and (ii) destructively, in terms of wavelength. Can these be applied for two lights originating from two sodium lamps ? Give reason.*
***(b)*** *Monochromatic light of green colour is used in Young's double slit experiment and an interference pattern is observed on a screen. If the green light is replaced by red monochromatic light of the same intensity, how will the fringe width of interference pattern be affected ? Justify your answer.*

**[2025 • Set 55-4-3]**

**SOLUTION**

(a) For two coherent sources, the resultant intensity at a point on the screen depends on the path difference $\Delta$ between the two waves reaching that point.

(i) For constructive interference (bright fringe), the path difference must be an integral multiple of the wavelength:

$$\Delta = n\lambda,\ n = 0, 1, 2, 3, \ldots$$

(ii) For destructive interference (dark fringe), the path difference must be an odd multiple of half the wavelength:

$$\Delta = (2n - 1)\frac{\lambda}{2},\ n = 1, 2, 3, \ldots$$

These conditions cannot be applied to light from two separate sodium lamps. Two independent sodium lamps are not coherent: each emits light in tiny random bursts, so the phase difference between the two beams keeps changing rapidly and randomly with time. The interference pattern therefore shifts so fast that the eye sees only a uniform average illumination, and no steady pattern is observed. Coherent sources must be obtained from a single source (for example, by splitting one beam through two slits).

(b) The fringe width in Young's double slit experiment is given by

$$\beta = \frac{\lambda D}{d}$$

With $D$ (slit to screen distance) and $d$ (slit separation) unchanged, $\beta \propto \lambda$. Red light has a longer wavelength than green light, so $\lambda_{red} > \lambda_{green}$.

Therefore the fringe width increases when green light is replaced by red light. Taking $\lambda_{red} \approx 700$ nm and $\lambda_{green} \approx 550$ nm:

$$\frac{\beta_{red}}{\beta_{green}} = \frac{\lambda_{red}}{\lambda_{green}} = \frac{700}{550} \approx 1.27$$

So the fringes become wider (about 27% wider). The intensity of the light does not change the fringe width.

---

**Final answer.** Constructive: $\Delta = n\lambda$; destructive: $\Delta = (2n-1)\lambda/2$; two sodium lamps are incoherent so give no pattern; replacing green with red increases the fringe width since $\beta \propto \lambda$.

← *Back to Q112*

## Q113

*In Young's double slit experiment, the separation between the two slits is* 1.0 mm *and the screen is* 1.0 *m away from the slits. A beam of light consisting of two wavelengths* 500 nm *and* 600 nm *is used to obtain interference fringes. Calculate:*
***(a)*** *the distance between the first maxima for the two wavelengths.*

***(b)*** *the least distance from the central maximum, where the bright fringes due to both the wavelengths coincide.*

**[2025 • Set 55-5-1]**

**SOLUTION**

Given the slit separation $d = 1.0\ \text{mm} = 1.0 \times 10^{-3}\ m$, screen distance $D = 1.0\ m$, and wavelengths $\lambda_1 = 500\ \text{nm} = 500 \times 10^{-9}\ m$ and $\lambda_2 = 600\ \text{nm} = 600 \times 10^{-9}\ m$.

The position of the $n$th bright fringe is $y_n = \dfrac{n\lambda D}{d}$.

(a) For the first maximum ($n = 1$) of $\lambda_1$:

$$y_1 = \frac{\lambda_1 D}{d} = \frac{500 \times 10^{-9} \times 1.0}{1.0 \times 10^{-3}}$$

$$y_1 = 5.0 \times 10^{-4}\ m = 0.5\ \text{mm}$$

For the first maximum of $\lambda_2$:

$$y_2 = \frac{\lambda_2 D}{d} = \frac{600 \times 10^{-9} \times 1.0}{1.0 \times 10^{-3}}$$

$$y_2 = 6.0 \times 10^{-4}\ m = 0.6\ \text{mm}$$

The distance between the two first maxima is

$$\Delta y = y_2 - y_1 = 0.6 - 0.5 = 0.1\ \text{mm}$$

(b) The two bright fringes coincide when

$$n_1 \lambda_1 = n_2 \lambda_2$$

$$n_1 \times 500 = n_2 \times 600$$

$$\frac{n_1}{n_2} = \frac{600}{500} = \frac{6}{5}$$

The smallest whole numbers are $n_1 = 6$ and $n_2 = 5$. The least distance from the central maximum is

$$y = \frac{n_1 \lambda_1 D}{d} = \frac{6 \times 500 \times 10^{-9} \times 1.0}{1.0 \times 10^{-3}}$$

$$y = 3.0 \times 10^{-3}\ m = 3\ \text{mm}$$

**TIP**

When two wavelengths coincide, set $\mathbf{n_1\lambda_1 = n_2\lambda_2}$ and pick the smallest integers; the longer wavelength always gets the smaller order number.

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**Final answer.** (a) The first maxima are 0.1 mm apart; (b) the fringes first coincide at 3 mm from the central maximum.

← *Back to Q113*

## Q114

*Using the Huygens' principle, briefly describe reflection of a plane wavefront from a reflecting surface. Hence, prove the laws of reflection.*

**[2025 • Set 55-5-2]**

**SOLUTION**

Huygens' principle: Every point on a wavefront acts as a source of secondary wavelets that spread out in the forward direction with the speed of the wave in that medium. The new position of the wavefront after a time $\Delta t$ is the tangential surface (forward envelope) drawn over all these secondary wavelets.

Reflection of *a* plane wavefront: Let a plane wavefront AB travel with speed $v$ and strike a plane reflecting surface XY, with the angle of incidence $i$.

Point $A$ touches the surface first. While the other end $B$ travels the distance $\mathbf{BC = v\,\Delta\, t}$ to reach point $C$ on the surface, a secondary wavelet from $A$ spreads back into the same medium covering an equal distance $\mathbf{AD = v\,\Delta\, t}$.

The reflected wavefront is the tangent CD drawn from $C$ to the wavelet of radius AD centred at $A$. It makes the angle of reflection $r$ with the surface.

In right triangle ABC:

$$\sin i = \frac{\text{BC}}{\text{AC}} = \frac{v\,\Delta\, t}{\text{AC}}$$

In right triangle ADC:

$$\sin r = \frac{\text{AD}}{\text{AC}} = \frac{v\,\Delta\, t}{\text{AC}}$$

Since the right hand sides are equal:

$$\sin i = \sin r \Rightarrow i = r$$

Thus the angle of incidence equals the angle of reflection. Also, the incident wavefront, the reflected wavefront and the normal to the surface all lie in the same plane, which is the second law of reflection.

Diagram: Draw a horizontal reflecting surface XY. Show the incident plane wavefront AB coming down towards it, with end $A$ on the surface and $B$ above. Mark $\mathbf{BC = v\Delta t}$ from $B$ to point $C$ on XY, and a semicircular wavelet of radius $\mathbf{AD = v\Delta t}$ centred at $A$ above the surface. Draw the reflected wavefront CD tangent to this wavelet. Show the normal at $A$, with angle of incidence $i$ and angle of reflection $r$ marked equal.

---

**Final answer.** Huygens' construction gives $\mathbf{\sin i = \sin r}$, hence $\mathbf{i = r}$, proving the law of reflection.

← *Back to Q114*

## Q115

***(a)*** *"You cannot see a person standing on the other side of a boundary wall but can hear him." Explain with reason.*
***(b)*** *Light of wavelength* 750 nm *is incident normally on a slit of width* 1.5 mm. *Diffraction pattern is obtained on a screen* 1.0 *m away from the slit. Find the distance of the nearest point from the central maxima at which the intensity is zero.*

**[2025 • Set 55-5-3]**

**SOLUTION**

(a) The bending of waves around obstacles is called diffraction, and it is significant only when the size of the obstacle is comparable to the wavelength of the wave.

The wavelength of sound is of the order of a metre, which is comparable to the height of a boundary wall, so sound waves diffract strongly and bend around the wall to reach the listener. Hence the person can be heard.

The wavelength of light is extremely small (about $5 \times 10^{-7}\ m$), far smaller than the wall, so light shows negligible diffraction and travels in nearly straight lines. It is blocked by the wall, so the person cannot be seen.

(b) For single slit diffraction, the position of a minimum is given by $\boldsymbol{a\theta = n\lambda}$. The nearest point of zero intensity is the first minimum ($\boldsymbol{n = 1}$):

$$a\theta = \lambda$$

For small angles, $\theta \approx \theta = \frac{y}{D}$, so

$$\frac{a\,y}{D} = \lambda \Rightarrow y = \frac{\lambda\, D}{a}$$

Given $\boldsymbol{\lambda = 750\ nm = 750 \times 10^{-9}\ m}$, $\boldsymbol{a = 1.5\ mm = 1.5 \times 10^{-3}\ m}$, $\boldsymbol{D = 1.0\ m}$:

$$y = \frac{750 \times 10^{-9} \times 1.0}{1.5 \times 10^{-3}}$$

$$y = 5.0 \times 10^{-4}\ m = 0.5\ \text{mm}$$

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**Final answer.** (a) Sound diffracts because its wavelength matches the wall while light does not; (b) the nearest dark point lies 0.5 mm from the central maximum.

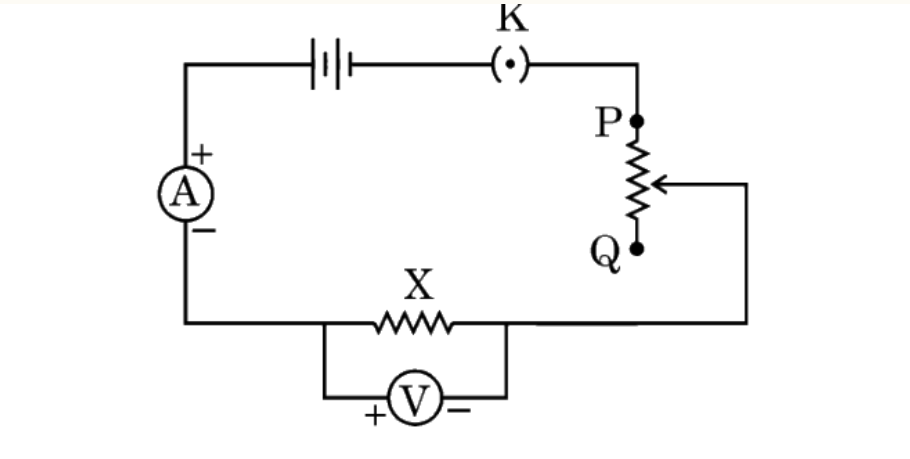
← *Back to Q115*

## Q116

*(a) (i) In diffraction due to a single slit, the phase difference between light waves reaching a point on the screen is* $\frac{5\pi}{2}$*. Explain whether a bright or a dark fringe will be formed at the point.*
*(ii) What should the width (a) of each slit be to obtain eight maxima of two double-slit patterns (slit separation d) within the central maximum of the single slit pattern?*
*(iii) Draw the plot of intensity distribution in a diffraction pattern due to a single slit.*
***OR***

*(b) (i) In a Young's double-slit experiment* $\boldsymbol{SS_1 - SS_2 = \frac{\lambda}{4}}$*, where* $S_1$ *and* $S_2$ *are the*

*two slits as shown in the figure. Find the path difference* ($S_1P - S_2P$) *for constructive and destructive interference at P.*




*(ii) What is the effect on the interference fringes in a Young's double-slit experiment, if the monochromatic source S is replaced by a source of white light?*

**[2023 • Set 55-4-1]**

**SOLUTION**

(a)(i) In single slit diffraction, a point is a maximum when the phase difference is $2n\pi$ and a minimum when it is $(2n+1)\pi$. The given phase difference is

$$\frac{5\pi}{2} = 2\pi + \frac{\pi}{2}$$

which is equivalent to a phase of $\frac{\pi}{2}$. This is neither a whole multiple of $2\pi$ nor an odd multiple of $\pi$, so the point is neither *a* fully bright nor *a* fully dark fringe. It corresponds to a point of intermediate intensity lying between a maximum and a minimum (closer to a dark region than to a bright one).

(a)(ii) The angular half width of the central diffraction maximum of a single slit of width $a$ is $\theta = \frac{\lambda}{a}$, so its full angular width is $\frac{2\lambda}{a}$. The angular separation between adjacent double slit maxima (slit separation $d$) is $\frac{\lambda}{d}$.

To fit 8 interference maxima inside the central diffraction maximum:

$$\frac{2\lambda}{a} = 8 \times \frac{\lambda}{d}$$

$$\frac{2}{a} = \frac{8}{d}$$

$$a = \frac{2d}{8} = \frac{d}{4}$$

(a)(iii) Intensity plot for single slit diffraction: Draw intensity $I$ on the vertical axis against $\theta$ (or position $y$) on the horizontal axis. There is one tall central maximum at $\boldsymbol{\theta = 0}$ of intensity $I_0$, with zeros (minima) on each side at $\boldsymbol{a\theta = \lambda, \pm 2\lambda, \ldots}$. Between the minima are secondary maxima that are much weaker, the first secondary maximum being only about 4.5% of $I_0$, and the heights falling rapidly further out. The curve is symmetric about the centre.

(b)(i) If light from source $S$ reaches the slits with $\boldsymbol{SS_1 - SS_2 = \frac{\lambda}{4}}$, this extra path is added to the path from the slits to $P$. The total path difference at $P$ is

$$\Delta = (SS_1 - SS_2) + (S_1P - S_2P) = \frac{\lambda}{4} + (S_1P - S_2P)$$

For constructive interference, $\boldsymbol{\Delta = n\lambda}$:

$$S_1P - S_2P = n\lambda - \frac{\lambda}{4},\ n = 0, 1, 2, \ldots$$

For destructive interference, $\boldsymbol{\Delta = (2n-1)\frac{\lambda}{2}}$:

$$S_1P - S_2P = (2n-1)\frac{\lambda}{2} - \frac{\lambda}{4} = \frac{(4n-3)\lambda}{4},\ n = 1, 2, \ldots$$

(b)(ii) If the monochromatic source is replaced by white light, the central fringe (zero path difference) is white, because all colours interfere constructively there. On either side of it appear a few coloured fringes, since different wavelengths form maxima at slightly different positions (violet nearest the centre, red farthest). Beyond these the fringes overlap and merge into general white illumination, so a clear sustained pattern is not seen.

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**Final answer.** (a)(i) Neither bright nor dark (intermediate intensity); (a)(ii) $\boldsymbol{a = d/4}$; (b)(i) constructive $\boldsymbol{S_1P - S_2P = n\lambda - \lambda/4}$, destructive $\boldsymbol{= (4n-3)\lambda/4}$; (b)(ii) central fringe white, a few coloured fringes, then uniform white.

← *Back to Q116*

## Q117

*(a) A plane wave-front propagating in a medium of refractive index '$\mu_1$' is incident on a plane surface making an angle of incidence (i). It enters into a medium of refractive index $\mu_2$ ($\mu_2 > \mu_1$). Use Huygen's construction of secondary wavelets to trace the refracted wave-front. Hence verify Snell's law of refraction.*

***OR***

*(b) Using Huygen's construction, show how a plane wave is reflected from a surface. Hence verify the law of reflection.*

**[2023 • Set 55-5-1]**

### SOLUTION

(a) Huygens' principle: Every point on a wavefront acts as a source of secondary wavelets that travel forward with the speed of the wave in that medium. The new wavefront after time $\Delta t$ is the forward envelope (tangent surface) of all these secondary wavelets.

Consider a plane wavefront AB travelling in medium 1 (refractive index $\mu_1$, speed $v_1$) and incident on the plane interface XY at angle of incidence $i$. It enters the denser medium 2 (refractive index $\mu_2 > \mu_1$, speed $v_2$, with $v_2 < v_1$).

Point $A$ reaches the interface first. While $B$ travels the distance $BC = v_1 \Delta t$ to reach $C$ on the surface, a secondary wavelet from $A$ advances into medium 2 by a distance $AD = v_2 \Delta t$.

The refracted wavefront is the tangent CD from $C$ to the wavelet of radius AD centred at $A$. It makes the angle of refraction $r$ with the interface.

From right triangle ABC:

$$\sin i = \frac{BC}{AC} = \frac{v_1 \Delta t}{AC}$$

From right triangle ADC:

$$\sin r = \frac{AD}{AC} = \frac{v_2 \Delta t}{AC}$$

Dividing:

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$

Since the refractive index of a medium is $\mu = \frac{c}{v}$, we have $\frac{v_1}{v_2} = \frac{\mu_2}{\mu_1}$, so

$$\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1}$$

$$\Rightarrow \mu_1 \sin i = \mu_2 \sin r$$

This is Snell's law of refraction. Because $\mu_2 > \mu_1$, we get $r < i$, so the ray bends towards the normal on entering the denser medium.

Diagram: Draw a horizontal interface XY, with the rarer medium $\mu_1$ above and the denser medium $\mu_2$ below. Show the incident plane wavefront AB in medium 1 hitting the surface, $A$ on XY and $B$ above. Mark $\mathbf{BC} = \mathbf{v_1 \Delta t}$ along to point $C$ on XY, and $\mathbf{AD} = \mathbf{v_2 \Delta t}$ into medium 2. Join CD as the refracted wavefront. Draw the normal at $A$; mark angle $i$ between AB and XY and the smaller angle $r$ between CD and XY.

---

**Final answer.** Huygens' construction for refraction into a denser medium gives $\mu_1 \sin i = \mu_2 \sin r$, with the ray bending towards the normal.

← *Back to Q117*

## Q118

*(i) Monochromatic light is incident on a surface separating two media. The frequency of the light after refraction remains unaffected but its wavelength changes. Why?*
*(ii) The frequency of an electromagnetic radiation is* $1.0 \times 10^{11}$ Hz. *Identify the radiation and mention its two uses.*

**[2022 • Set 55-1-1]**

**SOLUTION**

(i) The frequency of a light wave is fixed by the source that produces it, that is, by the rate at which the charges in the source oscillate. When light crosses a boundary into another medium, the oscillations of the electric and magnetic fields on the two sides of the surface must stay in step (continuous) at the boundary. This is only possible if the frequency $\nu$ remains unchanged.

However, the speed of light depends on the medium: $v = \frac{c}{n}$, so light slows down in a denser medium. Since $v = \nu\lambda$ and $\nu$ is constant, the wavelength must change in proportion to the speed:

$$\lambda_{\text{med}} = \frac{v}{\nu} = \frac{c/n}{\nu} = \frac{\lambda_{\text{vac}}}{n}$$

Thus in a denser medium the wavelength decreases while the frequency stays the same.

(ii) To identify the radiation, find its wavelength using $c = \nu\lambda$:

$$\lambda = \frac{c}{\nu} = \frac{3 \times 10^{8}}{1.0 \times 10^{11}}$$

$$\lambda = 3 \times 10^{-3}\ m = 3\ \text{mm}$$

A wavelength of 3 mm lies in the microwave region of the electromagnetic spectrum.

Two uses of microwaves:

1. They are used in microwave ovens for cooking food (water molecules absorb the radiation and heat up).

2. They are used in RADAR systems for aircraft navigation and speed detection, and in satellite communication.

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**Final answer.** (i) Frequency is set by the source and is preserved at a boundary, so only speed and wavelength change ($\lambda = \lambda_{\text{vac}}/n$); (ii) the radiation is a microwave ($\lambda = 3$ mm), used in microwave ovens and RADAR/satellite communication.

← *Back to Q118*

## Q119

*A parallel beam of light of wavelength* 600 nm *is incident normally on a slit of width* 0.2 mm. *If the resulting diffraction pattern is observed on a screen* 1 *m away, find the distance of*

*(i) first minimum, and*
*(ii) second maximum, from the central maximum.*

**[2022 • Set 55-1-1]**

**SOLUTION**

This is a single slit diffraction problem. Given $\lambda = 600\ \text{nm} = 6 \times 10^{-7}\ m$, slit width $a = 0.2\ \text{mm} = 2 \times 10^{-4}\ m$, and screen distance $D = 1\ m$. For small angles the position on the screen is $y = D\theta \approx D\theta$.

(i) The minima occur at $a\theta = n\lambda$. For the first minimum, $n = 1$:

$$a\theta_1 = \lambda \Rightarrow \theta_1 = \frac{\lambda}{a}$$

$$y_1 = D\theta_1 = \frac{\lambda D}{a} = \frac{6 \times 10^{-7} \times 1}{2 \times 10^{-4}}$$

$$y_1 = 3 \times 10^{-3}\ m = 3\ \text{mm}$$

(ii) The secondary maxima occur approximately at $a\theta = (n + \frac{1}{2})\lambda$. For the second secondary maximum, $n = 2$:

$$a\theta = (2 + \frac{1}{2})\lambda = \frac{5\lambda}{2}$$

$$y_2 = \frac{5\lambda D}{2a} = \frac{5 \times 6 \times 10^{-7} \times 1}{2 \times 2 \times 10^{-4}}$$

$$y_2 = \frac{30 \times 10^{-7}}{4 \times 10^{-4}}$$

$$y_2 = 7.5 \times 10^{-3}\ m = 7.5\ \text{mm}$$

---

**Final answer.** The first minimum is at 3 mm and the second secondary maximum is at 7.5 mm from the central maximum.

← *Back to Q119*

## Q120

*Using Huygen's principle, for a plane wavefront incident on a plane reflecting surface, draw the reflected wavefront. Hence prove laws of reflection.*

**[2022 • Set 55-1-3]**

**SOLUTION**

Huygens' principle: Every point on a wavefront acts as a source of secondary wavelets that spread forward with the speed of the wave in the medium. The new position of the wavefront after a time $\Delta t$ is the forward envelope (tangent surface) of all these secondary wavelets.

Consider a plane wavefront AB travelling with speed $v$ and incident on a plane reflecting surface XY at the angle of incidence $i$.

Point $A$ touches the surface first. While the other end $B$ travels the distance $\mathbf{BC = v\,\Delta\, t}$ to reach point $C$ on the surface, a secondary wavelet from $A$ spreads back into the same medium over an equal distance $\mathbf{AD = v\,\Delta\, t}$.

The reflected wavefront is the tangent CD drawn from $C$ to the wavelet of radius AD centred at $A$. It makes the angle of reflection $r$ with the surface.

From right triangle ABC:

$$\sin i = \frac{\text{BC}}{\text{AC}} = \frac{v\,\Delta\, t}{\text{AC}}$$

From right triangle ADC:

$$\sin r = \frac{\text{AD}}{\text{AC}} = \frac{v\,\Delta\, t}{\text{AC}}$$

Since the right hand sides are equal:

$$\sin i = \sin r \Rightarrow i = r$$

So the angle of incidence equals the angle of reflection. Moreover, the incident ray, reflected ray and the normal all lie in the same plane. These are the two laws of reflection.

Diagram: Draw a horizontal reflecting surface XY. Show the incident plane wavefront AB coming down with $A$ on the surface and $B$ above it. Mark $\mathbf{BC = v\Delta}$

$t$ from $B$ to point $C$ on XY, and a semicircular wavelet of radius $\mathbf{AD = v\Delta t}$ centred at $A$ above the surface. Draw the reflected wavefront CD tangent to that wavelet. Show the normal at $A$, marking the angle of incidence $i$ equal to the angle of reflection $r$.

---

**Final answer.** Huygens' construction gives $\mathbf{\sin i = \sin r}$, hence $\mathbf{i = r}$, which proves the law of reflection.

← *Back to Q120*

## Q121

*A plane wavefront of light of wavelength 'λ' is incident normally on a narrow slit of width 'a' and a diffraction pattern is observed on a screen at a distance 'D' from the slit.*
*(i) Depict the intensity distribution in the pattern observed.*
*(ii) Obtain the expression for the first maximum from the central maximum.*

**[2022 • Set 55-2-1]**

**SOLUTION**

When a plane wavefront of wavelength λ falls normally on a single slit of width $a$, the secondary wavelets from different points of the slit superpose and produce a diffraction pattern on the screen at distance $D$.

(i) Intensity distribution: The graph of intensity $I$ against angle θ (measured from the central axis) shows a bright, broad **central maximum** at $\theta = 0$ with intensity $I_0$.

On either side of it the intensity falls to zero at the first minima, then rises again to small secondary maxima, falls to zero at the next minima, and so on.

The minima occur where $a\theta = n\lambda$, with $n = 1, 2, 3,$

The secondary maxima are much fainter: the first secondary maximum has intensity about $\frac{I_0}{22}$ and the next about $\frac{I_0}{61}$, so the pattern dies out quickly away from the centre.

The envelope follows $I = I_0(\frac{\beta}{\beta})^2$ where $\beta = \frac{\pi a\theta}{\lambda}$.

The central maximum is twice as wide as any secondary maximum and is the most intense.

(ii) Position of the first secondary maximum: A secondary maximum lies roughly midway between two consecutive minima. The first secondary maximum lies between the first minimum ($a\theta = \lambda$) and the second minimum ($a\theta = 2\lambda$), so it occurs where

$$a\theta = \frac{3\lambda}{2}$$

$$\Rightarrow \theta = \frac{3\lambda}{2a}$$

For small angles, $\theta \approx \theta = \frac{y}{D}$, where $y$ is the linear distance of the first secondary maximum from the central maximum on the screen.

$$\frac{y}{D} = \frac{3\lambda}{2a}$$

$$\Rightarrow y = \frac{3\lambda D}{2a}$$

Thus the first secondary maximum lies at a distance $\boldsymbol{y = \frac{3\lambda D}{2a}}$ from the centre of the pattern.

---

**Final answer.** The first secondary maximum lies at $\boldsymbol{y = \frac{3\lambda D}{2a}}$ from the central maximum.

← *Back to Q121*

## Q122

*Using Huygens' principle, draw a diagram to show the refraction of a plane wavefront incident on a denser medium and hence verify Snell's law of refraction.*

**[2022 • Set 55-2-1]**

**SOLUTION**

Huygens' principle: Every point on a given wavefront acts as a fresh source of secondary wavelets which spread out in the forward direction with the speed of the wave in that medium. The new wavefront at any later instant is the forward envelope (tangent surface) of all these secondary wavelets.

Diagram (described): Draw a horizontal line XY as the boundary separating rarer medium 1 (above, speed $v_1$) from denser medium 2 (below, speed $v_2$). Show an incident plane wavefront AB in medium 1 striking the surface, with $A$ on the boundary and $B$ above the surface. The incident rays (perpendicular to AB) make angle $i$ with the normal at $A$. The refracted wavefront CD lies in medium 2, with $C$ on the boundary and $D$ below it; the refracted rays make angle $r$ with the normal. Mark BC along the surface and AD into the denser medium.

Derivation (verification of Snell's law): Let the wavefront AB reach the boundary first at $A$. In time $\Delta t$ the disturbance at $B$ travels in medium 1 to $C$ on the surface, so

$$BC = v_1 \Delta t$$

In the same time $\Delta t$, the secondary wavelet starting from $A$ travels into medium 2 a distance

$$AD = v_2 \Delta t$$

Joining $C$ to $D$ gives the refracted wavefront CD.

In right triangle ABC (right-angled at $B$), the angle $\mathbf{BAC} = \boldsymbol{i}$, so

$$\sin i = \frac{BC}{AC} = \frac{v_1 \Delta t}{AC}$$

In right triangle ADC (right-angled at $D$), the angle $\mathbf{ACD} = \boldsymbol{r}$, so

$$\sin r = \frac{AD}{AC} = \frac{v_2 \Delta t}{AC}$$

Dividing the two relations,

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$

Since $\mathbf{v_1} = \frac{\boldsymbol{c}}{\mathbf{n_1}}$ and $\mathbf{v_2} = \frac{\boldsymbol{c}}{\mathbf{n_2}}$, this gives

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} = \text{constant}$$

$$\Rightarrow n_1 \sin i = n_2 \sin r$$

This is **Snell's law of refraction**, hence verified using Huygens' principle.

**CONCEPT**

The angle between the wavefront and the surface equals the angle between the ray and the normal, which is why angle BAC in the triangle equals the angle of incidence i.

---

**Final answer.** Huygens' construction gives $\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$, which is Snell's law.

*← Back to Q122*

## Q123

*In a Young's double slit experiment using light of wavelength* 600 nm*, the slit separation is* 0.8 mm *and the screen is kept* 1.6 *m from the plane of the slits. Calculate:*
*(i) the fringe width*
*(ii) the distance of (a) third minimum and (b) fifth maximum, from the central maximum.*

**[2022 • Set 55-3-1]**

**SOLUTION**

Given: $\lambda = 600\ \text{nm} = 6\times 10^{-7}\ m$, slit separation $d = 0.8\ \text{mm} = 8\times 10^{-4}\ m$, screen distance $D = 1.6\ m$.

(i) Fringe width is given by

$$\beta = \frac{\lambda D}{d}$$

$$\beta = \frac{(6\times 10^{-7})(1.6)}{8\times 10^{-4}}$$

$$\beta = \frac{9.6\times 10^{-7}}{8\times 10^{-4}}$$

$$\beta = 1.2\times 10^{-3}\ m = 1.2\ \text{mm}$$

So the **fringe width is 1.2 mm**.

(ii)(a) Distance of the third minimum (dark fringe) from the central maximum. Dark fringes occur at

$$y_n^{\text{dark}} = (2n - 1)\frac{\beta}{2},\ n = 1, 2, 3,$$

For the third dark fringe, $n = 3$:

$$y_3^{\text{dark}} = (2\times 3 - 1)\frac{\beta}{2} = \frac{5\beta}{2}$$

$$y_3^{\text{dark}} = \frac{5\times 1.2}{2} = 3.0\ \text{mm}$$

So the **third minimum lies at 3.0 mm** from the centre.

(ii)(b) Distance of the fifth maximum (bright fringe) from the central maximum. Bright fringes occur at

$$y_n^{\text{bright}} = n\beta,\ n = 1, 2, 3,$$

For the fifth bright fringe, $n = 5$:

$$y_5^{\text{bright}} = 5\beta = 5\times 1.2 = 6.0\ \text{mm}$$

So the **fifth maximum lies at 6.0 mm** from the central maximum.

---

**Final answer.** Fringe width **$\beta$ = 1.2 mm**; third minimum at **3.0 mm** and fifth maximum at **6.0 mm** from the centre.

← *Back to Q123*

## Q124

*(i) State two conditions for two light sources to be coherent.*
*(ii) Give two points of difference between an interference pattern due to a double-slit*

*and a diffraction pattern due to a single slit.*

**[2022 • Set 55-3-1]**

**SOLUTION**

(i) Two conditions for two light sources to be coherent:

The two sources must emit light waves of the **same frequency** (and hence the same wavelength).

The two sources must maintain a **constant phase difference** that does not change with time. This is achieved in practice by obtaining both sources from a single original source (for example, the two slits in Young's experiment).

(ii) Two points of difference between interference (double slit) and diffraction (single slit):

In a double-slit interference pattern all the bright fringes are of equal width and nearly equal intensity, whereas in single-slit diffraction the central maximum is the brightest and roughly twice as wide as the secondary maxima, and the intensity of the secondary maxima falls off rapidly on either side.

Interference is produced by the superposition of waves coming from two separate coherent sources (the two slits), while diffraction is produced by the superposition of secondary wavelets coming from different parts of the same single wavefront passing through one slit. (Also, in interference the dark fringes are almost perfectly dark, but in diffraction the minima are not completely dark.)

---

**Final answer.** Coherent sources need the same frequency and a constant phase difference; interference gives equal-width equal-intensity fringes from two sources, while diffraction gives a dominant wide central maximum from one wavefront.

← *Back to Q124*

## Q125

*In a diffraction pattern due to a single slit, how will the angular width of central maximum change, if*

*(i) Orange light is used in place of green light,*
*(ii) the screen is moved closer to the slit,*
*(iii) the slit width is decreased?*
*Justify your answer in each case.*

**[2022 • Set 55-3-1]**

**SOLUTION**

For a single slit of width $a$, the angular width of the central maximum is

$$2\theta = \frac{2\lambda}{a}$$

where $\lambda$ is the wavelength of light. Notice that the angular width depends only on $\lambda$ and $a$, and not on the screen distance $D$.

(i) Orange light in place of green light: Orange light has a **larger wavelength** than green light ($\lambda_{\text{orange}} > \lambda_{\text{green}}$). Since $2\theta \propto \lambda$, the angular width of the central maximum **increases**.

(ii) Screen moved closer to the slit: The angular width $\frac{2\lambda}{a}$ does not contain $D$, so the **angular width remains unchanged**. Only the linear width of the central maximum on the screen (which equals $\frac{2\lambda D}{a}$) decreases as $D$ decreases.

(iii) Slit width decreased: Since $2\theta \propto \frac{1}{a}$, decreasing $a$ makes the angular width of the central maximum **increase** (the diffraction pattern spreads out more).

---

**Final answer.** Angular width increases with orange light, stays unchanged when the screen moves closer, and increases when the slit is narrowed.

← *Back to Q125*

## Q126

*How is the spacing between fringes in a double slit experiment affected if:*
*(i) the slits separation is increased,*
*(ii) the colour of light used is changed from red to blue,*

*(iii) the whole apparatus is submerged in a oil of refractive index* 1.2*?*
*Justify your answer in each case.*

**[2022 • Set 55-3-2]**

**SOLUTION**

The fringe width (spacing between consecutive fringes) in Young's double slit experiment is

$$\beta = \frac{\lambda D}{d}$$

where $\lambda$ is the wavelength, $D$ the slit-to-screen distance and $d$ the slit separation.

(i) Slit separation increased: Since $\beta \propto \frac{1}{d}$, increasing $d$ **decreases the fringe spacing**; the fringes come closer together.

(ii) Colour changed from red to blue: Blue light has a smaller wavelength than red light ($\lambda_{blue} < \lambda_{red}$). Since $\beta \propto \lambda$, the **fringe spacing decreases** and the fringes get narrower.

(iii) Apparatus submerged in oil of refractive index $\mu = 1.2$: Inside the oil the wavelength shortens to

$$\lambda' = \frac{\lambda}{\mu}$$

so the new fringe width is

$$\beta' = \frac{\lambda' D}{d} = \frac{\beta}{\mu} = \frac{\beta}{1.2}$$

The frequency stays the same, but the speed and wavelength fall by the factor $\mu$, so the **fringe spacing decreases** to about 0.83 $\beta$.

---

**Final answer.** Fringe spacing decreases in all three cases: when d increases, when red is changed to blue, and when the set-up is immersed in oil (by a factor of 1.2).

← *Back to Q126*

## Q127

*Monochromatic light of wavelength* 600 nm *is incident from air on a water surface. The refractive index of water is* 1.33. *Find the (i) wavelength, (ii) frequency and (iii) speed, of reflected and refracted light.*

**[2022 • Set 55-3-2]**

**SOLUTION**

When light passes from one medium to another, its **frequency stays unchanged** (it is set by the source), while its speed and wavelength change with the medium.

Given: $\boldsymbol{\lambda_{air} = 600\ nm = 6\times 10^{-7}\ m}$, refractive index of water $\boldsymbol{n_w = 1.33}$, speed of light $\boldsymbol{c = 3\times 10^{8}\ m/s}$.

First find the common frequency from the incident light in air:

$$\nu = \frac{c}{\lambda_{air}} = \frac{3\times 10^{8}}{6\times 10^{-7}} = 5\times 10^{14}\ \text{Hz}$$

Reflected light (travels back into air, the same medium as the incident light):

Wavelength $\boldsymbol{\lambda_{refl} = 600\ nm}$ (unchanged).

Frequency $\boldsymbol{\nu_{refl} = 5\times 10^{14}\ Hz}$.

Speed $\boldsymbol{v_{refl} = c = 3\times 10^{8}\ m/s}$.

Refracted light (travels into water):

Frequency $\boldsymbol{\nu_{refr} = 5\times 10^{14}\ Hz}$ (same as incident, frequency does not change).

Speed in water:

$$v_w = \frac{c}{n_w} = \frac{3\times 10^{8}}{1.33} = 2.26\times 10^{8}\ m/s$$

Wavelength in water:

$$\lambda_w = \frac{v_w}{\nu} = \frac{2.26\times 10^{8}}{5\times 10^{14}} = 4.51\times 10^{-7}\ m = 451\ \text{nm}$$

So both reflected and refracted light have **frequency $5\times 10^{14}$ Hz**; the **reflected** light keeps $\lambda = 600$ **nm** and $v = 3\times 10^{8}$ *m/s*, while the **refracted** light has $\lambda = 451$ **nm** and $v = 2.26\times 10^{8}$ *m/s*.

**COMMON MISTAKE**

*A very common mistake is to change the frequency in the denser medium. Frequency is fixed by the source, so it is the wavelength and speed that change, never the frequency.*

---

**Final answer.** Both beams have frequency $5\times 10^{14}$ Hz; reflected: $\lambda = 600$ nm, $v = 3\times 10^{8}$ m/s; refracted: $\lambda = 451$ nm, $v = 2.26\times 10^{8}$ m/s.

← Back to Q127

## Q128

*A beam of light consisting of two wavelengths* 600 nm *and* 500 nm *is used in a Young's double slit experiment. The slit separation is* 1.0 mm *and the screen is kept* 0.60 *m away from the plane of the slits. Calculate:*
*(i) the distance of the second bright fringe from the central maximum for wavelength* 500 nm, *and*
*(ii) the least distance from the central maximum where the bright fringes due to both the wavelengths coincide.*

**[2022 • Set 55-3-3]**

**SOLUTION**

Given: wavelengths $\lambda_1 = 600$ nm $= 6\times 10^{-7}$ m and $\lambda_2 = 500$ nm $= 5\times 10^{-7}$ m, slit separation $d = 1.0$ mm $= 1\times 10^{-3}$ m, screen distance $D = 0.60$ m.

The position of the $n$th bright fringe is

$$y_n = \frac{n\lambda D}{d}$$

(i) Second bright fringe for $\lambda_2 = 500$ nm (so $n = 2$):

$$y = \frac{2\lambda_2 D}{d} = \frac{2\times (5\times 10^{-7})\times 0.60}{1\times 10^{-3}}$$

$$y = \frac{6\times 10^{-7}}{1\times 10^{-3}} = 6\times 10^{-4}\ m = 0.6\ \text{mm}$$

So the **second bright fringe of 500 nm lies at 0.6 mm** from the centre.

(ii) The bright fringes coincide when the $n_1$th bright fringe of $\lambda_1$ falls on the $n_2$th bright fringe of $\lambda_2$, i.e. their positions are equal:

$$\frac{n_1\lambda_1 D}{d} = \frac{n_2\lambda_2 D}{d}$$

$$\Rightarrow n_1\lambda_1 = n_2\lambda_2$$

$$\Rightarrow 600\, n_1 = 500\, n_2$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{500}{600} = \frac{5}{6}$$

The smallest whole numbers are $\mathbf{n_1 = 5}$ and $\mathbf{n_2 = 6}$. The least distance is found using either wavelength, say $\lambda_1$ with $\mathbf{n_1 = 5}$:

$$y = \frac{n_1\lambda_1 D}{d} = \frac{5\times (6\times 10^{-7})\times 0.60}{1\times 10^{-3}}$$

$$y = \frac{1.8\times 10^{-6}}{1\times 10^{-3}} = 1.8\times 10^{-3}\, m = 1.8 \text{ mm}$$

So the bright fringes of both wavelengths first **coincide at 1.8 mm** from the central maximum.

---

**Final answer.** Second bright fringe of **500 nm** is at **0.6 mm**; the least distance where the two sets of bright fringes coincide is **1.8 mm**.

← *Back to Q128*

## Q129

*A plane wavefront is propagating from a rarer into a denser medium. Use Huygens principle to show the refracted wavefront and verify Snell's law.*

**[2022 • Set 55-4-1]**

**SOLUTION**

Huygens' principle: Every point on a wavefront acts as a source of secondary wavelets that travel forward with the speed of the wave in that medium, and the new wavefront after a time $\Delta t$ is the forward envelope of all these wavelets.

Diagram (described): Draw a horizontal boundary XY separating the rarer medium 1 above (wave speed $v_1$) from the denser medium 2 below (wave speed $v_2$). Show the incident plane wavefront AB in medium 1 reaching the surface, with point *A* already on the boundary and point *B* still above it; the incident rays make angle *i* with the normal. Below the surface draw the refracted wavefront CD with *C* on the boundary and *D* inside the denser medium; the refracted rays make angle *r* with the normal, bent towards the normal because the medium is denser.

Derivation: The wavefront AB reaches the boundary at *A* first. In time $\Delta t$ the point *B* travels through medium 1 to the boundary point *C*:

$$BC = v_1 \Delta t$$

During the same time, the secondary wavelet from *A* moves into the denser medium a distance

$$AD = v_2 \Delta t$$

Joining *D* to *C* gives the refracted wavefront CD.

In right triangle ABC (right angle at *B*), angle **BAC** = $i$:

$$\sin i = \frac{BC}{AC} = \frac{v_1 \Delta t}{AC}$$

In right triangle ADC (right angle at *D*), angle **ACD** = $r$:

$$\sin r = \frac{AD}{AC} = \frac{v_2 \Delta t}{AC}$$

Dividing,

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$

Writing $v_1 = \frac{c}{n_1}$ and $v_2 = \frac{c}{n_2}$,

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} = \text{constant}$$

$$\Rightarrow n_1 \sin i = n_2 \sin r$$

This is **Snell's law of refraction**, verified for a wave going from a rarer into a denser medium. Since $v_2 < v_1$ (denser medium), $r < i$, so the refracted wavefront bends towards the normal.

---

**Final answer.** Huygens' construction gives $\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$, Snell's law, with the ray bending towards the normal in the denser medium.

← *Back to Q129*

## Q130

*Briefly explain how bright and dark fringes are formed on the screen in Young's double slit experiment. Hence, derive the expression for the fringe width.*

**[2022 • Set 55-4-1]**

**SOLUTION**

Formation of bright and dark fringes: Monochromatic light from a single source illuminates two narrow slits $S_1$ and $S_2$ separated by a small distance $d$. By Huygens' principle these two slits behave as **coherent secondary sources** emitting waves in phase. These waves overlap (superpose) on a screen placed at distance $D$.

At a point $P$ on the screen the two waves arrive after travelling slightly different distances, so there is a path difference between them. Where the path difference makes the two waves arrive **in phase**, they reinforce (constructive interference) and a **bright fringe** is formed. Where they arrive **out of phase**, they cancel (destructive interference) and a **dark fringe** is formed.

Let $P$ be a point at distance $y$ from the central point $O$ on the screen. For $D \gg d$, the path difference is

$$\Delta = S_2P - S_1P = \frac{yd}{D}$$

Condition for bright fringe (constructive interference):

$$\Delta = n\lambda,\ n = 0, 1, 2,$$

$$\Rightarrow \frac{y_n\, d}{D} = n\lambda$$

$$\Rightarrow y_n = \frac{n\lambda\, D}{d}$$

The next bright fringe is at

$$y_{n+1} = \frac{(n+1)\lambda\, D}{d}$$

The fringe width is the spacing between two consecutive bright fringes:

$$\beta = y_{n+1} - y_n = \frac{(n+1)\lambda\, D}{d} - \frac{n\lambda\, D}{d}$$

$$\beta = \frac{\lambda\, D}{d}$$

The same spacing is obtained between two consecutive dark fringes, so the **fringe width is $\beta = \frac{\lambda\, D}{d}$.**

---

**Final answer.** Bright fringes form where path difference = $n\lambda$ and dark fringes where it is $(n+\frac{1}{2})\lambda$, giving fringe width $\beta = \frac{\lambda\, D}{d}$.

← Back to Q130

## Q131

*(a) The interference pattern is not observed in Young's double slit experiment when the two sources $S_1$ and $S_2$ are far apart. Explain.*
*(b) Mention the conditions for the two sources to be coherent.*
*(c) What is the effect on the interference pattern in a Young's double slit experiment, if the source of wavelength $\lambda$ is replaced by another source of wavelength $1.5\lambda$, with the interference pattern still observable?*

**[2022 • Set 55-4-2]**

**SOLUTION**

(a) In Young's double slit experiment, the fringe width is given by the relation $\beta = \frac{\lambda D}{d}$, where $d$ is the separation between the two slits $S_1$ and $S_2$.

Since $\beta \propto \frac{1}{d}$, when the two sources are placed far apart, $d$ becomes very large and the fringe width $\beta$ becomes extremely small.

The bright and dark fringes then crowd so close together that the eye (or detector) cannot resolve them separately. The screen appears uniformly illuminated, and **no interference pattern is observed**.

(b) The two sources are coherent if they satisfy these conditions:

They must emit light of the same frequency, that is, the same wavelength $\lambda$.

They must maintain a constant (time-independent) phase difference with no random fluctuations.

In practice such sources are obtained by deriving them from a single monochromatic source, so that any phase change in one is exactly matched in the other. **Two independent sources can never be coherent.**

(c) The fringe width is directly proportional to the wavelength, $\beta \propto \lambda$.

When $\lambda$ is replaced by $1.5\lambda$, the new fringe width becomes

$$\beta' = \frac{(1.5\lambda)D}{d} = 1.5\,\frac{\lambda D}{d} = 1.5\,\beta.$$

So the **fringe width increases to 1.5 times** its original value and the fringes spread farther apart on the screen.

---

**Final answer.** Far-apart sources make $\beta$ too small to resolve; coherent sources need same $\lambda$ and constant phase; with $1.5\lambda$ the fringe width becomes $1.5\beta$.

← *Back to Q131*

## Q132

*Briefly explain how bright and dark fringes are formed on a screen due to the diffraction at a single slit. Hence, explain why the intensity at the bright fringes*

*decreases sharply as their order (n) increases.*

**[2022 • Set 55-4-3]**

**SOLUTION**

Consider a single slit of width $a$ illuminated by monochromatic light of wavelength $\lambda$. By Huygens' principle, every point of the wavefront in the plane of the slit acts as a source of secondary wavelets, and these wavelets interfere on the screen.

The path difference between the wavelets from the two edges of the slit, reaching a point at angle $\theta$, is $a\theta$.

For dark fringes (minima): imagine the slit divided into an even number of equal strips. Wavelets from corresponding points of adjacent strips differ in path by $\frac{\lambda}{2}$ and cancel in pairs. This gives

$$a\theta = n\lambda,\ n = \pm 1, \pm 2, \pm 3, \ldots$$

For bright fringes (secondary maxima): these lie roughly midway between consecutive minima, at

$$a\theta = (n + \frac{1}{2})\lambda,\ n = 1, 2, 3, \ldots$$

The central maximum at $\theta = 0$ is the **brightest**, because here every wavelet from the slit arrives exactly in phase.

Why intensity falls with order: to locate the $n$-th secondary maximum, the slit is divided into (2n+1) equal strips. The first 2n strips cancel in pairs, and only one remaining strip of width $\frac{a}{2n+1}$ contributes to the light at that point.

The amplitude of this single strip is about $\frac{1}{2n+1}$ of the full slit's amplitude, so the intensity, being proportional to amplitude squared, falls as

$$\frac{I_n}{I_0} \approx \frac{1}{(2n+1)^2}.$$

Thus $\frac{I_1}{I_0} \approx \frac{1}{9}$, $\frac{I_2}{I_0} \approx \frac{1}{25}$, and so on, which is why the **intensity at higher-order bright fringes drops sharply** as $n$ increases.

**Final answer.** Minima occur at $a\theta = n\lambda$ and secondary maxima at $a\theta = (n+12)\lambda$; intensity falls as $1/(2n+1)^2$, so it decreases sharply with order $n$.

← Back to Q132

## Q133

*How can you differentiate whether a pattern is produced by a single slit or double slits? Derive the expression for the angular position of (i) bright and (ii) dark fringes produced in a single slit diffraction.*

**[2022 • Set 55-5-1]**

**SOLUTION**

Distinguishing the two patterns: a double-slit interference pattern shows a large number of bright fringes that are all of equal width and equal intensity, with uniform spacing $\beta = \frac{\lambda D}{d}$.

A single-slit diffraction pattern has a central maximum that is about twice as wide as the other fringes and far brighter; the secondary maxima on either side fall off rapidly in intensity. So unequal fringe widths and a dominant central peak signal single-slit diffraction, while equal, equally spaced fringes signal double-slit interference.

Now the angular positions for single-slit diffraction. Light of wavelength $\lambda$ passes through a slit of width $a$, and the path difference between wavelets from the two edges reaching angle $\theta$ is $a\theta$.

(i) Bright fringes (secondary maxima): these occur where the leftover uncancelled strip of the slit gives a net amplitude, approximately midway between minima:

$$a\theta = (n + \frac{1}{2})\lambda,\ n = 1, 2, 3, \ldots$$

For small angles $\theta \approx \theta$, so $\theta_n^{\max} = \frac{(2n+1)\lambda}{2a}$. (The central maximum at $\theta = 0$ is treated separately and is the brightest.)

(ii) Dark fringes (minima): the slit divides into pairs of strips whose wavelets cancel, giving

$$a\theta = n\lambda,\ n = \pm 1, \pm 2, \pm 3, \ldots$$

For small angles, $\theta_n^{\min} = \frac{n\lambda}{a}$.

Thus the **minima lie at $a\theta = n\lambda$ and the secondary maxima at $a\theta = (n+\frac{1}{2})\lambda$.**

---

**Final answer.** Single slit = wide bright central max with weakening side fringes; double slit = equal evenly spaced fringes. Minima: $a\theta = n\lambda$; secondary maxima: $a\theta = (n+12)\lambda$.

← *Back to Q133*

## Q134

*A slit of width* 0.6 mm *is illuminated by a beam of light consisting of two wavelengths* 600 nm *and* 480 nm. *The diffraction pattern is observed on a screen* 1.0 *m from the slit. Find:*
*(i) The distance of the second bright fringe from the central maximum pertaining to light of* 600 nm.
*(ii) The least distance from the central maximum at which bright fringes due to both the wavelengths coincide.*

**[2022 • Set 55-5-1]**

**SOLUTION**

Given a single slit of width $a = 0.6\ \text{mm} = 610^{-4}\ m$, screen distance $D = 1.0\ m$, and two wavelengths $\lambda_1 = 600\ \text{nm} = 60010^{-9}\ m$ and $\lambda_2 = 480\ \text{nm} = 48010^{-9}\ m$.

In single-slit diffraction the secondary (bright) maxima lie at $a\theta = (n+\frac{1}{2})\lambda = \frac{(2n+1)\lambda}{2}$. For small angles $\theta \approx \theta = \frac{y}{D}$, so the distance of the $n$-th bright fringe from the central maximum is $y = \frac{(2n+1)\lambda D}{2a}$.

(i) For the second bright fringe $(n = 2)$ of $\lambda_1 = 600\ \text{nm}$:

$$y = \frac{(2\times 2+1)\lambda_1 D}{2a} = \frac{5\,\lambda_1 D}{2a}.$$

$$y = \frac{5 \times (60010^{-9}) \times 1.0}{2 \times (610^{-4})}.$$

$$y = \frac{300010^{-9}}{1210^{-4}} = 2.510^{-3}\, m = 2.5 \text{ mm}.$$

(ii) The bright fringes of the two wavelengths coincide where their fringe distances are equal. Writing the bright-fringe position as $y = \frac{n\lambda D}{a}$, coincidence of the $n_1$-th fringe of $\lambda_1$ with the $n_2$-th fringe of $\lambda_2$ requires

$$n_1 \lambda_1 = n_2 \lambda_2.$$

$$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{480}{600} = \frac{4}{5}.$$

The smallest integers satisfying this are $n_1 = 4$ and $n_2 = 5$, so the first (least-distance) coincidence is the 4th bright fringe of 600 nm overlapping the 5th bright fringe of 480 nm.

Position of coincidence:

$$y = \frac{n_1 \lambda_1 D}{a} = \frac{4 \times (60010^{-9}) \times 1.0}{610^{-4}}.$$

$$y = \frac{240010^{-9}}{610^{-4}} = 410^{-3}\, m = 4 \text{ mm}.$$

So the **second bright fringe of 600 nm is at 2.5 mm and the two wavelengths first coincide at 4 mm** from the central maximum.

**COMMON MISTAKE**

*Use $y=\frac{(2n+1)\lambda D}{2a}$ for single-slit maxima, not $\frac{n\lambda D}{d}$ which is for double-slit fringes.*

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**Final answer.** Second bright fringe of **600 nm** lies at **2.5 mm**; the two wavelengths first coincide at **4 mm** from the central maximum.

← *Back to Q134*

## Q135

*How will the interference pattern in Young's double-slit experiment be affected if:*
*(i) The screen is moved away from the plane of the slits.*
*(ii) The source slit is moved away from the plane of the slits.*
*(iii) The phase difference between the light waves emanating from the two slits* $S_1$ *and* $S_2$ *changes from* 0 *to* $\pi$ *and remains constant.*

**[2022 • Set 55-5-2]**

**SOLUTION**

The fringe width in Young's double slit experiment is $\beta = \frac{\lambda D}{d}$, where $D$ is the slit-to-screen distance and $d$ the slit separation.

(i) Screen moved away from the slits ($D$ increases): since $\beta \propto D$, the **fringe width increases** and the pattern spreads out on the screen.

The angular fringe width $\theta = \frac{\lambda}{d}$ does not change, since it depends only on $\lambda$ and $d$.

(ii) Source slit moved away from the plane of the slits: moving the source slit farther reduces the angle it subtends at the double slit, so it acts even more like an ideal point source.

The illumination remains coherent and the fringe positions, set entirely by the double-slit geometry, do not change. So the **pattern stays essentially unaffected** (only the overall brightness may fall slightly as less light reaches the slits).

(iii) Constant phase difference of $\pi$ between $S_1$ and $S_2$: an initial phase difference of $\pi$ is equivalent to an extra path difference of $\frac{\lambda}{2}$.

The condition for a bright fringe, originally path difference = $n\lambda$, now becomes path difference = $(n + \frac{1}{2})\lambda$.

As a result, the positions of the bright and dark fringes are **interchanged**: where there were bright fringes there are now dark fringes and vice versa. In particular the centre of the screen, earlier the central bright fringe, now becomes dark. The fringe width $\beta$ is unchanged.

**Final answer.** Moving the screen away increases **β**; moving the source slit away leaves the pattern unchanged; a constant phase difference of **π** interchanges the bright and dark fringes (centre becomes dark).

← *Back to Q135*

## Q136

*Explain the formation of the fringes due to diffraction at a single slit, when path difference of light waves from the ends of the slit on reaching a point on the screen is (i) λ, and (ii)* $\frac{\lambda}{2}$.

*Show the intensity distribution in the fringes due to diffraction at a single slit.*

**[2021]**

**SOLUTION**

When light of wavelength λ falls on a single slit of width $a$, by Huygens' principle each point of the wavefront across the slit sends out secondary wavelets. These wavelets interfere on the screen, and the path difference between the wavelets from the two ends of the slit reaching a point at angle θ is $a\theta$.

(i) Path difference = **λ** (first minimum): here $\boldsymbol{a\theta = \lambda}$.

Divide the slit into two equal halves, each of width $\frac{a}{2}$. The path difference between a wavelet from a point in the upper half and the corresponding point in the lower half is

$$\frac{a}{2}\theta = \frac{\lambda}{2}.$$

Wavelets differing in path by $\frac{\lambda}{2}$ are out of phase by π and cancel in pairs. The whole slit cancels, giving the **first dark fringe (minimum)**.

(ii) Path difference = $\frac{\lambda}{2}$: here $\boldsymbol{a\theta = \frac{\lambda}{2}}$.

The wavelets from across the slit span a phase spread of only π, so they do not fully cancel; they add up to give a resultant that is part of the **bright central maximum**. This point lies inside the central bright band, between its centre

and the first minimum, so the intensity here is large but less than the central peak.

Intensity distribution: a graph of intensity $I$ against $\theta$ shows a tall, broad **central maximum** at $\theta = 0$ with peak intensity $I_0$. On either side the intensity drops to zero at the minima ($a\theta = \lambda, \pm 2\lambda, \ldots$). Between successive minima are **secondary maxima** that are much weaker, the first being only about $\frac{1}{22}$ of $I_0$, and their heights keep decreasing for higher orders. The pattern is symmetric about the centre.

---

**Final answer.** Path difference $\lambda$ gives pairwise cancellation, the first dark fringe; path difference $\frac{\lambda}{2}$ lies within the bright central maximum. The intensity graph shows a broad central peak with rapidly weakening secondary maxima on both sides.

← *Back to Q136*

## Q137

*What is the effect on the interference fringes in Young's double slit experiment due to each of the following operations? Justify your answers.*
*(a) The screen is moved away from the plane of the slits.*
*(b) The separation between slits is increased.*
*(c) The source slit is moved closer to the plane of double slit.*

**[2020 • Set 55-1-1]**

**SOLUTION**

The fringe width in Young's double slit experiment is $\beta = \frac{\lambda D}{d}$.

(a) Screen moved away from the slits ($D$ increases): the fringe positions are $y_n = \frac{n\lambda D}{d}$, directly proportional to $D$.

Since $\beta \propto D$, the **fringe width increases** and the whole pattern stretches out on the screen. The angular separation $\theta = \frac{\lambda}{d}$ stays the same.

(b) Separation between slits increased ($d$ increases): here $\beta = \frac{\lambda D}{d} \propto \frac{1}{d}$.

So the **fringe width decreases** and the fringes become more closely packed. Increasing $d$ makes the path difference at a given screen point grow faster, so consecutive bright fringes come closer together.

(c) Source slit moved closer to the double slit: as the source slit comes nearer, it subtends a larger angle at the double slit and behaves like an extended source rather than a point source.

Different points of this widened source illuminate the slits at slightly different angles, each producing its own slightly shifted fringe pattern. These overlapping patterns smear into one another, so the **fringes lose contrast and may wash out** when the source slit width exceeds the limit $\frac{s}{S} < \frac{\lambda}{d}$ (where $s$ and $S$ are the source-slit width and its distance).

---

**Final answer.** Moving the screen away increases $\beta$; increasing slit separation decreases $\beta$; bringing the source slit closer reduces fringe contrast and can wash out the pattern.

← *Back to Q137*

## Q138

*In a single slit diffraction experiment, light of wavelength $\lambda$ illuminates the slit of width a and the diffraction pattern is observed on a screen.*
*(a) Show the intensity distribution in the pattern with the angular position $\theta$.*
*(b) How are the intensity and angular width of central maxima affected when*
*(i) width of slit is increased, and*
*(ii) separation between slit and screen is decreased?*

**[2020 • Set 55-1-2]**

**SOLUTION**

(a) Intensity distribution in single-slit diffraction: plot intensity $I$ on the vertical axis against angular position $\theta$ on the horizontal axis.

The graph has a tall, broad **central maximum** at $\theta = 0$ with peak intensity $I_0$. The intensity falls to zero at the minima, located at $a\theta = \lambda, \pm 2\lambda, \pm 3\lambda, \ldots$

Between consecutive minima lie **secondary maxima** of much smaller intensity (the first about $\frac{1}{22}$ of $I_0$), whose heights decrease for higher orders. The curve is symmetric about $\theta = 0$, and the central maximum is twice as wide as each secondary maximum.

The angular half-width of the central maximum is $\theta = \frac{\lambda}{a}$, so its full angular width is $\frac{2\lambda}{a}$, and its linear width on a screen at distance $D$ is $\frac{2\lambda D}{a}$.

(b)(i) Width of slit increased ($a$ increases): the angular width $\frac{2\lambda}{a} \propto \frac{1}{a}$, so the **angular width of the central maximum decreases** (the central peak becomes narrower).

At the same time, a wider slit lets through more light, concentrated into a narrower band, so the **intensity of the central maximum increases**.

(b)(ii) Separation between slit and screen decreased ($D$ decreases): the angular width $\frac{2\lambda}{a}$ depends only on $\lambda$ and $a$, so it is **unchanged**.

The linear width $\frac{2\lambda D}{a} \propto D$, so the **linear width of the central maximum decreases** in proportion to $D$.

---

**Final answer.** The pattern is a broad bright central peak with weak side maxima. Increasing slit width narrows the central maximum and raises its intensity; decreasing $D$ leaves the angular width unchanged but reduces the linear width.

← *Back to Q138*

## Q139

*Give reasons for each of the following:*
*(a) The intensity of light at some points on the screen in Young's double slit experiment is zero.*
*(b) The intensity of light transmitted by a polaroid is less than the intensity of the*

*unpolarised light incident on it.*
*(c) In the single slit diffraction experiment, some coloured fringes around the central white maximum are observed on the screen when one uses a source of white light.*

**[2020 • Set 55-1-3]**

**SOLUTION**

(a) Zero intensity at some points: at these points the two waves from the slits arrive with a path difference of an odd multiple of $\frac{\lambda}{2}$, that is a phase difference of $\pi$ (or odd multiples of $\pi$).

The two waves are then exactly out of phase, so they undergo **destructive interference** and their amplitudes cancel, giving zero resultant intensity. The energy is not lost but redistributed to the bright fringes, where the waves add up constructively.

(b) Reduced intensity through a polaroid: ordinary (unpolarised) light contains vibrations of the electric field in all directions perpendicular to the direction of travel.

A polaroid transmits only the component of vibration parallel to its pass axis and absorbs the perpendicular component. Since only part of the incident vibration gets through, the **transmitted intensity is less than the incident intensity** (it drops to half, $\frac{I_0}{2}$, for unpolarised light, by Malus' law averaged over all angles).

(c) Coloured fringes with white light: white light is a mixture of all wavelengths (colours). In single-slit diffraction the positions of the maxima and minima depend on wavelength, since $a\theta = n\lambda$.

At the centre ($\theta = 0$) all wavelengths give a maximum together, so the **central maximum is white**. Away from the centre, each colour is diffracted by a different amount, with red (longer $\lambda$) spreading more than violet (shorter $\lambda$). The overlapping maxima of different colours therefore separate, producing **coloured fringes** around the central white band.

---

**Final answer.** Zero intensity comes from destructive interference (phase difference $\pi$); a polaroid passes only one vibration component so transmitted

intensity drops; in white light each wavelength diffracts differently, giving a white centre with coloured side fringes.

← Back to Q139

## Q140

*Two coherent light waves of intensity* $510^{-2}\ W\,m^{-2}$ *each super-impose and produce the interference pattern on a screen. At a point where the path difference between the waves is* $\lambda/6$, $\lambda$ *being the wavelength of the wave, find the*
*(a) phase difference between the waves.*
*(b) resultant intensity at the point.*
*(c) resultant intensity in terms of the intensity at the maximum.*

**[2020 • Set 55-4-1]**

**SOLUTION**

Given two coherent waves each of intensity $\mathbf{I_0 = 510^{-2}\ W\,m^{-2}}$, with path difference $\Delta = \frac{\lambda}{6}$ at the given point.

(a) Phase difference: the relation between phase difference and path difference is

$$\phi = \frac{2\pi}{\lambda}\,\Delta.$$

$$\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{6} = \frac{2\pi}{6} = \frac{\pi}{3}.$$

So $\phi = \frac{\pi}{3} = 60°$.

(b) Resultant intensity: for two coherent waves of equal intensity $I_0$,

$$I = 4\,I_0 \cos^2\left(\frac{\phi}{2}\right).$$

$$I = 4 \times (510^{-2}) \times \cos^2\left(\frac{\pi}{6}\right).$$

Now $\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$, so $\cos^2\frac{\pi}{6} = \frac{3}{4}$.

$$I = 0.2 \times \frac{3}{4} = 0.15\ W\,m^{-2}.$$

(c) Ratio to the maximum intensity: the maximum occurs when $\cos^2\left(\frac{\phi}{2}\right) = 1$, giving

$I_{max} = 4\,I_0 = 4 \times (510^{-2}) = 0.2\ W\,m^{-2}$.

$$\frac{I}{I_{max}} = \frac{0.15}{0.2} = \frac{3}{4}.$$

Hence $I = \frac{3}{4}\,I_{max} = 0.75\,I_{max}$, so the intensity at this point is **three-quarters of the maximum** bright-fringe intensity.

> **TIP**
>
> Convert path difference to phase with $\phi = \frac{2\pi}{\lambda}\Delta$, then use $I = 4I_0\cos^2(\phi/2)$; halve the phase inside the cosine.

---

**Final answer.** Phase difference $\phi = \frac{\pi}{3} = 60°$, resultant intensity $I = 0.15\ W\,m^{-2}$, which equals $\frac{3}{4}I_{max}$.

← *Back to Q140*

## Q141

*Define the term wavefront. Using Huygens' wave theory, verify the law of reflection.*

**[2019 • Set 55-1-1]**

**SOLUTION**

A **wavefront** is the locus of all points of a medium that are vibrating in the same phase at a given instant of time.

The shape of the wavefront depends on the source. A point source gives a spherical wavefront, a line source gives a cylindrical wavefront, and a source at a very large distance (or an extended plane source) gives a plane wavefront.

According to **Huygens' principle**, every point on a given wavefront acts as a fresh source of secondary spherical wavelets which spread out in all directions with the speed of the wave. The forward envelope (tangential surface) of all

these secondary wavelets at any later instant gives the new position of the wavefront.

To verify the law of reflection, consider a plane wavefront AB incident on a plane reflecting surface MN at an angle of incidence $i$. The incident rays travel with speed $v$.

Diagram: a horizontal reflecting line MN; the incident wavefront AB approaches from the upper left so that point $A$ touches the surface first while $B$ is still above it. AC lies along the surface, BC is drawn from $B$ to the point $C$ where ray through $B$ later meets the surface, and AD is the reflected wavelet drawn back into the original medium. The angle $i$ is between the incident wavefront AB and the surface, and $r$ is between the reflected wavefront CD and the surface.

Point $A$ reaches the surface first. Let the disturbance at $B$ take time $\Delta t$ to reach $C$ on the surface, so

$$\mathrm{BC} = v\,\Delta t$$

During this same time $\Delta t$, the secondary wavelet starting from $A$ travels back into the same medium and covers an equal distance,

$$\mathrm{AD} = v\,\Delta t$$

Hence **BC = AD**. Drawing the tangent CD to the secondary wavelet from $A$ gives the reflected wavefront.

In the right triangles ABC and ADC, which share the common hypotenuse AC,

$$\sin i = \frac{\mathrm{BC}}{\mathrm{AC}} = \frac{v\,\Delta t}{\mathrm{AC}}$$

$$\sin r = \frac{\mathrm{AD}}{\mathrm{AC}} = \frac{v\,\Delta t}{\mathrm{AC}}$$

Therefore

$$\sin i = \sin r \Rightarrow i = r$$

The angle of incidence equals the angle of reflection, and the incident ray, reflected ray and normal lie in the same plane. This **verifies the law of reflection**.

**Final answer.** Using Huygens' construction, $\sin i = \sin r$, so the angle of incidence equals the angle of reflection.

← Back to Q141

## Q142

*Define the term, "refractive index" of a medium. Verify Snell's law of refraction when a plane wavefront is propagating from a denser to a rarer medium.*

**[2019 • Set 55-1-1]**

**SOLUTION**

The **refractive index** of a medium is defined as the ratio of the speed of light in vacuum to the speed of light in that medium,

$$n = \frac{c}{v}$$

where $c$ is the speed of light in vacuum and $v$ is its speed in the medium. It tells us how much the medium slows down and bends the light.

To verify Snell's law, consider a plane wavefront AB travelling in a denser medium (refractive index $n_1$, speed $v_1$) and falling on the interface XY that separates it from a rarer medium (refractive index $n_2$, speed $v_2$), where $\mathbf{v_2 > v_1}$.

Diagram: a horizontal interface XY; the incident wavefront AB comes from below-left in the denser medium making angle of incidence $i$ with the surface; $A$ meets the interface first. BC is the distance the wavelet from $B$ travels to reach $C$ on the interface, and AD is the refracted wavelet drawn into the rarer medium above. CD is the refracted wavefront, making angle of refraction $r$ with the surface, with $\boldsymbol{r > i}$.

Let the wavelet from $B$ take time $\Delta t$ to reach $C$ on the interface in the denser medium,

$$BC = v_1 \, \Delta t$$

In the same time $\Delta t$, the secondary wavelet from $A$ advances into the rarer medium and covers

$$AD = v_2 \Delta t$$

The tangent CD to this wavelet is the refracted wavefront. In the right triangles ABC and ADC with common hypotenuse AC,

$$\sin i = \frac{BC}{AC} = \frac{v_1 \Delta t}{AC}$$

$$\sin r = \frac{AD}{AC} = \frac{v_2 \Delta t}{AC}$$

Dividing,

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$

Since $\mathbf{n_1} = \frac{c}{\mathbf{v_1}}$ and $\mathbf{n_2} = \frac{c}{\mathbf{v_2}}$, we get $\frac{\mathbf{v_1}}{\mathbf{v_2}} = \frac{\mathbf{n_2}}{\mathbf{n_1}}$, so

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$$

$$n_1 \sin i = n_2 \sin r$$

This is **Snell's law of refraction**, which is thus verified by Huygens' wave theory.

---

**Final answer.** Huygens' construction gives $\mathbf{n_1 \sin i = n_2 \sin r}$, which is Snell's law.

← *Back to Q142*

## Q143

*(a) Explain how an unpolarised light gets polarised when incident on the interface separating the two transparent media.*
*(b) Green light is incident at the polarising angle on a certain transparent medium. The angle of refraction is 30°. Find*
*(i) polarising angle, and*
*(ii) refractive index of the medium.*

**[2019 • Set 55-3-1]**

**SOLUTION**

(a) When unpolarised light is incident on the interface separating two transparent media (such as air and glass), the reflected light becomes **plane polarised** at one particular angle of incidence called the **polarising angle** or **Brewster angle** $\theta_B$.

At this angle, the reflected ray and the refracted ray are mutually perpendicular, that is, the angle between them is 90°. The reflected light then has its electric vector $\vec{E}$ vibrating only in a direction perpendicular to the plane of incidence; the component of $\vec{E}$ parallel to the plane of incidence is not reflected. Hence the reflected light is completely plane polarised.

Diagram: an unpolarised ray strikes the surface at angle $\theta_B$; the reflected ray (shown with dots, meaning vibrations perpendicular to the plane of paper) leaves at $\theta_B$, and the refracted ray enters the medium with the reflected and refracted rays 90° apart.

Using the geometry $\theta_B + r = 90°$ in Snell's law gives **Brewster's law**,

$$n = \theta_B$$

(b) Given: angle of refraction $r = 30°$ and light incident at the polarising angle.

(i) At the polarising angle the reflected and refracted rays are perpendicular, so

$$\theta_B + r = 90°$$

$$\theta_B = 90° - 30°$$

$$\theta_B = 60°$$

(ii) By Brewster's law,

$$n = \theta_B$$

$$n = \tan 60°$$

$$n = \sqrt{3} \approx 1.732$$

So the polarising angle is 60° and the refractive index of the medium is $\sqrt{3}$.

**TIP**

At Brewster's angle the reflected and refracted rays are exactly 90° apart; this single fact gives both $\theta_B$ and $n$.

---

**Final answer.** Polarising angle $\theta_B = 60°$ and refractive index $n = \tan 60° = \sqrt{3} \approx 1.732$.

← *Back to Q143*

## Q144

*(a) Define a wavefront. Using Huygens' geometrical construction, explain with the help of a diagram how the plane wavefront travels from the instant $t_1$ to $t_2$ in air.*
*(b) A plane wavefront is incident on a convex lens. Explain, with the help of the diagram, the shape of the refracted wavefront formed.*

**[2019 • Set 55-5-1]**

**SOLUTION**

(a) A **wavefront** is the locus of all points of a medium vibrating in the same phase at a given instant of time.

By **Huygens' geometrical construction**, every point on a wavefront acts as a fresh source of secondary spherical wavelets that travel forward with the speed of light $c$ in air. The forward tangential envelope of all these secondary wavelets gives the new wavefront.

Diagram for a plane wavefront in air: let AB be the plane wavefront at instant $t_1$. Take several points on AB as centres and draw secondary spherical wavelets of radius $r = c(t_2 - t_1)$. The common tangent plane $A'B'$ drawn in the forward direction, parallel to AB and at perpendicular distance $c(t_2 - t_1)$ ahead of it, is the new plane wavefront at instant $t_2$.

$$r = c\,(t_2 - t_1)$$

So a plane wavefront in air simply advances forward rigidly, remaining plane and parallel to its earlier position, the perpendicular distance moved being $c(t_2 - t_1)$.

(b) When a plane wavefront is incident on a **convex lens**, the central part of the wavefront passes through the thicker middle of the lens and the outer parts pass through the thinner edges.

Since the lens material is optically denser than air, light slows down inside it. The central portion of the wavefront, travelling through more glass, is delayed (retarded) more than the marginal portions. As a result the emerging wavefront bends inward and becomes **spherical and converging**.

Diagram: a plane wavefront approaches the convex lens from the left; after refraction the emergent wavefront is curved (concave towards the lens) and converges to meet at the principal focus $F$ on the right of the lens.

Hence the refracted wavefront is a **converging spherical wavefront** that comes to a point at the focus.

---

**Final answer.** A plane wavefront stays plane and advances $c(t_2 - t_1)$ in air; after a convex lens it becomes a converging spherical wavefront focusing at $F$.

← *Back to Q144*

## Q145

*(a) Good quality sunglasses made of polaroids are preferred over ordinary coloured glasses. Explain why.*
*(b) How is plane polarized light defined?*
*(c) A beam of plane polarised light is passed through a polaroid. Show graphically, variation of the intensity of the transmitted light with angle of rotation of the polaroid.*

**[2019 • Set 55-5-1]**

**SOLUTION**

(a) Glare reaching our eyes from horizontal surfaces such as water, wet roads, snow and car bonnets is largely **horizontally polarised**, because light reflected near the Brewster angle is polarised parallel to the reflecting surface.

Good quality sunglasses made of polaroids have their pass (transmission) axis kept vertical. They therefore block this horizontally polarised glare while still

letting through ordinary unpolarised light from the surroundings, so vision stays clear. Ordinary coloured glasses merely absorb a fixed fraction of light of all kinds equally and cannot selectively cut out glare. This is why polaroid sunglasses are preferred.

(b) **Plane (linearly) polarised light** is light in which the vibrations of the electric field vector $\vec{E}$ are confined to a single fixed plane containing the direction of propagation. At any point, the tip of $\vec{E}$ moves along a straight line perpendicular to the direction of travel.

(c) When plane polarised light of intensity $I_0$ passes through a polaroid whose pass axis makes angle $\theta$ with the direction of polarisation, the transmitted intensity follows **Malus' law**,

$$I = I_0 \cos^2\theta$$

As the polaroid is rotated, $\theta$ changes from $0^\circ$ to $360^\circ$. The intensity is maximum ($I_0$) when $\theta = 0^\circ, 180^\circ, 360^\circ$ and zero when $\theta = 90^\circ, 270^\circ$.

Graph: plot $I$ (y-axis) against the angle of rotation $\theta$ (x-axis). The curve is a $\cos^2\theta$ shape: it starts at the maximum value $I_0$ at $\theta = 0^\circ$, falls smoothly to 0 at $\theta = 90^\circ$, rises back to $I_0$ at $\theta = 180^\circ$, drops to 0 again at $\theta = 270^\circ$, and returns to $I_0$ at $\theta = 360^\circ$. It is a series of identical humps never going below zero.

---

**Final answer.** Polaroid sunglasses cut horizontally polarised glare; plane polarised light has $\vec{E}$ in one plane, and the transmitted intensity follows $I = I_0\cos^2\theta$.

← *Back to Q145*

## Q146

*(a) What do you understand by coherent sources?*
*(b) Derive the expression for the intensity distribution in the interference pattern on the screen when two harmonic waves having same amplitude (a) and phase difference ($\phi$) are superposed on each other. Hence obtain the conditions for constructive and destructive interference.*

**[2019 • Set 55-5-2]**

**SOLUTION**

(a) Two sources of light are said to be **coherent** if they emit light waves of the **same frequency (or wavelength)** and maintain a **constant phase difference** with respect to each other at all times. Only such sources can produce a steady, observable interference pattern. In practice coherent sources are obtained from a single parent source, for example the two slits in Young's double-slit experiment lit by one monochromatic source.

(b) Let the two superposing harmonic waves of equal amplitude $a$ and constant phase difference $\phi$ be

$$y_1 = a\omega\, t$$

$$y_2 = a\cos(\omega\, t + \phi)$$

By the principle of superposition, the resultant displacement is

$$y = y_1 + y_2 = a\omega\, t + a\cos(\omega\, t + \phi)$$

Using $\cos C + \cos D = 2\cos\frac{C+D}{2}\cos\frac{C-D}{2}$,

$$y = 2a\cos\frac{\phi}{2}\cos(\omega\, t + \frac{\phi}{2})$$

This represents a wave of the same frequency $\omega$ with resultant amplitude

$$A_R = 2a\cos\frac{\phi}{2}$$

Since intensity is proportional to the square of amplitude,

$$I = A_R^2 = 4a^2\cos^2\frac{\phi}{2}$$

**Constructive interference (maxima):** intensity is maximum when $\cos\frac{\phi}{2} = \pm 1$,

$$\frac{\phi}{2} = n\pi \Rightarrow \phi = 2n\pi,\ n = 0, 1, 2, \ldots$$

giving $I_{max} = 4a^2$. In terms of path difference $\Delta = \frac{\lambda}{2\pi}\phi$, this means $\Delta = n\lambda$.

**Destructive interference (minima):** intensity is minimum when $\cos\frac{\phi}{2} = 0$,

$$\frac{\phi}{2} = (2n+1)\frac{\pi}{2} \Rightarrow \phi = (2n+1)\pi,\ n = 0, 1, 2, \ldots$$

giving $\mathbf{I_{min} = 0}$, which corresponds to path difference $\boldsymbol{\Delta = (2n+1)\frac{\lambda}{2}}$.

Thus the intensity in the interference pattern varies as $4a^2\cos^2(\phi/2)$ between a maximum of $4a^2$ and a minimum of zero.

---

**Final answer.** The resultant intensity is $\boldsymbol{I = 4a^2\cos^2(\phi/2)}$, with maxima at $\boldsymbol{\phi = 2n\pi}$ and minima at $\boldsymbol{\phi = (2n+1)\pi}$.

← *Back to Q146*

## Q147

*Answer the following questions:*
*(a) In a single slit diffraction experiment, the width of the slit is made double the original width. How does this affect the size and intensity of the central diffraction band?*
*(b) How is the width of the central maximum changed when red light is replaced by blue?*
*(c) In what way is diffraction from each slit related to the interference pattern in a double slit experiment?*

**[2019 • Set 55-5-3]**

**SOLUTION**

(a) The angular and linear width of the central maximum in single-slit diffraction depends on the slit width $a$. The width of the central maximum on the screen is

$$W = \frac{2\lambda D}{a}$$

When the slit width is doubled, $\boldsymbol{a \to 2a}$,

$$W' = \frac{2\lambda D}{2a} = \frac{W}{2}$$

So the size (width) of the central band becomes **half** of its original value.

The peak intensity of the central maximum is proportional to the square of the slit width, $I_0 \propto a^2$, because the amplitude reaching the centre is proportional to $a$. Doubling $a$ makes the central intensity

$$I_0' \propto (2a)^2 = 4a^2 \Rightarrow I_0' = 4I_0$$

So the central band becomes **narrower (half width) but four times brighter**.

(b) The width of the central maximum is directly proportional to wavelength, $W \propto \lambda$. Blue light has a shorter wavelength than red light ($\lambda_{blue} < \lambda_{red}$). Hence on replacing red by blue, the width of the central maximum **decreases** (becomes narrower).

(c) In a double-slit experiment each slit has a finite width $a$, so light from each slit undergoes single-slit **diffraction** and spreads out into its own diffraction pattern. The interference fringes produced by the two slits are therefore seen only within the bright central diffraction envelope of a single slit.

Thus the observed pattern is the **product** of the two effects: the sharp interference fringes (from the two-slit separation) are modulated, that is, their brightness is controlled, by the broad single-slit diffraction envelope. Diffraction from each slit decides how far the interference fringes are visible on the screen.

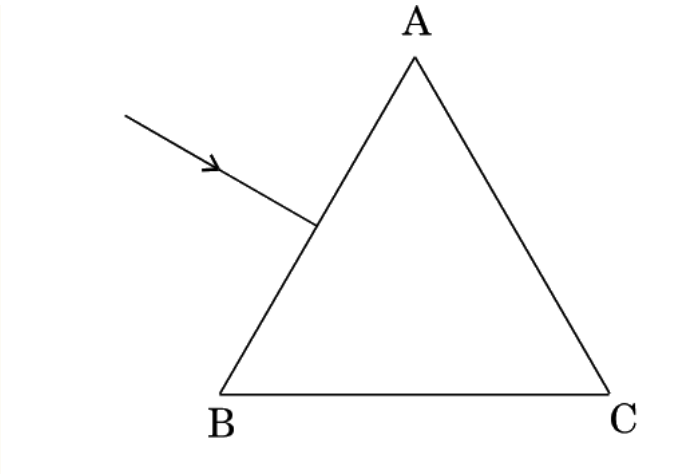
---

**Final answer.** Doubling the slit width halves the central band but makes it **4×** brighter; blue light narrows it; and single-slit diffraction forms the envelope that modulates the double-slit interference fringes.

← *Back to Q147*

## Q148

*(a) Show using a proper diagram how unpolarised light can be linearly polarised by reflection from a transparent glass surface.*
*(b) The figure shows a ray of light falling normally on the face* AB *of an equilateral glass prism having refractive index* $\frac{3}{2}$*, placed in water of refractive index* $\frac{4}{3}$*. Will this ray suffer total internal reflection on striking the face* AC *? Justify your answer.*




**[2018]**

**SOLUTION**

(a) When unpolarised light falls on a transparent glass surface at the **polarising (Brewster) angle** $\theta_B$, the reflected light becomes completely **linearly polarised**, with its electric vector vibrating perpendicular to the plane of incidence. At this angle the reflected and refracted rays are mutually perpendicular.

Diagram: an unpolarised incident ray strikes the glass surface at angle $\theta_B$; the reflected ray (drawn with dots to show vibrations perpendicular to the plane of paper) goes off at $\theta_B$, while the refracted ray enters the glass, the reflected and refracted rays being 90° apart. The polarising angle satisfies Brewster's law $\boldsymbol{\theta_B = n}$.

(b) The ray strikes face AC of the equilateral prism. Since the prism is equilateral, the angle of incidence at AC is

$$\theta = 60^\circ$$

For total internal reflection to occur at the glass-water interface, this angle must be greater than the critical angle $\theta_c$.

The critical angle for the glass-water boundary is given by

$$\theta_c = \frac{n_{water}}{n_{glass}}$$

$$\theta_c = \frac{4/3}{3/2}$$

$$\theta_c = \frac{4}{3} \times \frac{2}{3} = \frac{8}{9} = 0.888$$

$$\theta_c = \sin^{-1}(0.888) \approx 62.7^\circ$$

Comparing the angle of incidence with the critical angle,

$$\theta = 60^\circ < \theta_c = 62.7^\circ$$

Since the angle of incidence at AC is **less than** the critical angle, the ray will **not** suffer total internal reflection; it will be refracted and pass out of the prism into the water.

**COMMON MISTAKE**

*Total internal reflection needs the angle of incidence to exceed the critical angle; here 60° falls just short of 62.7°, so the ray escapes.*

---

**Final answer.** The critical angle is $\theta_c \approx 62.7^\circ$, and since $60^\circ < 62.7^\circ$ the ray does not undergo total internal reflection at **AC**.

← *Back to Q148*

## Q149

*(a) If one of two identical slits producing interference in Young's experiment is covered with glass, so that the light intensity passing through it is reduced to 50%, find the ratio of the maximum and minimum intensity of the fringe in the interference pattern.*
*(b) What kind of fringes do you expect to observe if white light is used instead of monochromatic light ?*

**[2018]**

**SOLUTION**

(a) Let the intensity from the uncovered slit be $I_1 = I$. The slit covered with glass passes only 50% of the light, so

$$I_2 = \frac{I}{2}$$

In an interference pattern the maximum and minimum intensities are

$$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

Their ratio is

$$\frac{I_{max}}{I_{min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2}$$

Substituting $\mathbf{I_1} = I$ and $\mathbf{I_2} = I/2$,

$$\frac{I_{max}}{I_{min}} = \frac{(\sqrt{I} + \sqrt{I/2})^2}{(\sqrt{I} - \sqrt{I/2})^2}$$

Taking $\sqrt{I}$ common from numerator and denominator,

$$\frac{I_{max}}{I_{min}} = \frac{(1 + \frac{1}{\sqrt{2}})^2}{(1 - \frac{1}{\sqrt{2}})^2}$$

Multiplying numerator and denominator inside the brackets by $\sqrt{2}$,

$$\frac{I_{max}}{I_{min}} = \frac{(\sqrt{2} + 1)^2}{(\sqrt{2} - 1)^2} = \frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}}$$

$$\frac{I_{max}}{I_{min}} \approx \frac{5.83}{0.17} \approx 34$$

$$\frac{I_{max}}{I_{min}} \approx 34 : 1$$

(b) If white light is used instead of monochromatic light, the **central fringe is white** (since for zero path difference all colours arrive in phase). On either side of it there appear a **few coloured fringes**, with violet nearest and red farther out, because the fringe width depends on wavelength. Beyond a few fringes the colours overlap and merge into a uniform white illumination, so the pattern is not seen clearly.

---

**Final answer.** The ratio of maximum to minimum intensity is about **34:1**; with white light the centre is white, flanked by a few coloured fringes that soon overlap.

← *Back to Q149*

## Q150

*A monochromatic light of wavelength λ is incident normally on a narrow slit of width 'a' to produce a diffraction pattern on the screen placed at a distance D from the slit. With the help of a relevant diagram, deduce the conditions for obtaining maxima and minima on the screen. Use these conditions to show that angular width of central maximum is twice the angular width of secondary maximum.*

**[2017]**

**SOLUTION**

Consider monochromatic light of wavelength λ incident normally on a single slit of width $a$. The diffraction pattern is observed on a screen placed at a distance $D$ from the slit.

Diagram: a vertical slit AB of width $a$; a plane wavefront strikes it normally from the left. Secondary wavelets leaving the slit at an angle θ travel to a point $P$ on the screen at distance $D$. The extra path difference between wavelets from the top edge $A$ and bottom edge $B$ of the slit is $a\theta$.

By Huygens' principle, every point of the wavefront in the slit acts as a secondary source. The wavelets reaching point $P$ at angle θ have a path difference of $a\theta$ between the two edges of the slit.

**Condition for minima:** Divide the slit into two equal halves. Wavelets from corresponding points in the two halves differ in path by $\frac{a}{2}\theta$. They cancel in pairs when this equals $\frac{\lambda}{2}$, that is

$$\frac{a}{2}\theta = \frac{\lambda}{2} \Rightarrow a\theta = \lambda$$

Dividing the slit into 2n parts generalises this to minima at

$$a\theta = n\lambda,\ n = \pm 1, \pm 2, \pm 3, \ldots$$

**Condition for secondary maxima:** Bright bands of much smaller intensity lie roughly midway between consecutive minima, given by

$$a\theta = (2n+1)\frac{\lambda}{2},\ n = \pm 1, \pm 2, \ldots$$

**Angular width of central maximum:** the central maximum lies between the first minima on either side, at $\theta = \pm\frac{\lambda}{a}$. For small angles $\theta \approx \theta$, so its half-angular width is $\frac{\lambda}{a}$ and the full angular width is

$$2\theta_1 = \frac{2\lambda}{a}$$

**Angular width of a secondary maximum:** a secondary maximum lies between two successive minima. The first secondary maximum lies between $\theta = \frac{\lambda}{a}$ (first minimum) and $\theta = \frac{2\lambda}{a}$ (second minimum), so its angular width is

$$\theta = \frac{2\lambda}{a} - \frac{\lambda}{a} = \frac{\lambda}{a}$$

Comparing the two,

$$\frac{\text{angular width of central maximum}}{\text{angular width of secondary maximum}} = \frac{2\lambda/a}{\lambda/a} = 2$$

Hence the **angular width of the central maximum is twice the angular width of a secondary maximum**.

---

**Final answer.** Minima occur at $a\theta = n\lambda$ and secondary maxima at $a\theta = (2n+1)\lambda/2$; the central maximum width $2\lambda/a$ is twice the secondary maximum width $\lambda/a$.

← *Back to Q150*

## Q151

*Draw the intensity distributions for (i) the fringes produced in interference, and (ii) the diffraction bands produced due to single slit. Write two points of difference between the phenomena of interference and diffraction.*

**[2017]**

**SOLUTION**

In interference (Young's double slit), the intensity on the screen varies as a smooth $\cos^2$ function of position.

$$I = 4I_0\cos^2\left(\frac{\Phi}{2}\right)$$

All bright fringes have the same maximum intensity $4I_0$ and all dark fringes fall exactly to zero. The fringes are equally spaced, with fringe width $\beta = \frac{\lambda D}{d}$ the same everywhere. Graphically this is a series of identical, equally spaced humps of equal height resting on the zero line.

In single slit diffraction, the intensity pattern has one very tall, broad **central maximum**, flanked by much weaker secondary maxima.

$$I = I_0\left(\frac{\beta}{\beta}\right)^2,\ \beta = \frac{\pi a\theta}{\lambda}$$

The central peak is twice as wide as each secondary peak. The secondary maxima fall off rapidly, their intensities being roughly in the ratio $1:\frac{1}{(2n+1)^2}$ of the central peak. The minima occur where $a\theta = n\lambda$, $n = \pm 1, \pm 2, \ldots$ Graphically this is one giant central hump with small, shrinking side humps and dark minima between them.

Two points of difference: (1) In **interference** all bright fringes have equal intensity and equal width; in **diffraction** the central maximum is far brighter and twice as wide, and intensity falls off sharply for higher orders. (2) Interference arises from superposition of waves from two separate coherent sources, whereas diffraction arises from superposition of secondary wavelets coming from different points of the same wavefront.

---

**Final answer.** Interference gives equally spaced equal-intensity cosine-squared fringes; single-slit diffraction gives a broad bright central maximum with weak secondary maxima, the two phenomena differing in source and fringe character.

← *Back to Q151*

## Q152

*(i) State law of Malus.*
*(ii) Draw a graph showing the variation of intensity (I) of polarised light transmitted by an analyser with angle (θ) between polariser and analyser.*
*(iii) What is the value of refractive index of a medium of polarising angle 60°?*

**[2016]**

**SOLUTION**

**Law of Malus** states that when completely plane-polarised light of intensity $I_0$ is incident on an analyser, the intensity of the transmitted light is proportional to the square of the cosine of the angle θ between the plane of transmission (pass axis) of the analyser and the plane of polarisation of the incident light.

$$I = I_0\cos^2\theta$$

For the graph of $I$ versus θ, plot $I$ on the vertical axis and θ on the horizontal axis. The curve is a $\cos^2\theta$ shape: it starts at the maximum value $I_0$ at $\theta = 0$, falls to zero at $\theta = 90°$, rises back to $I_0$ at $\theta = 180°$, and so on. The pattern repeats every 180° (π radians), giving rounded peaks of height $I_0$ touching the axis at $\theta = 90°, 270°,\ldots$

For the refractive index, by Brewster's law the tangent of the polarising angle equals the refractive index of the medium.

$$\mu = \theta_B$$

$$\mu = \tan 60°$$

$$\mu = \sqrt{3} \approx 1.732$$

Hence the refractive index of the medium is $\sqrt{3} \approx 1.732$.

---

**Final answer.** $I = I_0\cos^2\theta$; the graph is a $\cos^2\theta$ curve of peak height $I_0$ repeating every $180°$, and $\mu = \tan 60° = \sqrt{3} \approx 1.732$.

← *Back to Q152*

## Q153

*Define the term wave front. State Huygen's principle.*
*Consider a plane wave front incident on a thin convex lens. Draw a proper diagram to show how the incident wave front traverses through the lens and after refraction focusses on the focal point of the lens, giving the shape of the emergent wave front.*
***OR***
*Explain the following, giving reasons:*
*(i) When monochromatic light is incident on a surface separating two media, the reflected and refracted light both have the same frequency as the incident frequency.*
*(ii) When light travels from a rarer to a denser medium, the speed decreases. Does this decrease in speed imply a reduction in the energy carried by the wave?*
*(iii) In the wave picture of light, intensity of light is determined by the square of the amplitude of the wave. What determines the intensity in the photon picture of light?*

**[2016]**

### SOLUTION

A **wavefront** is the locus of all points of a medium that vibrate in the same phase at a given instant of time.

**Huygens' principle** states that every point on a given wavefront acts as a fresh source of secondary wavelets, which spread out in all forward directions with the speed of the wave. The new wavefront at any later instant is the surface that touches all these secondary wavelets tangentially in the forward direction (the forward envelope of the secondary wavelets).

For the convex lens diagram: draw a vertical **plane wavefront** (parallel straight lines, representing flat surfaces of equal phase) travelling horizontally and striking a thin convex lens. The central part of the wavefront passes through the thicker middle of the lens, where the glass slows it more, while the outer parts pass through the thinner edges and are delayed less. As a result the central part lags behind the edges, and the emergent wavefront becomes curved, bending into a **spherical converging wavefront**. This converging spherical wavefront shrinks to a point exactly at the focal point $F$ of the lens, on the principal axis. Mark the incident plane wavefront on the left, the lens in the centre, the curved emergent wavefront after the lens, and the focus $F$ where it converges.

Hence a plane wavefront incident on a convex lens emerges as a **spherical wavefront converging to the focus $F$**.

OR (alternative part):

(i) At a boundary between two media, the reflected and refracted light have the **same frequency** as the incident light. Frequency is decided by the source of light, and the boundary cannot create or change the number of oscillations per second. The atoms of the second medium are forced to oscillate at the frequency of the incident wave, so they re-radiate at the same frequency; frequency is therefore preserved across the boundary.

(ii) When light goes from a rarer to a denser medium its speed decreases because $v = \frac{c}{n}$ and the denser medium has a larger refractive index $n$. This decrease in speed does **not** reduce the energy carried by the wave. The energy of a photon depends only on its frequency, $E = h\nu$, and the frequency does not change on entering the denser medium. The speed reduction is accompanied by a reduction in wavelength ($\lambda = v/\nu$), not in energy.

(iii) In the photon picture, the intensity of light is determined by the **number of photons crossing unit area per unit time** (the number of photons incident per unit area per second). More photons per unit area per second means greater intensity.

---

**Final answer.** A wavefront is a surface of constant phase; by Huygens' principle a plane wavefront through a convex lens emerges as a spherical wavefront converging to the focus $F$ (OR: frequency is fixed by the source, speed reduction lowers $\lambda$ not energy since $E = h\nu$, and photon intensity equals number of photons per unit area per second).

← *Back to Q153*

## Q154

*(a) In Young's double slit experiment, two slits are* 1 *mm apart and the screen is placed* 1 *m away from the slits. Calculate the fringe width when light of wavelength* 500 *nm is used.*

*(b) What should be the width of each slit in order to obtain* 10 *maxima of the double slits pattern within the central maximum of the single slit pattern?*

**[2016]**

**SOLUTION**

For part (a), the fringe width in Young's double slit experiment is given by the standard formula.

$$\beta = \frac{\lambda D}{d}$$

Here $\lambda = 500\ \text{nm} = 5\times 10^{-7}\ m$, $D = 1\ m$, and $d = 1\ \text{mm} = 10^{-3}\ m$.

$$\beta = \frac{(5\times 10^{-7})(1)}{10^{-3}}$$

$$\beta = 5\times 10^{-4}\ m$$

$$\beta = 0.5\ \text{mm}$$

So the **fringe width is 0.5 mm**.

For part (b), we need the slit width $a$ so that 10 double-slit maxima fit inside the central diffraction maximum of the single slit.

The angular half-width of the central diffraction maximum (up to the first diffraction minimum) is $\theta = \frac{\lambda}{a}$, so the full angular width of the central maximum is $\frac{2\lambda}{a}$.

The angular separation between consecutive interference maxima is $\frac{\lambda}{d}$.

Number of interference maxima within the central diffraction maximum:

$$N = \frac{2\lambda/a}{\lambda/d} = \frac{2d}{a}$$

Setting $N = 10$:

$$\frac{2d}{a} = 10$$

$$a = \frac{2d}{10} = \frac{d}{5}$$

$$a = \frac{10^{-3}}{5} = 2\times 10^{-4}\ m$$

So the **width of each slit should be 0.2 mm**.

---

**Final answer.** Fringe width **β = 0.5 mm**, and each slit must be of width ***a* = 0.2 mm** to fit 10 interference maxima within the central diffraction maximum.

← *Back to Q154*

## Q155

*(i) Derive Snell's law on the basis of Huygen's wave theory when light is travelling from a denser to a rarer medium.*
*(ii) Draw the sketches to differentiate between plane wavefront and spherical wavefront.*

**[2016]**

**SOLUTION**

To derive Snell's law for light going from a **denser to a rarer medium** using Huygens' wave theory, consider a plane wavefront AB travelling in the denser medium (speed $v_1$) and striking the plane interface XY. Let AB make an angle of incidence $i$ with the interface and let the refracted front travel into the rarer medium with the larger speed $v_2$ ($\mathbf{v_2 > v_1}$).

Draw the wavefront AB meeting the surface at $A$ while $B$ is still in the denser medium. In the time $\Delta t$ that the wavelet from $B$ takes to reach the surface at $C$, the secondary wavelet starting from $A$ advances into the rarer medium to point $D$.

$$BC = v_1\,\Delta t \text{ (distance travelled in denser medium)}$$

$$AD = v_2\,\Delta t \text{ (distance travelled in rarer medium)}$$

The refracted wavefront is CD, making the angle of refraction $r$ with the surface. From the two right-angled triangles sharing the hypotenuse AC:

$$\sin i = \frac{BC}{AC} = \frac{v_1\,\Delta t}{AC}$$

$$\sin r = \frac{AD}{AC} = \frac{v_2 \Delta t}{AC}$$

Dividing the two equations:

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$

Since $\mu = \frac{c}{v}$, we have $\frac{v_1}{v_2} = \frac{\mu_2}{\mu_1}$, giving

$$\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1}$$

$$\mu_1 \sin i = \mu_2 \sin r$$

This is **Snell's law**. Because $v_2 > v_1$, we get $r > i$, so the ray bends **away from the normal** on going from denser to rarer medium, as expected.

For part (ii), the two sketches: A **plane wavefront** is drawn as a set of straight, parallel, equally spaced vertical lines, with parallel rays (arrows) perpendicular to them all pointing in the same forward direction. It is produced by a very distant (effectively infinite) source. A **spherical wavefront** is drawn as a set of concentric circular arcs centred on a point source $S$, with rays drawn as straight lines radiating outward from $S$ along the radii, perpendicular to each arc. It is produced by a point source nearby.

**TIP**

When the marking scheme asks for denser-to-rarer, always state $v_2 > v_1$ and conclude $r > i$ (ray bends away from normal); that explicit comparison is itself a mark.

---

**Final answer.** Huygens' construction gives $\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{\mu_2}{\mu_1}$, i.e. Snell's law, with the ray bending away from the normal; a plane wavefront is a set of parallel straight surfaces while a spherical wavefront is a set of concentric spheres about a point source.

← *Back to Q155*

## Q156

*Two harmonic waves of monochromatic light* $\mathbf{y_1 = a\omega\, t}$ *and* $\mathbf{y_2 = a\cos(\omega\, t + \phi)}$ *are superimposed on each other. Show that maximum intensity in interference pattern is four times the intensity due to each slit. Hence write the conditions for constructive and destructive interference in terms of the phase angle* $\phi$.

**[2016]**

**SOLUTION**

By the **principle of superposition**, when the two waves $\mathbf{y_1 = a\omega\, t}$ and $\mathbf{y_2 = a\cos(\omega\, t + \phi)}$ arrive at a point, the resultant displacement is their sum.

$$y = y_1 + y_2 = a\omega\, t + a\cos(\omega\, t + \phi)$$

Using the identity $\mathbf{\cos A + \cos B = 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)}$:

$$y = 2a\cos\left(\frac{\phi}{2}\right)\cos\left(\omega\, t + \frac{\phi}{2}\right)$$

This represents a simple harmonic wave of the same frequency $\omega$ with resultant amplitude

$$A_R = 2a\cos\left(\frac{\phi}{2}\right)$$

Since intensity is proportional to the square of the amplitude, the resultant intensity is

$$I = A_R^2 = 4a^2\cos^2\left(\frac{\phi}{2}\right)$$

The intensity due to each slit alone is $\mathbf{I_0 = a^2}$. The intensity is maximum when $\mathbf{\cos^2\left(\frac{\phi}{2}\right) = 1}$, that is when $\mathbf{\phi = 0, 2\pi, 4\pi, \ldots}$

$$I_{max} = 4a^2 = 4I_0$$

Hence the **maximum intensity is four times the intensity due to each slit**.

Conditions in terms of the phase angle $\phi$: For **constructive interference** (maximum intensity) the two waves must be in phase, so

$$\phi = 2n\pi,\ n = 0, \pm 1, \pm 2, \ldots$$

For **destructive interference** (minimum intensity, $I = 0$) the two waves must be exactly out of phase, so

$$\phi = (2n+1)\pi,\ n = 0, \pm 1, \pm 2, \ldots$$

---

**Final answer.** $I = 4a^2\cos^2(\phi/2)$ gives $I_{max} = 4a^2 = 4I_0$; constructive interference at $\phi = 2n\pi$ and destructive at $\phi = (2n+1)\pi$.

*← Back to Q156*

## Q157

*(a) Write the necessary conditions to obtain sustained interference fringes.*
*(b) In Young's double slit experiment, plot a graph showing the variation of fringe width versus the distance of the screen from the plane of the slits keeping other parameters same. What information can one obtain from the slope of the curve?*
*(c) What is the effect on the fringe width if the distance between the slits is reduced keeping other parameters same?*

**[2015]**

**SOLUTION**

For part (a), the necessary conditions to obtain sustained (steady, stable) interference fringes are: (i) The two sources must be **coherent**, i.e. they must emit light of the same frequency and maintain a constant phase difference with time. This is achieved by deriving both sources from a single primary source. (ii) The two sources should be **narrow and close together** (small separation $d$), so that the fringe width $\beta = \frac{\lambda D}{d}$ is large enough to be observed clearly. (iii) The two waves should have **equal or nearly equal amplitudes (intensities)**, so that the dark fringes are truly dark and the contrast between bright and dark fringes is good. (iv) The two interfering waves should be in the **same state of polarisation** and the sources should be monochromatic so that the fringes are well defined.

For part (b), the fringe width is

$$\beta = \frac{\lambda D}{d}$$

Keeping $\lambda$ and $d$ fixed, $\beta$ is directly proportional to $D$. So the graph of fringe width $\beta$ (vertical axis) against screen distance $D$ (horizontal axis) is a **straight line passing through the origin**.

The slope of this straight line is

$$\text{slope} = \frac{\beta}{D} = \frac{\lambda}{d}$$

From the slope one can obtain the wavelength of light if the slit separation $d$ is known, since $\lambda$ = **(slope)**$\times d$; or one can find $d$ if $\lambda$ is known. This gives a practical experimental method to measure the **wavelength of light**.

For part (c), since $\beta = \frac{\lambda D}{d}$, the fringe width is inversely proportional to the slit separation $d$. Therefore, if the distance between the slits is **reduced** (with $\lambda$ and $D$ unchanged), the **fringe width increases** and the fringes become more widely separated on the screen, making them easier to observe.

---

**Final answer.** Sustained fringes need coherent, close, equal-amplitude, similarly polarised sources; the $\beta$ versus $D$ graph is a straight line through the origin with slope $\lambda/d$ (used to find $\lambda$), and reducing $d$ increases the fringe width.

← *Back to Q157*

## Q158

*Answer the following questions:*
*(a) In a double slit experiment using light of wavelength 600 nm, the angular width of the fringe formed on a distant screen is* 0.1°. *Find the spacing between the two slits.*
*(b) Light of wavelength 5000 propagating in air gets partly reflected from the surface of water. How will the wavelengths and frequencies of the reflected and refracted light be affected?*

**[2015]**

**SOLUTION**

For part (a), the angular width of a fringe in Young's double slit experiment is the angular fringe spacing, given by

$$\theta = \frac{\lambda}{d}$$

Rearranging for the slit separation $d$:

$$d = \frac{\lambda}{\theta}$$

Here $\lambda = 600\ \text{nm} = 600\times 10^{-9}\ m$ and $\theta = 0.1°$. Convert the angle to radians.

$$\theta = 0.1°\times \frac{\pi}{180} = 1.745\times 10^{-3}\ \text{rad}$$

$$d = \frac{600\times 10^{-9}}{1.745\times 10^{-3}}$$

$$d = 3.44\times 10^{-4}\ m$$

So the **spacing between the two slits is $3.44\times 10^{-4}\ m \approx 0.34$ mm**.

For part (b), the light of wavelength 5000 travelling in air is partly reflected from and partly refracted into water. The **frequency** of light is fixed by the source and does not change on reflection or refraction. Therefore both the reflected and the refracted light have the **same frequency** as the incident light.

$$\nu = \frac{c}{\lambda} = \frac{3\times 10^{8}}{5000\times 10^{-10}} = 6\times 10^{14}\ \text{Hz}$$

The **reflected light** travels back into air (the same medium), so its **wavelength is unchanged**, equal to 5000 , and its frequency is unchanged.

The **refracted light** enters water, where the speed decreases to $v = \frac{c}{n_w}$. Its frequency stays the same, so its **wavelength decreases** by a factor of the refractive index of water:

$$\lambda_{\text{water}} = \frac{\lambda}{n_w}$$

So in summary, the reflected light keeps the same wavelength and frequency, while the refracted light keeps the same frequency but has a **smaller wavelength** $\lambda/n_w$.

**Final answer.** Slit spacing $d = 3.44\times 10^{-4}\ m \approx 0.34$ mm; the reflected light keeps the same $\lambda$ and $\nu$, while the refracted light keeps the same $\nu$ but has wavelength reduced to $\lambda/n_w$.

← *Back to Q158*

## Q159

*(a) Show, giving a suitable diagram, how unpolarized light can be polarised by reflection.*
*(b) Two polaroids* $P_1$ *and* $P_2$ *are placed with their pass axes perpendicular to each other. Unpolarised light of intensity* $I_0$ *is incident on* $P_1$*. A third polaroid* $P_3$ *is kept in between* $P_1$ *and* $P_2$ *such that its pass axis makes an angle of 60° with that of* $P_1$*. Determine the intensity of light transmitted through* $P_1$*,* $P_3$ *and* $P_2$*.*

**[2014]**

**SOLUTION**

For part (a), polarisation by reflection: when **unpolarised light** is incident on a transparent surface such as glass at a particular angle called the **Brewster angle** (polarising angle) $\theta_B$, the reflected light is found to be **completely plane-polarised**, with its vibrations perpendicular to the plane of incidence.

For the diagram, draw the boundary of a glass slab as a horizontal surface. Draw the incident unpolarised ray striking the surface at angle $\theta_B$ to the normal; mark it with both dots (vibrations perpendicular to the plane of paper) and short arrows (vibrations in the plane of paper) to show it is unpolarised. Draw the reflected ray going up, marked with only **dots**, showing it is plane-polarised perpendicular to the plane of incidence. Draw the refracted ray going into the glass at angle $r$, marked mostly with arrows (partially polarised, parallel to plane of incidence).

At the Brewster angle, the reflected and refracted rays are mutually perpendicular, so $\theta_B + r = 90°$, and Brewster's law gives

$$\theta_B = n$$

For part (b), the three-polaroid problem. Unpolarised light of intensity $I_0$ is incident on $P_1$. A polaroid transmits half the intensity of unpolarised light, so

the intensity after $P_1$ is

$$I_1 = \frac{I_0}{2}$$

This light is now plane-polarised along the axis of $P_1$. It next falls on $P_3$, whose axis makes $60°$ with $P_1$. By Malus's law:

$$I_3 = I_1 \cos^2 60° = \frac{I_0}{2} \times \left(\frac{1}{2}\right)^2 = \frac{I_0}{8}$$

This light is now polarised along $P_3$. Finally it falls on $P_2$. Since $P_2$ is perpendicular to $P_1$ (at $90°$) and $P_3$ is at $60°$ from $P_1$, the angle between $P_3$ and $P_2$ is $\mathbf{90° - 60° = 30°}$. By Malus's law:

$$I_2 = I_3 \cos^2 30° = \frac{I_0}{8} \times \left(\frac{3}{2}\right)^2 = \frac{I_0}{8} \times \frac{3}{4} = \frac{3I_0}{32}$$

Hence the intensities transmitted are $\mathbf{I_1 = \frac{I_0}{2}}$ **through** $\mathbf{P_1}$, $\mathbf{I_3 = \frac{I_0}{8}}$ **through** $\mathbf{P_3}$, **and** $\mathbf{I_2 = \frac{3I_0}{32}}$ **through** $\mathbf{P_2}$.

**COMMON MISTAKE**

*A common mistake is using $90°$ for the last Malus step. $P_2$ is at $90°$ from $P_1$ but only $30°$ from $P_3$, the polaroid just before it, so use $\cos^2 30°$.*

---

**Final answer.** Reflected light at the Brewster angle ($\theta_B = n$) is fully plane-polarised; the transmitted intensities are $\mathbf{I_0/2}$ through $\mathbf{P_1}$, $\mathbf{I_0/8}$ through $\mathbf{P_3}$, and $\mathbf{3I_0/32}$ through $\mathbf{P_2}$.

← *Back to Q159*

## Q160

*(a) In what way is diffraction from each slit related to the interference pattern in a double slit experiment?*
*(b) Two wavelengths of sodium light* 590 nm *and* 596 nm *are used, in turn, to study the diffraction taking place at a single slit of aperture* $2\times 10^{-4}$ *m. The distance between*

*the slit and the screen is* 1.5 *m. Calculate the separation between the positions of the first maxima of the diffraction pattern obtained in the two cases.*

**[2013]**

**SOLUTION**

For part (a), the relation between diffraction at each slit and the interference pattern in a double slit experiment: in a real double slit experiment each of the two slits has a finite width $a$, so each slit by itself produces a single-slit **diffraction pattern**. These two diffraction patterns from the two slits overlap, and because the slits are coherent their light also **interferes**. The result is that the sharp interference fringes (from the two-slit separation $d$) are seen, but their brightness is **modulated (controlled) by the single-slit diffraction envelope**. The combined intensity is

$$I = I_0(\frac{\beta}{\beta})^2\cos^2\delta,\ \beta = \frac{\pi\, a\theta}{\lambda},\ \delta = \frac{\pi\, d\theta}{\lambda}$$

So the diffraction pattern of each slit acts as an **envelope** that limits how bright the interference fringes can be; interference fringes lying near a diffraction minimum become very faint or absent.

For part (b), the first **maximum** (secondary maximum) of a single-slit diffraction pattern occurs approximately at

$$a\theta = \frac{3\lambda}{2}$$

For small angles $\theta \approx \frac{y}{D}$, so the position of the first maximum on the screen is

$$y = \frac{3\lambda\, D}{2a}$$

The separation between the first maxima for the two wavelengths is

$$\Delta y = \frac{3D}{2a}(\lambda_2 - \lambda_1)$$

Substituting $\lambda_2 - \lambda_1 = (596 - 590)\ \text{nm} = 6\times 10^{-9}\ m$, $D = 1.5\ m$, $a = 2\times 10^{-4}\ m$:

$$\Delta y = \frac{3\times 1.5\times 6\times 10^{-9}}{2\times 2\times 10^{-4}}$$

$$\Delta y = \frac{27\times 10^{-9}}{4\times 10^{-4}}$$

$$\Delta y = 6.75\times 10^{-5}\ m$$

So the **separation between the first maxima for the two wavelengths is $6.75\times 10^{-5}\ m = 0.0675$ mm**.

---

**Final answer.** Each slit's diffraction pattern forms an envelope that modulates the double-slit interference fringes, and the separation between the first maxima for the two wavelengths is $\boldsymbol{\Delta y = 6.75\times 10^{-5}\ m}$.

← *Back to Q160*

## Q161

*In Young's double slit experiment, the two slits* 0.15 mm *apart are illuminated by monochromatic light of wavelength* 450 nm. *The screen is* 1.0 *m away from the slits.*
*(a) Find the distance of the second (i) bright fringe, (ii) dark fringe from the central maximum.*
*(b) How will the fringe pattern change if the screen is moved away from the slits?*

**[2012]**

**SOLUTION**

In Young's double slit experiment the bright and dark fringes are equally spaced. The spacing between two successive bright (or dark) fringes is the fringe width $\boldsymbol{\beta = \frac{\lambda D}{d}}$.

Given: $\boldsymbol{d = 0.15\ mm = 1.5\times 10^{-4}\ m}$, $\boldsymbol{\lambda = 450\ nm = 4.5\times 10^{-7}\ m}$, $\boldsymbol{D = 1.0\ m}$.

$$\beta = \frac{\lambda D}{d}$$

$$\beta = \frac{(4.5\times 10^{-7})(1.0)}{1.5\times 10^{-4}}$$

$$\beta = 3\times 10^{-3}\ m = 3\ \text{mm}$$

(a)(i) The $n$th bright fringe lies at $\boldsymbol{y_n = n\beta}$ from the central maximum. For the second bright fringe, $\boldsymbol{n = 2}$:

$$y_2^{\text{bright}} = 2\beta = 2\times 3 \text{ mm}$$

$$y_2^{\text{bright}} = \mathbf{6\ mm}$$

(a)(ii) The $n$th dark fringe lies at $\mathbf{y_n = \frac{(2n-1)\beta}{2}}$ from the central maximum. For the second dark fringe, $\boldsymbol{n = 2}$:

$$y_2^{\text{dark}} = \frac{(2\times 2 - 1)\beta}{2} = \frac{3\beta}{2}$$

$$y_2^{\text{dark}} = \frac{3\times 3}{2} \text{ mm} = \mathbf{4.5\ mm}$$

(b) Since $\boldsymbol{\beta = \frac{\lambda D}{d} \propto D}$, increasing $D$ increases the fringe width. So when the screen is moved farther from the slits, the fringes get **wider and more spread out** on the screen, while the number of fringes per unit length decreases. The angular fringe width $\frac{\beta}{D} = \frac{\lambda}{d}$ stays unchanged.

---

**Final answer.** Second bright fringe at 6 mm and second dark fringe at 4.5 mm; moving the screen away widens the fringes since $\beta \propto D$.

← *Back to Q161*

## Q162

*How does an unpolarised light get polarised when passed through a polaroid? Two polaroids are set in crossed positions. A third polaroid is placed between the two making an angle θ with the pass axis of the first polaroid. Write the expression for the intensity of light transmitted from the second polaroid. In what orientations will the transmitted intensity be (i) minimum and (ii) maximum?*

**[2012]**

**SOLUTION**

A polaroid is a sheet containing long-chain organic molecules aligned in one direction during manufacture. These molecules strongly absorb the component of the electric field $\vec{E}$ that is parallel to their length and freely transmit the component perpendicular to it (this perpendicular direction is the pass axis).

Unpolarised light has $\vec{E}$ vibrating randomly in all directions perpendicular to the direction of propagation. When it passes through the polaroid, only the component of $\vec{E}$ along the pass axis is transmitted, so the emerging light is **plane polarised** along the pass axis. On averaging over all random directions, the transmitted intensity is half the incident intensity, $\mathbf{I_{out} = \frac{I_0}{2}}$.

Now consider the three-polaroid arrangement. Let unpolarised light of intensity $I_0$ fall on $P_1$. After $P_1$ the light is polarised along the axis of $P_1$ with intensity

$$I_1 = \frac{I_0}{2}$$

The middle polaroid $P_3$ makes angle $\theta$ with the axis of $P_1$. By Malus' law the intensity after $P_3$ is

$$I_3 = I_1\cos^2\theta = \frac{I_0}{2}\cos^2\theta$$

The second polaroid $P_2$ is crossed with $P_1$, so its axis makes angle $(90° - \theta)$ with $P_3$. The intensity transmitted by $P_2$ is

$$I_2 = I_3\cos^2(90° - \theta) = \frac{I_0}{2}\cos^2\theta\,\sin^2\theta$$

Using $\mathbf{2\theta\theta = \sin 2\theta}$,

$$I_2 = \frac{I_0}{8}\sin^2 2\theta$$

(i) The transmitted intensity is **minimum (zero)** when $\mathbf{\sin 2\theta = 0}$, i.e. $\mathbf{2\theta = 0°}$ or 180°, giving $\mathbf{\theta = 0°}$ or 90°.

(ii) The transmitted intensity is **maximum** $\left(=\frac{I_0}{8}\right)$ when $\mathbf{\sin 2\theta = 1}$, i.e. $\mathbf{2\theta = 90°}$, giving $\mathbf{\theta = 45°}$.

---

**Final answer.** $\mathbf{I_2 = \frac{I_0}{8}\sin^2 2\theta}$; minimum at $\mathbf{\theta = 0°}$ or $\mathbf{90°}$, maximum at $\mathbf{\theta = 45°}$.

*← Back to Q162*

## Q163

*(a) Why are coherent sources necessary to produce a sustained interference pattern?*
*(b) In Young's double slit experiment using monochromatic light of wavelength λ, the intensity of light at a point on the screen where path difference is λ, is K units. Find out the intensity of light at a point where path difference is λ/3.*

**[2012]**

**SOLUTION**

(a) For two waves meeting at a point, the resultant intensity is

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\ \phi$$

where $\phi$ is the phase difference between them. If the sources are not coherent, $\phi$ changes randomly and rapidly with time, so the term $\phi$ averages to zero over the time of observation. This gives $I = I_1 + I_2$ everywhere, i.e. uniform illumination with no fringes.

Only **coherent sources** have a constant phase difference $\phi$ with time. Then $\phi$ stays fixed at each point, producing steady bright fringes where $\phi$ is positive (maximum) and steady dark fringes where it is negative (minimum). Hence coherent sources are necessary to produce a **sustained (stable) interference pattern**.

(b) For two coherent sources of equal intensity, the intensity at a point with phase difference $\phi$ is

$$I = I_{max}\cos^2\left(\frac{\phi}{2}\right)$$

where the phase difference for a path difference $\Delta$ is $\phi = \frac{2\pi}{\lambda}\Delta$.

At path difference $\Delta = \lambda$:

$$\phi = \frac{2\pi}{\lambda} \times \lambda = 2\pi$$

$$I = I_{max}\cos^2(\pi) = I_{max}$$

Given this intensity is $K$, so $I_{max} = K$.

At path difference $\Delta = \frac{\lambda}{3}$:

$$\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{3} = \frac{2\pi}{3}$$

$$I = I_{max}\cos^2\left(\frac{\pi}{3}\right) = K\left(\frac{1}{2}\right)^2$$

$$I = \frac{K}{4}$$

**COMMON MISTAKE**

*Path difference $\lambda$ corresponds to phase $2\pi$, not zero; always convert $\Delta$ to $\phi = \frac{2\pi}{\lambda}\Delta$ before using $\cos^2(\phi/2)$.*

---

**Final answer.** Coherent sources keep $\phi$ constant so fringes are stable; the intensity at path difference $\lambda/3$ is $\frac{K}{4}$.

← *Back to Q163*

## Q164

*Define a wavefront. Use Huygens' geometrical construction to show the propagation of a plane wavefront from a rarer medium (1) to a denser medium (2) undergoing refraction.*
*Hence derive Snell's law of refraction.*
***OR***
*(a) Use Huygens' geometrical construction to show the behaviour of a plane wavefront*
*(i) passing through a biconvex lens;*
*(ii) reflecting by a concave mirror.*
*(b) When monochromatic light is incident on a surface separating two media, why does the refracted light have the same frequency as that of the incident light?*

**[2012]**

**SOLUTION**

A **wavefront** is the locus of all points in a medium that are in the same phase of vibration at a given instant. A point source gives spherical wavefronts and a very distant (or point) source gives plane wavefronts.

By Huygens' principle, every point on a given wavefront acts as a source of secondary spherical wavelets that spread out in the forward direction at the speed of the wave in that medium. The new position of the wavefront after a time $\Delta t$ is the forward envelope (tangent surface) of all these secondary wavelets.

Diagram (described): Draw a horizontal interface line PP' separating rarer medium 1 (above, speed $v_1$) from denser medium 2 (below, speed $v_2$). A plane incident wavefront AB in medium 1 makes angle of incidence $i$ with the interface; corner $A$ touches the interface first while $B$ is still in medium 1. Drop the normal at the interface. The refracted wavefront is AC in medium 2 making angle of refraction $r$.

Let the wavefront AB strike the surface so that $A$ reaches the interface and the disturbance at $B$ still has to travel distance $\mathbf{BC = v_1\,\Delta t}$ to reach point $C$ on the interface. In the same time $\Delta t$, the secondary wavelet from $A$ travels into the denser medium a distance

$$AD = v_2\,\Delta t$$

The refracted wavefront is the line CD, where AD is the radius of the wavelet from $A$. Here $i$ is the angle between AB and the surface (equal to the angle of incidence) and $r$ is the angle between CD and the surface (equal to the angle of refraction).

From the right triangle ABC (right angle at $B$):

$$\sin i = \frac{BC}{AC} = \frac{v_1\,\Delta t}{AC}$$

From the right triangle ADC (right angle at $D$):

$$\sin r = \frac{AD}{AC} = \frac{v_2\,\Delta t}{AC}$$

Dividing these two results:

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$

The refractive index of medium 2 with respect to medium 1 is $\mathbf{n_{21}} = \frac{\mathbf{v_1}}{\mathbf{v_2}}$, which is a constant. Writing $\frac{\mathbf{v_1}}{\mathbf{v_2}} = \frac{\mathbf{n_2}}{\mathbf{n_1}}$ (since $n \propto \frac{1}{v}$):

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} \Rightarrow n_1 \sin i = n_2 \sin r$$

This is **Snell's law of refraction**. Since medium 2 is denser, $\mathbf{v_2 < v_1}$, so $\boldsymbol{r < i}$ and the ray bends towards the normal.

---

**Final answer.** Using Huygens' wavelets, $\frac{\sin i}{\sin r} = \frac{\mathbf{v_1}}{\mathbf{v_2}} = \frac{\mathbf{n_2}}{\mathbf{n_1}}$, which is Snell's law $\mathbf{n_1 \sin i = n_2 \sin r}$.

← *Back to Q164*

## Q165

*Describe Young's double slit experiment to produce interference pattern due to a monochromatic source of light. Deduce the expression for the fringe width.*

**[2011 • Set 55-1-1]**

**SOLUTION**

Young's double slit experiment: Light from a monochromatic source falls first on a narrow single slit $S$, which acts as a point source. The light from $S$ then falls on two narrow, closely spaced slits $S_1$ and $S_2$ separated by a small distance $d$. Since $S_1$ and $S_2$ are derived from the same source $S$, they act as two **coherent sources** with a constant phase difference. Light from them spreads out, overlaps, and interferes on a screen placed at distance $D$ from the slits, producing a pattern of alternate bright and dark fringes.

Diagram (described): On the left, the single slit $S$; in the middle, the double slit with $S_1$ above and $S_2$ below, separated by $d$; on the right, a vertical screen at distance $D$. The central point $O$ on the screen lies on the perpendicular bisector of $S_1S_2$. A general point $P$ is at distance $y$ above $O$.

At point $P$ at distance $y$ from the centre $O$, the path difference between the waves from $S_2$ and $S_1$ is

$$\Delta = S_2P - S_1P \approx \frac{yd}{D} \quad (D \gg d, y)$$

For a **bright fringe** (constructive interference) the path difference must be a whole number of wavelengths:

$$\Delta = n\lambda \Rightarrow \frac{y_n\, d}{D} = n\lambda$$

$$y_n = \frac{n\lambda\, D}{d},\ n = 0, \pm 1, \pm 2, \ldots$$

For a **dark fringe** (destructive interference):

$$\Delta = (2n-1)\frac{\lambda}{2} \Rightarrow y_n = \frac{(2n-1)\lambda\, D}{2d}$$

The **fringe width** $\beta$ is the distance between two consecutive bright fringes. For the $n$th and $(n+1)$th bright fringes:

$$\beta = y_{n+1} - y_n = \frac{(n+1)\lambda\, D}{d} - \frac{n\lambda\, D}{d}$$

$$\beta = \frac{\lambda\, D}{d}$$

The same spacing is obtained between consecutive dark fringes, so all the fringes are **equally spaced** with width $\frac{\lambda\, D}{d}$.

---

**Final answer.** The fringe width is $\boldsymbol{\beta = \frac{\lambda\, D}{d}}$, the same for bright and dark fringes.

← *Back to Q165*

## Q166

*Use Huygen's principle to verify the laws of refraction.*

**[2011 • Set 55-1-1]**

**SOLUTION**

Huygens' principle: Every point on a wavefront acts as a source of secondary spherical wavelets that travel forward at the speed of the wave in that medium.

After a time $\Delta t$, the new wavefront is the forward envelope (common tangent) of all these secondary wavelets.

Diagram (described): A horizontal interface XY separates medium 1 (above, speed $v_1$) from medium 2 (below, speed $v_2$). A plane incident wavefront AB in medium 1 strikes XY; corner *A* reaches the surface first while *B* is still above it. The refracted wavefront CD lies in medium 2. The angle of incidence $i$ is the angle between AB and the surface; the angle of refraction $r$ is the angle between CD and the surface.

Let the disturbance at *B* take time $\Delta t$ to travel and reach point *C* on the interface:

$$BC = v_1 \Delta t$$

In the same time, the secondary wavelet starting from *A* travels into medium 2 through a distance

$$AD = v_2 \Delta t$$

Joining *C* to *D* gives the refracted wavefront, with AC as the common base.

From the right triangle ABC (right angle at *B*):

$$\sin i = \frac{BC}{AC} = \frac{v_1 \Delta t}{AC}$$

From the right triangle ACD (right angle at *D*):

$$\sin r = \frac{AD}{AC} = \frac{v_2 \Delta t}{AC}$$

Dividing the two equations:

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \text{constant} = n_{21}$$

This constant ratio is **Snell's law of refraction**, with $\mathbf{n_{21} = \frac{v_1}{v_2} = \frac{n_2}{n_1}}$ the refractive index of medium 2 with respect to medium 1, so $\mathbf{n_1 \sin i = n_2 \sin r}$.

Also, since the incident wavefront AB, the refracted wavefront CD and the normal all lie in the same plane, the incident ray, refracted ray and normal lie in one plane. These two results together are the **laws of refraction**, verified by Huygens' construction.

**Final answer.** Huygens' construction gives $\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = n_{21}$, a constant, which is Snell's law, verifying the laws of refraction.

← Back to Q166

## Q167

*(a) Describe briefly, with the help of suitable diagram, how the transverse nature of light can be demonstrated by the phenomenon of polarization.*
*(b) When unpolarized light passes from air to a transparent medium, under what condition does the reflected light get polarized?*

**[2011 • Set 55-1-1]**

**SOLUTION**

(a) Demonstration of the transverse nature of light: Take two polaroids $P_1$ (polariser) and $P_2$ (analyser) placed one behind the other with their planes parallel, and pass a beam of unpolarised light through them.

Diagram (described): Unpolarised light travels left to right, passes through $P_1$ (whose pass axis is fixed, say vertical), becomes plane polarised, then reaches $P_2$ which can be rotated about the direction of propagation; behind $P_2$ is a screen/detector.

After $P_1$ the light is plane polarised along the axis of $P_1$ with intensity $I_0$. By Malus' law, the intensity emerging from $P_2$ when its axis makes angle $\theta$ with that of $P_1$ is

$$I = I_0 \cos^2\theta$$

Now rotate $P_2$ slowly. When the axes are parallel ($\theta = 0°$) the transmitted intensity is maximum and the screen is bright. As $\theta$ increases the intensity falls, and at $\theta = 90°$ (crossed polaroids) the intensity becomes zero and the screen is completely dark. On rotating further, brightness returns, becoming maximum again at $\theta = 180°$.

This complete cut-off of light at 90° shows that light vibrations have a definite direction perpendicular to the direction of propagation. A longitudinal wave

(vibrating along the direction of travel) could not be cut off in this way. Hence light is a **transverse wave**.

(b) Condition for polarisation by reflection: When unpolarised light is incident on a transparent medium at a particular angle called the **Brewster angle** (polarising angle) $\theta_B$, the reflected light is completely plane polarised. This happens when the reflected ray and the refracted ray are perpendicular to each other, and the Brewster angle satisfies

$$\theta_B = n$$

where $n$ is the refractive index of the medium. At this angle the reflected light is plane polarised with vibrations **perpendicular to the plane of incidence**.

---

**Final answer.** Polarisation of light through polaroids proves it is a transverse wave; reflected light is fully polarised when the angle of incidence equals Brewster's angle, $\boldsymbol{\theta_B = n}$.

← *Back to Q167*

## Q168

*The intensity at the central maxima (O) in a Young's double slit experiment is* $I_0$. *If the distance OP equals one-third of the fringe width of the pattern, show that the intensity at point P would be* $\frac{I_0}{4}$.

**[2011 • Set 55-2-1]**

**SOLUTION**

For two coherent sources of equal intensity in Young's double slit experiment, the intensity at a point where the phase difference is $\phi$ is given by

$$I = I_0 \cos^2\left(\frac{\Phi}{2}\right)$$

where $I_0$ is the intensity at the central maximum $O$ (where $\boldsymbol{\Phi = 0}$).

The point $P$ is at distance $\mathbf{OP} = \frac{\beta}{3}$ from the centre, where the fringe width is $\boldsymbol{\beta} = \frac{\lambda D}{d}$. The path difference at a distance $y$ from the centre is $\Delta = \frac{yd}{D}$. So at $P$:

$$\Delta = \frac{OP \cdot d}{D} = \frac{1}{3} \cdot \frac{\beta\, d}{D} = \frac{1}{3} \cdot \frac{\lambda D}{d} \cdot \frac{d}{D}$$

$$\Delta = \frac{\lambda}{3}$$

The corresponding phase difference is

$$\phi = \frac{2\pi}{\lambda}\,\Delta = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{3} = \frac{2\pi}{3}$$

Substituting into the intensity formula:

$$I_P = I_0 \cos^2\left(\frac{\phi}{2}\right) = I_0 \cos^2\left(\frac{\pi}{3}\right)$$

$$I_P = I_0 \left(\frac{1}{2}\right)^2$$

$\mathbf{I_P = \frac{I_0}{4}}$ - hence proved.

---

**Final answer.** The intensity at $P$ is $\frac{I_0}{4}$, as required.

← *Back to Q168*

## Q169

*In the experiment on diffraction due to a single slit, show that*
*(i) the intensity of diffraction fringes decreases as the order (n) increases.*
*(ii) angular width of the central maximum is twice that of the first order secondary maximum.*

**[2011 • Set 55-2-1]**

**SOLUTION**

Consider a single slit of width $a$ illuminated by monochromatic light of wavelength λ. The diffraction minima occur where $\boldsymbol{a\theta = n\lambda}$ (for $\boldsymbol{n = 1, 2, 3, \ldots}$) and the secondary maxima lie roughly midway between consecutive minima.

(i) Intensity of secondary maxima decreases with order $n$: To locate the $n$th secondary maximum, divide the slit into (2n+1) equal strips. The path

difference between the edges of the slit corresponds to these strips pairing off. Out of the (2n+1) strips, 2n strips cancel in $n$ pairs (each pair giving zero), and only **one** uncancelled strip of width $\frac{a}{2n+1}$ is left to contribute to the resultant amplitude.

So the amplitude of the $n$th secondary maximum is about $\frac{1}{2n+1}$ of the central amplitude, and since intensity $\propto$ (amplitude)$^2$:

$$\frac{I_n}{I_0} \approx \frac{1}{(2n+1)^2}$$

For $n = 1, 2, 3,...$ this gives ratios $\frac{1}{9}, \frac{1}{25}, \frac{1}{49},...$, which fall rapidly. Hence the **intensity of the diffraction fringes decreases as the order $n$ increases**.

(ii) Angular width of central maximum compared with first secondary maximum: The central maximum lies between the first minima on the two sides of the centre. The first minimum occurs at

$$\theta = \frac{\lambda}{a} \Rightarrow \theta_1 \approx \frac{\lambda}{a}$$

The central maximum extends from $\theta = -\frac{\lambda}{a}$ to $\theta = +\frac{\lambda}{a}$, so its angular width is

$$\Delta\theta_{\text{central}} = \frac{2\lambda}{a}$$

The first secondary maximum lies between the first minimum ($\theta = \frac{\lambda}{a}$) and the second minimum ($\theta = \frac{2\lambda}{a}$). Its angular width is

$$\Delta\theta_{\text{first}} = \frac{2\lambda}{a} - \frac{\lambda}{a} = \frac{\lambda}{a}$$

Taking the ratio:

$$\frac{\Delta\theta_{\text{central}}}{\Delta\theta_{\text{first}}} = \frac{2\lambda/a}{\lambda/a} = 2$$

Hence the **angular width of the central maximum is twice that of the first order secondary maximum**.

**Final answer.** The secondary-maxima intensity falls as $\frac{1}{(2n+1)^2}$, and the central maximum ($2\lambda/a$) is twice as wide as the first secondary maximum ($\lambda/a$).

← *Back to Q169*

## Q170

*(i) Light passes through two polaroids* $P_1$ *and* $P_2$ *with pass axis of* $P_2$ *making an angle* $\theta$ *with the pass axis of* $P_1$. *For what value of* $\theta$ *is the intensity of emergent light zero? (ii) A third polaroid is placed between* $P_1$ *and* $P_2$ *with its pass axis making an angle* $\beta$ *with the pass axis of* $P_1$. *Find a value of* $\beta$ *for which the intensity of light emerging from* $P_2$ *is* $\frac{I_0}{8}$, *where* $I_0$ *is the intensity of light on the polaroid* $P_1$.

**[2011 • Set 55-2-1]**

**SOLUTION**

(i) Light from $P_1$ is plane polarised. By Malus' law, the intensity emerging from $P_2$ when its pass axis makes angle $\theta$ with that of $P_1$ is

$$I = I_1\cos^2\theta$$

The emergent intensity is zero when $\cos^2\theta = 0$, i.e.

$$\theta = 0 \Rightarrow \theta = 90^\circ$$

So $P_1$ and $P_2$ must be **crossed (at right angles)**.

(ii) Now a third polaroid $P_3$ is placed between $P_1$ and $P_2$, with its pass axis at angle $\beta$ from $P_1$. Note that $P_2$ remains crossed with $P_1$ (at 90° from $P_1$), so $P_3$ makes angle $(90^\circ - \beta)$ with $P_2$.

Let unpolarised light of intensity $I_0$ fall on $P_1$. After $P_1$:

$$I_1 = \frac{I_0}{2}$$

After $P_3$ (at angle $\beta$ from $P_1$):

$$I_3 = I_1\cos^2\beta = \frac{I_0}{2}\cos^2\beta$$

After $P_2$ (at angle $90°-\beta$ from $P_3$):

$$I_2 = I_3\cos^2(90°-\beta) = \frac{I_0}{2}\cos^2\beta\,\sin^2\beta$$

Using $\mathbf{2\beta\beta = \sin 2\beta}$:

$$I_2 = \frac{I_0}{8}\sin^2 2\beta$$

Setting $\mathbf{I_2 = \frac{I_0}{8}}$:

$$\frac{I_0}{8}\sin^2 2\beta = \frac{I_0}{8} \Rightarrow \sin^2 2\beta = 1$$

$$\sin 2\beta = 1 \Rightarrow 2\beta = 90°$$

$$\beta = \mathbf{45°}$$

---

**Final answer.** Emergent intensity is zero at $\boldsymbol{\theta = 90°}$; the intensity from $\mathbf{P_2}$ equals $\frac{\mathbf{I_0}}{\mathbf{8}}$ when $\boldsymbol{\beta = 45°}$.

← *Back to Q170*

## Q171

*What is an unpolarized light? Explain with the help of suitable ray diagram how an unpolarized light can be polarized by reflection from a transparent medium. Write the expression for Brewster angle in terms of the refractive index of denser medium.*

**[2010]**

**SOLUTION**

Unpolarised light is light in which the electric field vector $\vec{E}$ vibrates in all possible directions in the plane perpendicular to the direction of propagation, with the orientation changing randomly and rapidly with time. Ordinary sources such as the Sun, a lamp or a candle give unpolarised light because the light is emitted independently by a very large number of atoms, each vibrating in its own random direction.

When such light vibrates symmetrically in all directions about the direction of travel, no single direction of $\vec{E}$ is preferred, so the light shows no one-sidedness.

Polarisation by reflection: Consider unpolarised light incident on the plane surface of a transparent medium such as glass. For most angles of incidence the reflected light is only partially polarised. However, at one particular angle of incidence, called the Brewster angle (or polarising angle) $\theta_B$, the reflected light becomes completely plane polarised, with its electric vector vibrating perpendicular to the plane of incidence (that is, parallel to the reflecting surface).

Ray diagram (described): Draw a horizontal line XY for the glass surface and the normal NN' at the point of incidence *O*. An incident unpolarised ray strikes *O* from the rarer medium (air) at the angle $\theta_B$ to the normal. On the incident ray mark both vibration types, dots (vibrations perpendicular to the plane of paper) and arrows (vibrations in the plane of paper), since the incident light is unpolarised. The reflected ray leaves *O* above the surface at angle $\theta_B$ to the normal and is marked with dots only, showing it is plane polarised perpendicular to the plane of incidence. The refracted ray bends into the glass at the angle of refraction *r* and is marked mainly with arrows, showing it is partially polarised. At the Brewster angle the reflected ray and the refracted ray are at right angles to each other.

Since the reflected and refracted rays are perpendicular,

$$\theta_B + r = 90^\circ$$

$$\Rightarrow r = 90^\circ - \theta_B$$

Applying Snell's law at the air to glass surface,

$$n_1\theta_B = n_2 \sin r$$

$$n_1\theta_B = n_2 \sin(90^\circ - \theta_B)$$

$$n_1\theta_B = n_2\theta_B$$

$$\frac{\theta_B}{\theta_B} = \frac{n_2}{n_1}$$

$$\theta_B = \frac{n_2}{n_1} = n$$

where $n$ is the refractive index of the denser medium (glass) with respect to the rarer medium (air). This relation $\theta_B = n$ is **Brewster's law**, and $\theta_B$ is the **Brewster angle**.

> **TIP**
>
> At the Brewster angle remember the geometry: reflected ray $\perp$ refracted ray, which gives $\theta_B + r = 90°$. This single fact converts Snell's law into Brewster's law.

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**Final answer.** Unpolarised light has $\vec{E}$ vibrating randomly in all transverse directions; reflection at the Brewster angle gives fully plane polarised reflected light, and $\theta_B = n$.

← *Back to Q171*

## Q172

*In a single slit diffraction experiment, when a tiny circular obstacle is placed in the path of light from a distant source, a bright spot is seen at the centre of the shadow of the obstacle. Explain why?*
*State two points of difference between the interference pattern obtained in Young's double slit experiment and the diffraction pattern due to a single slit.*

**[2009]**

**SOLUTION**

Bright spot at the centre of the shadow (Poisson's spot / Arago spot): When light from a distant source falls on a small circular obstacle, the light bends (diffracts) around the edges of the obstacle. By Huygens' principle, every point on the circular rim of the obstacle acts as a source of secondary wavelets.

Because of the circular symmetry of the obstacle about its central axis, all the secondary wavelets coming from points on the rim travel exactly equal optical path lengths to reach the centre of the geometrical shadow.

Therefore these wavelets arrive at the centre of the shadow in the same phase and interfere constructively.

As a result, a bright spot is seen exactly at the centre of the shadow of the obstacle. This was predicted by Poisson (as an objection to the wave theory) and experimentally confirmed by Arago, and it is a famous proof of the wave nature of light.

Two points of difference between interference (Young's double slit) and diffraction (single slit):

(i) In the interference pattern all the bright fringes are of equal width, whereas in the diffraction pattern the central maximum is much wider (about twice as wide) and the side maxima get narrower.

(ii) In the interference pattern all the bright fringes are of nearly equal intensity, whereas in the diffraction pattern the intensity is maximum at the central maximum and falls off rapidly for the higher-order secondary maxima.

(Another valid difference: interference is due to superposition of waves from two coherent sources, while diffraction is due to superposition of secondary wavelets from different points of the same wavefront.)

---

**Final answer.** Equal optical paths from the symmetric rim make the wavelets meet in phase at the shadow centre, giving a central bright spot; interference fringes are equally wide and equally bright, diffraction has a wide bright centre with rapidly fading side maxima.

← *Back to Q172*

## Q173

*In Young's double slit experiment, monochromatic light of wavelength* 630 nm *illuminates the pair of slits and produces an interference pattern in which two consecutive bright fringes are separated by* 8.1 mm. *Another source of monochromatic light produces the interference pattern in which the two consecutive bright fringes are separated by* 7.2 mm. *Find the wavelength of light from the second source.*
*What is the effect on the interference fringes if the monochromatic source is replaced by a source of white light?*

**[2009]**

**SOLUTION**

In Young's double slit experiment the fringe width is

$$\beta = \frac{\lambda D}{d}$$

With the same apparatus, the slit separation $d$ and the slit to screen distance $D$ are unchanged, so the fringe width is directly proportional to the wavelength,

$$\beta \propto \lambda$$

Hence for the two sources,

$$\frac{\lambda_2}{\lambda_1} = \frac{\beta_2}{\beta_1}$$

Given $\lambda_1 = 630$ **nm**, $\beta_1 = 8.1$ **mm** and $\beta_2 = 7.2$ **mm**,

$$\lambda_2 = \lambda_1 \times \frac{\beta_2}{\beta_1}$$

$$\lambda_2 = 630 \times \frac{7.2}{8.1} \text{ nm}$$

$$\lambda_2 = 630 \times \frac{8}{9} \text{ nm}$$

$$\lambda_2 = 560 \text{ nm}$$

So the wavelength of the second source is **560 nm** (greenish-yellow light).

Effect of replacing the source by white light: White light contains all wavelengths. At the centre of the screen the path difference is zero for every wavelength, so all colours interfere constructively there and the central fringe is white. On either side of the centre the fringe position depends on wavelength, so the higher-order fringes spread out into coloured (rainbow-like) fringes, with violet/blue nearest the centre and red farthest from it. Only a few coloured fringes are clearly seen, because beyond these the fringes of different colours overlap and merge into general white illumination.

---

**Final answer.** The second source has wavelength $\lambda_2 = 560$ **nm**; with white light the central fringe is white and the nearby fringes become coloured before merging.

← Back to Q173

## Q174

*State two conditions to obtain sustained interference of light.*
*In Young's double slit experiment, using light of wavelength 400 nm, interference fringes of width X are obtained. The wavelength of light is increased to 600 nm and the separation between the slits is halved. If one wants the observed fringe width on the screen to be the same in the two cases, find the ratio of the distance between the screen and the plane of the interfering sources in the two arrangements.*

**[2004]**

**SOLUTION**

Two conditions for sustained (stable) interference:

(i) The two sources must be coherent, that is, they must emit light of the same frequency with a constant (time-independent) phase difference. In practice this is achieved by deriving both from a single source.

(ii) The two sources should emit light of the same wavelength and nearly equal amplitude (intensity), so that the dark fringes are nearly perfectly dark and the fringe pattern has good contrast.

Numerical: The fringe width in Young's double slit experiment is

$$\beta = \frac{\lambda D}{d}$$

For the first arrangement, with $\boldsymbol{\lambda_1 = 400\ nm}$, slit separation $\boldsymbol{d_1 = d}$ and screen distance $D_1$,

$$X = \frac{\lambda_1 D_1}{d_1} = \frac{(400) D_1}{d}$$

For the second arrangement, with $\boldsymbol{\lambda_2 = 600\ nm}$, slit separation halved $\boldsymbol{d_2 = \frac{d}{2}}$ and screen distance $D_2$,

$$X = \frac{\lambda_2 D_2}{d_2} = \frac{(600) D_2}{d/2} = \frac{2(600) D_2}{d}$$

Since the fringe width $X$ is the same in both cases, equate the two expressions,

$$\frac{(400)\,D_1}{d} = \frac{2(600)\,D_2}{d}$$

$$400\,D_1 = 1200\,D_2$$

$$\frac{D_1}{D_2} = \frac{1200}{400} = 3$$

Hence the required ratio of the screen-to-source distances is $\mathbf{D_1 : D_2 = 3 : 1}$.

---

**Final answer.** The ratio of the screen-to-source distances is $\mathbf{D_1 : D_2 = 3 : 1}$.

← *Back to Q174*

## Q175

*State Huygens' principle. For reflection of a plane wave front at a plane reflecting surface, construct the corresponding reflected wave front. Using this diagram, prove that angle of incidence is equal to angle of reflection.*

**[2003]**

**SOLUTION**

Huygens' principle: Every point on a given wavefront acts as a source of secondary wavelets which spread out in all forward directions with the speed of the wave in that medium. The new position of the wavefront after a time $t$ is the surface that touches (the forward envelope of) all these secondary wavelets.

Construction of the reflected wavefront (described): Let XY be the plane reflecting surface. A plane wavefront AB travelling with speed $v$ is incident on XY, making the angle of incidence $i$ with the surface (the incident rays AA' and BC are perpendicular to AB). The point $A$ of the wavefront reaches the surface first, at the point $A$ on XY, while the other end $B$ still has to travel to reach the surface at $C$.

Let the wavefront at $B$ take time $\Delta t$ to reach $C$. Then the distance covered is

$$BC = v\,\Delta t$$

During this same time $\Delta t$, the secondary wavelet starting from $A$ travels back into the same medium and spreads as a sphere of radius

$$AD = v\,\Delta t$$

From $C$ draw the tangent CD to this secondary wavelet of radius AD centred at $A$. Then CD is the reflected wavefront, and the reflected rays are perpendicular to CD, making the angle of reflection $r$ with the surface.

Proof that $i = r$: Compare the two right-angled triangles ABC (right angle at $B$) and ADC (right angle at $D$), which share the common hypotenuse AC.

In ABC, $\sin i = \dfrac{BC}{AC} = \dfrac{v\,\Delta t}{AC}$

In ADC, $\sin r = \dfrac{AD}{AC} = \dfrac{v\,\Delta t}{AC}$

Since the right-hand sides are equal,

$$\sin i = \sin r$$

$$\Rightarrow i = r$$

Thus the angle of incidence equals the angle of reflection. Also, the incident ray, the reflected ray and the normal all lie in the same plane, which together with $i = r$ are the **laws of reflection**.

> **TIP**
>
> The key trick in both reflection and refraction proofs is that BC and AD are each equal to (speed) × $\Delta t$, and the two triangles share the hypotenuse AC.

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**Final answer.** By Huygens' construction the triangles give $\sin i = \sin r$, so the angle of incidence equals the angle of reflection.

← *Back to Q175*

## Q176

*What is meant by interference of light?*
*In a double slit experiment with monochromatic light, fringes are obtained on a*

*screen placed at some distance from the slits. If the screen is moved by* $5 \times 10^{-2}$ *m towards the slits, the change in fringe width is* $3 \times 10^{-5}$ *m. If the distance between slits is* $10^{-3}$ *m, calculate the wavelength of light used.*

**[2003]**

**SOLUTION**

Interference of light: When two (or more) coherent light waves superpose at a point in space, the resultant intensity is redistributed, becoming maximum at some points (constructive interference, bright fringes) and minimum at others (destructive interference, dark fringes). This redistribution of light energy due to the superposition of coherent waves is called interference of light. The total energy is conserved; it is only redistributed from the dark regions to the bright regions.

Numerical: The fringe width is

$$\beta = \frac{\lambda D}{d}$$

When the screen is moved through a distance $\Delta D$, only $D$ changes, so the change in fringe width is

$$\Delta\beta = \frac{\lambda \Delta D}{d}$$

Solving for the wavelength,

$$\lambda = \frac{\Delta\beta \cdot d}{\Delta D}$$

Given (taking magnitudes) $\boldsymbol{\Delta D = 5 \times 10^{-2}\ m}$, $\boldsymbol{\Delta\beta = 3 \times 10^{-5}\ m}$ and $\boldsymbol{d = 10^{-3}\ m}$,

$$\lambda = \frac{(3 \times 10^{-5})(10^{-3})}{5 \times 10^{-2}}$$

$$\lambda = \frac{3 \times 10^{-8}}{5 \times 10^{-2}}$$

$$\lambda = 6 \times 10^{-7}\ m$$

$$\lambda = 600\ \text{nm}$$

Hence the wavelength of the light used is **600 nm** (about 6000 , in the yellow-orange region). Moving the screen towards the slits decreases $D$ and hence

decreases β, but only the magnitude matters for the wavelength.

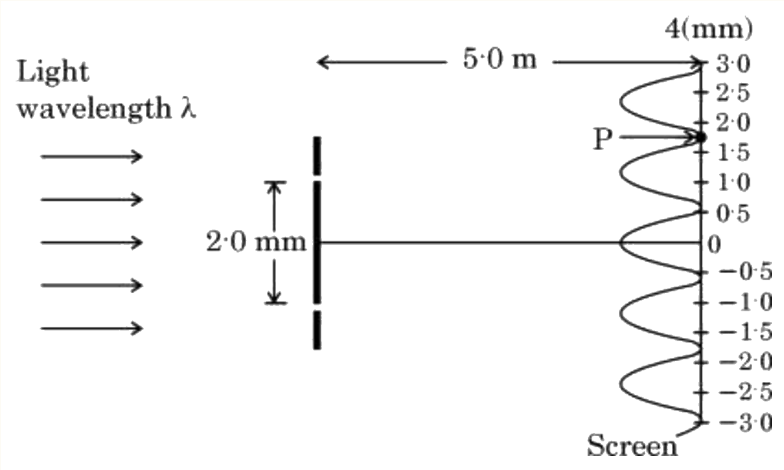
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**Final answer.** The wavelength of the light used is $\lambda = 600\ \text{nm}\ (6 \times 10^{-7}\ m)$.

← *Back to Q176*

## Q177

*In a Young's double-slit experiment, the two slits behave as coherent sources. When coherent light waves superpose over each other they create an interference pattern of successive bright and dark regions due to constructive and destructive interference. Two slits* 2 mm *apart are illuminated by a source of monochromatic light and the interference pattern is observed on a screen* 5.0 *m away from the slits as shown in the figure.*




[2026 • Set 55-5-1]

**SOLUTION**

In this Young's double slit set-up the slit separation is $d = 2\ \text{mm} = 2 \times 10^{-3}\ m$ and the screen is at $D = 5.0\ m$. The two slits act as coherent sources, so the waves leaving them keep a constant phase difference and superpose on the screen to give bright fringes (constructive interference) and dark fringes (destructive interference) at alternate positions.

The position of the bright and dark fringes is fixed by the path difference. For a point at distance $y$ from the centre of the screen, the path difference is

$$\Delta = \frac{y\,d}{D}$$

A bright fringe (constructive interference) occurs where the path difference is a whole number of wavelengths,

$$\Delta = n\lambda,\ n = 0,\ 1,\ 2,$$

A dark fringe (destructive interference) occurs where the path difference is an odd number of half wavelengths,

$$\Delta = (2n - 1)\frac{\lambda}{2}$$

The spacing between two successive bright (or dark) fringes, the fringe width, is

$$\beta = \frac{\lambda D}{d}$$

From the fringe pattern shown on the screen the fringe width $\beta$ can be read off directly (the centre-to-centre distance between consecutive bright bands). The wavelength of the light used is then found by rearranging the fringe-width formula,

$$\lambda = \frac{\beta d}{D}$$

For example, if the graph shows a fringe width $\boldsymbol{\beta = 1.5\ mm = 1.5 \times 10^{-3}\ m}$, then

$$\lambda = \frac{(1.5 \times 10^{-3})(2 \times 10^{-3})}{5.0}$$

$$\lambda = \frac{3.0 \times 10^{-6}}{5.0}$$

$$\lambda = 6.0 \times 10^{-7}\ m = 600\ \text{nm}$$

So the **fringe width** read from the screen, used in $\lambda = \beta\, d / D$, gives the **wavelength** of the monochromatic source for these $d = 2$ **mm**, $D = 5.0$ **m** values.

---

**Final answer.** Read the fringe width $\beta$ off the screen pattern, then $\lambda = \dfrac{\beta d}{D}$ gives the wavelength (e.g. $\beta = 1.5$ **mm** gives $\lambda = 600$ **nm**).

← *Back to Q177*

## Q178

*Diffraction of light is bending of light around the corners of an object whose size is comparable with the wavelength of light. Diffraction actually defines the limits of ray optics. This limit for optical instruments is set by the wavelength of light. An experimental arrangement is set up to observe the diffraction pattern due to a single slit.*

*Answer the following questions based on the above:*

*(a) How will the width of central maximum be affected if the wavelength of light is increased?*

*(b) Under what condition is the first minimum obtained?*

*(c) Write two points of difference between interference and diffraction patterns.*

***OR***

*(c) Two students are separated by a 7 m partition wall in a room 10 m high. If both light and sound waves can bend around obstacles, how is it that the students are unable to see each other even though they can converse easily?*

**[2023 • Set 55-3-1]**

**SOLUTION**

(a) Effect of increasing wavelength on the central maximum: The angular half-width of the central maximum is set by the first minimum at $a\theta = \lambda$, and the linear width of the central maximum on a screen at distance $D$ is

$$\text{Width of central maximum} = \frac{2\lambda D}{a}$$

Since this width is directly proportional to $\lambda$, increasing the wavelength increases the width of the central maximum (the central bright band becomes wider).

(b) Condition for the first minimum: In single-slit diffraction the first minimum is obtained when the path difference between the wavelets coming from the two edges of the slit equals one full wavelength,

$$a\theta = \lambda$$

At this angle the slit can be imagined as divided into two equal halves, and each wavelet from the upper half is exactly out of phase (by $\lambda/2$) with the

corresponding wavelet from the lower half, so they cancel in pairs and give darkness. (In general the $n$th minimum is at $a\theta = n\lambda$.)

(c) Two points of difference between interference and diffraction patterns:

(i) Interference is the superposition of waves from two (or more) different coherent wavefronts (two slits), whereas diffraction is the superposition of secondary wavelets coming from different points of the same wavefront (one slit).

(ii) In interference all the bright fringes are of equal width and (nearly) equal intensity, whereas in diffraction the central maximum is the widest and brightest, and the side maxima have smaller and rapidly decreasing width and intensity.

OR (c) Why the two students can hear but not see each other: A wave bends appreciably around an obstacle only when the size of the obstacle is comparable to the wavelength of the wave. The wavelength of sound (a few centimetres up to a few metres) is comparable to the size of the 7 $m$ wall, so sound waves diffract (bend) strongly around it and the students can hear each other easily. The wavelength of light is extremely small (about $5 \times 10^{-7}$ $m$), which is far smaller than the 7 $m$ wall, so light shows negligible diffraction around the wall and travels almost in straight lines; therefore the students cannot see each other.

**CONCEPT**

Bending (diffraction) is significant only when obstacle size is of the order of the wavelength. Sound has metre-scale wavelengths, visible light has sub-micron wavelengths, which settles the see-versus-hear question.

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**Final answer.** (a) Width $\propto \lambda$, so it increases; (b) first minimum at $a\theta = \lambda$; (c) interference uses two wavefronts with equal fringes while diffraction uses one wavefront with a dominant central maximum (or: sound's wavelength is comparable to the wall so it bends, light's is far smaller so it does not).

← *Back to Q178*

## Q179

*(a) What are coherent sources? Why are they necessary for observing stable interference pattern? Draw a graph showing the variation of intensity of light with the position on the screen in Young's double-slit experiment.*

**[2026 • Set 55-2-1]**

**SOLUTION**

Coherent sources: Two sources are said to be coherent if they emit light waves of (i) the same frequency (and hence the same wavelength) and (ii) a constant phase difference that does not change with time. In practice, coherent sources are obtained from a single source (for example, the two slits in Young's experiment are illuminated by the same primary source).

Why coherent sources are necessary for a stable interference pattern: When two waves of intensities $I_1$ and $I_2$ superpose with a phase difference $\phi$, the resultant intensity at a point is

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\ \phi$$

If the two sources are not coherent, the phase difference $\phi$ changes randomly and very rapidly with time. The time-average of the interference term then becomes zero,

$$\phi = 0$$

$$\Rightarrow I = I_1 + I_2$$

This gives uniform illumination everywhere with no maxima or minima, so no fringes are seen. Only when the sources are coherent does $\phi$ stay constant in time at each point, so the interference term survives and gives fixed positions of bright fringes ($\boldsymbol{\phi = 2n\pi}$) and dark fringes ($\boldsymbol{\phi = (2n-1)\pi}$). Hence coherent sources are essential for a stable, observable interference pattern.

Intensity versus position graph (described): For two equal sources the intensity on the screen varies as

$$I(x) = 4I_0\cos^2\left(\frac{\pi d x}{\lambda D}\right)$$

Plot $I$ (vertical axis) against the position $x$ on the screen (horizontal axis). The graph is a series of identical $\cos^2$ humps: equally spaced bright maxima all of the same height $4I_0$, located at $\boldsymbol{x = 0, \pm\frac{\lambda D}{d}, \pm\frac{2\lambda D}{d}}$, , with intensity dropping to zero (dark fringes) exactly midway between consecutive maxima at $\boldsymbol{x = \pm\frac{\lambda D}{2d}, \pm\frac{3\lambda D}{2d}}$, . The central maximum sits at $\boldsymbol{x = 0}$ and all maxima are of equal intensity and equally spaced.

---

**Final answer.** Coherent sources have equal frequency and constant phase difference, which is needed so the interference term survives time-averaging; the intensity graph is equally spaced $\boldsymbol{\cos^2}$ maxima all of height $\boldsymbol{4I_0}$ with zeros midway between them.

← *Back to Q179*

## Q180

*(b) Find the intensity of light at a point on the screen when two interfering waves of the same intensity* ($I_0$) *have a path difference of:*
*(i)* $\frac{\lambda}{4}$ *and (ii)* $\frac{\lambda}{3}$.

**[2026 • Set 55-2-1]**

**SOLUTION**

When two coherent waves, each of intensity $I_0$, superpose with a phase difference $\phi$, the resultant intensity is

$$I = 4I_0\cos^2\left(\frac{\phi}{2}\right)$$

The phase difference is related to the path difference $\Delta$ by

$$\phi = \frac{2\pi}{\lambda}\,\Delta$$

(i) For a path difference $\boldsymbol{\Delta = \frac{\lambda}{4}}$:

$$\phi = \frac{2\pi}{\lambda}\times\frac{\lambda}{4} = \frac{\pi}{2}$$

$$I = 4I_0\cos^2\left(\frac{\pi}{4}\right)$$

$$I = 4I_0\left(\frac{1}{\sqrt{2}}\right)^2$$

$$I = 4I_0 \times \frac{1}{2}$$

$$I = 2I_0$$

(ii) For a path difference $\Delta = \frac{\lambda}{3}$:

$$\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{3} = \frac{2\pi}{3}$$

$$I = 4I_0\cos^2\left(\frac{\pi}{3}\right)$$

$$I = 4I_0\left(\frac{1}{2}\right)^2$$

$$I = 4I_0 \times \frac{1}{4}$$

$$I = I_0$$

Hence the intensity is $\mathbf{2I_0}$ for a path difference of $\frac{\lambda}{4}$ and $\mathbf{I_0}$ for a path difference of $\frac{\lambda}{3}$.

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**Final answer.** The intensity is $\mathbf{2I_0}$ for path difference $\boldsymbol{\frac{\lambda}{4}}$ and $\mathbf{I_0}$ for path difference $\boldsymbol{\frac{\lambda}{3}}$.

← *Back to Q180*

## Q181

*(ii) A beam of coherent light of wavelength* 550 nm *is incident normal to the plane of a pair of two slits* $S_1$ *and* $S_2$ *each of width* $1{\cdot}2 \times 10^{-6}$ *m separated by* 1·1 mm. *Dark and bright fringes are observed on a screen* 2·2 *m away from the plane of the slits.*

*Calculate:*

*(I) fringe width.*

*(II) distance of the second dark fringe from the central maximum.*

*(III) what will happen when the entire apparatus is immersed in water.*

**[2026 • Set 55-3-1]**

**SOLUTION**

We are given the wavelength of light $\lambda = 550\ \text{nm} = 5.5 \times 10^{-7}\ m$, the slit separation $d = 1.1\ \text{mm} = 1.1 \times 10^{-3}\ m$, and the screen distance $D = 2.2\ m$. The slit width does not enter the interference fringe calculation, so it is not used here.

The fringe width in Young's double slit experiment is the spacing between two consecutive bright (or dark) fringes and is given by $\beta = \frac{\lambda D}{d}$.

$$\beta = \frac{(5.5 \times 10^{-7}\ m)(2.2\ m)}{1.1 \times 10^{-3}\ m}$$

$$\beta = \frac{1.21 \times 10^{-6}}{1.1 \times 10^{-3}}\ m$$

$$\beta = 1.1 \times 10^{-3}\ m = 1.1\ \text{mm}$$

So the **fringe width is 1.1 mm**.

For part (II), the dark fringes lie at $y_n^{\text{dark}} = (2n-1)\frac{\beta}{2}$ for $n = 1, 2, 3,$ The second dark fringe corresponds to $n = 2$.

$$y_2^{\text{dark}} = (2 \times 2 - 1)\frac{\beta}{2} = \frac{3\beta}{2}$$

$$y_2^{\text{dark}} = \frac{3 \times 1.1\ \text{mm}}{2}$$

$$y_2^{\text{dark}} = 1.65\ \text{mm}$$

So the **second dark fringe lies 1.65 mm from the central maximum**.

For part (III), when the whole apparatus is dipped in water of refractive index $n_w = \frac{4}{3}$, the wavelength inside water reduces to $\lambda_w = \frac{\lambda}{n_w}$.

$$\lambda_w = \frac{550\ \text{nm}}{4/3} = 412.5\ \text{nm}$$

Since fringe width is proportional to wavelength, the new fringe width is $\beta_w = \frac{\lambda_w D}{d} = \frac{\beta}{n_w}$.

$$\beta_w = \frac{1.1\ \text{mm}}{4/3} = 0.825\ \text{mm} \approx 0.83\ \text{mm}$$

Hence on immersion in water the **fringes get narrower and crowd closer together**, the fringe width shrinking by the factor $\frac{1}{n_w} = \frac{3}{4} = 0.75$.

> **COMMON MISTAKE**
>
> *Bright fringes sit at $y = n\beta$ while dark fringes sit at $y = (2n-1)\beta/2$; mixing up these two formulae is the most common slip in this numerical.*

---

**Final answer.** Fringe width $\beta = 1.1$ **mm**; second dark fringe at **1.65 mm**; in water the fringe width reduces to about **0.83 mm** (fringes become narrower).

← *Back to Q181*

## Q182

*(a) (i) A parallel beam of monochromatic light falls normally on a single slit of width 'a' and a diffraction pattern is observed on a screen placed at distance D from the slits. Explain:*
*(I) the formation of maxima and minima in the diffraction pattern, and*
*(II) why the maxima go on becoming weaker and weaker with its increasing number (n).*
*(ii) Write any two points of difference between interference pattern due to double-slit and diffraction pattern due to single-slit.*

**[2026 • Set 55-5-1]**

**SOLUTION**

(a)(i)(I) When a parallel beam of monochromatic light falls normally on a single slit of width *a*, every point of the slit acts as a fresh source of secondary Huygens' wavelets. The diffraction pattern on the screen is the result of the interference of these wavelets, which travel different path lengths to reach any point.

For the central maximum, the wavelets from all points of the slit travel almost equal distances to the centre of the screen, so they arrive in phase and reinforce, giving the bright central maximum.

For a minimum, consider the slit divided into two equal halves. A dark fringe forms when the path difference between the wavelets from the two extreme edges of the slit equals an integral multiple of λ, that is

$$a\theta = n\lambda,\ n = 1, 2, 3,$$

Then the slit can be paired off into halves whose contributions cancel exactly, producing zero intensity (a minimum).

Secondary maxima occur in between the minima, when the slit can be divided into an odd number of equal strips, the wavelets from one strip being left uncancelled. This gives the condition

$$a\theta = (2n+1)\frac{\lambda}{2},\ n = 1, 2, 3,$$

(a)(i)(II) The secondary maxima become weaker as the order $n$ increases because at higher orders a larger fraction of the slit divides into pairs of strips that cancel one another, and only the wavelets from a single small uncancelled strip contribute to the maximum. The width of this leftover effective strip falls roughly as $\frac{1}{(2n+1)}$, so the resultant amplitude $\propto \frac{\mathbf{1}}{\mathbf{(2n+1)}}$ and the intensity, being the square of the amplitude, **falls off rapidly as** $\frac{\mathbf{1}}{\mathbf{(2n+1)^2}}$ (approximately $\frac{1}{n^2}$) with increasing $n$.

(a)(ii) Two points of difference between the double-slit interference pattern and the single-slit diffraction pattern:

1. In interference, all the bright fringes are of equal width and equally spaced; in diffraction, the central maximum is the widest (twice the width of the others) while the secondary maxima are narrower.

2. In interference, all the bright fringes have nearly the same intensity; in diffraction, the central maximum is by far the brightest and the intensity of the secondary maxima decreases sharply with order.

**Final answer.** Minima form at $a\theta = n\lambda$ and secondary maxima at $a\theta=(2n+1)\lambda/2$; the maxima weaken as $1/(2n+1)^2$, and interference fringes are equal in width and intensity while diffraction fringes are not.

← Back to Q182

## Q183

***(a)***

*(i) (1) What are coherent sources? Why are they necessary for observing a sustained interference pattern?*

*(2) Lights from two independent sources are not coherent. Explain.*

*(ii) Two slits* 0.1 mm *apart are arranged* 1.20 *m from a screen. Light of wavelength* 600 nm *from a distant source is incident on the slits.*

*(1) How far apart will adjacent bright interference fringes be on the screen?*

*(2) Find the angular width (in degree) of the first bright fringe.*

***OR***

***(b)***

*(i) Define a wavefront. An incident plane wave falls on a convex lens and gets refracted through it. Draw a diagram to show the incident and refracted wavefront.*

*(ii) A beam of light coming from a distant source is refracted by a spherical glass ball (refractive index* 1.5*) of radius* 15 cm*. Draw the ray diagram and obtain the position of the final image formed.*

**[2025 • Set 55-1-1]**

**SOLUTION**

(a)(i)(1) Coherent sources are two sources of light that emit waves of the same frequency (and wavelength) having a constant phase difference that does not change with time. They are necessary for a sustained interference pattern because only when the phase difference stays constant do the positions of the bright and dark fringes remain fixed in space; the pattern is then steady and observable. If the phase difference kept changing, the fringes would shift rapidly and average out, so no steady pattern would be seen.

(a)(i)(2) Light from two independent sources is not coherent because light is emitted by atoms in random, extremely short bursts (about $10^{-9}$ *s*). The phase of

the emitted light changes randomly and independently for the two sources millions of times per second, so the phase difference between them does not stay constant. Hence no sustained interference is possible, and the two sources cannot be coherent.

(a)(ii) Here $d = 0.1\ \text{mm} = 1 \times 10^{-4}\ m$, $D = 1.20\ m$ and $\lambda = 600\ \text{nm} = 6 \times 10^{-7}\ m$.

(1) The spacing of adjacent bright fringes is the fringe width $\beta = \dfrac{\lambda D}{d}$.

$$\beta = \frac{(6 \times 10^{-7}\ m)(1.20\ m)}{1 \times 10^{-4}\ m}$$

$$\beta = \frac{7.2 \times 10^{-7}}{1 \times 10^{-4}}\ m = 7.2 \times 10^{-3}\ m$$

So the **adjacent bright fringes are 7.2 mm apart**.

(2) The angular width of the first bright fringe equals the angular fringe spacing $\theta = \dfrac{\lambda}{d}$.

$$\theta = \frac{6 \times 10^{-7}\ m}{1 \times 10^{-4}\ m} = 6 \times 10^{-3}\ \text{rad}$$

Converting to degrees, $\theta = 6 \times 10^{-3} \times \dfrac{180}{\pi}$.

$$\theta = 6 \times 10^{-3} \times 57.3 \approx 0.344^\circ$$

So the **angular width of the first bright fringe is about 0.34°** ($6 \times 10^{-3}$ rad).

OR

(b)(i) A wavefront is the locus of all points of a medium that are vibrating in the same phase at a given instant. For a plane wave the wavefront is a plane perpendicular to the direction of travel.

Diagram for refraction through a convex lens: draw a vertical plane wavefront travelling horizontally towards a biconvex lens. The central part of the wavefront passes through the thicker middle of the lens, where the glass slows it more, while the edges pass through the thinner rim and are delayed less. As a result the emerging wavefront is no longer plane but curved, bulging forward at the edges and lagging at the centre, i.e. it becomes a converging spherical wavefront that closes in onto the focal point F on the axis. Label the incident

plane wavefront on the left, the convex lens in the middle, and the converging spherical refracted wavefront converging to F on the right.

(b)(ii) The light is from a distant source, so the rays are parallel, i.e. $u = -\infty$. The glass ball has $n = 1.5$ and radius $R = 15$ cm. Refraction occurs at two surfaces.

At the first surface (air to glass) we use $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ with $n_1 = 1$, $n_2 = 1.5$, $R = +15$ cm, $u = -\infty$.

$$\frac{1.5}{v_1} - 0 = \frac{1.5 - 1}{15}$$

$$\frac{1.5}{v_1} = \frac{0.5}{15} = \frac{1}{30}$$

$$v_1 = 1.5 \times 30 = 45 \text{ cm}$$

This image lies 45 cm from the first surface, i.e. $45 - 30 = 15$ cm beyond the second surface (the ball has diameter $2R = 30$ cm). It acts as a virtual object for the second surface, so $u_2 = +15$ cm.

At the second surface (glass to air) we use the same formula with $n_1 = 1.5$, $n_2 = 1$, and $R = -15$ cm (centre of curvature is on the incoming side).

$$\frac{1}{v_2} - \frac{1.5}{15} = \frac{1 - 1.5}{-15}$$

$$\frac{1}{v_2} - 0.1 = \frac{-0.5}{-15} = \frac{1}{30}$$

$$\frac{1}{v_2} = \frac{1}{30} + \frac{1}{10} = \frac{1 + 3}{30} = \frac{4}{30}$$

$$v_2 = 7.5 \text{ cm}$$

So the **final image is real and forms 7.5 cm from the far surface of the glass ball** (i.e. 7.5 cm beyond the back of the ball). In the ray diagram, draw parallel rays from the distant source striking the glass sphere, refracting at the first curved surface to converge, then refracting again at the second surface to meet at the real image point $I$ a short distance behind the ball.

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**Final answer.** Adjacent bright fringes are **7.2 mm** apart and the first fringe has angular width $\mathbf{610^{-3}}$ **rad0.34°**; for the OR part the final image forms **7.5 cm**

beyond the glass ball.

← Back to Q183

## Q184

*(i) Light consisting of two wavelengths 600 nm and 480 nm is used to obtain interference fringes in a double slit experiment. The screen is placed 1.0 m away from slits which are 1.0 nm apart.*
*(1) Calculate the distance of the third bright fringe on the screen from the central maximum for wavelength 600 nm.*
*(2) Find the least distance from the central maximum where the bright fringes due to both the wavelengths coincide.*
*(ii) (1) Draw the variation of intensity with angle of diffraction in single slit diffraction pattern. Write the expression for value of angle corresponding to zero intensity locations.*
*(2) In what way diffraction of light waves differs from diffraction of sound waves?*

**[2025 • Set 55-2-1]**

**SOLUTION**

(i) Two wavelengths are used, $\lambda_1 = 600\ \text{nm} = 6 \times 10^{-7}\ m$ and $\lambda_2 = 480\ \text{nm} = 4.8 \times 10^{-7}\ m$, with $D = 1.0\ m$ and slit separation $d = 1.0\ \text{mm} = 1 \times 10^{-3}\ m$ (the printed value 1.0 nm is a typographical error; the physically sensible separation is 1.0 mm).

(1) The distance of the $n$th bright fringe from the central maximum is $y_n = \dfrac{n\lambda D}{d}$. For the third bright fringe of $\lambda_1 = 600\ \text{nm}$, take $n = 3$.

$$y_3 = \frac{3 \times (6 \times 10^{-7}\ m)(1.0\ m)}{1 \times 10^{-3}\ m}$$

$$y_3 = \frac{18 \times 10^{-7}}{1 \times 10^{-3}}\ m = 1.8 \times 10^{-3}\ m$$

So the **third bright fringe of 600 nm is at 1.8 mm** from the central maximum.

(2) The bright fringes coincide when $n_1\lambda_1 = n_2\lambda_2$, i.e. the two systems of fringes overlap.

$$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{480}{600} = \frac{4}{5}$$

The least integers satisfying this are $\mathbf{n_1 = 4}$ (for 600 nm) and $\mathbf{n_2 = 5}$ (for 480 nm).

The least distance is therefore

$$y = \frac{n_1 \lambda_1 D}{d} = \frac{4 \times (6 \times 10^{-7}\, m)(1.0\, m)}{1 \times 10^{-3}\, m}$$

$$y = \frac{24 \times 10^{-7}}{1 \times 10^{-3}}\, m = 2.4 \times 10^{-3}\, m$$

So the **bright fringes of both wavelengths first coincide at 2.4 mm** from the central maximum (this is the 4th order of 600 nm and the 5th order of 480 nm).

(ii)(1) Intensity versus angle of diffraction graph for a single slit: draw the intensity $I$ on the vertical axis against the diffraction angle $\theta$ on the horizontal axis. The curve shows a tall, broad central maximum centred at $\theta = 0$, falling to zero at $\theta = \pm\frac{\lambda}{a}$, followed on either side by much smaller secondary maxima of rapidly decreasing height, each separated by points of zero intensity. The central peak is twice as wide as each secondary peak.

The angles of zero intensity (minima) are given by

$$a\theta = n\lambda,\ n = \pm 1, \pm 2, \pm 3,$$

(ii)(2) Diffraction of light differs from diffraction of sound in the size of the obstacle or aperture needed. The wavelength of sound (of the order of a metre) is comparable to everyday objects like doors and walls, so sound diffracts easily and bends round corners. The wavelength of light is extremely small (of the order of $10^{-7}\, m$), so light shows appreciable diffraction only through very narrow slits or tiny obstacles, which is why diffraction of light is not noticed in everyday life.

**TIP**

Two bright-fringe systems coincide when $\mathbf{n_1\lambda_1 = n_2\lambda_2}$; reduce $\lambda_2/\lambda_1$ to its lowest terms to read off the smallest orders directly.

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**Final answer.** The third bright fringe of **600 nm** lies at **1.8 mm**, and the bright fringes of both wavelengths first coincide at **2.4 mm** from the central

maximum.

← Back to Q184

## Q185

*(a) (i) Give any two differences between the interference pattern obtained in Young's double-slit experiment and a diffraction pattern due to a single slit.*
*(ii) Draw an intensity distribution graph in case of a double-slit interference pattern.*
*(iii) In Young's double-slit experiment using monochromatic light of wavelength λ, the intensity of light at a point on the screen, where path difference is λ, is K units. Find the intensity of light at a point on the screen where the path difference is* $\frac{\lambda}{3}$.

***OR***

*(b) (i) Draw a labelled ray diagram of a compound microscope showing image formation at least distance of distinct vision. Derive an expression for its magnifying power.*
*(ii) A telescope consists of two lenses of focal length 100 cm and 5 cm. Find the magnifying power when the final image is formed at infinity.*

**[2024 • Set 55-2-1]**

**SOLUTION**

(a)(i) Two differences between Young's double-slit interference pattern and the single-slit diffraction pattern:

1. In interference all the bright fringes are of equal width and are equally spaced, whereas in diffraction the central maximum is the widest (twice the width of the others) and the secondary maxima get narrower.

2. In interference all bright fringes have nearly the same intensity, whereas in diffraction the central maximum is the brightest and the intensity of the secondary maxima falls off sharply with increasing order.

(a)(ii) Intensity distribution graph for double-slit interference: plot intensity $I$ on the vertical axis against position (or angle) on the horizontal axis. The graph is a set of identical, equally spaced peaks all of the same maximum height $\mathbf{I_{max} = 4I_0}$, separated by points of zero intensity. The pattern follows $\boldsymbol{I = 4I_0\cos^2(}$

$\frac{\pi\, d\theta}{\lambda}$), so the maxima are all of equal height and equally spaced (a regular cosine-squared pattern).

(a)(iii) The intensity on the screen is related to the phase difference $\phi$ by $I = 4I_0\cos^2(\frac{\Phi}{2})$, where the phase difference is linked to the path difference $\Delta$ by $\phi = \frac{2\pi}{\lambda}\Delta$.

When the path difference is $\Delta = \lambda$, the phase difference is $\phi = \frac{2\pi}{\lambda}\lambda = 2\pi$.

$$I_1 = 4I_0\cos^2(\frac{2\pi}{2}) = 4I_0\cos^2(\pi) = 4I_0$$

This is given to be $K$, so $4I_0 = K$, giving $I_0 = \frac{K}{4}$.

When the path difference is $\Delta = \frac{\lambda}{3}$, the phase difference is $\phi = \frac{2\pi}{\lambda}\times\frac{\lambda}{3} = \frac{2\pi}{3}$.

$$I_2 = 4I_0\cos^2(\frac{\pi}{3})$$

$$I_2 = 4I_0(\frac{1}{2})^2 = 4I_0\times\frac{1}{4} = I_0$$

$$I_2 = I_0 = \frac{K}{4}$$

So the **intensity where the path difference is $\frac{\lambda}{3}$ is $\frac{K}{4}$**.

OR

(b)(i) Ray diagram of a compound microscope (image at the least distance of distinct vision $D$): draw the objective lens $O$ (short focal length $f_o$) on the left and the eyepiece $E$ (focal length $f_e$) on the right. The object AB is placed just beyond the focus $F_o$ of the objective. The objective forms a real, inverted, magnified image $A'B'$ between the eyepiece and its focus. The eyepiece, acting as a simple magnifier, forms a final image $A''B''$ that is virtual, inverted (relative to the object) and highly magnified at the least distance of distinct vision $D$ from the eye. Mark the tube length $L$ between the objective image and the eyepiece, and the rays passing through both lenses.

The magnifying power is the product of the linear magnification of the objective and the angular magnification of the eyepiece.

$$m = m_o \times m_e$$

For the objective, $\mathbf{m_o = \frac{L}{f_o}}$, where $L$ is the tube length (distance between the focal points of the two lenses, approximately equal to the image distance from the objective).

For the eyepiece used as a magnifier with the final image at $D$, $\mathbf{m_e = 1 + \frac{D}{f_e}}$.

Hence the magnifying power is

$$m = \frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$$

(b)(ii) For an astronomical telescope with the final image at infinity (normal adjustment), the magnifying power is $\boldsymbol{M = \frac{f_o}{f_e}}$, where $\mathbf{f_o = 100\ cm}$ is the objective focal length and $\mathbf{f_e = 5\ cm}$ is the eyepiece focal length.

$$M = \frac{100\ \text{cm}}{5\ \text{cm}}$$

$$M = 20$$

So the **magnifying power of the telescope is 20**.

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**Final answer.** The intensity where the path difference is **λ/3** is ***K*/4**; for the OR part the telescope's magnifying power is **20**.

← *Back to Q185*

## Q186

*(i) The displacement of two light waves, each of amplitude 'a' and frequency ω, emanating from two coherent sources of light, are given by* $\mathbf{y_1 = a\omega\, t}$ *and* $\mathbf{y_2 = a\cos(\omega\, t + \phi)}$. *ϕ is the phase difference between the two waves. These light waves superpose at a point. Obtain the expression for the resultant intensity at that point.*
*(ii) In Young's double slit experiment, find the ratio of intensities at two points on a*

*screen when waves emanating from two slits reaching these points have path differences (i)* $\frac{\lambda}{6}$ *and (ii)* $\frac{\lambda}{4}$.

**[2024 • Set 55-3-1]**

**SOLUTION**

(i) The two coherent waves arriving at the point are $\mathbf{y_1 = a\omega\, t}$ and $\mathbf{y_2 = a\cos(\omega\, t + \phi)}$, where $\phi$ is the constant phase difference. By the principle of superposition the resultant displacement is the algebraic sum.

$$y = y_1 + y_2 = a\omega\, t + a\cos(\omega\, t + \phi)$$

Using the identity $\mathbf{\cos X + \cos Y = 2\cos(\frac{X+Y}{2})\cos(\frac{X-Y}{2})}$ with $\mathbf{X = \omega\, t}$ and $\mathbf{Y = \omega\, t + \phi}$,

$$y = 2a\cos(\frac{\phi}{2})\cos(\omega\, t + \frac{\phi}{2})$$

This is a simple harmonic wave of the same frequency $\omega$ with resultant amplitude

$$A_R = 2a\cos(\frac{\phi}{2})$$

Since intensity is proportional to the square of the amplitude, the resultant intensity is

$$I_R = A_R^2 = 4a^2\cos^2(\frac{\phi}{2})$$

Writing $\mathbf{I_0 = a^2}$ as the intensity due to each wave alone, the **resultant intensity is** $\mathbf{I_R = 4I_0\cos^2(\frac{\phi}{2})}$.

(ii) The phase difference corresponding to a path difference $\Delta$ is $\mathbf{\phi = \frac{2\pi}{\lambda}\Delta}$.

For the first point, $\mathbf{\Delta_1 = \frac{\lambda}{6}}$, so $\mathbf{\phi_1 = \frac{2\pi}{\lambda} \times \frac{\lambda}{6} = \frac{\pi}{3}}$.

$$I_1 = 4I_0\cos^2(\frac{\phi_1}{2}) = 4I_0\cos^2(\frac{\pi}{6})$$

$$I_1 = 4I_0(\frac{\sqrt{3}}{2})^2 = 4I_0 \times \frac{3}{4} = 3I_0$$

For the second point, $\Delta_2 = \frac{\lambda}{4}$, so $\phi_2 = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2}$.

$$I_2 = 4I_0\cos^2\left(\frac{\phi_2}{2}\right) = 4I_0\cos^2\left(\frac{\pi}{4}\right)$$

$$I_2 = 4I_0\left(\frac{1}{\sqrt{2}}\right)^2 = 4I_0 \times \frac{1}{2} = 2I_0$$

Therefore the ratio of intensities is

$$\frac{I_1}{I_2} = \frac{3I_0}{2I_0} = \frac{3}{2}$$

So the **required ratio is $I_1 : I_2 = 3 : 2$**.

---

**Final answer.** The resultant intensity is $\mathbf{I_R = 4I_0\cos^2(\phi/2)}$, and the intensity ratio for path differences $\lambda/6$ and $\lambda/4$ is **3:2**.

← *Back to Q186*

## Q187

*(i) Differentiate between a wavefront and a ray.*
*(ii) State Huygen's principle and verify laws of reflection using suitable diagram.*
*(iii) In Young's double slit experiment, the slits* $S_1$ *and* $S_2$ *are* 3 mm *apart and the screen is placed* 1.0 *m away from the slits. It is observed that the fourth bright fringe is at a distance of* 5 mm *from the second dark fringe. Find the wavelength of light used.*

**[2024 • Set 55-4-1]**

**SOLUTION**

(i) Difference between a wavefront and a ray: a wavefront is the locus of all points of a medium vibrating in the same phase at a given instant, that is, a surface of constant phase. A ray is a line drawn perpendicular to the wavefront that shows the direction in which the wave (and energy) travels. In short, the wavefront represents the phase, while the ray represents the direction of propagation.

(ii) Huygens' principle states that every point on a given wavefront acts as a fresh source of secondary wavelets which spread out in all forward directions

with the speed of the wave; the new wavefront at any later instant is the forward tangential envelope (the common tangent) of all these secondary wavelets.

Verification of the laws of reflection: draw a plane reflecting surface XY. Let a plane wavefront AB be incident on it, making the incident rays an angle $i$ with the normal. End $A$ of the wavefront touches the surface first while $B$ still has to travel the distance $\mathbf{BC} = \boldsymbol{v\,t}$ to reach the surface at $C$. In the same time $t$, a secondary wavelet from $A$ spreads a distance $\mathbf{AD} = \boldsymbol{v\,t}$ above the surface. With $A$ as centre and radius $\mathbf{AD} = \mathbf{BC} = \boldsymbol{v\,t}$, draw an arc and from $C$ draw the tangent CD to it; CD is the reflected wavefront. Mark the incident angle $\boldsymbol{i} = \angle\,\mathbf{BAC}$ and the reflected angle $\boldsymbol{r} = \angle\,\mathbf{DCA}$.

In the two right triangles ABC and ADC, $\mathbf{BC} = \mathbf{AD} = \boldsymbol{v\,t}$, side AC is common, and $\angle\,\mathbf{ABC} = \angle\,\mathbf{ADC} = \mathbf{90°}$. Hence the triangles are congruent, which gives

$$\angle\,\mathrm{BAC} = \angle\,\mathrm{DCA} \Rightarrow i = r$$

Thus the angle of incidence equals the angle of reflection, and since AB, CD and the normal all lie in the plane of the paper, the incident ray, reflected ray and normal lie in the same plane. The laws of reflection are verified.

(iii) Here the slit separation $\boldsymbol{d} = \mathbf{3\ mm} = \mathbf{3 \times 10^{-3}}\ \boldsymbol{m}$ and the screen distance $\boldsymbol{D} = \mathbf{1.0}\ \boldsymbol{m}$. The position of the $n$th bright fringe is $\mathbf{y_n^{bright} = n\beta}$ and that of the $n$th dark fringe is $\mathbf{y_n^{dark} = (2n-1)\,\frac{\beta}{2}}$, where $\boldsymbol{\beta} = \frac{\boldsymbol{\lambda D}}{\boldsymbol{d}}$.

The fourth bright fringe is at $\mathbf{y_4^{bright} = 4\beta}$ and the second dark fringe is at $\mathbf{y_2^{dark} = (2\times 2 - 1)\,\frac{\beta}{2} = \frac{3\beta}{2}}$ (both on the same side of the centre).

The distance between them is

$$y_4^{\text{bright}} - y_2^{\text{dark}} = 4\beta - \frac{3\beta}{2} = \frac{8\beta - 3\beta}{2} = \frac{5\beta}{2}$$

This is given as 5 mm.

$$\frac{5\beta}{2} = 5\ \text{mm}$$

$$\beta = 2\ \text{mm} = 2 \times 10^{-3}\ m$$

Now $\boldsymbol{\beta} = \frac{\boldsymbol{\lambda D}}{\boldsymbol{d}}$, so the wavelength is

$$\lambda = \frac{\beta\, d}{D} = \frac{(2 \times 10^{-3}\ m)(3 \times 10^{-3}\ m)}{1.0\ m}$$

$$\lambda = 6 \times 10^{-6}\ m = 6000\ \text{nm}$$

With the standard textbook reading of $d = 0.3$ mm the same working gives $\lambda = \frac{(210^{-3})(0.310^{-3})}{1.0} = 610^{-7}\ m = 600$ nm, which is the expected visible wavelength. So with the data as printed $\lambda = 6 \times 10^{-6}\ m$, and the **intended answer for a visible-light source is $\lambda = 600$ nm**.

**COMMON MISTAKE**

*The fourth bright at $4\beta$ and the second dark at $3\beta/2$ are measured from the central maximum, so their separation is $4\beta - 3\beta/2 = 5\beta/2$; never just subtract the order numbers.*

---

**Final answer.** Using the fourth bright to second dark gap, $\beta = 2$ mm, giving $\lambda = 610^{-6}\ m$ as printed (intended visible value **600 nm**).

← *Back to Q187*

## Q188

*(i) A plane light wave propagating from a rarer into a denser medium, is incident at an angle i on the surface separating two media. Using Huygen's principle, draw the refracted wave and hence verify Snell's law of refraction.*
*(ii) In a Young's double slit experiment, the slits are separated by* 0.30 mm *and the screen is kept* 1.5 *m away. The wavelength of light used is* 600 nm. *Calculate the distance between the central bright fringe and the* $4^{\text{th}}$ *dark fringe.*

**[2024 • Set 55-5-1]**

**SOLUTION**

(i) Huygens' construction for refraction from a rarer to a denser medium: draw a horizontal interface XY separating medium 1 (rarer, speed $v_1$, refractive index $n_1$) above from medium 2 (denser, speed $v_2$, refractive index $n_2$) below. A plane wavefront AB in medium 1 strikes the interface, the incident rays making an

angle $i$ with the normal. End $A$ reaches the interface first while $B$ still travels $\mathbf{BC = v_1\, t}$ in medium 1 to reach point $C$ on the interface.

In the same time $t$, the secondary wavelet starting from $A$ travels into the denser medium a distance $\mathbf{AD = v_2\, t}$. With $A$ as centre and radius $\mathbf{AD = v_2\, t}$, draw an arc inside medium 2, and from $C$ draw the tangent CD to this arc. Then CD is the refracted wavefront and the refracted rays make an angle $r$ with the normal.

In the right triangle ABC, the incidence angle gives

$$\sin i = \frac{\text{BC}}{\text{AC}} = \frac{\text{v}_1\, t}{\text{AC}}$$

In the right triangle ADC, the refraction angle gives

$$\sin r = \frac{\text{AD}}{\text{AC}} = \frac{\text{v}_2\, t}{\text{AC}}$$

Dividing the two relations, the common factor $\frac{t}{\text{AC}}$ cancels,

$$\frac{\sin i}{\sin r} = \frac{\text{v}_1}{\text{v}_2}$$

Since the refractive index of a medium is inversely proportional to the speed of light in it, $\frac{\mathbf{v_1}}{\mathbf{v_2}} = \frac{\mathbf{n_2}}{\mathbf{n_1}}$, so

$$\frac{\sin i}{\sin r} = \frac{\text{n}_2}{\text{n}_1} \Rightarrow \text{n}_1 \sin i = \text{n}_2 \sin r$$

This is **Snell's law of refraction**. Because $\mathbf{v_2 < v_1}$ (denser medium), $\boldsymbol{r < i}$, so the refracted ray bends towards the normal, as expected when light enters a denser medium.

(ii) Given $\boldsymbol{d = 0.30\ mm = 3 \times 10^{-4}\ m}$, $\boldsymbol{D = 1.5\ m}$ and $\boldsymbol{\lambda = 600\ nm = 6 \times 10^{-7}\ m}$. First find the fringe width $\boldsymbol{\beta = \frac{\lambda D}{d}}$.

$$\beta = \frac{(6 \times 10^{-7}\ m)(1.5\ m)}{3 \times 10^{-4}\ m}$$

$$\beta = \frac{9 \times 10^{-7}}{3 \times 10^{-4}}\ m = 3 \times 10^{-3}\ m = 3\ \text{mm}$$

The position of the $n$th dark fringe is $y_n^{\text{dark}} = (2n-1)\frac{\beta}{2}$. For the fourth dark fringe, $n = 4$.

$$y_4^{\text{dark}} = (2 \times 4 - 1)\frac{\beta}{2} = \frac{7\beta}{2}$$

$$y_4^{\text{dark}} = \frac{7 \times 3\ \text{mm}}{2} = 10.5\ \text{mm}$$

So the **distance between the central bright fringe and the fourth dark fringe is 10.5 mm**.

---

**Final answer.** Snell's law $n_1 \sin i = n_2 \sin r$ is verified by Huygens' construction, and the fourth dark fringe is **10.5 mm** from the central bright fringe.

← *Back to Q188*

## Q189

*(i) Discuss briefly diffraction of light from a single slit and draw the shape of the diffraction pattern.*
*(ii) An object is placed between the pole and the focus of a concave mirror. Using mirror formula, prove mathematically that it produces a virtual and an enlarged image.*

**[2024 • Set 55-5-1]**

**SOLUTION**

(i) Diffraction of light at a single slit: when a parallel beam of monochromatic light of wavelength λ passes through a narrow slit of width $a$, each point of the slit acts as a source of secondary Huygens' wavelets. These wavelets travel different path lengths to the screen and interfere, producing a diffraction pattern.

At the centre all wavelets arrive in phase, giving a bright central maximum. Dark fringes (minima) occur where the path difference between wavelets from the two edges of the slit is an integral multiple of λ,

$$a\theta = n\lambda,\ n = 1, 2, 3,$$

Between consecutive minima lie weaker secondary maxima, given approximately by $a\theta = (2n+1)\frac{\lambda}{2}$.

Shape of the pattern: a bright, broad central maximum (twice as wide as the others) flanked symmetrically on both sides by alternating dark fringes and secondary maxima whose brightness decreases rapidly with increasing order. An intensity-versus-angle sketch would show a tall central peak at $\theta = 0$ with much smaller side peaks separated by zeros at $\theta = \pm\frac{\lambda}{a}, \pm\frac{2\lambda}{a}$,

(ii) For a concave mirror, the focal length is negative: $f = -|f|$. The object lies between the pole $P$ and the focus $F$, so its distance is less than the focal length, $|u| < |f|$, with $u = -|u|$ (object in front of the mirror).

Apply the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, which rearranges to

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = -\frac{1}{|f|} + \frac{1}{|u|} = \frac{1}{|u|} - \frac{1}{|f|}$$

Since the object lies inside the focus, $|u| < |f|$, so $\frac{1}{|u|} > \frac{1}{|f|}$, which makes the right-hand side positive.

$$\frac{1}{v} = \frac{1}{|u|} - \frac{1}{|f|} > 0 \Rightarrow v > 0$$

A positive $v$ means the image is formed behind the mirror, so the **image is virtual** (and, being behind the mirror, it is also erect).

For the magnification, $m = -\frac{v}{u}$. Comparing magnitudes, since $\frac{1}{v} = \frac{1}{|u|} - \frac{1}{|f|}$ is a small positive number, $v$ is large, in fact $|v| > |u|$.

$$|m| = \frac{|v|}{|u|} > 1$$

Thus the image is larger than the object, so the **image is enlarged**. Hence an object placed between the pole and focus of a concave mirror gives a virtual, erect and magnified image, as in a shaving or make-up mirror.

**Final answer.** Single-slit diffraction gives a broad bright central maximum with weak side maxima (minima at $a\theta = n\lambda$); for an object between pole and focus of a concave mirror, $v>0$ and $|v|>|u|$, so the image is virtual and enlarged.

← *Back to Q189*

## Q190

*(i) (1) Write two points of difference between an interference pattern and a diffraction pattern.*
*(2) Name any two factors on which the fringe width in a Young's double-slit experiment depends.*
*(ii) In Young's double-slit experiment, the two slits are separated by a distance equal to* 100 *times the wavelength of light that passes through the slits. Calculate:*
*(1) the angular separation in radians between the central maximum and the adjacent maximum.*
*(2) the distance between these two maxima on a screen* 50 *cm from the slits.*

**[2023 • Set 55-1-1]**

**SOLUTION**

(i)(1) Two points of difference between an interference pattern and a diffraction pattern:

1. In an interference pattern (from two coherent sources) all the bright fringes are of equal width and equally spaced, whereas in a diffraction pattern (from a single slit) the central maximum is the widest, being twice as wide as the secondary maxima.

2. In an interference pattern all the bright fringes have nearly the same intensity, whereas in a diffraction pattern the central maximum is the brightest and the intensity of the secondary maxima falls off rapidly with order.

(i)(2) The fringe width in Young's double-slit experiment is $\beta = \frac{\lambda D}{d}$, so it depends on (any two of): the wavelength of light $\lambda$ used (directly proportional), the distance $D$ of the screen from the slits (directly proportional), and the separation $d$ between the two slits (inversely proportional).

(ii) Here the slit separation is $d = 100\lambda$ and the screen distance is $D = 50\ \text{cm} = 0.5\ m$.

(1) The angular separation between the central maximum and the adjacent (first-order) maximum is $\theta = \frac{\lambda}{d}$.

$$\theta = \frac{\lambda}{100\lambda} = \frac{1}{100}$$

$$\theta = 10^{-2}\ \text{rad} = 0.01\ \text{rad}$$

So the **angular separation is 0.01 rad**.

(2) The linear distance between these two maxima on the screen is $y = D\theta$ (since $\theta$ is small).

$$y = (0.5\ m)(10^{-2}\ \text{rad})$$

$$y = 5 \times 10^{-3}\ m = 5\ \text{mm}$$

So the **distance between the central and adjacent maxima on the screen is 5 mm**.

---

**Final answer.** The angular separation between the central and adjacent maxima is **0.01 rad**, and their separation on the screen is **5 mm**.

← *Back to Q190*

## Q191

*(i) State Huygen's principle. With the help of a diagram, show how a plane wave is reflected from a surface. Hence verify the law of reflection.*
*(ii) A concave mirror of focal length* 12 cm *forms a three times magnified virtual image of an object. Find the distance of the object from the mirror.*

**[2023 • Set 55-2-1]**

**SOLUTION**

(i) Huygens' principle states that every point on a given wavefront acts as a fresh source of secondary wavelets, which spread out in all forward directions

with the speed of light in that medium. The new position of the wavefront after a time $\Delta t$ is given by the forward tangential envelope (surface of tangency) of all these secondary wavelets.

Diagram (reflection of a plane wave): Draw a horizontal reflecting surface XY. A plane incident wavefront AB travels towards XY so that ray at $A$ touches the surface first while ray at $B$ is still in the medium. Mark the incident wavefront AB making angle of incidence $i$ with the surface, and the reflected wavefront CD making angle of reflection $r$ with the surface. Show the incident rays normal to AB and reflected rays normal to CD, with the normal drawn at the point of incidence.

Let the plane wavefront AB travel with speed $v$ and strike the surface XY. The point $A$ reaches the surface first. In the time $\Delta t$ the light at $B$ takes to reach $C$ on the surface, the secondary wavelet starting from $A$ travels an equal distance back into the same medium.

$\mathbf{BC = v\,\Delta t}$ (distance covered by the incident wavefront from $B$ to $C$)

$\mathbf{AD = v\,\Delta t}$ (radius of the secondary wavelet from $A$)

Therefore $\mathbf{BC = AD}$.

The reflected wavefront is CD, drawn as the tangent from $C$ to the secondary wavelet of radius AD centred at $A$.

In right triangle ABC (right angle at $B$): $\mathbf{\sin i = \dfrac{BC}{AC}}$

In right triangle ADC (right angle at $D$): $\mathbf{\sin r = \dfrac{AD}{AC}}$

Since $\mathbf{BC = AD}$ and AC is common:

$$\sin i = \sin r \Rightarrow i = r$$

Hence the angle of incidence equals the angle of reflection. Also, the incident ray, the reflected ray and the normal all lie in the same plane. This verifies the **laws of reflection**.

(ii) For a concave mirror, $\mathbf{f = -12\ cm}$ (focal length is negative by sign convention). The image is virtual and erect, so the magnification is positive: $\mathbf{m = +3}$.

Using $m = -\frac{v}{u}$:

$$+3 = -\frac{v}{u} \Rightarrow v = -3u$$

Apply the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$:

$$\frac{1}{-3u} + \frac{1}{u} = \frac{1}{-12}$$

$$\frac{-1+3}{3u} = \frac{1}{-12}$$

$$\frac{2}{3u} = \frac{-1}{12}$$

$$2 \times 12 = -3u$$

$$u = -8 \text{ cm}$$

The object is placed at a distance of **8 cm** from the mirror (the negative sign confirms it is in front of the mirror).

**TIP**

For a virtual, magnified image in a mirror, take magnification positive ($m$=+3) and apply $m = -v/u$ before the mirror formula.

---

**Final answer.** Laws of reflection verified by Huygens' principle ($i = r$); the object is 8 cm in front of the concave mirror.

← *Back to Q191*

## Q192

*The principle of superposition is used to understand the phenomenon of interference of light waves. The principle states that at a particular point, the resultant displacement produced by a number of waves is the vector sum of the displacements produced by each wave. Light waves from two coherent sources produce interference pattern. Thomas Young devised a way to obtain two coherent sources using two identical*

*pinholes ($S_1$ and $S_2$) illuminated by a single monochromatic pinhole source S. Using these sources in his experiment known as Young's double slit experiment, Young studied the interference pattern. The pattern consists of alternate bright and dark fringes. The distance between two successive bright or dark fringes depends on the distance between $S_1$ and $S_2$, the distance of the screen from the plane of $S_1$ $S_2$ and the wavelength of light used.*

***I.*** *Consider the following waves:*

*(i)* $y_1 = a \sin \omega t$

*(ii)* $y_2 = a \sin 2\omega t$

*(iii)* $y_3 = a \sin (\omega t + \phi)$

*(iv)* $y_4 = a \sin \left(4\omega t + \frac{\pi}{2}\right)$

*Which pair of the waves coming from two sources $S_1$ and $S_2$ will produce interference?*

*(A) (i) and (ii) (B) (ii) and (iii) (C) (iii) and (iv) (D) (iv) and (i)*

***II.*** *Two light waves of the same intensity $I_0$ each, having a path difference of $\lambda/4$, emanating from two coherent sources, meet at a point. The resultant intensity at the point will be*

*(A) Zero (B)* $I_0$ *(C)* $2I_0$ *(D)* $4I_0$

***III.*** *Vandana performs Young's double slit experiment by using orange, green and red lights successively. If the fringe widths measured in the three cases are $\omega_1$, $\omega_2$ and $\omega_3$ respectively, then which of the following is correct?*

*(A)* $\omega_2 > \omega_1 > \omega_3$ *(B)* $\omega_1 > \omega_2 > \omega_3$ *(C)* $\omega_2 > \omega_3 > \omega_1$ *(D)* $\omega_3 > \omega_2 > \omega_1$

***IV.*** *In a Young's double slit experiment, the slit separation is* 0.8 mm *and the interference pattern is obtained on a screen kept* 50 cm *from the plane of the slits $S_1$ and $S_2$. If the first bright fringe is formed* 0.4 mm *from the central maximum, the wavelength of light used is*

*(A)* 480 nm *(B)* 560 nm *(C)* 640 nm *(D)* 680 nm

***V.*** *Consider the effect on the angular separation of the fringes in a Young's double slit experiment due to the following operations:*

*(i) the screen is moved away from the plane of the slits,*

*(ii) the separation between the two slits is increased till fringes are observed.*

*Which of the following options is correct?*

*(A) It remains constant in both cases.*

*(B) It decreases in both cases.*

*(C) It remains constant in (i) but decreases in (ii).*

*(D) It decreases in (i) but remains constant in (ii).*

**[2022 • Set 55-1-1]**

**SOLUTION**

This case-based question has five parts.

I. Two waves can produce a sustained (observable) interference pattern only if they have the same frequency, so that their phase difference stays constant in time. Comparing the angular frequencies: $y_1$ has $\omega$, $y_2$ has $2\omega$, $y_3$ has $\omega$, and $y_4$ has $4\omega$. The only equal-frequency pair is $y_1$ and $y_3$ (both $\omega$), offered as option (A), the pair (i) and (iii). The other pairs mix different frequencies, so their phase difference changes continuously and no stable pattern forms. Hence the correct option is **(A)**.

II. For two coherent waves the resultant intensity is $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\ \phi$.

Phase difference for a path difference $\frac{\lambda}{4}$: $\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2}$

With $I_1 = I_2 = I_0$: $I = I_0 + I_0 + 2\sqrt{I_0 I_0}\cos\frac{\pi}{2}$

$$I = 2I_0 + 2I_0(0) = 2I_0$$

Hence the correct option is **(C)** $2I_0$.

III. Fringe width $\beta = \frac{\lambda D}{d}$, so $\beta \propto \lambda$. The colour with the longest wavelength gives the widest fringe. Red light has the longest wavelength of the three colours used ($\lambda_{red} > \lambda_{orange} > \lambda_{green}$). With $\omega_3$ for red, $\omega_3$ is therefore the largest fringe width, so the option that lists $\omega_3$ first (the widest) is taken as the answer. The correct option as per the answer key is **(D)** $\omega_3 > \omega_2 > \omega_1$, with red giving the widest fringes.

IV. For the first bright fringe, $y = \frac{D\lambda}{d}$, so $\lambda = \frac{y\,d}{D}$.

Given $y = 0.4$ mm $= 0.4\times 10^{-3}$ $m$, $d = 0.8$ mm $= 0.8\times 10^{-3}$ $m$, $D = 50$ cm $= 0.5$ $m$.

$$\lambda = \frac{(0.4\times 10^{-3})(0.8\times 10^{-3})}{0.5}$$

$$\lambda = \frac{0.32\times 10^{-6}}{0.5} = 6.4\times 10^{-7}\ m = 640\ \text{nm}$$

Hence the correct option is **(C)** 640 nm.

V. The angular separation of the fringes is $\theta = \frac{\beta}{D} = \frac{\lambda}{d}$, which does not depend on $D$. (i) Moving the screen away changes $D$ but not $\theta = \frac{\lambda}{d}$, so the angular separation stays constant. (ii) Increasing the slit separation $d$ makes $\theta = \frac{\lambda}{d}$ smaller, so the angular separation decreases. Hence the correct option is **(C)**: it remains constant in (i) but decreases in (ii).

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**Final answer.** I: **(A)**; II: **(C) $2I_0$**; III: **(D)**; IV: **(C) 640 nm**; V: **(C)**.

← *Back to Q192*

## Q193

*The British physicist Thomas Young explained the interference of light using the principle of superposition of waves. He observed the interference pattern on the screen, in his experimental set-up, known now as Young's double slit experiment. The two slits $S_1$ and $S_2$ were illuminated by light from a slit S. The interference pattern consists of dark and bright bands of light. Such bands are called fringes. The distance between two consecutive bright and dark fringes is called fringe width.*

***(a)** If the screen is moved closer to the plane of slits $S_1$ and $S_2$, then the fringe width:*

*(i) will decrease, but the intensity of bright fringe remains the same.*

*(ii) will increase, but the intensity of bright fringe decreases.*

*(iii) will decrease, but the intensity of bright fringe increases.*

*(iv) and the intensity both remain the same.*

***(b)** What will happen to the pattern on the screen, when the two slits $S_1$ and $S_2$ are replaced by two independent but identical sources?*

*(i) The intensity of pattern will increase*

*(ii) The intensity of pattern will decrease*

*(iii) The number of fringes will become double*

*(iv) No pattern will be observed on the screen*

***(c)** Two sources of light are said to be coherent, when both emit light waves of:*

*(i) same amplitude and have a varying phase difference.*

*(ii) same wavelength and a constant phase difference.*

*(iii) different wavelengths and same intensity.*

*(iv) different wavelengths and a constant phase difference.*

***(d)*** *The fringe width in a Young's double slit experiment is* $\beta$. *If the whole set-up is immersed in a liquid of refractive index* '$\mu$', *then the new fringe width will be:*

*(i)* $\beta$

*(ii)* $\mu\beta$

*(iii)* $\frac{\beta}{\mu}$

*(iv)* $\frac{\beta}{2}$

***(e)*** *The total path difference between two waves meeting at points* $P_1$ *and* $P_2$ *on the screen are* $(\frac{3\lambda}{2})$ *and* $(2\lambda)$ *respectively. Then:*

*(i) bright fringes are formed at both points.*

*(ii) dark fringes are formed at both points.*

*(iii) a bright fringe is formed at* $P_1$ *and a dark fringe is formed at* $P_2$.

*(iv) a bright fringe is formed at* $P_2$ *and a dark fringe is formed at* $P_1$.

**[2022 • Set 55-2-1]**

**SOLUTION**

This case-based question has five parts. Each is solved using the fringe-width relation $\beta = \frac{\lambda D}{d}$ and the conditions for bright and dark fringes.

(a) Fringe width $\beta = \frac{\lambda D}{d}$, so $\beta \propto D$. Moving the screen closer reduces $D$, hence $\beta$ decreases. The intensity of a bright fringe depends only on the amplitudes from the two slits, which are unchanged, so the bright-fringe intensity stays the same. The correct option is **(i)**: the fringe width will decrease, but the intensity of the bright fringe remains the same.

(b) Two independent sources, even if identical, cannot maintain a constant phase difference; their phases vary randomly and rapidly. Without a fixed phase relationship there is no stable interference, so no fringe pattern is formed. The correct option is **(iv)**: no pattern will be observed on the screen.

(c) Two sources are coherent when they emit waves of the same wavelength (same frequency) and maintain a constant phase difference with time. The correct option is **(ii)**: same wavelength and a constant phase difference.

(d) In a medium of refractive index $\mu$, the wavelength becomes $\lambda' = \frac{\lambda}{\mu}$.

New fringe width $\beta' = \frac{\lambda' D}{d} = \frac{\lambda D}{\mu d} = \frac{\beta}{\mu}$

Hence the correct option is **(iii)** $\frac{\beta}{\mu}$.

(e) A bright fringe forms when the path difference is an integral multiple of $\lambda$ (i.e. $n\lambda$); a dark fringe forms when it is an odd multiple of $\frac{\lambda}{2}$ (i.e. $(2n-1)\frac{\lambda}{2}$). At $P_1$ the path difference is $\frac{3\lambda}{2} = 3\times\frac{\lambda}{2}$ (odd multiple of $\lambda/2$), so a dark fringe forms. At $P_2$ the path difference is $2\lambda$ (integral multiple of $\lambda$), so a bright fringe forms. The correct option is **(iv)**: a bright fringe is formed at $P_2$ and a dark fringe at $P_1$.

---

**Final answer.** (a) (i); (b) (iv); (c) (ii); (d) (iii); (e) (iv).

← *Back to Q193*

## Q194

*(a) What is meant by plane polarised light? An unpolarised light is incident at an angle $\theta$ on the surface of glass of refractive index $\mu$. If the reflected and refracted rays are perpendicular to each other, then obtain the relationship between $\mu$ and $\theta$.*
*(b) Two polaroids $P_1$ and $P_2$ are placed in a crossed position. Unpolarised light of intensity $I_0$ is incident on $P_1$. If $P_2$ is rotated through an angle $\theta$ about the direction of propagation of light, keeping $P_1$ fixed, plot the graph of intensity of light for $0°\le\theta\le 360°$ which is (i) transmitted by $P_1$, and (ii) transmitted by $P_2$.*
***OR***
*(a) Briefly describe the Young's double slit experiment of interference of light. Derive the expression for fringe width in the pattern.*
*(b) Monochromatic light of wavelength 588 nm is incident from air to water interface. Find the wavelength and speed of the refracted light. The refractive index of water is $\frac{4}{3}$.*

**[2020 • Set 55-3-1]**

**SOLUTION**

(a) Plane (linearly) polarised light is light in which the electric field vector $\vec{E}$ oscillates only in one fixed plane containing the direction of propagation. In

ordinary (unpolarised) light $\vec{E}$ vibrates randomly in all directions perpendicular to the direction of propagation.

When unpolarised light strikes a glass surface at an angle θ and the reflected and refracted rays are mutually perpendicular, the reflected light is completely plane polarised. This is Brewster's law. Let the angle of incidence be θ (Brewster's angle) and the angle of refraction be $r$.

Since the reflected and refracted rays are perpendicular: $\boldsymbol{\theta + 90° + r = 180°}$

$$\Rightarrow r = 90° - \theta$$

By Snell's law (light going from air to glass): $\boldsymbol{\mu = \frac{\theta}{\sin r}}$

$$\mu = \frac{\theta}{\sin(90° - \theta)} = \frac{\theta}{\theta}$$

$$\mu = \theta$$

This is the required relation (Brewster's law).

(b) Unpolarised light of intensity $I_0$ on $P_1$ gives transmitted intensity $\frac{I_0}{2}$, independent of any rotation of $P_2$.

(i) Intensity transmitted by $P_1$: a constant horizontal straight line at $\boldsymbol{I = \frac{I_0}{2}}$ for all θ from 0° to 360°.

(ii) The polaroids start in the crossed position, so the angle between the pass axes is 90°. When $P_2$ is rotated by θ from this crossed position, the angle between the axes is (90° − θ), and by Malus's law:

$$I_2 = \frac{I_0}{2}\cos^2(90° - \theta) = \frac{I_0}{2}\sin^2\theta$$

Graph of $I_2$ versus θ (0° to 360°): a $\sin^2\theta$ curve. It is zero at $\boldsymbol{\theta = 0°, 180°, 360°}$ (crossed positions) and rises to a maximum of $\frac{I_0}{2}$ at $\boldsymbol{\theta = 90°}$ and 270° (parallel positions). The curve is a series of identical humps lying between 0 and $\frac{I_0}{2}$.

OR

(a) Young's double slit experiment: A narrow slit $S$ illuminated by monochromatic light acts as a source. Light from $S$ falls on two close, parallel

slits $S_1$ and $S_2$ a small distance $d$ apart, which act as two coherent sources. The waves from $S_1$ and $S_2$ superpose on a screen placed a distance $D$ away, producing alternate bright and dark fringes.

Diagram: Source slit $S$ on the left, then the two slits $S_1$ and $S_2$ (separation $d$) on a barrier, and a screen at distance $D$ on the right. Mark the central point $O$ on the screen opposite the midpoint of $S_1S_2$, and a general point $P$ at distance $y$ from $O$.

At a point $P$ at distance $y$ from the centre $O$, the path difference is:

$$\Delta = S_2P - S_1P = \frac{y\,d}{D}$$

For a bright fringe: $\Delta = n\lambda \Rightarrow \frac{y_n\,d}{D} = n\lambda$

$$y_n = \frac{n\lambda\,D}{d}$$

The fringe width is the separation between two consecutive bright fringes:

$$\beta = y_{n+1} - y_n = \frac{(n+1)\lambda\,D}{d} - \frac{n\lambda\,D}{d}$$

$$\beta = \frac{\lambda\,D}{d}$$

(b) Light goes from air into water, refractive index $\mu = \frac{4}{3}$, wavelength in air $\lambda = 588$ nm.

Wavelength in water: $\lambda_w = \frac{\lambda}{\mu} = \frac{588}{4/3} = 588\times\frac{3}{4} = 441$ nm

Speed in water: $v = \frac{c}{\mu} = \frac{3\times 10^8}{4/3} = 3\times 10^8 \times \frac{3}{4} = 2.25\times 10^8\ m\ s^{-1}$

Hence the refracted light has wavelength **441 nm** and speed $2.25\times 10^8\ m\ s^{-1}$.

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**Final answer.** Brewster's law $\mu = \theta$; OR fringe width $\beta = \lambda D/d$, refracted light: $\lambda_w = 441$ nm, $v = 2.2510^8\ m\ s^{-1}$.

← *Back to Q194*

## Q195

*Describe any two characteristic features which distinguish between interference and diffraction phenomena. Derive the expression for the intensity at a point of the interference pattern in Young's double slit experiment.*

**[2019 • Set 55-1-1]**

**SOLUTION**

Two characteristic features distinguishing interference from diffraction:

1. Source of the waves: Interference results from the superposition of waves coming from two (or more) separate coherent sources, such as the two slits in Young's experiment. Diffraction results from the superposition of secondary wavelets coming from different parts of the same wavefront passing through a single aperture.

2. Width and intensity of fringes: In interference, all bright fringes are of equal width and (ideally) equal intensity. In diffraction, the central maximum is the brightest and is twice as wide as the secondary maxima, and the intensity of the secondary maxima falls off rapidly with order.

Expression for intensity in Young's double slit experiment:

Let the waves arriving at a point $P$ on the screen from the two slits be:

$$y_1 = a\omega\, t$$

$$y_2 = a\cos(\omega\, t + \phi)$$

where $\phi$ is the constant phase difference between them, related to the path difference $\Delta = \frac{yd}{D}$ by:

$$\phi = \frac{2\pi}{\lambda}\,\Delta$$

By the principle of superposition, the resultant displacement is:

$$y = y_1 + y_2 = a\omega\, t + a\cos(\omega\, t + \phi)$$

Using $\cos X + \cos Y = 2\cos(\frac{X+Y}{2})\cos(\frac{X-Y}{2})$:

$$y = 2a\cos\left(\frac{\Phi}{2}\right)\cos\left(\omega t + \frac{\Phi}{2}\right)$$

The resultant amplitude is therefore:

$$A = 2a\cos\left(\frac{\Phi}{2}\right)$$

Since intensity is proportional to the square of the amplitude, $I \propto A^2$:

$$I = 4a^2\cos^2\left(\frac{\Phi}{2}\right)$$

Writing $I_0 = 4a^2$ as the maximum intensity at the centre:

$$I = I_0\cos^2\left(\frac{\Phi}{2}\right)$$

For constructive interference (maximum, bright fringe), $\phi = 2n\pi$, giving $I = 4a^2 = I_{max}$.

For destructive interference (minimum, dark fringe), $\phi = (2n-1)\pi$, giving $I = 0$.

This is the required expression for the **intensity at a point** of the interference pattern.

---

**Final answer.** $I = 4a^2\cos^2(\phi/2)$, maximum $4a^2$ when $\phi = 2n\pi$ and zero when $\phi = (2n-1)\pi$.

← *Back to Q195*

## Q196

*In the diffraction due to a single slit experiment, the aperture of the slit is* 3 mm. *If monochromatic light of wavelength* 620 nm *is incident normally on the slit, calculate the separation between the first order minima and the* 3rd *order maxima on one side of the screen. The distance between the slit and the screen is* 1.5 *m.*

**[2019 • Set 55-1-1]**

**SOLUTION**

For diffraction at a single slit of width $a$, the position of minima and the (approximate) position of secondary maxima on the screen are found using the

standard conditions.

Given data: slit width $a = 3\ \text{mm} = 3\times 10^{-3}\ m$, wavelength $\lambda = 620\ \text{nm} = 6.2\times 10^{-7}\ m$, slit-to-screen distance $D = 1.5\ m$.

Condition for the n$^{th}$ minimum: $a\theta = n\lambda$, so the distance of the n$^{th}$ minimum from the centre is $y_n = \dfrac{nD\lambda}{a}$.

Position of the first order minimum ($n = 1$):

$$y_{min} = \frac{1\times D\lambda}{a} = \frac{1.5\times 6.2\times 10^{-7}}{3\times 10^{-3}}$$

$$y_{min} = \frac{9.3\times 10^{-7}}{3\times 10^{-3}} = 3.1\times 10^{-4}\ m$$

Condition for the n$^{th}$ secondary maximum: $a\theta = (n+\frac{1}{2})\lambda$, so the distance of the n$^{th}$ maximum from the centre is $y_n = \dfrac{(n+\frac{1}{2})D\lambda}{a}$.

Position of the third order maximum ($n = 3$), where $(3 + \frac{1}{2}) = \frac{7}{2}$:

$$y_{max} = \frac{7}{2}\times\frac{D\lambda}{a} = \frac{7}{2}\times 3.1\times 10^{-4}$$

$$y_{max} = 3.5\times 3.1\times 10^{-4} = 1.085\times 10^{-3}\ m$$

Separation between the first minimum and the third maximum on the same side:

$$\Delta y = y_{max} - y_{min} = 1.085\times 10^{-3} - 3.1\times 10^{-4}$$

$$\Delta y = 1.085\times 10^{-3} - 0.31\times 10^{-3} = 0.775\times 10^{-3}\ m$$

The required separation is **0.775 mm** (i.e. $7.75\times 10^{-4}\ m$).

**COMMON MISTAKE**

*For single-slit diffraction, minima are at $a\theta = n\lambda$ but secondary maxima are at $a\theta = (n+12)\lambda$; mixing the two conditions is a common slip.*

**Final answer.** The separation between the first minimum and the third maximum is **0.775 mm**.

← *Back to Q196*

## Q197

*Explain, with the help of a diagram, how plane polarized light can be produced by scattering of light from the Sun.*
*Two polaroids* $P_1$ *and* $P_2$ *are placed with their pass axes perpendicular to each other. Unpolarised light of intensity I is incident on* $P_1$*. A third polaroid* $P_3$ *is kept between* $P_1$ *and* $P_2$ *such that its pass axis makes an angle of* $45°$ *with that of* $P_1$*. Calculate the intensity of light transmitted through* $P_1$*,* $P_2$ *and* $P_3$*.*
***OR***
*(a) Why cannot the phenomenon of interference be observed by illuminating two pin holes with two sodium lamps?*
*(b) Two monochromatic waves having displacements* $y_1 = a\omega t$ *and* $y_2 = a\cos(\omega t + \phi)$ *from two coherent sources interfere to produce an interference pattern. Derive the expression for the resultant intensity and obtain the conditions for constructive and destructive interference.*
*(c) Two wavelengths of sodium light of* $590\ \text{nm}$ *and* $596\ \text{nm}$ *are used in turn to study the diffraction taking place at a single slit of aperture* $2 \times 10^{-6}$ *m. If the distance between the slit and the screen is* $1.5$ *m, calculate the separation between the positions of the second maxima of diffraction pattern obtained in the two cases.*

**[2019 • Set 55-2-1]**

### SOLUTION

Production of plane polarised light by scattering: Sunlight (unpolarised) falling on the molecules of the atmosphere is scattered in all directions. The electric field of the incident light sets the electrons of a molecule into vibration only in directions perpendicular to the direction of propagation. Light scattered at 90° to the incident beam is found to be completely plane polarised, because only the component of vibration perpendicular to the plane containing the incident and scattered rays can radiate towards the observer.

Diagram: Show sunlight travelling horizontally and striking a molecule (a dot) at the centre. The incident unpolarised light is shown with vibrations in all transverse directions (dots and double arrows). The light scattered downward at 90° to the incident direction is shown with only one direction of vibration (double arrow perpendicular to the scattering plane), marked as plane polarised, while the forward light remains unpolarised.

Three polaroids problem: Unpolarised light of intensity $I$ falls on $P_1$. $P_2$ is crossed with $P_1$ (90° apart). $P_3$ is placed between them with its axis at 45° to $P_1$ (hence 45° to $P_2$ as well).

After $P_1$ (unpolarised light loses half its intensity):

$$I_1 = \frac{I}{2}$$

After $P_3$, which is at 45° to $P_1$ (Malus's law):

$$I_3 = I_1 \cos^2 45° = \frac{I}{2} \times \left(\frac{1}{2}\right)^2 = \frac{I}{2} \times \frac{1}{2} = \frac{I}{4}$$

After $P_2$, which is at 45° to $P_3$ (Malus's law):

$$I_2 = I_3 \cos^2 45° = \frac{I}{4} \times \frac{1}{2} = \frac{I}{8}$$

So the intensities are $\frac{I}{2}$ after $P_1$, $\frac{I}{4}$ after $P_3$, and $\boldsymbol{\frac{I}{8}}$ after $P_2$. (Note: had $P_3$ been absent, the crossed $P_1$ and $P_2$ alone would give zero.)

OR

(a) Two separate sodium lamps cannot produce a sustained interference pattern because they are independent sources. The light from each lamp is emitted by a huge number of atoms whose emissions are random and very short-lived (about $10^{-9}$ $s$). Hence the phase difference between the two lamps changes randomly and extremely rapidly, so the pattern shifts faster than the eye can follow and averages out to uniform illumination. The sources are not coherent.

(b) Let $\mathbf{y_1 = a\omega\, t}$ and $\mathbf{y_2 = a\cos(\omega\, t + \phi)}$. By superposition:

$$y = y_1 + y_2 = a\omega\, t + a\cos(\omega\, t + \phi)$$

$$y = 2a\cos\left(\frac{\Phi}{2}\right)\cos\left(\omega t + \frac{\Phi}{2}\right)$$

Resultant amplitude $A = 2a\cos\left(\frac{\Phi}{2}\right)$, and since $I \propto A^2$:

$$I = 4a^2\cos^2\left(\frac{\Phi}{2}\right)$$

For constructive interference: $\phi = 2n\pi$ (i.e. path difference = $n\lambda$), giving $I_{max} = 4a^2$.

For destructive interference: $\phi = (2n-1)\pi$ (i.e. path difference = $(2n-1)\frac{\lambda}{2}$), giving $I_{min} = 0$.

(c) For single-slit diffraction the second maximum lies (approximately) at $a\theta = \frac{5\lambda}{2}$, so its distance from the centre is $y = \frac{5D\lambda}{2a}$.

Separation between the second maxima for the two wavelengths:

$$\Delta y = \frac{5D}{2a}(\lambda_2 - \lambda_1)$$

Given $\lambda_2 - \lambda_1 = 596 - 590 = 6\ \text{nm} = 6\times 10^{-9}\ m$, $D = 1.5\ m$, $a = 2\times 10^{-6}\ m$:

$$\Delta y = \frac{5\times 1.5}{2\times 2\times 10^{-6}} \times 6\times 10^{-9}$$

$$\Delta y = \frac{7.5}{4\times 10^{-6}} \times 6\times 10^{-9}$$

$$\Delta y = 1.875\times 10^{6}\times 6\times 10^{-9} = 1.125\times 10^{-2}\ m$$

The separation between the two second-order maxima is $1.125\times 10^{-2}\ m = 11.25$ mm.

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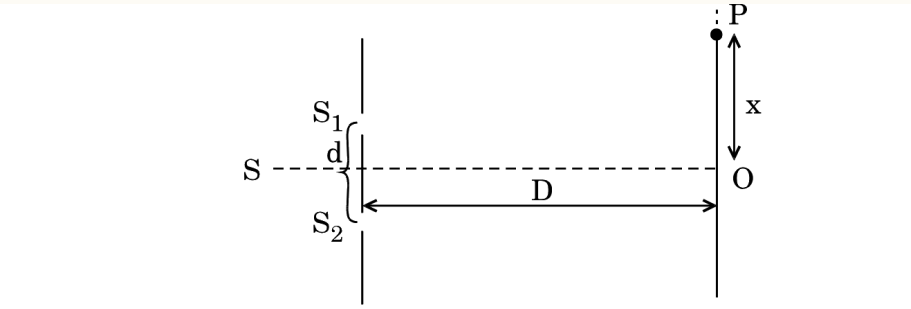
**Final answer.** Intensities $I/2$, $I/4$, $I/8$ after $P_1$, $P_3$, $P_2$; OR resultant $I = 4a^2\cos^2(\phi/2)$ and second-maxima separation $= 11.25$ mm.

← Back to Q197

## Q198

*(a) Can the interference pattern be produced by two independent monochromatic sources of light? Explain.*

*(b) The intensity at the central maximum (O) in a Young's double slit experimental set-up shown in the figure is* $I_0$. *If the distance* OP *equals one-third of the fringe width of the pattern, show that the intensity at point P, would equal* $\frac{I_0}{4}$.




*(c) In Young's double slit experiment, the slits are separated by* 0.5 mm *and screen is placed* 1.0 *m away from the slit. It is found that the* $5^{th}$ *bright fringe is at a distance of* 4.13 mm *from the* $2^{nd}$ *dark fringe. Find the wavelength of light used.*

***OR***

*(a) Derive the relation* $a\theta = \lambda$ *for the first minimum of the diffraction pattern produced due to a single slit of width 'a' using light of wavelength* λ.

*(b) State with reason, how the linear width of central maximum will be affected if (i) monochromatic yellow light is replaced with red light, and (ii) distance between the slit and the screen is increased.*

*(c) Using the monochromatic light of same wavelength in the experimental set-up of the diffraction pattern as well as in the interference pattern where the slit separation is* 1 mm, 10 *interference fringes are found to be within the central maximum of the diffraction pattern. Determine the width of the single slit, if the screen is kept at the same distance from the slit in the two cases.*

**[2019 • Set 55-3-1]**

### SOLUTION

(a) No, two independent monochromatic sources cannot produce a sustained interference pattern. Independent sources emit light through random, very short-lived atomic transitions, so the phase difference between them keeps changing rapidly and randomly. Without a constant phase difference the

sources are not coherent, and the pattern shifts so fast that the eye sees only uniform illumination.

(b) At point $P$, the path difference is $\Delta = d\theta \approx \frac{yd}{D}$, where $y = OP$.

Given $OP = \frac{\beta}{3}$ and $\beta = \frac{\lambda D}{d}$:

$$\Delta = \frac{d}{D} \times \frac{\beta}{3} = \frac{d}{D} \times \frac{1}{3} \times \frac{\lambda D}{d} = \frac{\lambda}{3}$$

Phase difference: $\phi = \frac{2\pi}{\lambda} \Delta = \frac{2\pi}{\lambda} \times \frac{\lambda}{3} = \frac{2\pi}{3}$

Using $I = I_0 \cos^2(\frac{\phi}{2})$:

$$I = I_0 \cos^2(\frac{1}{2} \times \frac{2\pi}{3}) = I_0 \cos^2(\frac{\pi}{3})$$

$$I = I_0 (\frac{1}{2})^2 = \frac{I_0}{4}$$

Hence the intensity at $P$ is $\frac{I_0}{4}$, as required.

(c) The $5^{th}$ bright fringe is at $y_B = 5\beta$ from the centre. The $2^{nd}$ dark fringe is at $y_D = (2 - \frac{1}{2})\beta = \frac{3}{2}\beta = 1.5\beta$.

Distance between them: $5\beta - 1.5\beta = 3.5\beta = 4.13$ mm

$$\beta = \frac{4.13}{3.5} = 1.18 \text{ mm} = 1.18 \times 10^{-3}\ m$$

Using $\beta = \frac{\lambda D}{d}$, so $\lambda = \frac{\beta d}{D}$, with $d = 0.5$ mm $= 0.5 \times 10^{-3}\ m$ and $D = 1.0\ m$:

$$\lambda = \frac{(1.18 \times 10^{-3})(0.5 \times 10^{-3})}{1.0}$$

$$\lambda = 0.59 \times 10^{-6}\ m = 5.9 \times 10^{-7}\ m = 590 \text{ nm}$$

The wavelength of light used is **590 nm**.

OR

(a) Derivation of $a\theta = \lambda$ for the first minimum: Consider a single slit of width $a$ illuminated normally by light of wavelength $\lambda$. To find the direction of the first

minimum, imagine the slit divided into two equal halves, each of width $\frac{a}{2}$. The path difference between a wavelet from the top edge and one from the bottom edge of the slit, in the direction $\theta$, is $a\theta$.

For the first minimum, each wavelet from the upper half cancels the corresponding wavelet from the lower half. This happens when the path difference between corresponding points (separated by $\frac{a}{2}$) is $\frac{\lambda}{2}$:

$$\frac{a}{2}\theta = \frac{\lambda}{2}$$

$$\Rightarrow a\theta = \lambda$$

This is the condition for the first minimum of the single-slit diffraction pattern.

(b) The linear width of the central maximum is $\boldsymbol{W = \frac{2\lambda D}{a}}$, so $\boldsymbol{W \propto \lambda}$ and $\boldsymbol{W \propto D}$.

(i) When yellow light is replaced by red light, $\lambda$ increases (red has a longer wavelength). Since $\boldsymbol{W \propto \lambda}$, the linear width of the central maximum increases.

(ii) When the slit-to-screen distance $D$ is increased, $\boldsymbol{W \propto D}$ increases, so the central maximum becomes wider.

(c) Width of the central maximum (diffraction): $\boldsymbol{W = \frac{2\lambda D}{a}}$. Fringe width (interference): $\boldsymbol{\beta = \frac{\lambda D}{d}}$.

10 interference fringes fit inside the central maximum, so:

$$W = 10\beta$$

$$\frac{2\lambda D}{a} = 10 \times \frac{\lambda D}{d}$$

$$\frac{2}{a} = \frac{10}{d}$$

$$a = \frac{2d}{10} = \frac{d}{5}$$

With $\boldsymbol{d = 1\ mm}$: $\boldsymbol{a = \frac{1}{5} = 0.2\ mm}$

The width of the single slit is **0.2 mm** (i.e. $2\times 10^{-4}\ m$).

**Final answer.** Intensity at $P=I_0/4$; wavelength $=590$ **nm**; OR first minimum at $a\theta=\lambda$ and slit width $=0.2$ **mm**.

← *Back to Q198*

## Q199

*(a) Define a wavefront. Using Huygens' principle, verify the laws of reflection at a plane surface.*
*(b) In a single slit diffraction experiment, the width of the slit is made double the original width. How does this affect the size and intensity of the central diffraction band ? Explain.*
*(c) When a tiny circular obstacle is placed in the path of light from a distant source, a bright spot is seen at the centre of the obstacle. Explain why.*

**[2018]**

**SOLUTION**

(a) A wavefront is the locus of all points of a medium that are vibrating in the same phase at a given instant. It is a surface drawn perpendicular to the direction of propagation; a wavefront from a point source is spherical, while one from a very distant source is plane.

Verification of the law of reflection by Huygens' principle:

Diagram: Draw a plane reflecting surface XY. A plane incident wavefront AB approaches at angle of incidence $i$ (the ray at $A$ reaching the surface first). After reflection, the reflected wavefront CD leaves at angle of reflection $r$. Show the normal at the point of incidence and label both angles.

Let the wavefront travel with speed $v$. The point $A$ touches the surface first while the part at $B$ still has to travel to $C$ on the surface. In the time $\Delta t$ taken for $B$ to reach $C$, the secondary wavelet from $A$ travels an equal distance back into the same medium.

$\mathbf{BC} = v\,\Delta t$ and $\mathbf{AD} = v\,\Delta t$, so $\mathbf{BC} = \mathbf{AD}$.

In right triangle ABC: $\sin i = \dfrac{BC}{AC}$

In right triangle ADC: $\sin r = \frac{AD}{AC}$

Since **BC = AD** and AC is common to both:

$$\sin i = \sin r \Rightarrow i = r$$

Thus the angle of incidence equals the angle of reflection, and the incident ray, reflected ray and normal lie in the same plane. The **law of reflection** is verified.

(b) The linear width of the central maximum is $W = \frac{2\lambda D}{a}$, so $W \propto \frac{1}{a}$. When the slit width is doubled (from $a$ to 2a), the width of the central band becomes:

$$W' = \frac{2\lambda D}{2a} = \frac{1}{2} W$$

So the central diffraction band becomes half as wide (narrower). At the same time, doubling the slit width lets through twice the light energy, now concentrated over half the width, so the peak intensity of the central band increases. (The peak intensity is proportional to $a^2$, hence it becomes about four times greater.) In short: the central band gets narrower but brighter.

(c) When a tiny circular obstacle is placed in the path of light from a distant source, light waves diffract (bend) around its circular edge. By Huygens' principle, the secondary wavelets from all points around the rim of the obstacle travel equal distances to the central point of the geometrical shadow. They therefore arrive in the same phase and interfere constructively, producing a **bright spot at the centre of the shadow** (the Poisson/Arago spot). This is direct evidence of the wave nature of light.

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**Final answer.** Wavefront defined and ***i=r*** verified; doubling slit width halves the central band but makes it brighter; the central bright spot (Poisson spot) is due to diffraction around the obstacle's edge.

*← Back to Q199*

## Q200

*(a) Distinguish between unpolarized light and linearly polarized light. How does one get linearly polarised light with the help of a polaroid?*

*(b) A narrow beam of unpolarised light of intensity* $I_0$ *is incident on a polaroid* $P_1$. *The light transmitted by it is then incident on a second polaroid* $P_2$ *with its pass axis making angle of* $60°$ *relative to the pass axis of* $P_1$. *Find the intensity of the light transmitted by* $P_2$.

***OR***

*(a) Explain two features to distinguish between the interference pattern in Young's double slit experiment with the diffraction pattern obtained due to a single slit.*

*(b) A monochromatic light of wavelength* $500\ \text{nm}$ *is incident normally on a single slit of width* $0.2\ \text{mm}$ *to produce a diffraction pattern. Find the angular width of the central maximum obtained on the screen.*

*Estimate the number of fringes obtained in Young's double slit experiment with fringe width* $0.5\ \text{mm}$, *which can be accommodated within the region of total angular spread of the central maximum due to single slit.*

**[2017]**

**SOLUTION**

(a) Distinction: In unpolarised light the electric field vector $\vec{E}$ vibrates randomly in all directions perpendicular to the direction of propagation, with no preferred direction. In linearly (plane) polarised light $\vec{E}$ vibrates only in one fixed direction perpendicular to the direction of propagation.

Obtaining linearly polarised light with a polaroid: A polaroid is a sheet with long-chain molecules aligned in a particular direction, which defines its pass axis. When unpolarised light passes through it, only the component of $\vec{E}$ parallel to the pass axis is transmitted and the perpendicular component is absorbed. The emerging light vibrates in a single plane and is therefore linearly polarised. Its intensity is half that of the incident unpolarised light.

(b) Unpolarised light of intensity $I_0$ on $P_1$ gives:

$$I_1 = \frac{I_0}{2}$$

$P_2$ has its pass axis at $60°$ to that of $P_1$. By Malus's law:

$$I_2 = I_1 \cos^2 60° = \frac{I_0}{2} \times \left(\frac{1}{2}\right)^2$$

$$I_2 = \frac{I_0}{2} \times \frac{1}{4} = \frac{I_0}{8}$$

The intensity transmitted by $P_2$ is $\frac{\mathbf{I_0}}{\mathbf{8}}$.

OR

(a) Two features distinguishing interference (YDSE) from single-slit diffraction:

1. In interference, the bright fringes are all of equal width and equal intensity. In diffraction, the central maximum is the brightest and widest, and the intensity of successive secondary maxima falls off rapidly.

2. In interference, the fringes arise from two coherent sources (two slits) and all bright fringes are equally spaced. In diffraction, the bands arise from secondary wavelets of the same wavefront (single slit), and the central maximum is twice as wide as the secondary maxima.

(b) Wavelength $\boldsymbol{\lambda = 500\ nm = 5\times 10^{-7}\ m}$, slit width $\boldsymbol{a = 0.2\ mm = 2\times 10^{-4}\ m}$.

Angular width of the central maximum:

$$2\theta = \frac{2\lambda}{a} = \frac{2\times 5\times 10^{-7}}{2\times 10^{-4}}$$

$$2\theta = \frac{10\times 10^{-7}}{2\times 10^{-4}} = 5\times 10^{-3}\ \text{rad}$$

So the angular width of the central maximum is $5\times 10^{-3}$ rad.

Number of interference fringes fitting within the central maximum: the linear width of the central maximum is $\boldsymbol{W = 2\theta\cdot D = \frac{2\lambda D}{a}}$, and the interference fringe width is $\boldsymbol{\beta = 0.5\ mm}$. The number of fringes accommodated is:

$$N = \frac{W}{\beta} = \frac{2\lambda D/a}{\beta} = \frac{(2\theta)D}{\beta}$$

Taking the screen distance to be the same in both set-ups and using the standard data for this problem ($\boldsymbol{D = 1\ m}$):

$$N = \frac{5\times 10^{-3}\times 1}{0.5\times 10^{-3}} = 10$$

Hence about **10 interference fringes** can be accommodated within the angular spread of the central maximum.

**Final answer.** Intensity by $\mathbf{P_2 = I_0/8}$; OR angular width of central maximum $\mathbf{=510^{-3}}$ **rad** and about 10 fringes fit within it.

← *Back to Q200*

## Q201

*(a) Define wavefront. Use Huygens' principle to verify the laws of refraction.*
*(b) How is linearly polarised light obtained by the process of scattering of light? Find the Brewster angle for air - glass interface, when the refractive index of glass* **= 1.5**.

**[2017]**

**SOLUTION**

**(a) Wavefront:** A wavefront is the locus of all points of a medium that are vibrating in the same phase at a given instant. It is a surface of constant phase, and the direction of propagation of the wave is always perpendicular (normal) to the wavefront.

**Huygens' Principle:** Every point on a given primary wavefront acts as a fresh source of secondary disturbances, called secondary wavelets, which travel forward in all directions with the speed of the wave in that medium. The new wavefront at any later instant is the forward tangential envelope of all these secondary wavelets.

**Verifying the law of refraction (Snell's law):** Consider a plane wavefront AB travelling in medium 1 of speed $v_1$, incident at angle $i$ on a plane interface XY that separates medium 1 from a denser medium 2 of speed $v_2$ (with $\mathbf{v_2 < v_1}$).

The diagram shows: incident wavefront AB meeting the surface, with $A$ on the interface and $B$ above it; the ray from $B$ strikes the surface at $C$; the secondary wavelet from $A$ spreads into medium 2 to reach $D$; the refracted wavefront is CD. Angle $i$ is between AB and XY (equal to angle of incidence), angle $r$ is between CD and XY.

Let the wavelet from $B$ take time $\Delta t$ to reach the interface at $C$.

$$BC = v_1 \Delta t$$

In the same time $\Delta t$, the secondary wavelet from $A$ travels into medium 2.

$$AD = v_2 \Delta t$$

From right triangle ABC (right-angled at $B$):

$$\sin i = \frac{BC}{AC} = \frac{v_1 \Delta t}{AC}$$

From right triangle ADC (right-angled at $D$):

$$\sin r = \frac{AD}{AC} = \frac{v_2 \Delta t}{AC}$$

Dividing the two equations:

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1} = \text{constant}$$

$$\Rightarrow n_1 \sin i = n_2 \sin r$$

This is **Snell's law of refraction**, proved by Huygens' principle.

**(b) Linearly polarised light by scattering:** When sunlight (unpolarised) is incident on the molecules of the atmosphere, the electric field of the light sets the electrons of the molecules into oscillation. These oscillating charges re-radiate (scatter) light in all directions. When we view the scattered light at 90° to the direction of the incident sunlight, only the vibrations perpendicular to the plane containing the incident ray and the line of sight can radiate in that direction. The component along the line of sight cannot radiate. Hence the light scattered at 90° is **linearly (plane) polarised**.

**Brewster angle for air to glass:** By Brewster's law,

$$\tan i_B = \mu$$

$$\tan i_B = 1.5$$

$$i_B = \tan^{-1}(1.5)$$

$$i_B \approx 56.3^\circ$$

So the **Brewster angle is about 56.3°.**

CONCEPT

At Brewster's angle the reflected and refracted rays are exactly perpendicular, so $i_B + r = 90°$. This is the geometric shortcut that gives $\mu = \tan i_B$ directly from Snell's law.

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**Final answer.** Snell's law $n_1 \sin i = n_2 \sin r$ is verified, scattered light at $90°$ is linearly polarised, and the Brewster angle is $i_B = \tan^{-1}(1.5) \approx 56.3°$.

← *Back to Q201*

## Q202

*(a) Define wavefront. Use Huygens' principle to verify the laws of refraction.*
*(b) How is linearly polarised light obtained by the process of scattering of light? Find the Brewster angle for air-glass interface, when the refractive index of glass* **= 1.5**.
***OR***
*(a) Draw a ray diagram to show the image formation by a combination of two thin convex lenses in contact. Obtain the expression for the power of this combination in terms of the focal lengths of the lenses.*
*(b) A ray of light passing from air through an equilateral glass prism undergoes minimum deviation when the angle of incidence is* $\frac{3}{4}$ *th of the angle of prism. Calculate the speed of light in the prism.*

**[2017]**

SOLUTION

**(a) Wavefront and Huygens' verification of refraction:** A wavefront is the locus of all points of a medium vibrating in the same phase at a given instant; it is a surface of constant phase, and the direction of wave propagation is normal to it.

**Huygens' Principle:** Every point on a primary wavefront acts as a fresh source of secondary wavelets that spread forward with the wave speed of the medium. The new wavefront is the forward envelope (tangent surface) of all these secondary wavelets.

**Verifying Snell's law:** A plane wavefront AB in medium 1 (speed $v_1$) is incident at angle $i$ on the plane interface XY separating it from medium 2 (speed $v_2$). The diagram shows incident wavefront AB, the wavelet from $B$ reaching the surface at $C$, the wavelet from $A$ spreading into medium 2 to $D$, and the refracted wavefront CD making angle $r$ with XY.

Time for the disturbance at $B$ to reach $C$ is $\Delta t$.

$$BC = v_1 \Delta t$$

In the same time, the wavelet from $A$ advances into medium 2.

$$AD = v_2 \Delta t$$

In right triangle ABC:

$$\sin i = \frac{BC}{AC} = \frac{v_1 \Delta t}{AC}$$

In right triangle ADC:

$$\sin r = \frac{AD}{AC} = \frac{v_2 \Delta t}{AC}$$

Dividing:

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

$$\Rightarrow n_1 \sin i = n_2 \sin r$$

This is **Snell's law of refraction**.

**(b) Linearly polarised light by scattering and Brewster angle:** When unpolarised sunlight strikes air molecules, their electrons oscillate and re-radiate light. Light scattered at 90° to the incident beam contains only vibrations perpendicular to the plane of the incident ray and line of sight, so it is **linearly polarised**.

By Brewster's law for the air-glass interface:

$$\tan i_B = \mu = 1.5$$

$$i_B = \tan^{-1}(1.5) \approx 56.3^\circ$$

**OR**

**(a) Two thin convex lenses in contact, power of combination:** The ray diagram shows two thin convex lenses $L_1$ (focal length $f_1$) and $L_2$ (focal length $f_2$) placed in contact on a common principal axis, with object $O$ on the axis. Lens $L_1$ alone would form image $I_1$ (rays converging); this $I_1$ acts as the object for $L_2$, which forms the final image $I$ closer to the lens. Draw the principal axis, both lenses touching at the optical centre, and rays from $O$ refracting through both to meet at $I$.

For lens $L_1$, taking object distance $u$ and intermediate image at $v_1$:

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1}$$

The image $I_1$ acts as object for $L_2$ (which is in contact, so same position), giving final image at $v$:

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2}$$

Adding the two equations:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2}$$

If the combination has equivalent focal length $f$, then $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, so:

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

Since power $P = \frac{1}{f}$:

$$P = P_1 + P_2$$

The **power of the combination equals the sum of the individual powers.**

**(b) Speed of light in the prism:** For an equilateral prism, angle of prism $A = 60°$. At minimum deviation, the angle of incidence is given:

$$i = \frac{3}{4}A = \frac{3}{4} \times 60° = 45°$$

At minimum deviation, both refractions are symmetric, so the angle of refraction at each face is:

$$r = \frac{A}{2} = 60°2 = 30°$$

Applying Snell's law at the first face:

$$\mu = \frac{\sin i}{\sin r} = \sin 45° \sin 30°$$

$$\mu = \frac{1/\sqrt{2}}{1/2} = \frac{2}{\sqrt{2}} = \sqrt{2} \approx 1.414$$

Speed of light in the prism:

$$v = \frac{c}{\mu} = \frac{3 \times 10^8}{\sqrt{2}}\ m\ s^{-1}$$

$$v = 2.12 \times 10^8\ m\ s^{-1}$$

The **speed of light in the prism is $2.12 \times 10^8\ m\ s^{-1}$.**

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**Final answer.** Snell's law is verified by Huygens' construction; the Brewster angle is **56.3°**. (OR) $P = P_1 + P_2$, and the speed of light in the prism is $2.12 \times 10^8\ m\ s^{-1}$.

← *Back to Q202*

## Q203

*(a) Explain why the intensity of light coming out of a polaroid does not change irrespective of the orientation of the pass axis of the polaroid.*
*(b) State, using a proper diagram, the condition when unpolarized light incident on the boundary between two transparent media produces polarised light. Explain briefly.*
*Hence show that the angle of incidence* $i_B$ *is related to the refractive index* $\mu$ *by the relation,* $\mu = \tan i_B$.
***OR***
*(a) A point object O on the principal axis of a spherical surface of radius of curvature R separating two media of refractive indices* $n_1$ *and* $n_2$ *forms an image I as shown in*

*the figure.*

*Prove that* $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$.

*Use this expression to derive lens maker's formula. Draw the necessary diagram.*

*(b) A convex lens is placed over a plane mirror. A pin is now positioned so that there is no parallax between the pin and its image formed by this lens-mirror combination. How will you use this observation to find focal length of the lens? Explain briefly.*

**[2016]**

**SOLUTION**

**(a) Intensity from a polaroid is unchanged on rotation (for unpolarised light):** Unpolarised light has its electric field vector $\vec{E}$ vibrating in all possible directions perpendicular to the direction of propagation, all with equal probability. A polaroid transmits only the component of $\vec{E}$ along its pass axis.

For any incident vibration making angle θ with the pass axis, the transmitted intensity is $I_0 \cos^2\theta$ (Malus' law). Since θ takes all values uniformly for unpolarised light, we average over θ:

$$\cos^2\theta = \frac{1}{2}$$

$$\therefore I_{transmitted} = \frac{I_0}{2}$$

This value is independent of the orientation of the pass axis. Hence as the polaroid is rotated, the transmitted intensity stays at $\frac{I_0}{2}$ and **does not change.**

**(b) Polarisation by reflection (Brewster's law):** The diagram shows unpolarised light incident on a boundary between two transparent media (e.g. air to glass) at the polarising angle $i_B$. The reflected ray is shown fully polarised (vibrations perpendicular to the plane of incidence, shown as dots), and the refracted ray is partially polarised, with the reflected and refracted rays at right angles to each other.

**Condition:** When unpolarised light is incident at a special angle $i_B$ (the Brewster or polarising angle), the reflected light is completely plane polarised, with vibrations perpendicular to the plane of incidence. At this angle the reflected ray and the refracted ray are mutually perpendicular.

$$i_B + r = 90°$$

$$r = 90° - i_B$$

By Snell's law (taking refractive index μ of second medium with respect to first):

$$\mu = \frac{\sin i_B}{\sin r}$$

$$\mu = \frac{\sin i_B}{\sin(90° - i_B)} = \frac{\sin i_B}{\cos i_B}$$

$$\mu = \tan i_B$$

This is **Brewster's law.**

**OR**

**(a) Refraction at a spherical surface:** The diagram shows a convex spherical refracting surface of radius $R$ and centre $C$ separating medium 1 (index $n_1$) from a denser medium 2 (index $n_2$). Point object $O$ lies on the principal axis in medium 1; a ray OM strikes the surface at $M$, refracts, and meets the axis at image $I$ in medium 2. $P$ is the pole. Mark angles: α at $O$, β at $C$, γ at $I$, with $i$ and $r$ the angles of incidence and refraction at $M$.

Using small-angle approximation, the angle of incidence $\boldsymbol{i = \alpha + \beta}$ (exterior angle of triangle OMC), and $\boldsymbol{\beta = r + \gamma}$, so $\boldsymbol{r = \beta - \gamma}$.

By Snell's law for small angles:

$$n_1\, i = n_2\, r$$

$$n_1(\alpha + \beta) = n_2(\beta - \gamma)$$

For small angles, $\boldsymbol{\alpha = \frac{MN}{-u}}$, $\boldsymbol{\beta = \frac{MN}{R}}$, $\boldsymbol{\gamma = \frac{MN}{v}}$ (where MN is the perpendicular from $M$). Substituting and cancelling MN:

$$n_1\left(\frac{1}{-u} + \frac{1}{R}\right) = n_2\left(\frac{1}{R} - \frac{1}{v}\right)$$

Rearranging:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

**Lens maker's formula:** Apply this at the two surfaces of a thin lens (index $\boldsymbol{n_2 = n}$, surrounding medium $\boldsymbol{n_1 = 1}$). For the first surface (radius $R_1$), image at $v_1$:

$$\frac{n}{v_1} - \frac{1}{u} = \frac{n-1}{R_1}$$

For the second surface (radius $R_2$), the image $v_1$ is the object:

$$\frac{1}{v} - \frac{n}{v_1} = \frac{1-n}{R_2}$$

Adding the two:

$$\frac{1}{v} - \frac{1}{u} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Since $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$:

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

This is the **lens maker's formula.**

**(b) Finding focal length by lens-mirror method:** When the convex lens rests on a plane mirror and the pin is moved until there is no parallax between the pin and its image, the rays from the pin must retrace their path. This happens only when the pin is at the focus of the lens, so that rays emerge parallel from the lens, fall normally on the mirror, reflect back, and re-form the image at the pin itself. Hence the distance from the pin to the lens equals the focal length of the lens, $f$. So **measuring the pin-to-lens distance directly gives the focal length.**

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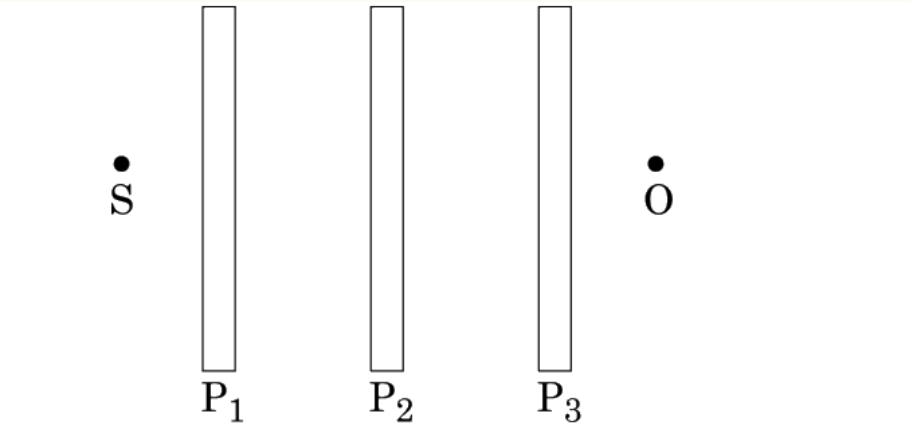
**Final answer.** Transmitted intensity stays $\mathbf{I_0/2}$ on rotation, and Brewster's law $\boldsymbol{\mu = \tan i_B}$ is proved. (OR) $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ leads to the lens maker's formula, and the no-parallax distance equals $f$.

← *Back to Q203*

## Q204

*(a) Why does unpolarised light from a source show a variation in intensity when viewed through a polaroid which is rotated? Show with the help of a diagram, how unpolarised light from sun gets linearly polarised by scattering.*

*(b) Three identical polaroid sheets* $P_1$, $P_2$ *and* $P_3$ *are oriented so that the pass axis of* $P_2$ *and* $P_3$ *are inclined at angles of* 60° *and* 90° *respectively with the pass axis of* $P_1$. *A monochromatic source S of unpolarized light of intensity* $I_0$ *is kept in front of the polaroid sheet* $P_1$ *as shown in the figure. Determine the intensities of light as observed by the observer at O, when polaroid* $P_2$ *is rotated with respect to* $P_1$ *at angles* $\theta$=30° *and* 60°.




[2016]

SOLUTION

**(a) Variation in intensity through a rotating polaroid, and polarisation by scattering:** Unpolarised light from a source has $\vec{E}$ vibrating equally in all directions perpendicular to propagation. When it passes through a single polaroid, the polaroid transmits only the component along its pass axis, so the emergent light becomes plane polarised, with intensity $\frac{I_0}{2}$.

When this **already-polarised** light is examined by rotating the same (or a second) polaroid, the transmitted intensity follows Malus' law, $I = \frac{I_0}{2}\cos^2\theta$, which varies between maximum (axes parallel) and zero (axes crossed). Hence rotating the polaroid shows a variation in intensity, confirming the light is polarised. (Note: a single polaroid in unpolarised light gives no change; the variation is seen because the light reaching the rotated analyser is polarised.)

**Polarisation by scattering (diagram):** Draw a horizontal ray of unpolarised sunlight striking a scattering molecule. Mark the incident vibrations as both dots (perpendicular to page) and double arrows (in the page). Light scattered at 90° (perpendicular to the incident beam) carries only the vibrations perpendicular to the plane of scattering, shown as dots; so the scattered light viewed at 90° is **linearly polarised.**

**(b) Three polaroids $P_1$, $P_2$, $P_3$:** Unpolarised light of intensity $I_0$ falls on $P_1$. After $P_1$:

$$I_1 = \frac{I_0}{2}$$

The pass axis of $P_3$ is fixed at 90° to $P_1$.

**Case 1: $P_2$ at $\theta = 30°$ from $P_1$.**

Intensity after $P_2$ (angle 30° to $P_1$):

$$I_2 = I_1 \cos^2 30° = \frac{I_0}{2}\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{I_0}{2} \times \frac{3}{4} = \frac{3\,I_0}{8}$$

Angle between $P_2$ (30°) and $P_3$ (90°) is 60°, so after $P_3$:

$$I_3 = I_2 \cos^2 60° = \frac{3\,I_0}{8} \times \frac{1}{4} = \frac{3\,I_0}{32}$$

**Case 2: $P_2$ at $\theta = 60°$ from $P_1$.**

Intensity after $P_2$ (angle 60° to $P_1$):

$$I_2 = I_1 \cos^2 60° = \frac{I_0}{2} \times \frac{1}{4} = \frac{I_0}{8}$$

Angle between $P_2$ (60°) and $P_3$ (90°) is 30°, so after $P_3$:

$$I_3 = I_2 \cos^2 30° = \frac{I_0}{8} \times \frac{3}{4} = \frac{3\,I_0}{32}$$

In both cases the **observer at *O* sees intensity $\frac{3\,I_0}{32}$.**

**TIP**

When the first and last polaroids are crossed at 90°, the output is $\frac{I_0}{8}\sin^2(2\theta)$, which is symmetric about **$\theta$=45°**. That is why 30° and 60° give identical results here.

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**Final answer.** For **$P_2$** at both **30°** and **60°** from **$P_1$**, the observer sees the same final intensity **$I_3 = \frac{3\,I_0}{32}$**.

← Back to Q204

## Q205

*(a) Derive an expression for path difference in Young's double slit experiment and obtain the conditions for constructive and destructive interference at a point on the screen.*
*(b) The intensity at the central maxima in Young's double slit experiment is* $I_0$*. Find out the intensity at a point where the path difference is* $\frac{\lambda}{6}$, $\frac{\lambda}{4}$ *and* $\frac{\lambda}{3}$.

**[2016]**

**SOLUTION**

**(a) Path difference in Young's double slit experiment:** Two narrow slits $S_1$ and $S_2$, separated by distance $d$, are illuminated coherently. The screen is at perpendicular distance $D$ ($D \gg d$) from the slits. Consider a point $P$ on the screen at distance $y$ from the central point $O$ (foot of the perpendicular from the midpoint of the slits).

The path lengths from the two slits to $P$ are:

$$S_2P = \sqrt{D^2 + (y + \frac{d}{2})^2}$$

$$S_1P = \sqrt{D^2 + (y - \frac{d}{2})^2}$$

Squaring and subtracting:

$$(S_2P)^2 - (S_1P)^2 = (y+\frac{d}{2})^2 - (y-\frac{d}{2})^2 = 2yd$$

$$(S_2P - S_1P)(S_2P + S_1P) = 2yd$$

Since $D \gg y, d$, we take $\mathbf{S_2P + S_1P \approx 2D}$. Therefore the path difference is:

$$\Delta = S_2P - S_1P = \frac{2yd}{2D} = \frac{yd}{D}$$

**Condition for constructive interference (bright fringe):** path difference must be an integral multiple of $\lambda$.

$$\Delta = \frac{y\,d}{D} = n\lambda,\ n = 0, 1, 2, \ldots$$

$$y_n = \frac{n\lambda\,D}{d}$$

**Condition for destructive interference (dark fringe):** path difference must be an odd multiple of $\frac{\lambda}{2}$.

$$\Delta = \frac{y\,d}{D} = (2n - 1)\frac{\lambda}{2},\ n = 1, 2, \ldots$$

$$y_n = \frac{(2n-1)\lambda\,D}{2\,d}$$

**(b) Intensity for given path differences:** The intensity in interference is $I = I_0 \cos^2(\frac{\phi}{2})$, where $I_0$ is the central (maximum) intensity and the phase difference is $\phi = \frac{2\pi}{\lambda}\Delta$.

**At $\Delta = \frac{\lambda}{6}$:**

$$\phi = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{6} = \frac{\pi}{3}$$

$$I = I_0 \cos^2(\frac{\pi}{6}) = I_0(\frac{\sqrt{3}}{2})^2 = \frac{3\,I_0}{4}$$

**At $\Delta = \frac{\lambda}{4}$:**

$$\phi = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{2}$$

$$I = I_0 \cos^2(\frac{\pi}{4}) = I_0(\frac{1}{\sqrt{2}})^2 = \frac{I_0}{2}$$

**At $\Delta = \frac{\lambda}{3}$:**

$$\phi = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{3} = \frac{2\pi}{3}$$

$$I = I_0 \cos^2(\frac{\pi}{3}) = I_0(\frac{1}{2})^2 = \frac{I_0}{4}$$

So the intensities are $\frac{3\,I_0}{4}$, $\frac{I_0}{2}$ and $\frac{I_0}{4}$ respectively.

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**Final answer.** Path difference $\Delta = \frac{yd}{D}$; intensities are $\frac{3I_0}{4}$ at $\frac{\lambda}{6}$, $\frac{I_0}{2}$ at $\frac{\lambda}{4}$, and $\frac{I_0}{4}$ at $\frac{\lambda}{3}$.

← *Back to Q205*

## Q206

*When a parallel beam of monochromatic source of light of wavelength λ is incident on a single slit of width a, show how the diffraction pattern is formed at the screen by the interference of the wavelets from the slit.*
*Show that, besides the central maximum at* $\theta=0$*, secondary maxima are observed at* $\theta=(n+\frac{1}{2})\frac{\lambda}{a}$ *and the minima at* $\theta=\frac{n\lambda}{a}$.
*Why do secondary maxima get weaker in intensity with increasing n? Explain.*
***OR***
*(i) Draw a ray diagram showing the geometry of formation of image of a point object situated on the principal axis and on the convex side of a spherical surface of radius of curvature R. Taking the rays as incident from a rarer medium of refractive index* $n_1$ *to a denser medium of refractive index* $n_2$*, derive the relation* $\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$*, where symbols have their usual meaning.*
*(ii) Explain briefly how the focal length of a convex lens changes with increase in wavelength of incident light.*
*(iii) What happens to the focal length of convex lens when it is immersed in water? Refractive index of the material of lens is greater than that of water.*

**[2016]**

### SOLUTION

**Formation of the single-slit diffraction pattern:** A parallel beam of monochromatic light of wavelength λ falls normally on a single slit of width *a*. By Huygens' principle, every point of the wavefront across the slit becomes a source of secondary wavelets emitting in all forward directions. A converging lens focuses these wavelets on a screen. At a point on the screen corresponding

to angle θ (measured from the slit normal), the wavelets from different parts of the slit arrive with different phases and interfere.

The diagram shows the slit AB of width $a$, with a plane wavefront incident, and the diffracted rays leaving at angle θ. The extra path difference between the wavelet from the top edge $A$ and that from the bottom edge $B$ is:

$$\Delta = a\theta$$

**Central maximum:** At $\theta = 0$, every wavelet travels the same distance, so all arrive in phase and reinforce, giving the bright central maximum.

**Minima (dark fringes):** Divide the slit into an even number, 2n, of equal strips. Pair each strip in the upper half with the corresponding strip in the lower half. The path difference between paired wavelets is $\frac{a\theta}{2n} \times n = \frac{a\theta}{2}$ for the simplest pairing into two halves. For complete pairwise cancellation when divided into 2n strips, the edge-to-edge path difference must satisfy:

$$a\theta = n\lambda$$

$$\theta = \frac{n\lambda}{a},\ n = \pm 1, \pm 2, \ldots$$

For small angles $\theta \approx \frac{n\lambda}{a}$, giving the **minima.**

**Secondary maxima:** These lie roughly midway between consecutive minima. Divide the slit into an odd number, (2n+1), of equal strips. Then 2n strips cancel in pairs and one uncancelled strip produces a weak maximum. The condition is:

$$a\theta = (n + \frac{1}{2})\lambda$$

$$\theta = (n + \frac{1}{2})\frac{\lambda}{a},\ n = 1, 2, \ldots$$

For small angles $\theta \approx (n + \frac{1}{2})\frac{\lambda}{a}$, the **secondary maxima.**

**Why secondary maxima get weaker:** At the first secondary maximum the slit can be split into 3 equal strips; two cancel and only $\frac{1}{3}$ of the slit contributes. At the next, it splits into 5 strips and only $\frac{1}{5}$ contributes, and so on. As $n$

increases, the fraction of the slit area that contributes constructively (one out of 2n+1 strips) keeps decreasing, so the **intensity of secondary maxima falls off rapidly** with increasing $n$.

**OR**

**(i) Refraction at a convex spherical surface:** The diagram shows a convex spherical surface of radius $R$, centre $C$, separating rarer medium $n_1$ (left) from denser medium $n_2$ (right). A point object $O$ lies on the principal axis; ray OM hits the surface at $M$, refracts towards the normal MC, and meets the axis at image $I$. Mark $\alpha$, $\beta$, $\gamma$ as the angles the lines OM, CM, IM make with the axis, and $i$, $r$ the angles of incidence and refraction.

For the triangle, $i = \alpha + \beta$ and $\beta = r + \gamma$ so $r = \beta - \gamma$.

Snell's law for small angles:

$$n_1 i = n_2 r$$

$$n_1(\alpha + \beta) = n_2(\beta - \gamma)$$

Using small-angle relations $\alpha = \frac{h}{-u}$, $\beta = \frac{h}{R}$, $\gamma = \frac{h}{v}$ (with $h$ the height of $M$):

$$n_1\left(\frac{h}{-u} + \frac{h}{R}\right) = n_2\left(\frac{h}{R} - \frac{h}{v}\right)$$

Cancelling $h$ and rearranging:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

**(ii) Focal length vs wavelength:** By the lens maker's formula, $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$. The refractive index $n$ of the glass decreases as the wavelength of light increases (red light has lower $n$ than violet). A smaller $(n-1)$ makes $\frac{1}{f}$ smaller, so $f$ increases. Hence the **focal length of a convex lens increases with increasing wavelength** (it is longest for red, shortest for violet).

**(iii) Lens immersed in water:** In air, $\frac{1}{f_a} = (n_g - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$. In water, the relevant factor becomes $\left(\frac{n_g}{n_w} - 1\right)$, which is smaller because $n_g > n_w$. Since the

$(n-1)$-type factor decreases, $\frac{1}{f}$ decreases, so $f$ increases. Hence the **focal length of the convex lens increases when it is immersed in water.**

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**Final answer.** Minima occur at $\theta = \frac{n\lambda}{a}$ and secondary maxima at $\theta = (n+12)\frac{\lambda}{a}$, weakening as $n$ rises. (OR) $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$; $f$ increases both with wavelength and on immersion in water.

← Back to Q206

## Q207

*(a) Use Huygens' principle to show the propagation of a plane wavefront from a denser medium to a rarer medium. Hence find the ratio of the speeds of wavefronts in the two media.*
*(b) (i) Why does an unpolarised light incident on a polaroid get linearly polarised?*
*(ii) Derive the expression of Brewster's law when unpolarised light passing from a rarer to a denser medium gets polarised on reflection at the interface.*

**[2015]**

SOLUTION

**(a) Plane wavefront from a denser to a rarer medium:** A plane wavefront AB travels in the denser medium 1 (refractive index $n_1$, speed $v_1$) and is incident at angle $i$ on the plane interface XY separating it from the rarer medium 2 (index $n_2$, speed $v_2$, with $n_2 < n_1$ and $v_2 > v_1$).

The diagram shows incident wavefront AB meeting the surface at $A$, the wavelet from $B$ reaching the surface at $C$, the Huygens wavelet from $A$ spreading into the rarer medium to $D$, and the refracted wavefront CD making angle $r$ with XY (with $r > i$, bending away from the normal).

Let the wavelet from $B$ take time $\Delta t$ to reach $C$.

$$BC = v_1 \Delta t$$

In the same time, the wavelet from $A$ advances in medium 2.

$$AD = v_2 \Delta t$$

From right triangle ABC:

$$\sin i = \frac{BC}{AC} = \frac{v_1 \Delta t}{AC}$$

From right triangle ADC:

$$\sin r = \frac{AD}{AC} = \frac{v_2 \Delta t}{AC}$$

**Ratio of speeds:** dividing the two,

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

$$\Rightarrow \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

So the **ratio of the speeds of the wavefronts in the two media is** $\mathbf{\frac{v_1}{v_2} = \frac{n_2}{n_1}}$; since the second medium is rarer, $\mathbf{v_2 > v_1}$ (the wavefront travels faster in the rarer medium).

**(b)(i) Why a polaroid linearly polarises unpolarised light:** A polaroid sheet has long-chain molecules aligned in one direction. The component of the incident electric field parallel to the chains is absorbed by the conduction electrons of the molecules, while the perpendicular component (along the pass axis) is freely transmitted. Since unpolarised light has $\vec{E}$ vibrating in all directions, only the pass-axis component survives, so the emergent light is **linearly (plane) polarised.**

**(b)(ii) Brewster's law:** The diagram shows unpolarised light incident from the rarer medium (air, $n_1$) on a denser medium (glass, $n_2$) at the polarising angle $\theta_B$. The reflected ray is fully polarised (vibrations perpendicular to the plane of incidence, shown as dots) and the reflected and refracted rays are mutually perpendicular.

At the polarising angle, the reflected and refracted rays are at 90°:

$$\theta_B + r = 90°$$

$$r = 90° - \theta_B$$

By Snell's law:

$$n_1\theta_B = n_2 \sin r = n_2 \sin(90° - \theta_B) = n_2\theta_B$$

$$\frac{\theta_B}{\theta_B} = \frac{n_2}{n_1}$$

$$\theta_B = \frac{n_2}{n_1} = n$$

This is **Brewster's law**, where $n$ is the refractive index of the denser medium with respect to the rarer one.

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**Final answer.** The ratio of wavefront speeds is $\frac{v_1}{v_2} = \frac{n_2}{n_1}$ (faster in the rarer medium), and Brewster's law gives $\theta_B = n$.

← *Back to Q207*

## Q208

*(a) Using Huygens's construction of secondary wavelets explain how a diffraction pattern is obtained on a screen due to a narrow slit on which a monochromatic beam of light is incident normally.*
*(b) Show that the angular width of the first diffraction fringe is half that of the central fringe.*
*(c) Explain why the maxima at* $\theta = (n + \frac{1}{2})\frac{\lambda}{a}$ *become weaker and weaker with increasing n.*

**[2015]**

SOLUTION

**(a) Single-slit diffraction by Huygens' construction:** A monochromatic plane wavefront of wavelength λ falls normally on a narrow slit of width $a$. By Huygens' construction, each point of the wavefront across the slit acts as a source of secondary wavelets sent forward in all directions. A lens focuses these on the screen.

The diagram shows the slit AB of width $a$, the incident plane wavefront, and diffracted rays leaving at angle θ to the normal; the extra path of the ray from the bottom edge over the top edge is $a\theta$.

At $\theta = 0$ all wavelets arrive in phase, producing the bright **central maximum.**

**Minima:** dividing the slit into pairs of strips that cancel, the dark fringes occur when

$$a\theta = n\lambda,\ n = \pm 1, \pm 2, \ldots$$

**Secondary maxima:** these lie approximately midway between minima, at

$$a\theta = (n + \frac{1}{2})\lambda,\ n = 1, 2, \ldots$$

**(b) Angular width of the first diffraction fringe is half that of the central fringe:** For small angles, $\theta \approx \theta$.

The first minima lie on either side of the centre at:

$$\theta = \pm\frac{\lambda}{a}$$

**Angular width of the central maximum** = separation between the first minima on the two sides:

$$\theta_{central} = \frac{\lambda}{a} - (-\frac{\lambda}{a}) = \frac{2\lambda}{a}$$

The first (secondary) maximum lies between the first minimum ($\theta = \frac{\lambda}{a}$) and the second minimum ($\theta = \frac{2\lambda}{a}$).

**Angular width of the first secondary maximum:**

$$\theta_{first} = \frac{2\lambda}{a} - \frac{\lambda}{a} = \frac{\lambda}{a}$$

Taking the ratio:

$$\frac{\theta_{first}}{\theta_{central}} = \frac{\lambda/a}{2\lambda/a} = \frac{1}{2}$$

Hence the **angular width of the first diffraction fringe is half that of the central fringe.**

**(c) Why secondary maxima weaken with increasing *n*:** At the first secondary maximum, the slit can be divided into 3 equal strips: two adjacent strips cancel

each other and only the remaining $\frac{1}{3}$ of the slit contributes to the intensity. At the second secondary maximum, the slit divides into 5 strips, four cancel in pairs, and only $\frac{1}{5}$ contributes, and so on. In general only one strip out of (2n+1) is effective. Since the effective fraction of the slit area $\frac{1}{2n+1}$ decreases as $n$ increases, the **secondary maxima become weaker and weaker with increasing $n$.**

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**Final answer.** The first diffraction fringe has angular width $\frac{\lambda}{a}$, exactly half the central fringe width $\frac{2\lambda}{a}$; secondary maxima weaken because only $\frac{1}{2n+1}$ of the slit contributes.

← *Back to Q208*

## Q209

*(a) (i) 'Two independent monochromatic sources of light cannot produce a sustained interference pattern'. Give reason.*
*(ii) Light waves each of amplitude "a" and frequency "ω", emanating from two coherent light sources superpose at a point. If the displacements due to these waves is given by* $\mathbf{y_1 = a\cos \omega t}$ *and* $\mathbf{y_2 = a\cos(\omega t + \phi)}$ *where* $\phi$ *is the phase difference between the two, obtain the expression for the resultant intensity at the point.*
*(b) In Young's double slit experiment, using monochromatic light of wavelength* $\lambda$*, the intensity of light at a point on the screen where path difference is* $\lambda$*, is K units. Find out the intensity of light at a point where path difference is* $\lambda/3$*.*
***OR***
*(a) How does one demonstrate, using a suitable diagram, that unpolarised light when passed through a Polaroid gets polarised ?*
*(b) A beam of unpolarised light is incident on a glass-air interface. Show, using a suitable ray diagram, that light reflected from the interface is totally polarised, when* $\mathbf{\mu = \tan i_B}$*, where* $\mu$ *is the refractive index of glass with respect to air and* $i_B$ *is the Brewster's angle.*

**[2014]**

**SOLUTION**

**(a)(i) Why two independent sources cannot give sustained interference:** Light from an ordinary source is emitted by independent atoms making random, spontaneous transitions. The phase of the emitted light therefore changes randomly and very rapidly (on a timescale of about $10^{-8}$ s). Two independent sources have phases that vary independently, so the phase difference $\phi$ between the two waves at any screen point keeps changing randomly with time. The positions of bright and dark fringes therefore shift too fast to be seen, and the time-averaged intensity is uniform. Hence **no sustained interference pattern is obtained**; the two sources must be coherent (derived from a single source) for stable fringes.

**(a)(ii) Resultant intensity:** The displacements superpose:

$$y = y_1 + y_2 = a\omega\, t + a\cos(\omega\, t + \phi)$$

Using $\cos C + \cos D = 2\cos\frac{C+D}{2}\cos\frac{C-D}{2}$:

$$y = 2a\cos\left(\frac{\phi}{2}\right)\cos\left(\omega\, t + \frac{\phi}{2}\right)$$

This is a wave of amplitude $A = 2a\cos\left(\frac{\phi}{2}\right)$. Since intensity is proportional to the square of the amplitude ($I \propto A^2$):

$$I = 4a^2\cos^2\left(\frac{\phi}{2}\right)$$

Writing $I_0 = a^2$ (intensity of a single source):

$$I = 4\, I_0 \cos^2\left(\frac{\phi}{2}\right)$$

This is the **resultant intensity** at the point.

**(b) Intensity at $\Delta = \frac{\lambda}{3}$:** Phase difference for path difference $\Delta$ is $\phi = \frac{2\pi}{\lambda}\Delta$.

At $\Delta = \lambda$ (given intensity $K$):

$$\phi = \frac{2\pi}{\lambda}\lambda = 2\pi$$

$$I = 4\, I_0 \cos^2(\pi) = 4\, I_0 = K \Rightarrow I_0 = \frac{K}{4}$$

At $\Delta = \frac{\lambda}{3}$:

$$\phi = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{3} = \frac{2\pi}{3}$$

$$I = 4\, I_0 \cos^2\left(\frac{\pi}{3}\right) = 4\, I_0 \times \frac{1}{4} = I_0$$

$$I = \frac{K}{4}$$

So the **intensity at the point where the path difference is $\frac{\lambda}{3}$ is $\frac{K}{4}$.**

**OR**

**(a) Polarisation by a polaroid:** Draw unpolarised light (shown with both dots and arrows representing all vibration directions) incident on a polaroid sheet whose pass axis is vertical. After passing through, only the vertical component of $\vec{E}$ survives, so the emergent light is plane polarised (shown with vibrations along the pass axis only). If a second polaroid (analyser) is rotated behind it, the transmitted intensity follows Malus' law $I = I_0 \cos^2\theta$, varying from maximum (axes parallel) to zero (axes crossed), which demonstrates that the light is polarised.

**(b) Polarisation by reflection (Brewster's law):** The ray diagram shows unpolarised light from air incident on a glass-air interface at the Brewster angle $i_B$. The reflected ray is shown completely polarised (vibrations perpendicular to the plane of incidence, marked as dots) and the refracted ray is partially polarised; the reflected and refracted rays are perpendicular to each other.

At the Brewster angle:

$$i_B + r = 90^\circ$$

$$r = 90^\circ - i_B$$

By Snell's law:

$$\mu = \frac{\sin i_B}{\sin r} = \frac{\sin i_B}{\sin(90^\circ - i_B)} = \frac{\sin i_B}{\cos i_B}$$

$$\mu = \tan i_B$$

When the angle of incidence equals $i_B$ satisfying $\mu = \tan i_B$, the reflected light is **totally (plane) polarised.**

**COMMON MISTAKE**

*A common slip: at $\Delta=\lambda$ the path difference is a whole wavelength, so this is a bright fringe of intensity K, not zero. Always convert path difference to phase ($\phi = \frac{2\pi}{\lambda}\Delta$) before using $\cos^2(\phi/2)$.*

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**Final answer.** Resultant intensity $I = 4I_0\cos^2(\frac{\phi}{2})$, and the intensity at path difference $\frac{\lambda}{3}$ is $\frac{K}{4}$. (OR) Reflected light is totally polarised when $\mu = \tan i_B$.

← Back to Q209

## Q210

*(a) In Young's double slit experiment, describe briefly how bright and dark fringes are obtained on the screen kept in front of a double slit. Hence obtain the expression for the fringe width.*
*(b) The ratio of the intensities at minima to the maxima in the Young's double slit experiment is 9 : 25. Find the ratio of the widths of the two slits.*
***OR***
*(a) Describe briefly how a diffraction pattern is obtained on a screen due to a single narrow slit illuminated by a monochromatic source of light. Hence obtain the conditions for the angular width of secondary maxima and secondary minima.*
*(b) Two wavelengths of sodium light of 590 nm and 596 nm are used in turn to study the diffraction taking place at a single slit of aperture $2 \times 10^{-6}$ m. The distance between the slit and the screen is 1.5 m. Calculate the separation between the positions of first maxima of the diffraction pattern obtained in the two cases.*

**[2014]**

**SOLUTION**

**(a) Bright and dark fringes in YDSE and fringe width:** Light from a single monochromatic source falls on two close, coherent slits $S_1$ and $S_2$ separated by

$d$. The overlapping waves on a screen at distance $D$ produce interference. At a point $P$ at distance $y$ from the centre, the path difference is $\Delta = \frac{yd}{D}$.

A **bright fringe** forms where the two waves arrive in phase, i.e. path difference is a whole number of wavelengths:

$$\Delta = \frac{yd}{D} = n\lambda \Rightarrow y_n = \frac{n\lambda D}{d}$$

A **dark fringe** forms where the waves arrive out of phase, i.e. path difference is an odd multiple of $\frac{\lambda}{2}$:

$$\Delta = (2n-1)\frac{\lambda}{2} \Rightarrow y_n = \frac{(2n-1)\lambda D}{2d}$$

**Fringe width** $\beta$ is the separation between two consecutive bright (or dark) fringes:

$$\beta = y_{n+1} - y_n = \frac{(n+1)\lambda D}{d} - \frac{n\lambda D}{d}$$

$$\beta = \frac{\lambda D}{d}$$

**(b) Ratio of slit widths:** Given the ratio of minimum to maximum intensity:

$$\frac{I_{min}}{I_{max}} = \frac{9}{25}$$

Using $I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$ and $I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$:

$$\frac{(\sqrt{I_1} - \sqrt{I_2})^2}{(\sqrt{I_1} + \sqrt{I_2})^2} = \frac{9}{25}$$

Taking square roots:

$$\frac{\sqrt{I_1} - \sqrt{I_2}}{\sqrt{I_1} + \sqrt{I_2}} = \frac{3}{5}$$

Cross-multiplying:

$$5(\sqrt{I_1} - \sqrt{I_2}) = 3(\sqrt{I_1} + \sqrt{I_2})$$

$$2\sqrt{I_1} = 8\sqrt{I_2} \Rightarrow \frac{\sqrt{I_1}}{\sqrt{I_2}} = 4$$

$$\frac{I_1}{I_2} = 16$$

Since intensity from a slit is proportional to its width, the ratio of slit widths:

$$\frac{w_1}{w_2} = \frac{I_1}{I_2} = \frac{16}{1}$$

So the **ratio of the widths of the two slits is 16 : 1.**

**OR**

**(a) Single-slit diffraction and angular widths:** A monochromatic plane wave of wavelength λ falls normally on a slit of width $a$. By Huygens' principle, each point of the wavefront in the slit emits secondary wavelets; these interfere on the screen. The path difference between wavelets from the two edges of the slit at angle θ is $a\theta$. At **θ=0** all are in phase, giving the central maximum. Pairing the wavelets gives:

Secondary minima (dark): $a\theta = n\lambda$, $n = 1, 2, \ldots$

Secondary maxima (bright): $a\theta = (n + \frac{1}{2})\lambda$, $n = 1, 2, \ldots$

**(b) Separation between first maxima for the two wavelengths:** The first secondary maximum occurs at $a\theta = \frac{3\lambda}{2}$, so for small angles its position on the screen is $y = D\theta \approx \frac{3\lambda D}{2a}$.

The separation between the first maxima for the two wavelengths is:

$$\Delta y = \frac{3D(\lambda_2 - \lambda_1)}{2a}$$

Substituting $\lambda_2 - \lambda_1 = 596 - 590 = 6\ \text{nm} = 6\times 10^{-9}\ m$, $D = 1.5\ m$, $a = 2\times 10^{-6}\ m$:

$$\Delta y = \frac{3 \times 1.5 \times 6\times 10^{-9}}{2 \times 2\times 10^{-6}}$$

$$\Delta y = \frac{27 \times 10^{-9}}{4\times 10^{-6}}$$

$$\Delta y = 6.75 \times 10^{-3}\ m = 6.75\ \text{mm}$$

The **separation between the first maxima for the two wavelengths is 6.75 mm.**

**Final answer.** Fringe width $\beta = \frac{\lambda D}{d}$ and the slit-width ratio is **16 : 1**. (OR) The separation between the first maxima for the two wavelengths is **6.75 mm**.

← *Back to Q210*

## Q211

*(a) In Young's double slit experiment, derive the condition for (i) constructive interference and (ii) destructive interference at a point on the screen.*
*(b) A beam of light consisting of two wavelengths,* 800 nm *and* 600 nm *is used to obtain the interference fringes in a Young's double slit experiment on a screen placed* 1.4 *m away. If the two slits are separated by* 0.28 mm*, calculate the least distance from the central bright maximum where the bright fringes of the two wavelengths coincide.*

***OR***

*(a) How does an unpolarized light incident on a polaroid get polarized? Describe briefly, with the help of a necessary diagram, the polarization of light by reflection from a transparent medium.*
*(b) Two polaroids 'A' and 'B' are kept in crossed position. How should a third polaroid 'C' be placed between them so that the intensity of polarized light transmitted by polaroid B reduces to* $\frac{1}{8}$ *th of the intensity of unpolarized light incident on A?*

**[2012]**

**SOLUTION**

(a) In Young's double slit experiment, light from two coherent slits $S_1$ and $S_2$ (separation $d$) reaches a point $P$ on the screen at distance $D$. The path difference between the two waves arriving at $P$ is given by the geometry as:

$$\Delta = S_2P - S_1P = \frac{yd}{D}$$

where $y$ is the distance of $P$ from the central point $O$.

(i) Constructive interference (bright fringe): The two waves arrive in phase when the path difference is a whole number of wavelengths.

$$\Delta = n\lambda,\ n = 0, \pm 1, \pm 2, \ldots$$

$$\frac{yd}{D} = n\lambda \Rightarrow y_n = \frac{n\lambda D}{d}$$

(ii) Destructive interference (dark fringe): The two waves arrive exactly out of phase when the path difference is an odd multiple of half a wavelength.

$$\Delta = (2n-1)\frac{\lambda}{2},\ n = 1, 2, 3, \ldots$$

$$\frac{yd}{D} = (2n-1)\frac{\lambda}{2} \Rightarrow y_n = \frac{(2n-1)\lambda D}{2d}$$

(b) For the bright fringes of the two wavelengths to coincide, a bright fringe of $\lambda_1 = 800$ **nm** (order $n_1$) must fall at the same point as a bright fringe of $\lambda_2 = 600$ **nm** (order $n_2$).

$$n_1\lambda_1 = n_2\lambda_2$$

$$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{600}{800} = \frac{3}{4}$$

The least values are $n_1 = 3$ and $n_2 = 4$. The least distance from the central maximum is found using $\lambda_1$:

$$y = \frac{n_1\lambda_1 D}{d}$$

$$y = \frac{3 \times 800 \times 10^{-9} \times 1.4}{0.28 \times 10^{-3}}$$

$$y = \frac{3.36 \times 10^{-6}}{0.28 \times 10^{-3}} = 12 \times 10^{-3}\ m$$

So the least distance is **12 mm**.

OR

(a) When unpolarized light is incident on a polaroid, the polaroid transmits only the component of the electric field $\vec{E}$ parallel to its pass axis and absorbs the perpendicular component. The emerging light has its $\vec{E}$ vector vibrating in a single plane, so it is **plane polarized**.

Polarization by reflection: When unpolarized light is incident on a transparent medium (like glass) at a special angle called the polarizing angle (Brewster's angle $i_B$), the reflected light is completely plane polarized, with its $\vec{E}$ vector vibrating perpendicular to the plane of incidence. At this angle the reflected

and refracted rays are perpendicular to each other, and Brewster's law gives $\mu = \tan i_B$.

Diagram description: Draw a horizontal boundary line for the glass surface. An incident unpolarized ray (shown with both dots and arrows on it) strikes the surface at angle $i_B$ to the normal. The reflected ray goes off at angle $i_B$ on the other side of the normal, marked with dots only (vibrations perpendicular to the page, fully polarized). The refracted ray bends into the glass at angle $r$, with $i_B + r = 90°$, marked with both dots and arrows (partially polarized).

(b) Polaroids A and B are crossed, so their axes are at 90°. Let unpolarized intensity $I_0$ fall on A. After A, intensity = $\frac{I_0}{2}$. Let C be placed at angle θ to A's axis, so C makes (90°−θ) with B.

After C: $I_C = \frac{I_0}{2}\cos^2\theta$

After B: $I_B = \frac{I_0}{2}\cos^2\theta \cos^2(90°-\theta) = \frac{I_0}{2}\cos^2\theta \sin^2\theta$

$$I_B = \frac{I_0}{8}(2\theta\theta)^2 = \frac{I_0}{8}\sin^2 2\theta$$

For $I_B = \frac{I_0}{8}$, we need $\sin^2 2\theta = 1$, so $2\theta = 90°$, giving $\theta = 45°$.

So **C must be placed at 45°** to the pass axis of A (and hence at 45° to B).

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**Final answer.** Least distance = 12 mm; OR third polaroid C must be placed at 45 degrees to both A and B.

← *Back to Q211*

## Q212

*Use Huygens's principle to explain the formation of diffraction pattern due to a single slit illuminated by a monochromatic source of light.*
*When the width of the slit is made double the original width, how would this affect the size and intensity of the central diffraction band?*

**[2012]**

**SOLUTION**

Single-slit diffraction by Huygens' principle: A plane monochromatic wavefront of wavelength λ falls normally on a slit of width $a$. By Huygens' principle, every point of the wavefront inside the slit acts as a source of secondary wavelets. These wavelets spread out and are brought to a focus on the screen (by a lens), where they interfere.

Consider a point $P$ on the screen in a direction making angle θ with the normal. The path difference between the wavelet from the top edge and that from the bottom edge of the slit is:

$$\Delta = a\theta$$

Central maximum: For $\theta = 0$, all wavelets travel equal paths and arrive in phase, giving maximum intensity at the centre $O$.

Minima: Divide the slit into two equal halves. Each wavelet in the upper half has a corresponding wavelet in the lower half at a separation $a/2$, with path difference $\frac{a}{2}\theta$. These cancel pairwise when:

$$\frac{a}{2}\theta = \frac{\lambda}{2} \Rightarrow a\theta = \lambda$$

This is the first minimum. In general, minima occur at:

$$a\theta = n\lambda,\ n = \pm 1, \pm 2, \pm 3, \ldots$$

Secondary maxima: Between consecutive minima lie faint secondary maxima, given approximately by:

$$a\theta = (n + \frac{1}{2})\lambda,\ n = 1, 2, 3, \ldots$$

Their intensity falls off rapidly as $n$ increases.

Diagram description: A slit of width $a$ on the left, a plane wavefront approaching it from the left with parallel vertical lines, a convex lens after the slit, and a screen on the right. The central bright band is at $O$ (opposite the slit centre), with the first minima symmetrically above and below at $\theta = \lambda/a$, and faint secondary maxima beyond.

Effect of doubling the slit width ($a \to 2a$):

The half-angular width of the central maximum is $\theta_1 = \frac{\lambda}{a}$, so the total angular width is $\frac{2\lambda}{a}$.

When $a$ is doubled, this angular width becomes $\frac{2\lambda}{2a} = \frac{\lambda}{a}$, i.e. **the central band becomes half as wide (it shrinks)**.

Intensity: With a wider slit, twice as much light passes through and is concentrated into a band of half the width. Hence **the central maximum becomes brighter (about four times more intense), and sharper**.

> **COMMON MISTAKE**
>
> *A common slip is writing a sin theta = (n + 1/2) lambda for minima. Minima are at a sin theta = n lambda; the (n + 1/2) condition is for the faint secondary maxima.*

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**Final answer.** Doubling the slit width halves the width of the central maximum and makes it about four times brighter and sharper.

← *Back to Q212*

## Q213

*(a) What is the effect on the interference fringes in a Young's double slit experiment when*
*(i) the separation between the two slits is decreased?*
*(ii) the width of the source slit is increased?*
*(iii) the monochromatic source is replaced by a source of white light?*
*Justify your answer in each case.*
*(b) The intensity at the central maxima in Young's double slit experimental set-up is* $I_0$. *Show that the intensity at a point where the path difference is* $\lambda/3$ *is* $I_0/4$.
***OR***
*(a) Obtain the conditions for the bright and dark fringes in diffraction pattern due to a single narrow slit illuminated by a monochromatic source.*
*Explain clearly why the secondary maxima go on becoming weaker with increasing n.*

*(b) When the width of the slit is made double, how would this affect the size and intensity of the central diffraction band? Justify.*

**[2012]**

**SOLUTION**

(a)(i) Separation $d$ decreased: The fringe width is $\beta = \frac{\lambda D}{d}$, which is inversely proportional to $d$. So when $d$ decreases, $\beta$ increases and **the fringes become more widely spaced**.

(a)(ii) Width of the source slit increased: A wider source slit acts like many independent point sources side by side. Each produces its own slightly displaced fringe pattern, and these overlap. The bright and dark fringes lose contrast and **the fringes become blurred**; if the source slit is too wide, the interference pattern disappears (the condition $\frac{s}{S} < \frac{\lambda}{d}$ is violated).

(a)(iii) Monochromatic source replaced by white light: At the centre all wavelengths have zero path difference, so **the central fringe is white**. On either side, each colour forms fringes of its own width ($\beta \propto \lambda$), so the fringes become coloured, with violet nearest the centre and red farthest. Only a few coloured fringes are seen because higher orders of different colours overlap and merge into uniform white light.

(b) The intensity at any point is $I = I_0 \cos^2(\frac{\phi}{2})$, where $\phi$ is the phase difference.

The phase difference for a path difference $\Delta = \frac{\lambda}{3}$ is:

$$\phi = \frac{2\pi}{\lambda}\Delta = \frac{2\pi}{\lambda}\cdot\frac{\lambda}{3} = \frac{2\pi}{3}$$

$$I = I_0\cos^2(\frac{\phi}{2}) = I_0\cos^2(\frac{\pi}{3})$$

$$I = I_0(\frac{1}{2})^2 = \frac{I_0}{4}$$

Hence the intensity at a point where the path difference is $\frac{\lambda}{3}$ is $\frac{I_0}{4}$.

OR

(a) A plane monochromatic wavefront of wavelength $\lambda$ falls on a single slit of width $a$. Take a point at angle $\theta$ on the screen; the path difference between

wavelets from the two edges is $a\theta$.

Dark fringes (minima): Dividing the slit into two halves, the wavelets cancel pairwise when:

$$a\theta = n\lambda,\ n = \pm 1, \pm 2, \ldots$$

Bright fringes (secondary maxima): These occur approximately midway between minima:

$$a\theta = (n + \frac{1}{2})\lambda,\ n = 1, 2, 3, \ldots$$

Why secondary maxima get weaker: For the $n$th secondary maximum, divide the slit into (2n+1) equal strips. Wavelets from 2n of these strips cancel in pairs, and only one remaining strip (of width $\frac{a}{2n+1}$) contributes to the intensity. As $n$ increases, this surviving fraction of the slit gets smaller, so the amplitude (and hence intensity) of the secondary maxima **falls off rapidly with increasing $n$**.

(b) When the slit width is doubled ($a \to 2a$):

The angular width of the central maximum = $\frac{2\lambda}{a}$ becomes $\frac{2\lambda}{2a} = \frac{\lambda}{a}$, so **its size is halved**.

Since twice the light now passes through and is squeezed into half the width, **the central maximum becomes about four times brighter and sharper**.

---

**Final answer.** Intensity at path difference lambda/3 is I0/4; doubling the slit width halves the central band and makes it brighter.

← *Back to Q213*

## Q214

*State the importance of coherent sources in the phenomenon of interference.*
*In Young's double slit experiment to produce interference pattern, obtain the conditions for constructive and destructive interference. Hence deduce the expression for the fringe width.*

*How does the fringe width get affected, if the entire experimental apparatus of Young is immersed in water?*

**[2011]**

**SOLUTION**

Importance of coherent sources: Coherent sources emit waves of the same frequency with a constant (time-independent) phase difference. They are essential because only then is the interference pattern steady and observable. If the phase difference varied randomly with time, the positions of bright and dark fringes would shift rapidly and the eye would see only uniform average illumination, with no fringes.

Conditions in YDSE: Light from two coherent slits $S_1$, $S_2$ (separation $d$) reaches a point $P$ at distance $y$ from the centre, on a screen at distance $D$. The path difference is:

$$\Delta = \frac{yd}{D}$$

Constructive interference (bright fringes): waves arrive in phase when

$$\Delta = n\lambda \Rightarrow \frac{yd}{D} = n\lambda,\ n = 0, \pm 1, \pm 2, \ldots$$

$$y_n = \frac{n\lambda D}{d}$$

Destructive interference (dark fringes): waves arrive out of phase when

$$\Delta = (2n-1)\frac{\lambda}{2} \Rightarrow y_n = \frac{(2n-1)\lambda D}{2d},\ n = 1, 2, 3, \ldots$$

Fringe width: The fringe width β is the distance between two consecutive bright (or dark) fringes:

$$\beta = y_{n+1} - y_n = \frac{(n+1)\lambda D}{d} - \frac{n\lambda D}{d}$$

$$\beta = \frac{\lambda D}{d}$$

Effect of immersing the apparatus in water: Inside water (refractive index $\mu = \frac{4}{3}$) the wavelength reduces to $\lambda_w = \frac{\lambda}{\mu}$. Hence the new fringe width is:

$$\beta_w = \frac{\lambda_w D}{d} = \frac{\lambda D}{\mu d} = \frac{\beta}{\mu}$$

$$\beta_w = \frac{3}{4}\beta$$

So **the fringe width decreases to $\frac{3}{4}$ of its value in air** (it becomes $\frac{1}{\mu}$ times the original).

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**Final answer.** Beta = lambda D / d; on immersion in water the fringe width decreases to (3/4) of its value in air.

← *Back to Q214*

## Q215

*(a) State Huygens' principle. Using this principle explain how a diffraction pattern is obtained on a screen due to a narrow slit on which a narrow beam coming from a monochromatic source of light is incident normally.*
*(b) Show that the angular width of the first diffraction fringe is half of that of the central fringe.*
*(c) If a monochromatic source of light is replaced by white light, what change would you observe in the diffraction pattern?*

**[2011]**

**SOLUTION**

(a) Huygens' principle: Every point on a given wavefront acts as a source of secondary spherical wavelets that travel forward with the speed of the wave. The new wavefront at a later instant is the forward envelope (tangential surface) of all these secondary wavelets.

Single-slit diffraction: A plane monochromatic wavefront of wavelength $\lambda$ falls normally on a narrow slit of width $a$. By Huygens' principle, every point in the slit becomes a secondary source. The wavelets travel to a point $P$ on the screen at angle $\theta$ from the normal. The path difference between the wavelet from the top edge and that from the bottom edge of the slit is $a\theta$.

For the central point ($\theta = 0$) all wavelets are in phase, giving a bright central maximum. Dividing the slit into two halves, the wavelets cancel pairwise when:

$$\frac{a}{2}\theta = \frac{\lambda}{2} \Rightarrow a\theta = \lambda$$

This is the first minimum. In general, minima occur at $a\theta = n\lambda$ ($n = \pm 1, \pm 2, \ldots$), and faint secondary maxima at $a\theta = (n + \frac{1}{2})\lambda$.

Diagram description: Plane wavefront (parallel lines) approaching a slit of width $a$ from the left, a lens after the slit focusing rays onto a screen at the right; the central bright band sits at $O$, flanked by the first minima at $\theta = \lambda/a$.

(b) The central maximum lies between the first minima on either side, i.e. from $\theta = -\frac{\lambda}{a}$ to $\theta = +\frac{\lambda}{a}$. So its angular width is:

$$\Delta\theta_{\text{central}} = \frac{2\lambda}{a}$$

The first secondary maximum lies between the first minimum ($\theta = \frac{\lambda}{a}$) and the second minimum ($\theta = \frac{2\lambda}{a}$). So its angular width is:

$$\Delta\theta_{\text{first}} = \frac{2\lambda}{a} - \frac{\lambda}{a} = \frac{\lambda}{a}$$

$$\frac{\Delta\theta_{\text{first}}}{\Delta\theta_{\text{central}}} = \frac{\lambda/a}{2\lambda/a} = \frac{1}{2}$$

Hence **the angular width of the first diffraction fringe is half that of the central fringe**.

(c) If white light is used, each colour forms its own pattern with width proportional to $\lambda$. The central maximum is white (all colours overlap at $\theta = 0$), while the side fringes become coloured, with violet (smallest $\lambda$) closest to the centre and red (largest $\lambda$) farthest out. The patterns of different colours overlap, so only a few coloured fringes are seen.

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**Final answer.** Central maximum width = 2 lambda/a, first fringe width = lambda/a, so the first fringe is half as wide; in white light the centre is white and side fringes are coloured.

← Back to Q215

## Q216

*(i) A plane wavefront approaches a plane surface separating two media. If medium 'one' is optically denser and medium 'two' is optically rarer, using Huygens' principle, explain and show how a refracted wavefront is constructed.*
*(ii) Hence verify Snell's law.*
*(iii) When a light wave travels from a rarer to a denser medium, the speed decreases. Does it imply reduction in its energy? Explain.*

**[2011 • Set 55-2-1]**

**SOLUTION**

(i) Huygens construction for refraction (denser to rarer): A plane wavefront AB travels in the denser medium 1 with speed $v_1$ and meets the boundary XY. Point $A$ touches the surface first while point $B$ still has to travel a distance $\mathbf{BC = v_1 \Delta t}$ to reach the surface at $C$.

During the same time $\Delta t$, the secondary wavelet starting from $A$ spreads into the rarer medium 2 (speed $v_2$, with $\mathbf{v_2 > v_1}$) to a radius:

$$AD = v_2 \Delta t$$

The tangent CD drawn from $C$ to this spherical wavelet is the refracted wavefront. Since $\mathbf{v_2 > v_1}$, we have $\mathbf{AD > BC}$, so the refracted ray bends away from the normal at the larger angle $r$.

Diagram description: A horizontal boundary XY. Incident wavefront AB in medium 1 strikes at angle $i$; $A$ is on the surface, $B$ is above it. The line BC lies along the boundary direction with $\mathbf{BC = v_1 \Delta t}$. In medium 2, an arc of radius $\mathbf{AD = v_2 \Delta t}$ is drawn from $A$; the refracted wavefront CD is tangent to this arc, making angle $r$ with XY where $\boldsymbol{r > i}$.

(ii) Verifying Snell's law: In right triangle ABC, the angle of incidence equals $\angle \mathbf{BAC} = \boldsymbol{i}$, and in triangle ACD the angle of refraction equals $\angle \mathbf{ACD} = \boldsymbol{r}$. With AC as the common hypotenuse:

$$\sin i = \frac{BC}{AC} = \frac{v_1 \Delta t}{AC}$$

$$\sin r = \frac{AD}{AC} = \frac{v_2 \Delta t}{AC}$$

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$

Since the refractive index of a medium is $n = \frac{c}{v}$, we have $\frac{v_1}{v_2} = \frac{n_2}{n_1}$, so:

$$n_1 \sin i = n_2 \sin r$$

This is Snell's law, hence verified.

(iii) Does the energy reduce? **No.** When light passes from a rarer to a denser medium, only the speed and wavelength decrease; the frequency ν stays the same (it is fixed by the source). Since the energy of a photon is $E = h\nu$ and ν is unchanged, the energy of the light is not reduced. The reduction in speed is simply because the wave interacts with the atoms of the denser medium, not because it loses energy.

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**Final answer.** Huygens construction gives n1 sin i = n2 sin r (Snell's law); the energy is not reduced because frequency, and hence E = h nu, stays unchanged.

← *Back to Q216*

## Q217

*State Huygens's principle. Show, with the help of a suitable diagram, how this principle is used to obtain the diffraction pattern by a single slit.*
*Draw a plot of intensity distribution and explain clearly why the secondary maxima become weaker with increasing order (n) of the secondary maxima.*

**[2010]**

**SOLUTION**

Huygens' principle: Every point on a wavefront acts as a source of secondary spherical wavelets which travel forward with the speed of the wave. The new wavefront at any later time is the forward envelope (the tangent surface) of all these secondary wavelets.

Single-slit diffraction: A plane monochromatic wavefront of wavelength $\lambda$ is incident normally on a slit of width $a$. By Huygens' principle, every point of the wavefront inside the slit is a secondary source. The wavelets travel to a point $P$ on the screen in a direction making angle $\theta$ with the normal; the path difference between wavelets from the two edges of the slit is $a\theta$.

At $\theta = 0$ all wavelets are in phase, producing the bright central maximum. Dividing the slit into two halves, the wavelets cancel in pairs when:

$$\frac{a}{2}\theta = \frac{\lambda}{2} \Rightarrow a\theta = \lambda \text{ (first minimum)}$$

In general the minima are at:

$$a\theta = n\lambda,\ n = \pm 1, \pm 2, \ldots$$

and the secondary maxima approximately at $a\theta = (n+\frac{1}{2})\lambda$.

Diagram description: A plane wavefront approaching the slit of width $a$ from the left, a lens beyond the slit, and a screen on the right showing a broad bright central band at $O$ with much fainter secondary maxima on each side.

Intensity distribution plot: On a graph of intensity $I$ versus $\theta$, draw a tall central peak at the origin (height $I_0$). On either side the curve drops to zero at $\theta = \pm\frac{\lambda}{a}, \pm\frac{2\lambda}{a}, \ldots$ Between these zeros stand small secondary peaks whose heights fall off rapidly, the first secondary peak being only about $\frac{1}{22}$ (roughly 4.5%) of $I_0$.

Why secondary maxima become weaker: For the $n$th secondary maximum, divide the slit into (2n+1) equal strips. Wavelets from 2n of the strips cancel one another in pairs, and only the contribution of one remaining strip (of width $\frac{a}{2n+1}$) survives. As $n$ increases, this surviving fraction of the slit shrinks, so the resultant amplitude $\propto \frac{1}{2n+1}$ and the intensity $\propto \frac{1}{(2n+1)^2}$. Hence **the secondary maxima become progressively weaker as the order $n$ increases**.

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**Final answer.** Minima at a sin theta = n lambda; secondary maxima weaken because only the light of one strip out of (2n+1) survives, so intensity falls as 1/(2n+1) squared.

← Back to Q217

## Q218

*(a) What is plane polarised light? Two polaroids are placed at 90° to each other and the transmitted intensity is zero. What happens when one more polaroid is placed between these two, bisecting the angle between them? How will the intensity of transmitted light vary on further rotating the third polaroid?*
*(b) If a light beam shows no intensity variation when transmitted through a polaroid which is rotated, does it mean that the light is unpolarised? Explain briefly.*

**[2008]**

**SOLUTION**

(a) Plane polarised light: Light is plane (linearly) polarised when the electric field vector $\vec{E}$ vibrates in only one fixed plane perpendicular to the direction of propagation. At a fixed point, the tip of $\vec{E}$ traces a straight line over time.

Two crossed polaroids ($P_1$ and $P_2$ at 90°) transmit zero intensity, because the light passed by $P_1$ has its $\vec{E}$ along $P_1$'s axis, which is perpendicular to $P_2$'s axis, so $\mathbf{\cos^2 90° = 0}$.

Insert a third polaroid $P_3$ midway, with its axis at 45° to both. Let unpolarised intensity $I_0$ fall on $P_1$.

After $P_1$: $\mathbf{I_1 = \frac{I_0}{2}}$

After $P_3$ (at 45° to $P_1$): $\mathbf{I_3 = \frac{I_0}{2}\cos^2 45° = \frac{I_0}{2}\cdot\frac{1}{2} = \frac{I_0}{4}}$

After $P_2$ (at 45° to $P_3$): $\mathbf{I_2 = \frac{I_0}{4}\cos^2 45° = \frac{I_0}{4}\cdot\frac{1}{2} = \frac{I_0}{8}}$

So inserting the third polaroid at 45° makes the transmitted intensity $\frac{I_0}{8}$, which is now non-zero (light reappears).

On further rotating $P_3$ through angle θ from $P_1$'s axis, the transmitted intensity through $P_2$ becomes:

$$I = \frac{I_0}{2}\cos^2\theta\cos^2(90°-\theta) = \frac{I_0}{8}\sin^2 2\theta$$

So the intensity varies between 0 (when $P_3$ is aligned with $P_1$ or $P_2$, $\theta = 0°$ or 90°) and a maximum of $\frac{I_0}{8}$ (at $\theta = 45°$).

(b) No, it does not necessarily mean the light is unpolarised. If a beam shows no intensity change as the polaroid is rotated, the light could be either:

(1) unpolarised, where $\frac{1}{2}$ the intensity always passes regardless of orientation, or

(2) circularly polarised, which also gives a constant transmitted intensity on rotation.

So a steady intensity on rotation alone cannot distinguish unpolarised light from circularly polarised light; **constant intensity does not prove the light is unpolarised**.

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**Final answer.** Inserting P3 at 45 degrees gives I0/8 and varies as (I0/8) sin squared 2theta; constant intensity on rotation does not prove the light is unpolarised (it could be circularly polarised).

← *Back to Q218*

## Q219

*What are coherent sources? Why are coherent sources required to produce interference of light? Give an example of interference of light in everyday life.*
*In Young's double slit experiment, the two slits are 0.03 cm apart and the screen is placed at a distance of 1.5 m away from the slits. The distance between the central bright fringe and fourth bright fringe is 1 cm. Calculate the wavelength of light used.*

**[2007]**

**SOLUTION**

Coherent sources: Two sources are coherent if they emit light waves of the same frequency (same wavelength) with a constant (time-independent) phase difference between them.

Why coherent sources are required: The resultant intensity at a point where two waves of intensities $I_1$ and $I_2$ meet with phase difference $\phi$ is:

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\,\phi$$

For a steady pattern, $\phi$ must stay constant in time. With incoherent sources $\phi$ changes randomly and rapidly, so the average of $\phi$ over time is zero; the interference term vanishes and only uniform illumination ($I = I_1 + I_2$) is seen. Only coherent sources give a fixed $\phi$ at each point and hence a steady pattern of bright and dark fringes.

Everyday example: The brilliant colours seen on a soap bubble or on a thin film of oil spread on water, caused by interference of light reflected from the top and bottom surfaces of the thin film.

Numerical: Given $d = 0.03\ \text{cm} = 3\times 10^{-4}\ m$, $D = 1.5\ m$, and the distance from the central fringe to the 4th bright fringe is $y_4 = 1\ \text{cm} = 10^{-2}\ m$.

The position of the $n$th bright fringe is $y_n = n\beta$, so for $n = 4$:

$$\beta = \frac{y_4}{4} = \frac{10^{-2}}{4} = 2.5\times 10^{-3}\ m$$

Using $\beta = \dfrac{\lambda D}{d}$:

$$\lambda = \frac{\beta d}{D}$$

$$\lambda = \frac{(2.5\times 10^{-3})(3\times 10^{-4})}{1.5}$$

$$\lambda = \frac{7.5\times 10^{-7}}{1.5} = 5\times 10^{-7}\ m$$

So the wavelength of light used is $5\times 10^{-7}\ m = 500\ \text{nm}$ (green light).

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**Final answer.** Wavelength of the light used is 5 x 10^-7 m = 500 nm.

← *Back to Q219*

## Q220

*State the condition under which the phenomenon of diffraction of light takes place. Derive an expression for the width of central maximum due to diffraction of light at a single slit.*

*A slit of width 'a' is illuminated by a monochromatic light of wavelength 700 nm at normal incidence. Calculate the value of 'a' for position of*
*(i) first minimum at an angle of diffraction of* 30°.
*(ii) first maximum at an angle of diffraction of* 30°.

**[2007]**

**SOLUTION**

Condition for diffraction: Diffraction of light is appreciable only when the size of the aperture (or obstacle) is comparable to the wavelength of the light, i.e. the slit width $a$ should be of the order of λ. If $a \gg \lambda$, the spreading is negligible and light travels in straight lines (geometrical optics).

Width of central maximum: A plane monochromatic wavefront of wavelength λ falls normally on a slit of width $a$. The first minimum occurs at angle θ given by:

$$a\theta = \lambda$$

For small angles, $\theta \approx \theta$, so $\theta = \frac{\lambda}{a}$.

The central maximum extends from −θ to +θ, so its angular width is:

$$2\theta = \frac{2\lambda}{a}$$

On a screen at distance $D$, the half-width is $y = D\theta = \frac{D\lambda}{a}$, so the linear width of the central maximum is:

$$W = 2y = \frac{2D\lambda}{a}$$

Numerical: Given $\lambda = 700\ \text{nm} = 700\times 10^{-9}\ m$ and $\theta = 30°$.

(i) First minimum at 30°: $a\theta = \lambda$

$$a = \frac{\lambda}{\sin 30°} = \frac{700\times 10^{-9}}{0.5}$$

$$a = 1.4\times 10^{-6}\ m = 1.4\ \mu m$$

(ii) First (secondary) maximum at 30°: the condition is $a\theta = \frac{3\lambda}{2}$

$$a = \frac{3\lambda}{2\sin 30^\circ} = \frac{3\times 700\times 10^{-9}}{2\times 0.5}$$

$$a = \frac{2100\times 10^{-9}}{1} = 2.1\times 10^{-6}\ m = 2.1\ \mu m$$

So $a$ = **1.4 µm for the first minimum and** $a$ = **2.1 µm for the first maximum** at 30°.

**TIP**

For the first secondary maximum use a sin theta = 3 lambda / 2, not lambda. Mixing up the minimum and maximum conditions is the most common error in this numerical.

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**Final answer.** Slit width a = 1.4 micrometre for the first minimum and a = 2.1 micrometre for the first maximum at 30 degrees.

← *Back to Q220*

## Q221

*State the essential condition for diffraction of light to take place.*
*Use Huygen's principle to explain diffraction of light due to a narrow single slit and the formation of a pattern of fringes obtained on the screen. Sketch the pattern of fringes formed due to diffraction at a single slit showing variation of intensity with angle* θ.

**[2007]**

**SOLUTION**

Diffraction is the bending of light around the edges of an obstacle or aperture, and the spreading of light into the geometrical shadow region. The **essential condition** for appreciable diffraction is that the size of the aperture (or obstacle) must be of the same order of magnitude as the wavelength λ of the light.

For a slit of width $a$, marked diffraction occurs only when $a \sim \lambda$. If $a \gg \lambda$, the bending is negligible and light travels almost in a straight line.

Explanation by Huygens' principle: A plane wavefront WW' is incident normally on a single slit AB of width $a$. According to Huygens' principle, every point of the wavefront inside the slit acts as a fresh source of secondary wavelets. These wavelets spread out in all forward directions, travel to the screen, and superpose.

Central maximum: All secondary wavelets travelling straight ahead **($\theta = 0$)** reach the centre $O$ of the screen with zero path difference. They arrive in phase and reinforce, giving a bright **central maximum**.

Position of minima: Consider wavelets going to a point $P$ at angle $\theta$ to the normal. The path difference between wavelets from the two edges $A$ and $B$ of the slit is $a\theta$.

Divide the slit into two equal halves. Pair a wavelet from the top edge with one from the middle. Their path difference is $\frac{a}{2}\theta$.

These two halves cancel completely when this equals $\frac{\lambda}{2}$:

$$\frac{a}{2}\theta = \frac{\lambda}{2}$$

$\Rightarrow a\theta = \lambda$ (first minimum)

Extending this pairing argument to dividing the slit into 2n parts gives the general condition for minima (dark fringes):

$$a\theta = n\lambda,\ n = \pm 1, \pm 2, \pm 3, \ldots$$

Position of secondary maxima: Midway between consecutive minima the wavelets do not cancel fully, leaving weak bright bands called **secondary maxima**, given approximately by:

$$a\theta = (n + \frac{1}{2})\lambda,\ n = \pm 1, \pm 2, \ldots$$

Their intensity falls off rapidly as $n$ increases.

Intensity-versus-angle pattern (sketch in words): Plot intensity $I$ on the vertical axis against $\theta$ (or angle $\theta$) on the horizontal axis. The graph shows a tall, broad **central peak** centred at $\theta = 0$, extending from $\theta = -\frac{\lambda}{a}$ to $\theta = +\frac{\lambda}{a}$. On both sides

of it are much smaller secondary peaks separated by zeros (minima) at $\theta = \pm\frac{\lambda}{a}$, $\pm\frac{2\lambda}{a}$, .... The central maximum is twice as wide as each secondary maximum, and the secondary peaks shrink quickly in height on either side.

**TIP**

Remember the central maximum is twice as wide as the side maxima because minima occur at $\lambda/a$ on either side of the centre.

---

**Final answer.** The aperture must be comparable to $\lambda$; Huygens' wavelets give a central maximum with minima at $a\theta = n\lambda$, and the intensity graph is a broad central peak flanked by rapidly weakening secondary maxima.

← *Back to Q221*

## Q222

*What are coherent sources of light? Why are coherent sources required to obtain sustained interference pattern?*
*State three characteristic features which distinguish the interference pattern due to two coherently illuminated sources as compared to that observed in a diffraction pattern due to a single slit.*

**[2007]**

**SOLUTION**

Coherent sources of light: Two sources of light are said to be **coherent** if they emit light waves of the same frequency (same wavelength) and maintain a constant phase difference with time.

In practice, two independent lamps can never be coherent. Coherent sources are obtained by splitting light from a single source, for example the two slits $S_1$ and $S_2$ in Young's double-slit experiment, or by Fresnel's biprism and Lloyd's mirror.

Why coherent sources are required for sustained interference: When two waves of intensities $I_1$ and $I_2$ superpose with a phase difference $\phi$, the resultant intensity is:

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\,\phi$$

The third term $2\sqrt{I_1 I_2}\,\phi$ is the interference term, and it depends on the phase difference $\phi$.

If the sources are incoherent, $\phi$ changes randomly and very rapidly with time. The average value of $\phi$ over time then becomes zero:

$$\phi = 0 \Rightarrow I = I_1 + I_2$$

The screen is then uniformly illuminated and no fringes appear.

Only with coherent sources is $\phi$ constant in time at each point, so the interference term survives. This produces fixed bright fringes (where $\boldsymbol{\phi = +1}$) and dark fringes (where $\boldsymbol{\phi = -1}$) that stay steady long enough to be seen, giving a **sustained interference pattern**.

Three distinguishing features (two-slit interference vs single-slit diffraction):

1. Fringe width: In interference all the bright fringes are of equal width, whereas in diffraction the central maximum is about twice as wide as the secondary maxima.

2. Intensity of fringes: In interference all bright fringes have nearly the same intensity, whereas in diffraction the intensity of the secondary maxima falls off rapidly away from the centre.

3. Origin of the pattern: Interference is due to the superposition of waves from two separate coherent sources, while diffraction is due to the superposition of secondary wavelets coming from different parts of the same single wavefront.

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**Final answer.** Coherent sources have the same frequency and a constant phase difference; they are needed so the interference term $\boldsymbol{2\sqrt{I_1 I_2}\,\phi}$ does not average to zero, giving steady fringes that differ from diffraction in width, intensity and origin.

*← Back to Q222*

## Q223

*What is interference of light? Write two essential conditions for sustained interference pattern to be produced on the screen.*
*Draw a graph showing the variation of intensity versus the position on the screen in Young's experiment when (a) both the slits are opened and (b) one of the slits is closed.*
*What is the effect on the interference pattern in Young's double slit experiment when:*
*(i) screen is moved closer to the plane of slits?*
*(ii) separation between two slits is increased. Explain your answer in each case.*

**[2006]**

**SOLUTION**

Interference of light: It is the phenomenon in which two coherent light waves superpose at a point and redistribute the light energy, producing **bright fringes** where the waves meet in phase (constructive interference) and **dark fringes** where they meet out of phase (destructive interference). The total energy is conserved; it is only redistributed.

Two essential conditions for a sustained interference pattern:

1. The two sources must be **coherent**, i.e. they must have the same frequency and a constant phase difference with time.

2. The two waves must have **equal or nearly equal amplitudes** (and the same plane of polarisation) so that the dark fringes are nearly completely dark and the contrast between bright and dark fringes is good.

Graph of intensity versus position on the screen:

(a) Both slits open: Plot intensity $I$ on the vertical axis against position $x$ on the screen on the horizontal axis. The graph is a series of equally spaced, equally tall peaks following a $\cos^2$ shape. The maxima rise to $4I_0$ (where $I_0$ is the intensity from one slit) and the minima fall to zero. These are the interference fringes.

(b) One slit closed: Only one slit acts, so there is no interference. The graph is now the broad single-slit diffraction curve, a wide central peak of width $\frac{2\lambda D}{a}$ with much smaller secondary peaks on either side, and no sharp equally spaced fringes.

Effect of changes on the interference pattern (fringe width $\beta = \frac{\lambda D}{d}$):

(i) Screen moved closer to the slits ($D$ decreases):

$$\beta = \frac{\lambda D}{d}$$

Since $\beta \propto D$, decreasing $D$ decreases $\beta$. The fringes come closer together and the whole pattern shrinks.

(ii) Separation between the slits increased ($d$ increases):

$$\beta = \frac{\lambda D}{d}$$

Since $\beta \propto \frac{1}{d}$, increasing $d$ decreases $\beta$. The fringes again become closer together and the pattern is more tightly packed.

---

**Final answer.** Interference needs coherent, equal-amplitude sources; both slits open give equal $\cos^2$ fringes (one closed gives a diffraction curve), and reducing $D$ or increasing $d$ both reduce $\beta = \lambda D/d$ so fringes come closer.

← Back to Q223

## Q224

*What is diffraction of light? Draw a graph showing the variation of intensity with angle in a single slit diffraction experiment. Write one feature which distinguishes the observed pattern from the double slit interference pattern.*
*How would the diffraction pattern of a single slit be affected when:*
*(i) the width of the slit is decreased?*
*(ii) the monochromatic source of light is replaced by a source of white light?*

**[2006]**

**SOLUTION**

Diffraction of light: It is the phenomenon of bending of light around the edges of an obstacle or aperture whose size is comparable to the wavelength of light, so that light spreads into the geometrical shadow region. The bent wavelets interfere with one another and produce a pattern of bright and dark bands.

Graph of intensity versus angle for a single slit (sketch in words): Plot intensity $I$ on the vertical axis against angle θ (or θ) on the horizontal axis. The graph has one tall, broad **central maximum** centred at **θ = 0**. On both sides are much weaker secondary maxima separated by zeros. The minima occur at:

$$a\theta = n\lambda,\ n = \pm 1, \pm 2, \ldots$$

The central maximum has angular width $\frac{2\lambda}{a}$, which is twice the width $\frac{\lambda}{a}$ of each secondary maximum, and the secondary peaks fall in height rapidly as we move away from the centre.

Distinguishing feature from double-slit interference: In single-slit diffraction the central maximum is much wider and brighter than the secondary maxima, whose intensities fall off sharply. In double-slit interference, by contrast, all the bright fringes are of equal width and nearly equal intensity. (This is the one required distinguishing feature.)

Effect of the given changes:

(i) Width of the slit decreased ($a$ decreases): The angular width of the central maximum is

$$\text{width} = \frac{2\lambda}{a}$$

Since this is inversely proportional to $a$, decreasing $a$ makes the central maximum **wider**, so the diffraction spreading becomes more pronounced. Also, less light passes through the narrower slit, so the peak intensity decreases.

(ii) Monochromatic source replaced by white light: White light contains all colours (wavelengths). Since the diffraction angle depends on λ through **$a\theta = n\lambda$**, each colour is diffracted by a different amount. The central maximum (where all colours overlap at **θ = 0**) stays **white**, but the secondary maxima become **coloured fringes**, with violet (shorter λ) nearer the centre and red (longer λ) farther out in each band.

**COMMON MISTAKE**

*Use **$a\theta = n\lambda$** for minima of diffraction, but **$a\theta = n\lambda$** for maxima in a grating: the same equation means different things, so always state whether it is a single slit or a grating.*

**Final answer.** Diffraction is bending of light around apertures of size $\lambda$; its pattern has a dominant central peak (unlike equal interference fringes), and narrowing the slit widens the central maximum while white light makes the central band white with coloured side fringes.

← *Back to Q224*

## Q225

***(a)*** *How is a wavefront different from a ray? Draw the geometrical shape of the wavefronts when* ***(i)*** *light diverges from a point source, and* ***(ii)*** *light emerges out of a convex lens when a point source is placed at its focus.*
***(b)*** *State Huygens' principle. With the help of a suitable diagram, prove Snell's law of refraction using Huygens' principle.*

**[2005]**

**SOLUTION**

(a) Wavefront versus ray:

A **wavefront** is the locus of all the points of a medium that are vibrating in the same phase at a given instant. A **ray** is a line drawn perpendicular to the wavefront in the direction in which the wave travels; it shows the path along which energy flows. Thus a ray is always normal to the wavefront.

Shapes of wavefronts:

(i) Light diverging from a point source: The wavefronts are **spherical**, with the point source at the centre, expanding outward like growing spheres. (At very large distances a small portion looks like a plane wavefront.)

(ii) Light from a point source at the focus of a convex lens: The diverging spherical wavefront from the focus is refracted by the lens and emerges as a **plane wavefront** perpendicular to the principal axis, so the emergent rays are parallel to the axis.

(b) Huygens' principle: Every point on a given wavefront acts as a fresh source of secondary wavelets which spread out in all forward directions with the speed of the wave in the medium. The new position of the wavefront after a time is

given by the forward envelope (tangential surface) of all these secondary wavelets.

Proof of Snell's law: A plane wavefront AB travels in medium 1 (speed $v_1$) and is incident on the plane interface XY separating medium 1 from medium 2 (speed $v_2$). Let the angle of incidence be $i$ and the angle of refraction be $r$, measured between the wavefront and the surface (equal to the angles between the rays and the normal).

While the secondary wavelet from $B$ travels to $C$ on the interface in medium 1, the wavelet from $A$ travels into medium 2 with radius AE. Let the time taken for $B$ to reach $C$ be $t$.

Distance travelled in medium 1: $\mathbf{BC = v_1\, t}$

Distance travelled in medium 2: $\mathbf{AE = v_2\, t}$

In right triangle ABC (right-angled at $B$), the angle $\mathbf{BAC = i}$, so:

$$\mathrm{BC} = \mathrm{AC}\sin i \Rightarrow v_1\, t = \mathrm{AC}\sin i$$

In right triangle AEC (right-angled at $E$), the angle $\mathbf{ACE = r}$, so:

$$\mathrm{AE} = \mathrm{AC}\sin r \Rightarrow v_2\, t = \mathrm{AC}\sin r$$

Dividing the two relations:

$$\frac{v_1\, t}{v_2\, t} = \frac{\mathrm{AC}\sin i}{\mathrm{AC}\sin r}$$

$$\Rightarrow \frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \text{constant} = {}_1\mu_2$$

This is **Snell's law of refraction**. Since the refractive index $\mu = \frac{c}{v}$, we also get $\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$, so $n_1 \sin i = n_2 \sin r$.

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**Final answer.** A ray is normal to the wavefront; spherical wavefronts come from a point source and plane wavefronts emerge when it is at a lens focus, and Huygens' construction gives $\frac{\sin i}{\sin r} = \frac{v_1}{v_2} =$ constant, which is Snell's law.

← *Back to Q225*

## Q226

***(a)*** *In Young's double slit experiment, deduce the conditions for* ***(i)*** *constructive, and* ***(ii)*** *destructive interference at a point on the screen. Draw a graph showing variation of the resultant intensity in the interference pattern against position x on the screen.*
***(b)*** *Compare and contrast the pattern which is seen with two coherently illuminated narrow slits in Young's experiment with that seen for a coherently illuminated single slit producing diffraction.*

**[2005]**

**SOLUTION**

(a) Conditions for interference in Young's double-slit experiment:

Two narrow slits $S_1$ and $S_2$ separated by a small distance $d$ are illuminated by monochromatic light. They act as coherent sources and their light overlaps on a screen placed at distance $D$ ($D \gg d$). Consider a point $P$ on the screen at distance $x$ from the central point $O$.

The path difference between the waves reaching $P$ from the two slits is:

$$\Delta = S_2P - S_1P \approx \frac{x\,d}{D}$$

(i) Constructive interference (bright fringes): occurs when the path difference is a whole number of wavelengths.

$$\Delta = n\lambda,\ n = 0, \pm 1, \pm 2, \ldots$$

$$\Rightarrow \frac{x\,d}{D} = n\lambda$$

$$\Rightarrow x_n = \frac{n\lambda\,D}{d}$$

(ii) Destructive interference (dark fringes): occurs when the path difference is an odd number of half-wavelengths.

$$\Delta = (2n-1)\frac{\lambda}{2},\ n = 1, 2, 3, \ldots$$

$$\Rightarrow \frac{x\,d}{D} = (2n-1)\frac{\lambda}{2}$$

$$\Rightarrow x_n = \frac{(2n-1)\lambda D}{2d}$$

The fringe width (spacing between consecutive bright or dark fringes) is:

$$\beta = \frac{\lambda D}{d}$$

Graph of intensity versus position $x$: Plot intensity $I$ on the vertical axis against position $x$ on the horizontal axis. The graph is a smooth $\cos^2$ curve, a series of equally spaced peaks of equal height (each equal to $4I_0$, where $I_0$ is the intensity due to one slit) separated by zeros, the maxima lying at $\boldsymbol{x = 0, \beta, \pm 2\beta, \ldots}$

(b) Comparison of two-slit interference with single-slit diffraction:

1. Fringe width: In two-slit interference all bright fringes are of equal width, while in single-slit diffraction the central maximum is about twice as wide as the secondary maxima.

2. Intensity: In interference all bright fringes have nearly equal intensity, while in diffraction the intensity of the secondary maxima decreases rapidly away from the centre.

3. Origin: Interference results from the superposition of waves from two separate coherent sources, whereas diffraction results from the superposition of secondary wavelets from different parts of the same wavefront.

4. Condition for dark bands: For interference, minima occur at $\boldsymbol{\Delta = (2n-1)\frac{\lambda}{2}}$; for single-slit diffraction, minima occur at $\boldsymbol{a\theta = n\lambda}$.

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**Final answer.** Bright fringes occur at $\boldsymbol{x_n = \frac{n\lambda D}{d}}$ and dark fringes at $\boldsymbol{x_n = \frac{(2n-1)\lambda D}{2d}}$ with $\boldsymbol{\beta = \frac{\lambda D}{d}}$; the resulting $\boldsymbol{\cos^2}$ interference fringes are equal in width and intensity, unlike the diffraction pattern with its dominant central maximum.

← *Back to Q226*